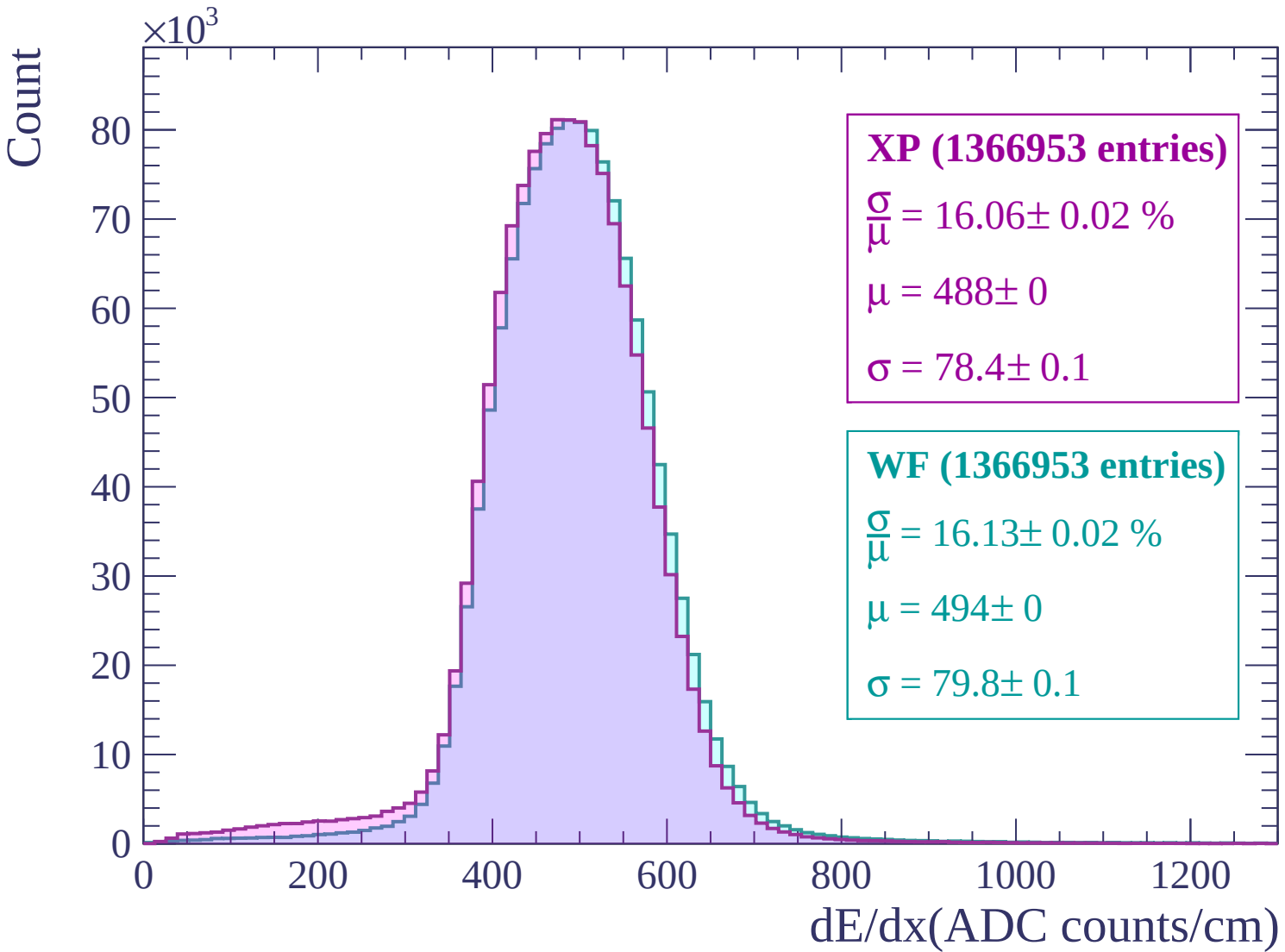
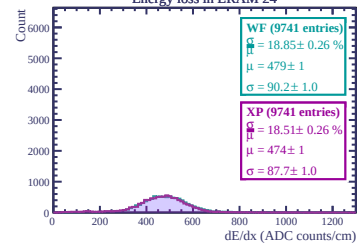
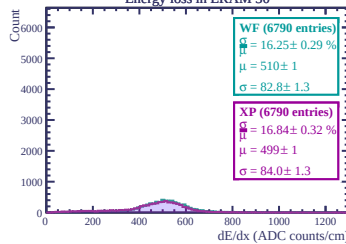




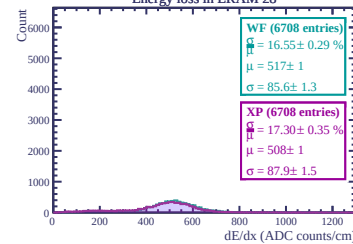
Energy loss in ERAM 14



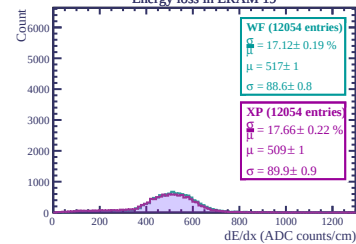
Energy loss in ERAM 30



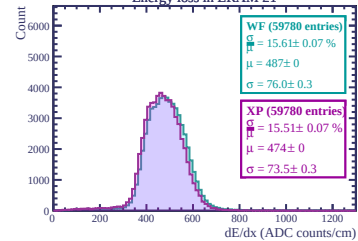
Energy loss in ERAM 33



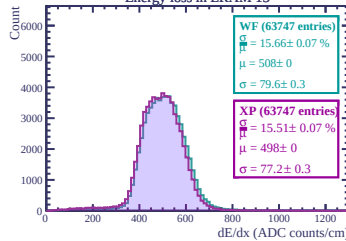
Energy loss in ERAM 36



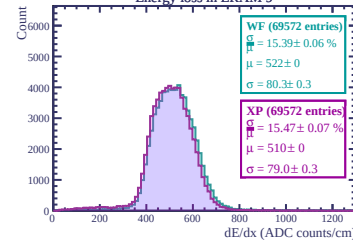
Energy loss in ERAM 21



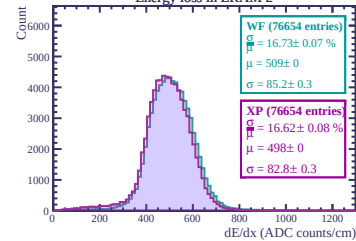
Energy loss in ERAM 13



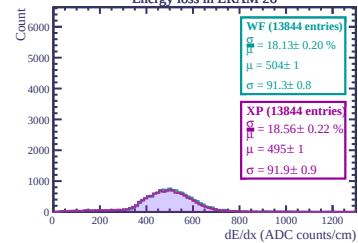
Energy loss in ERAM 9



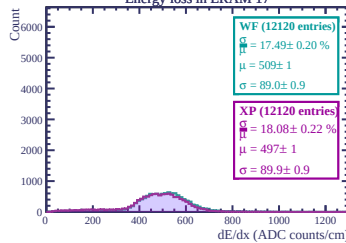
Energy loss in ERAM 2



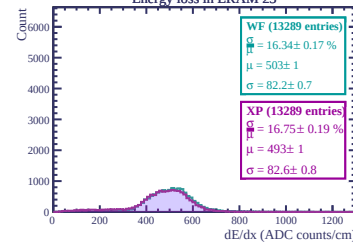
Energy loss in ERAM 26



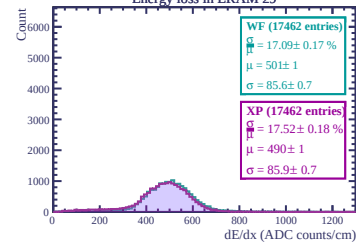
Energy loss in ERAM 17



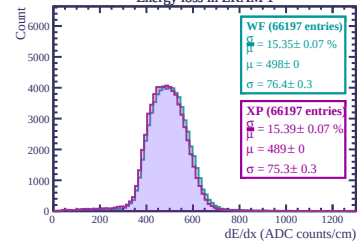
Energy loss in ERAM 23



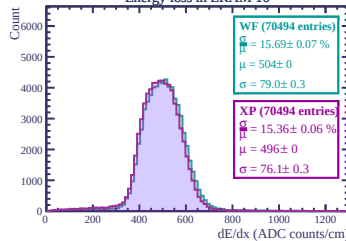
Energy loss in ERAM 29



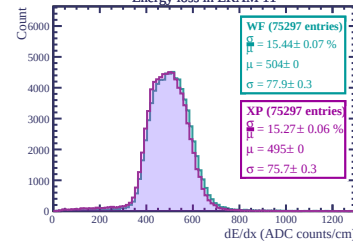
Energy loss in ERAM 1



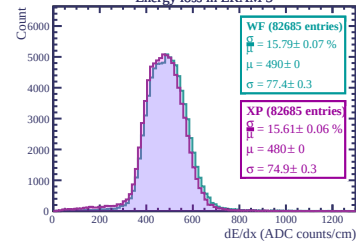
Energy loss in ERAM 10

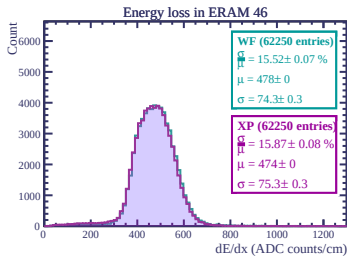
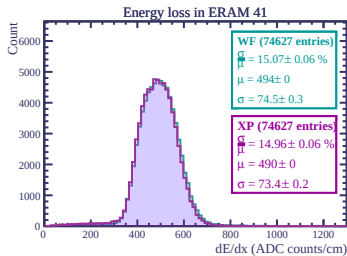
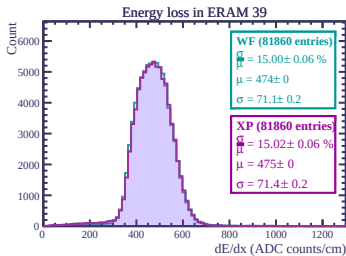
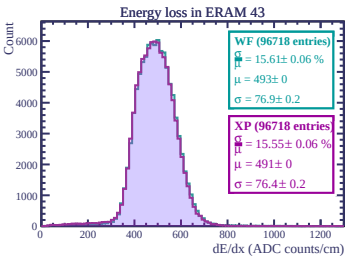
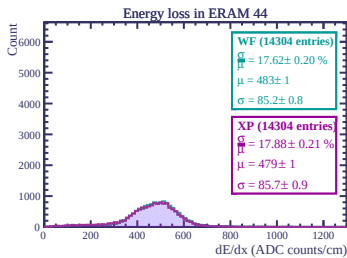
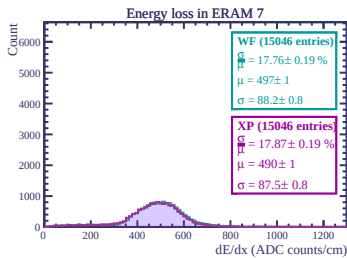
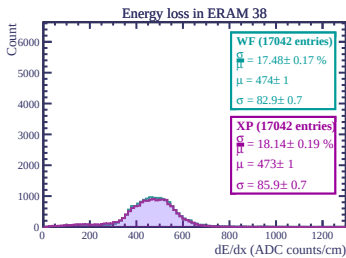
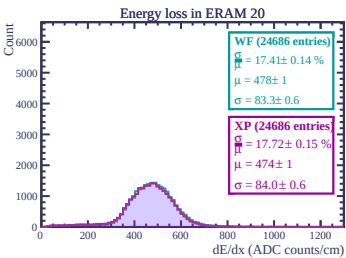
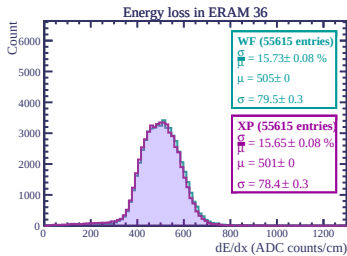
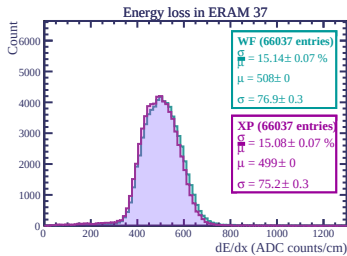
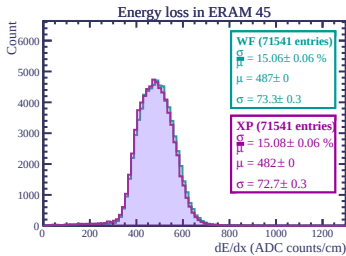
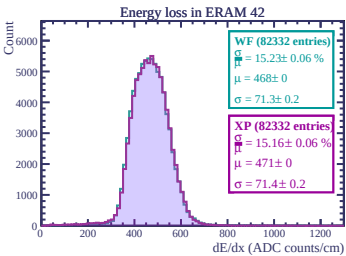
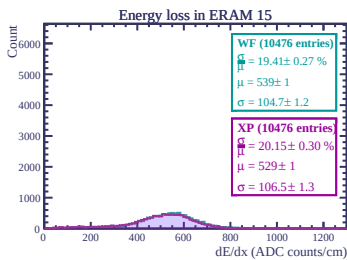
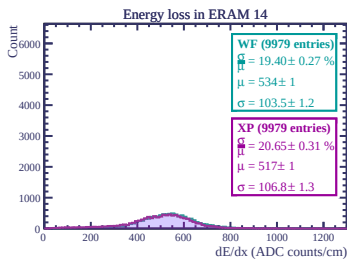
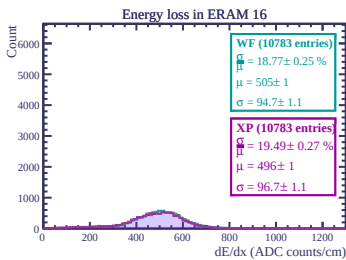
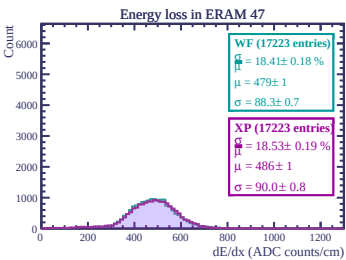


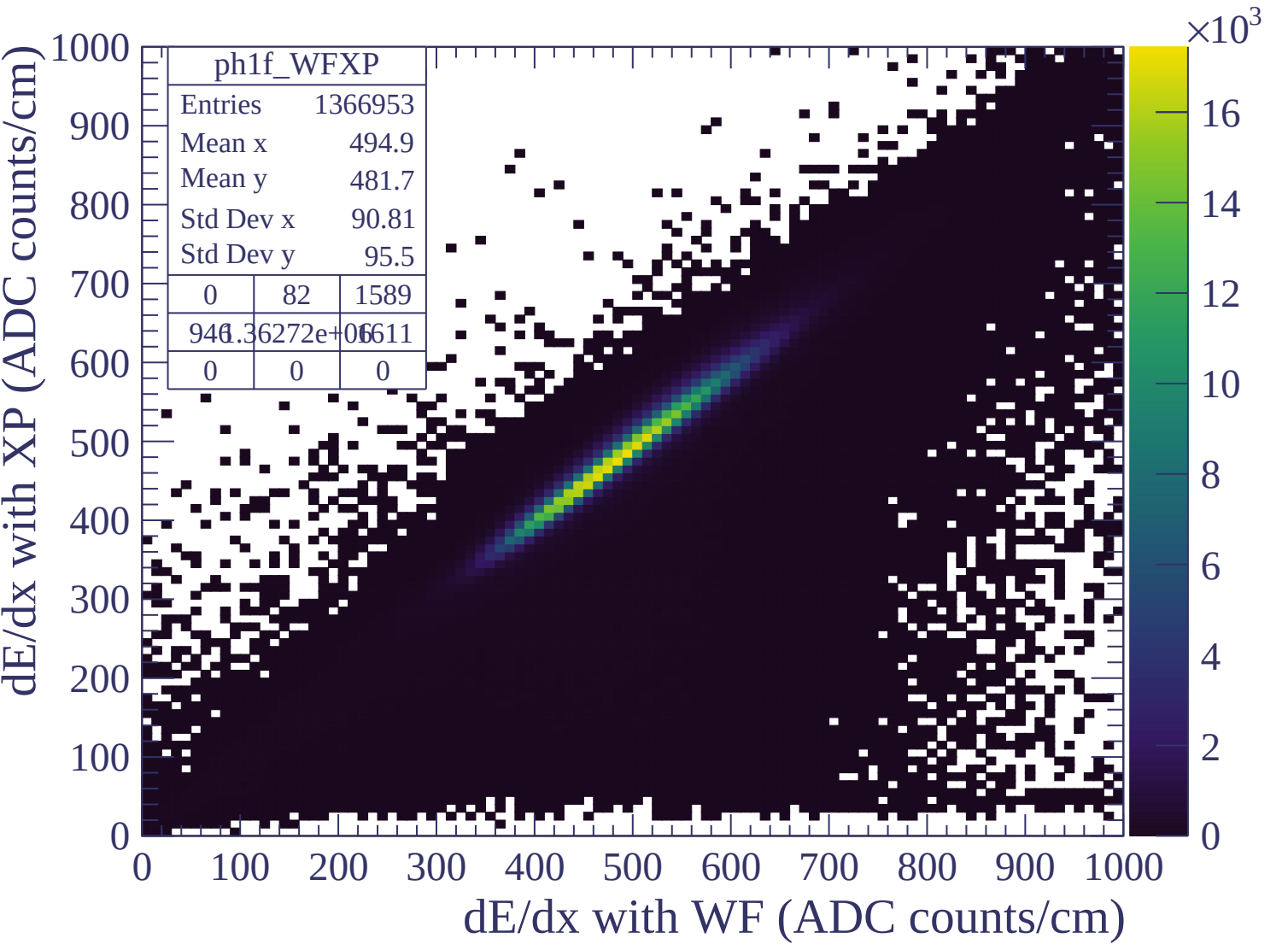
Energy loss in ERAM 11

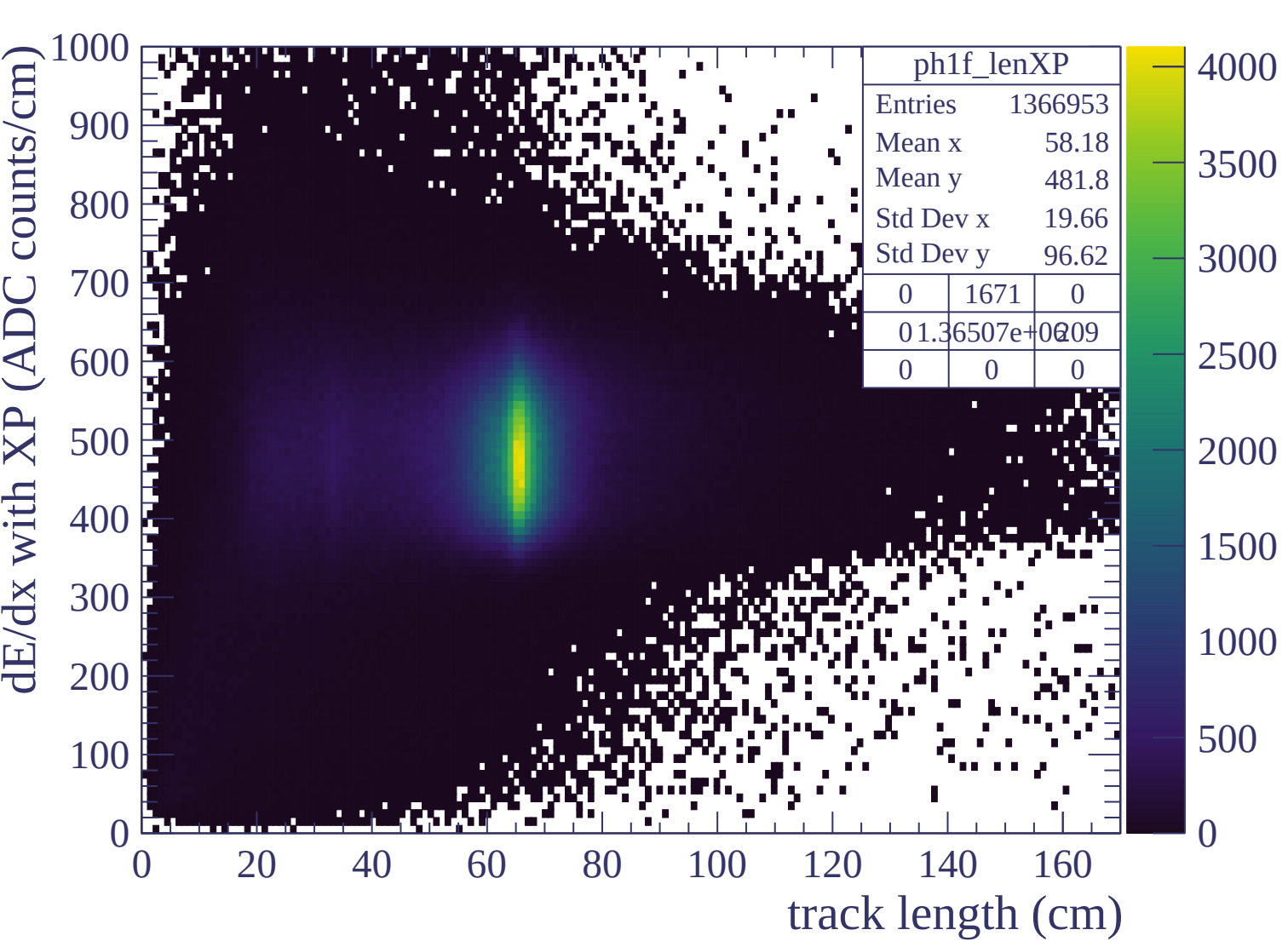


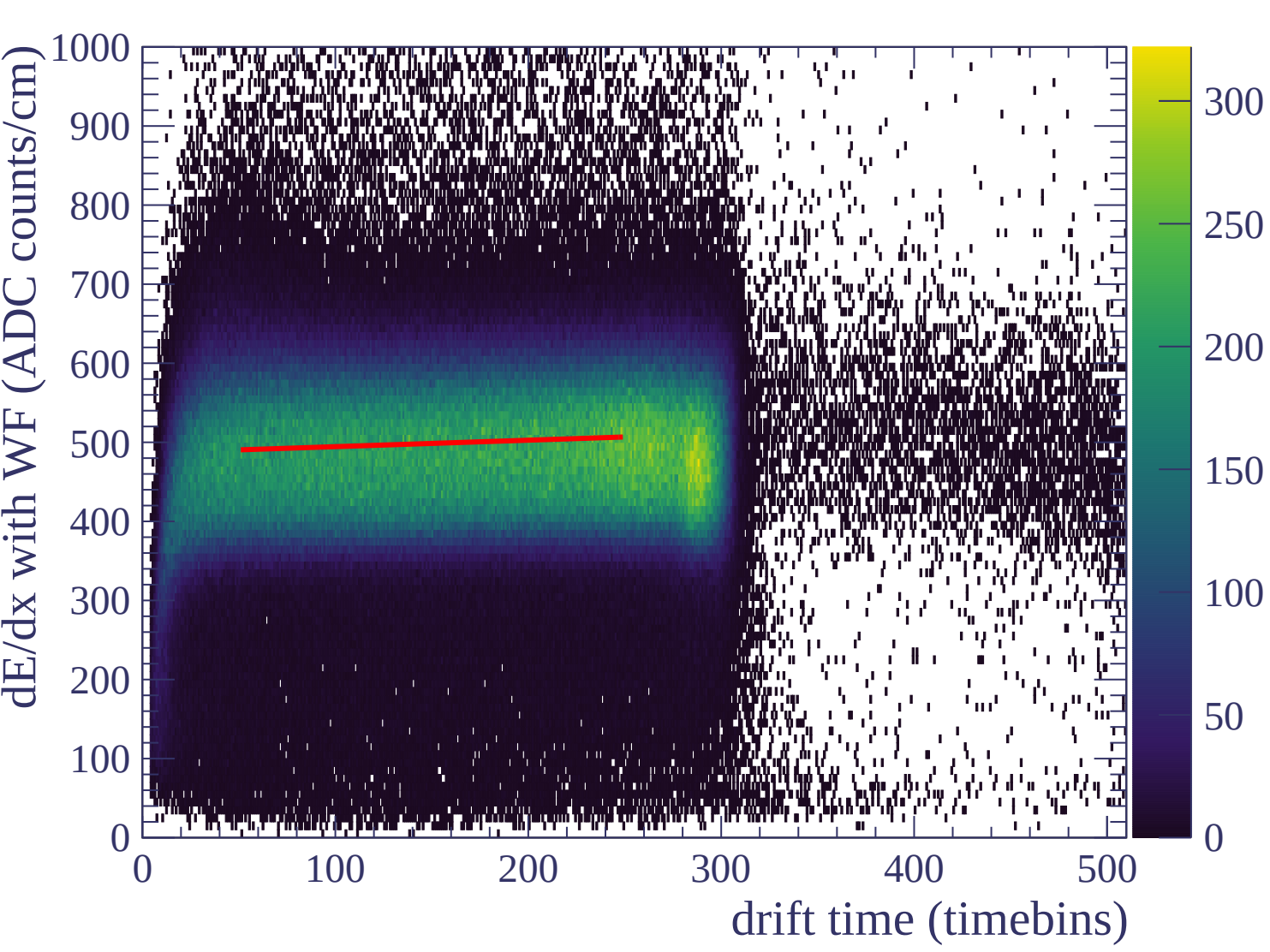
Energy loss in ERAM 3

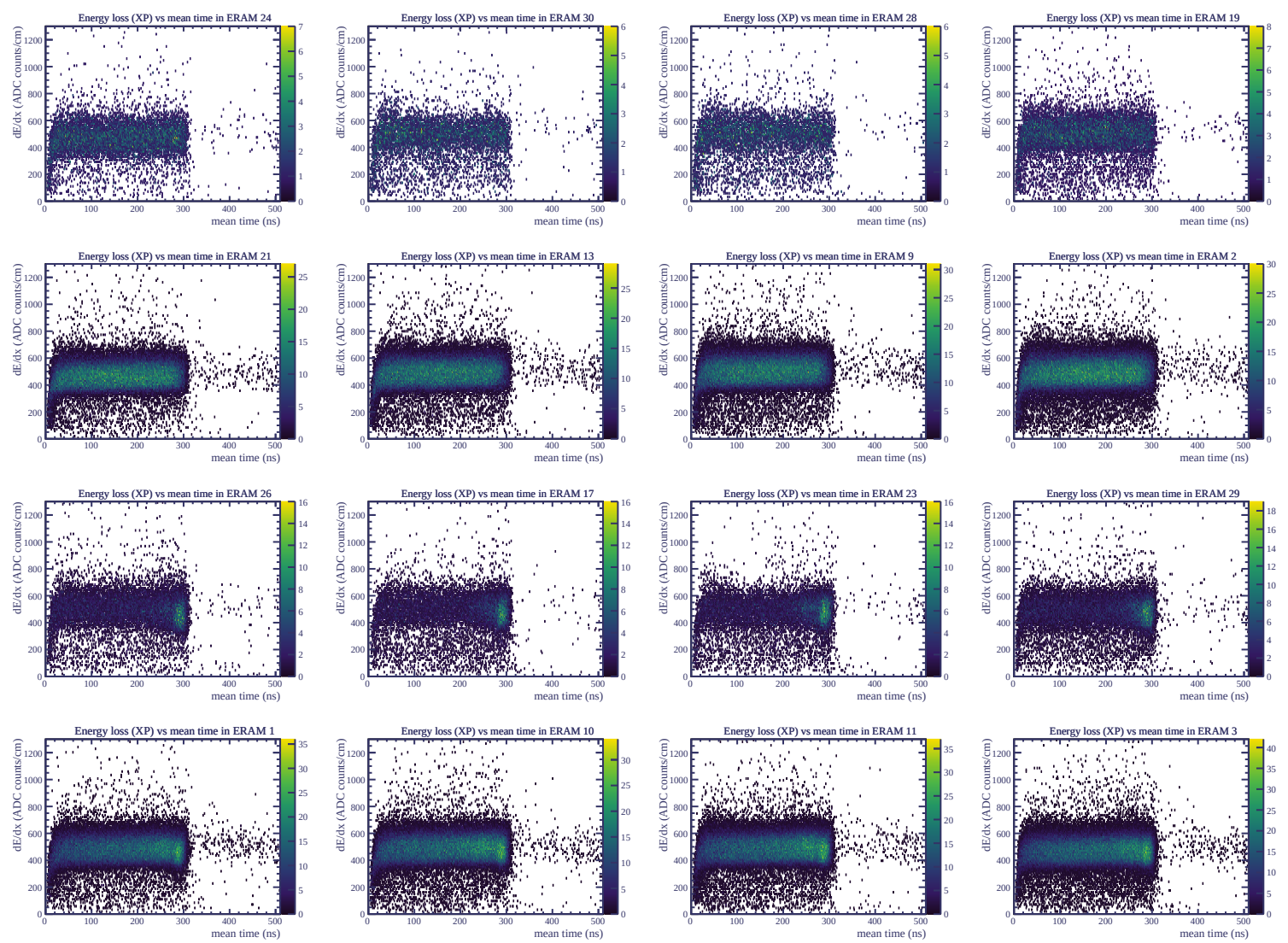


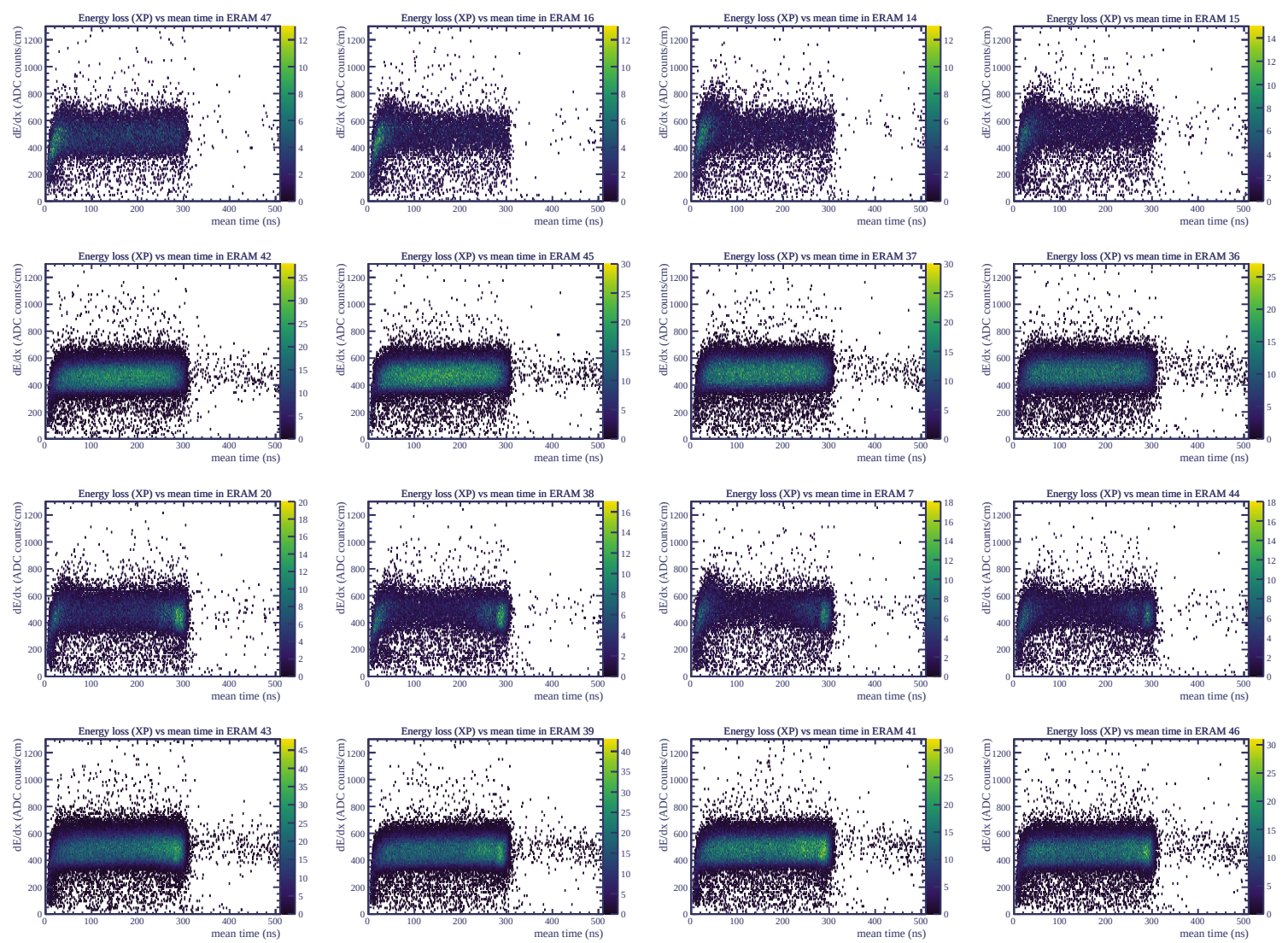




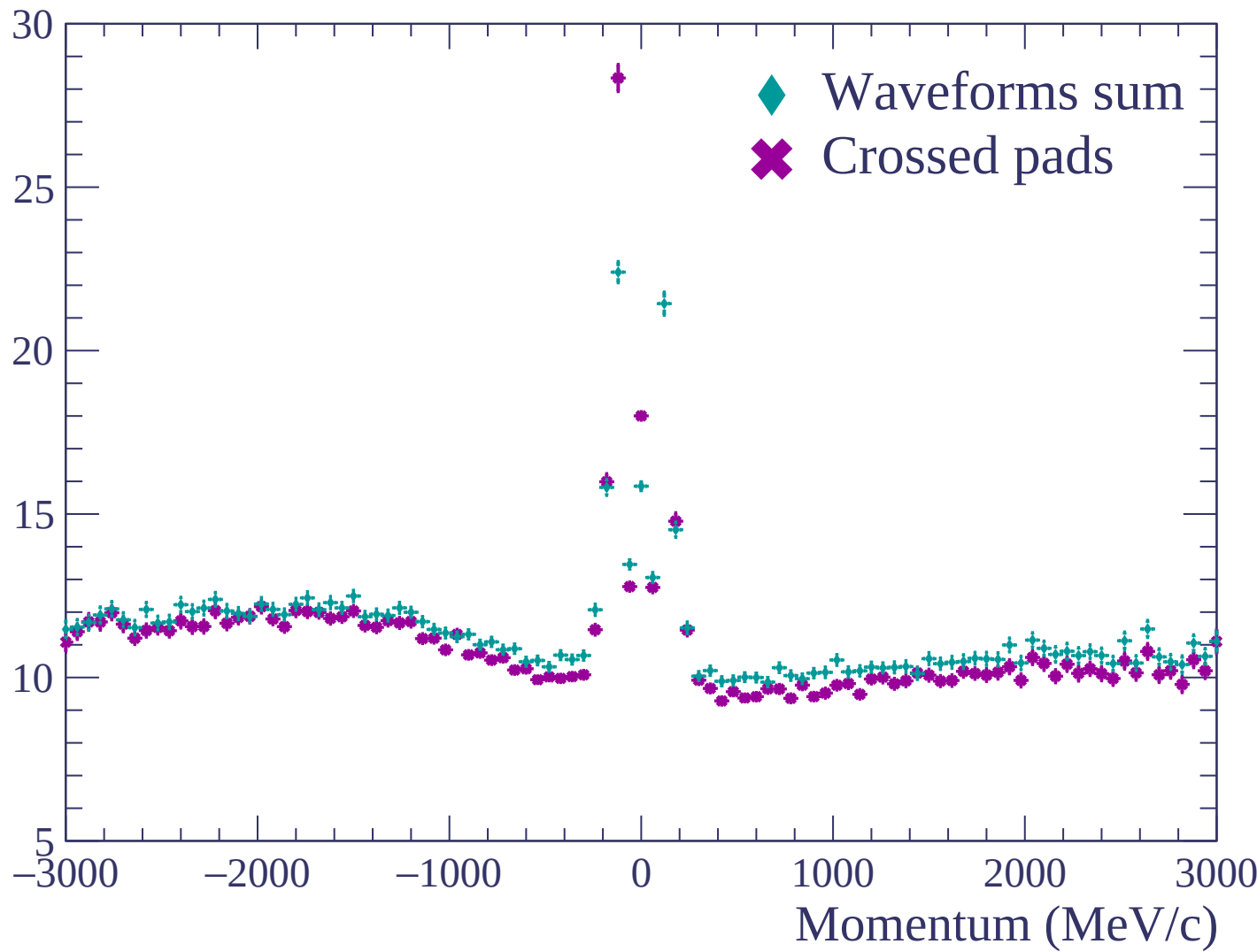


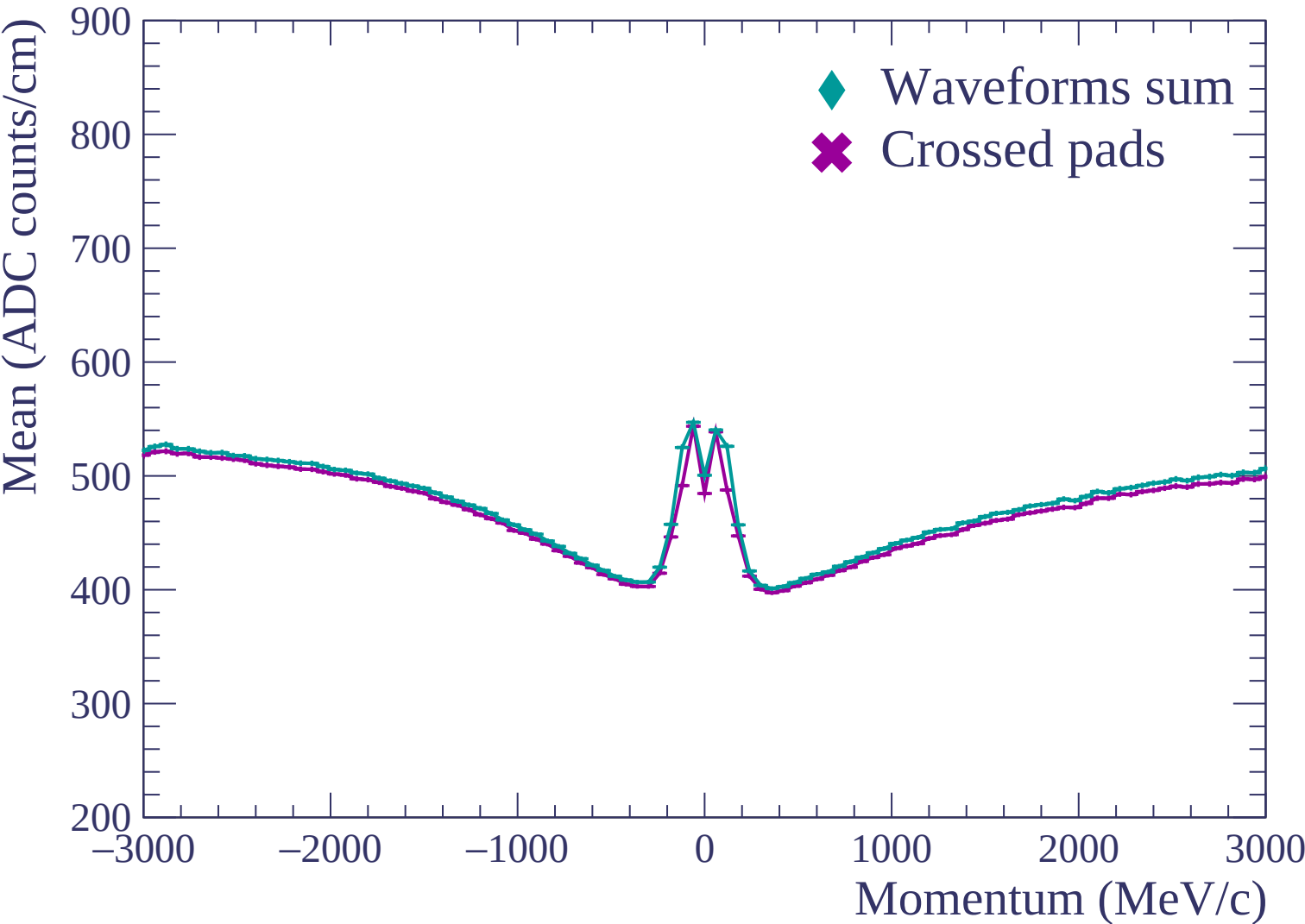


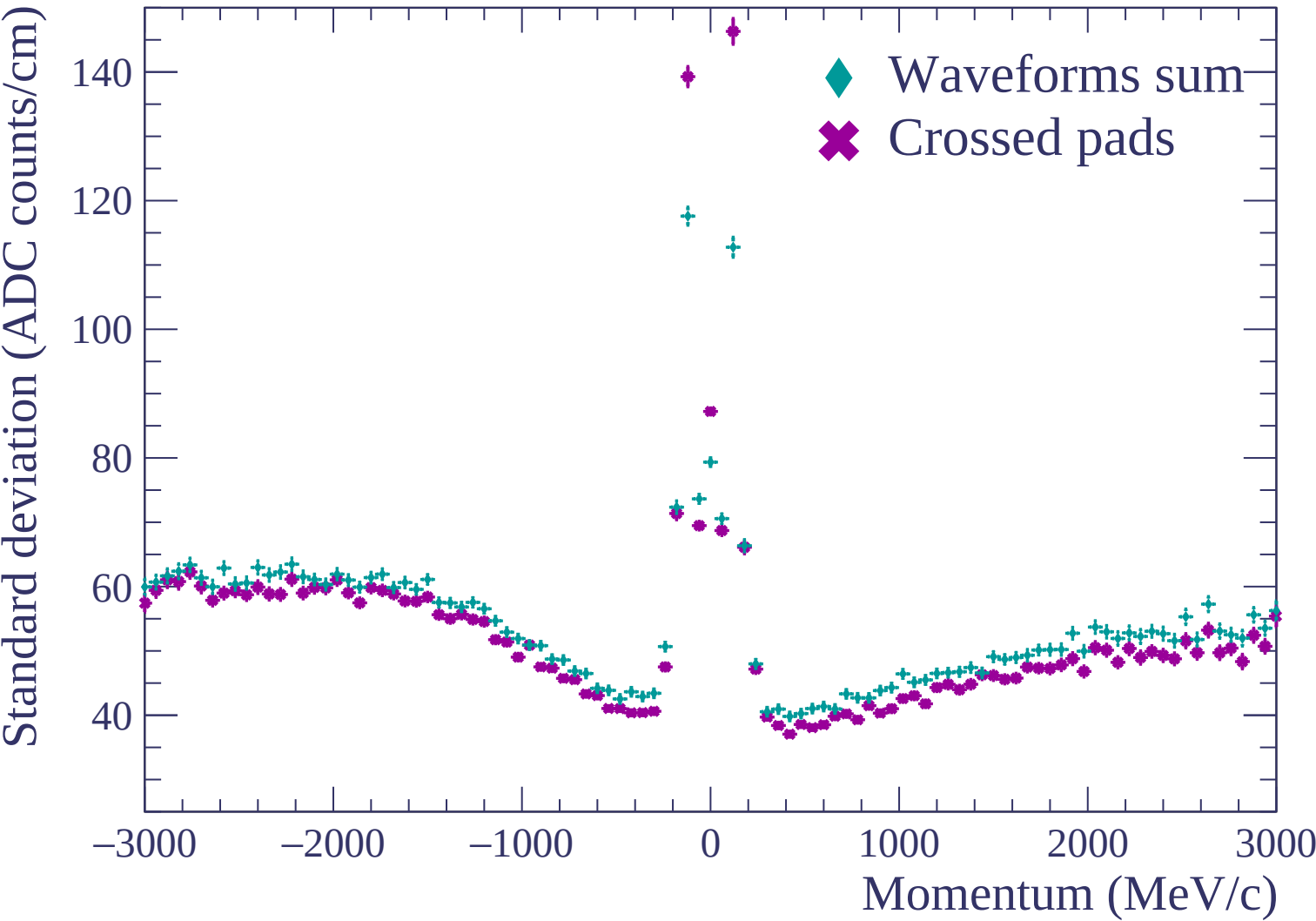


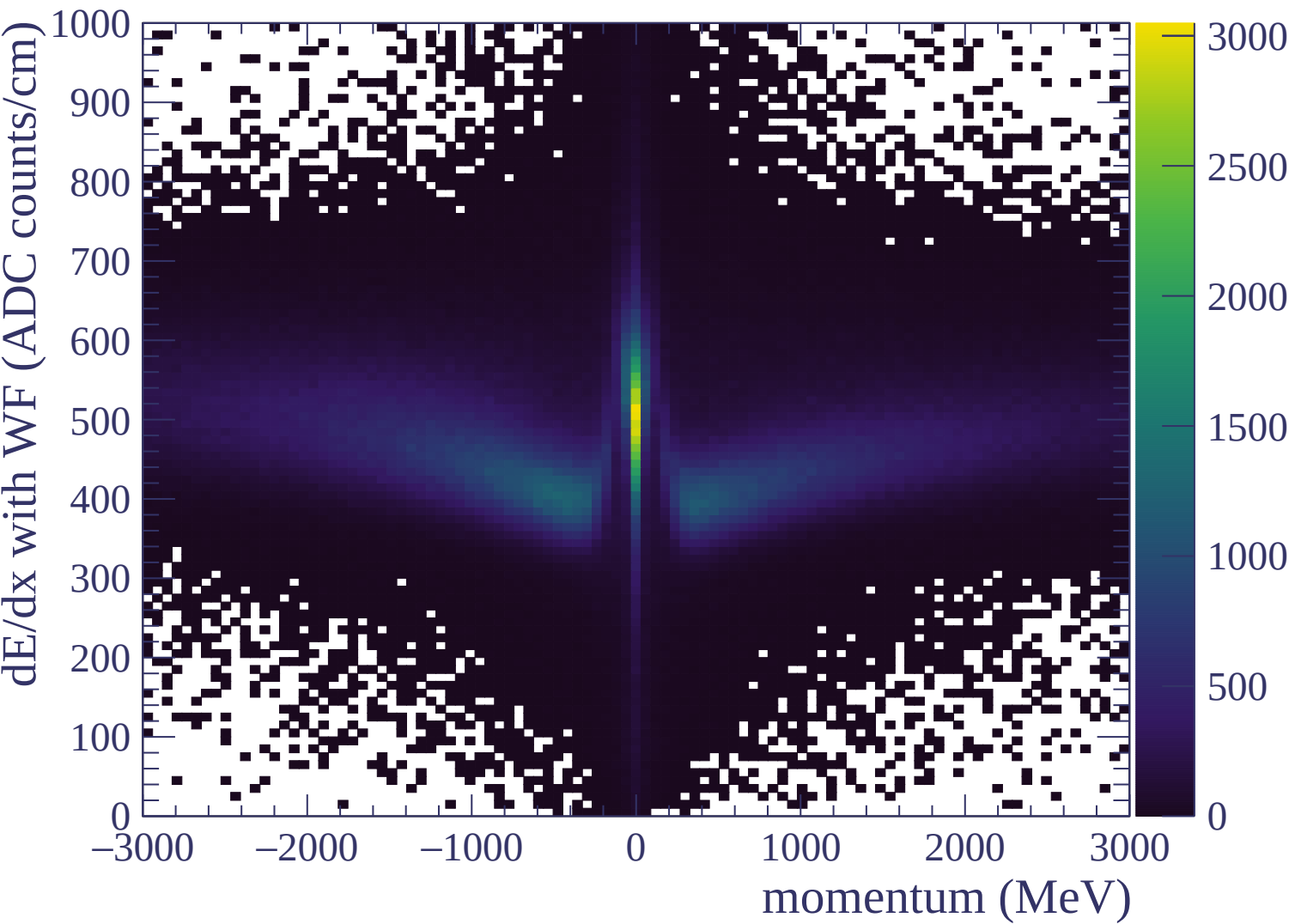


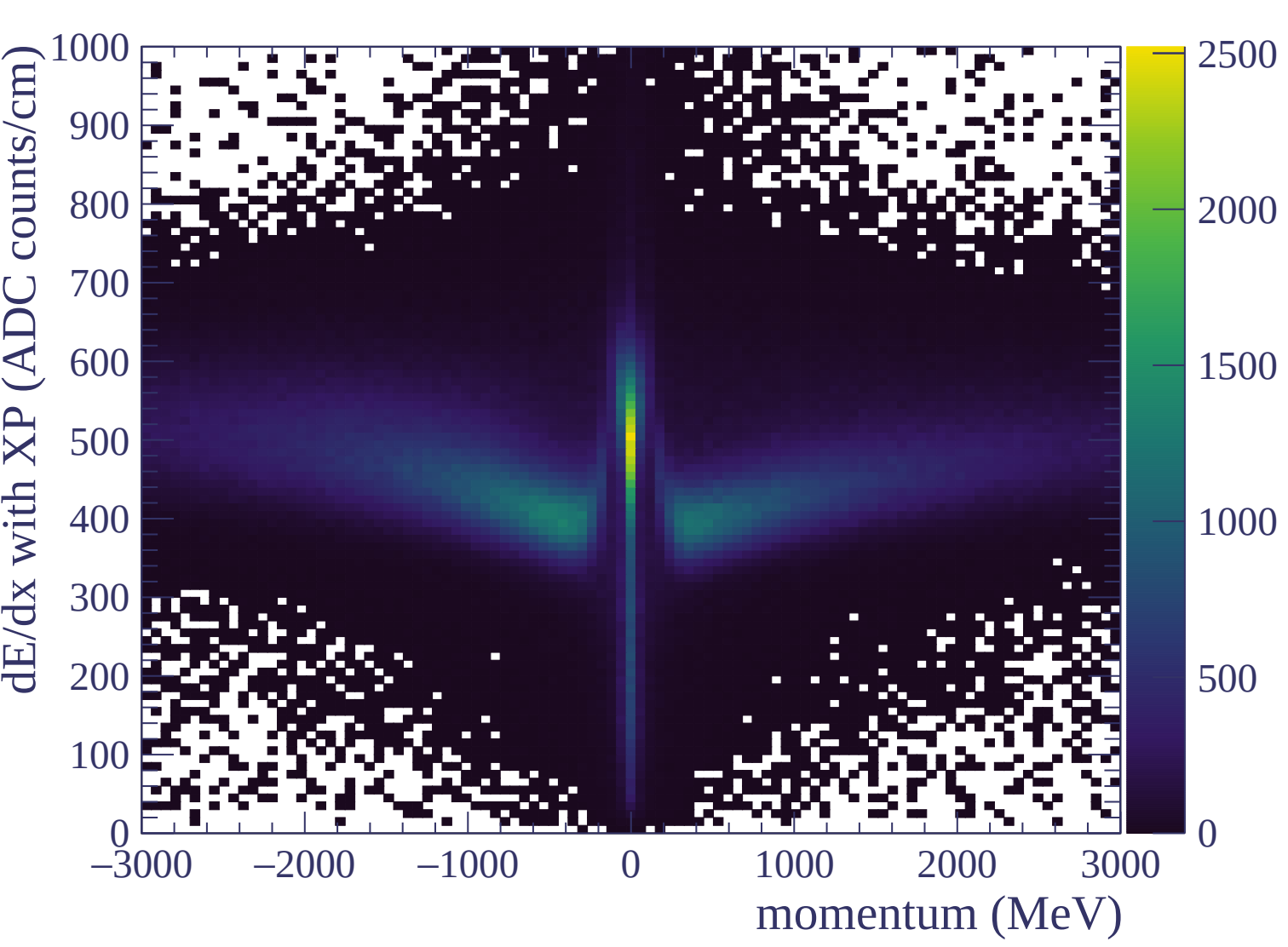
Resolution (%)

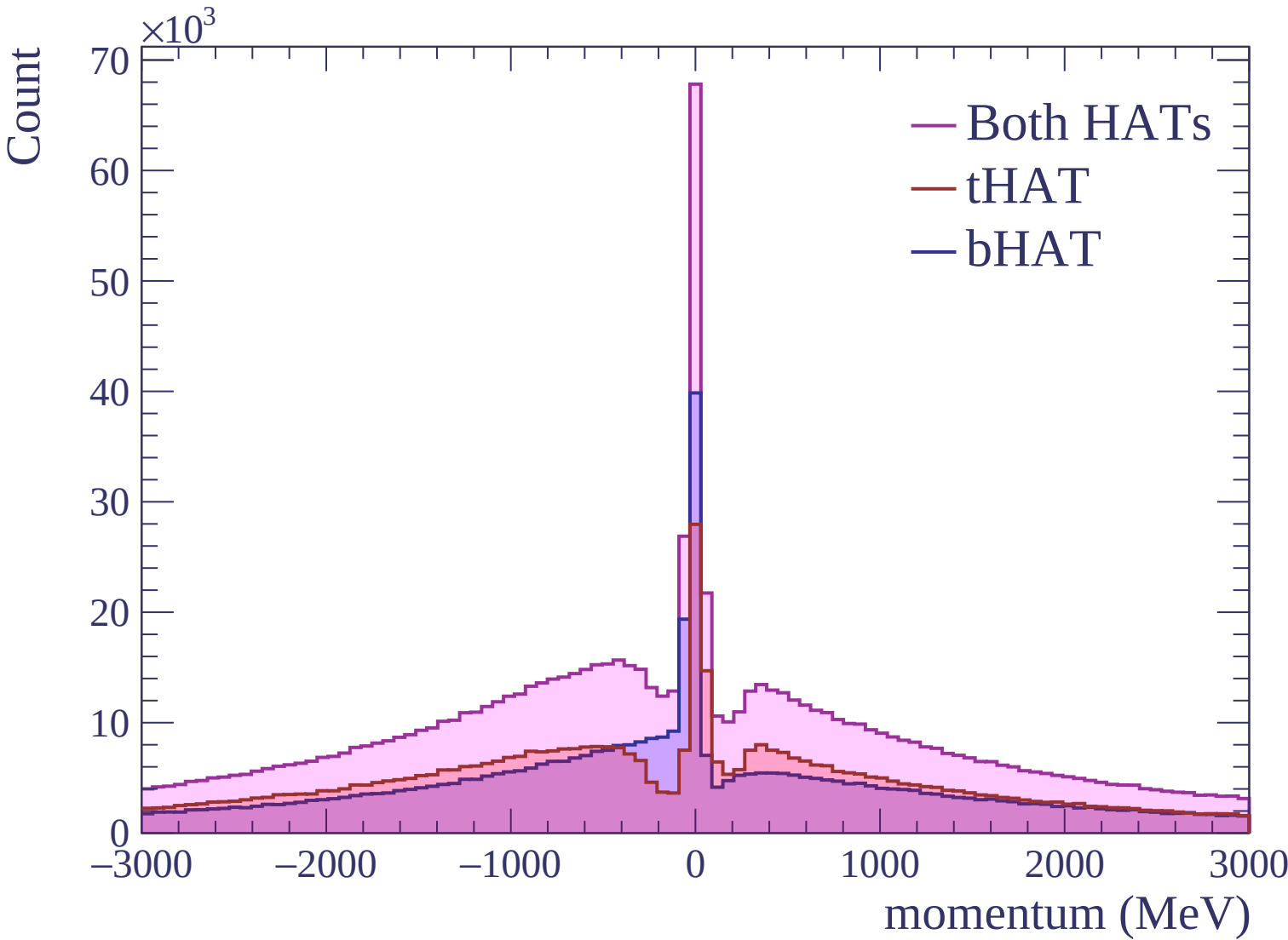






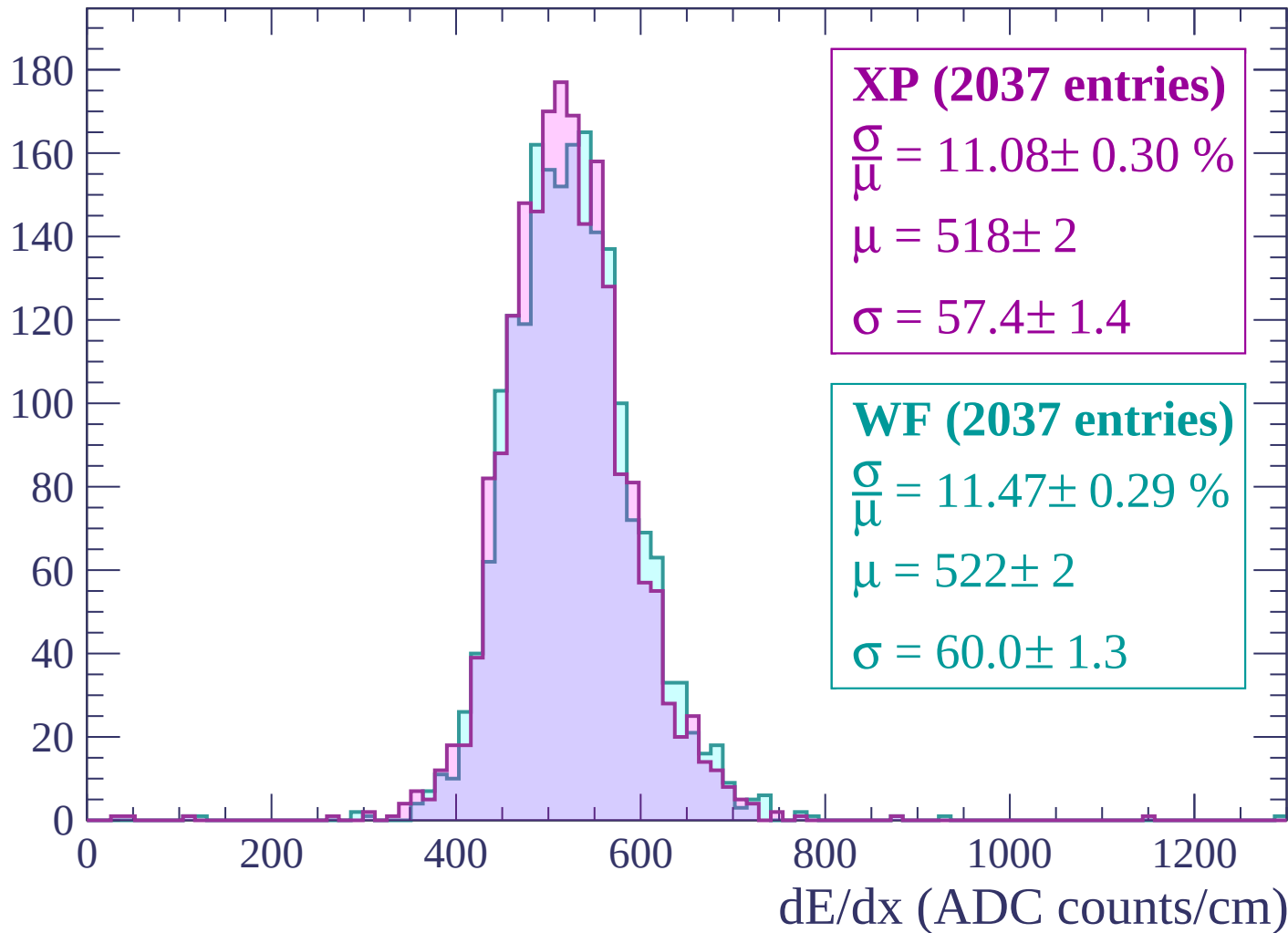




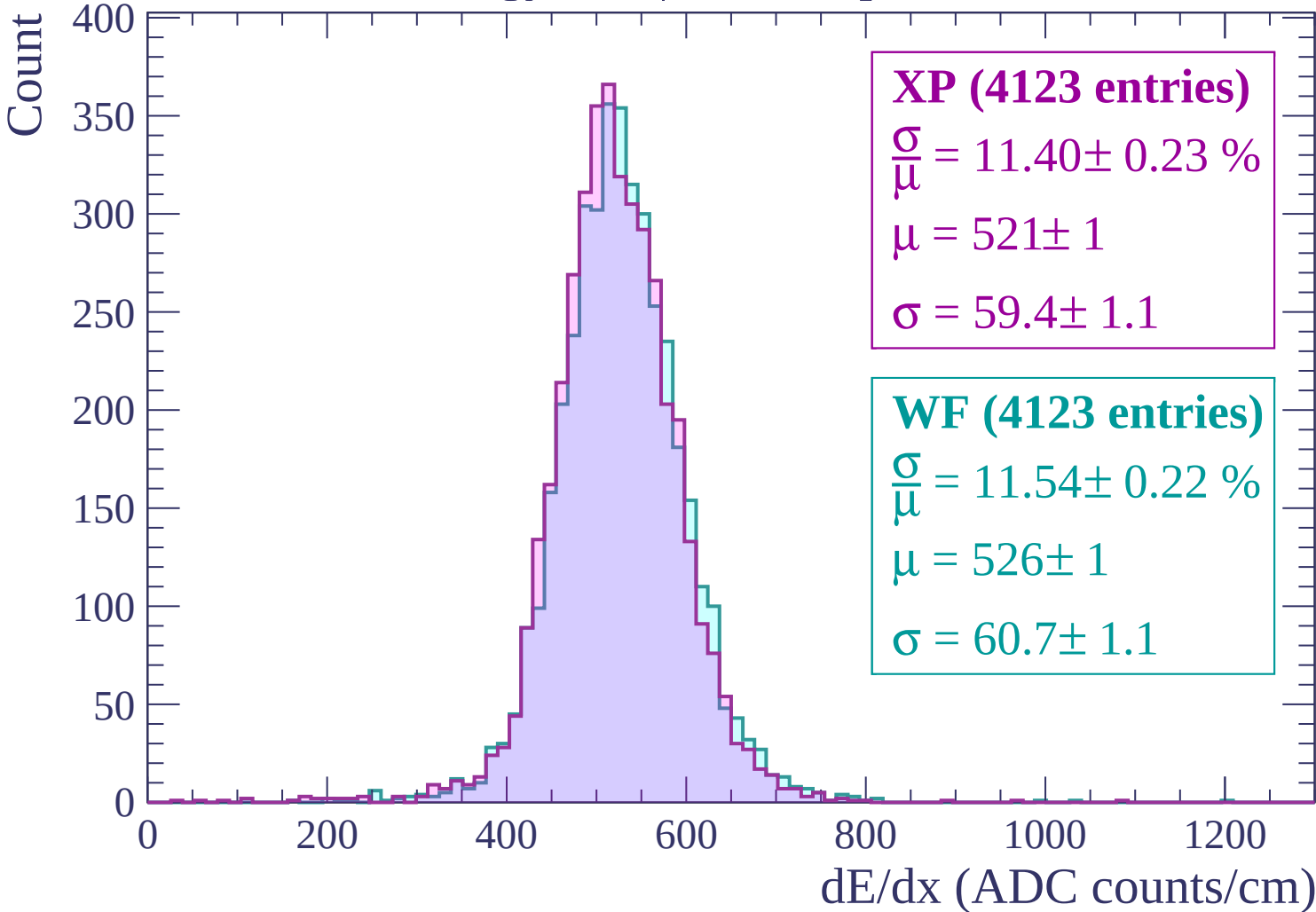


Energy loss | $-3000 < p < -2940$

Count

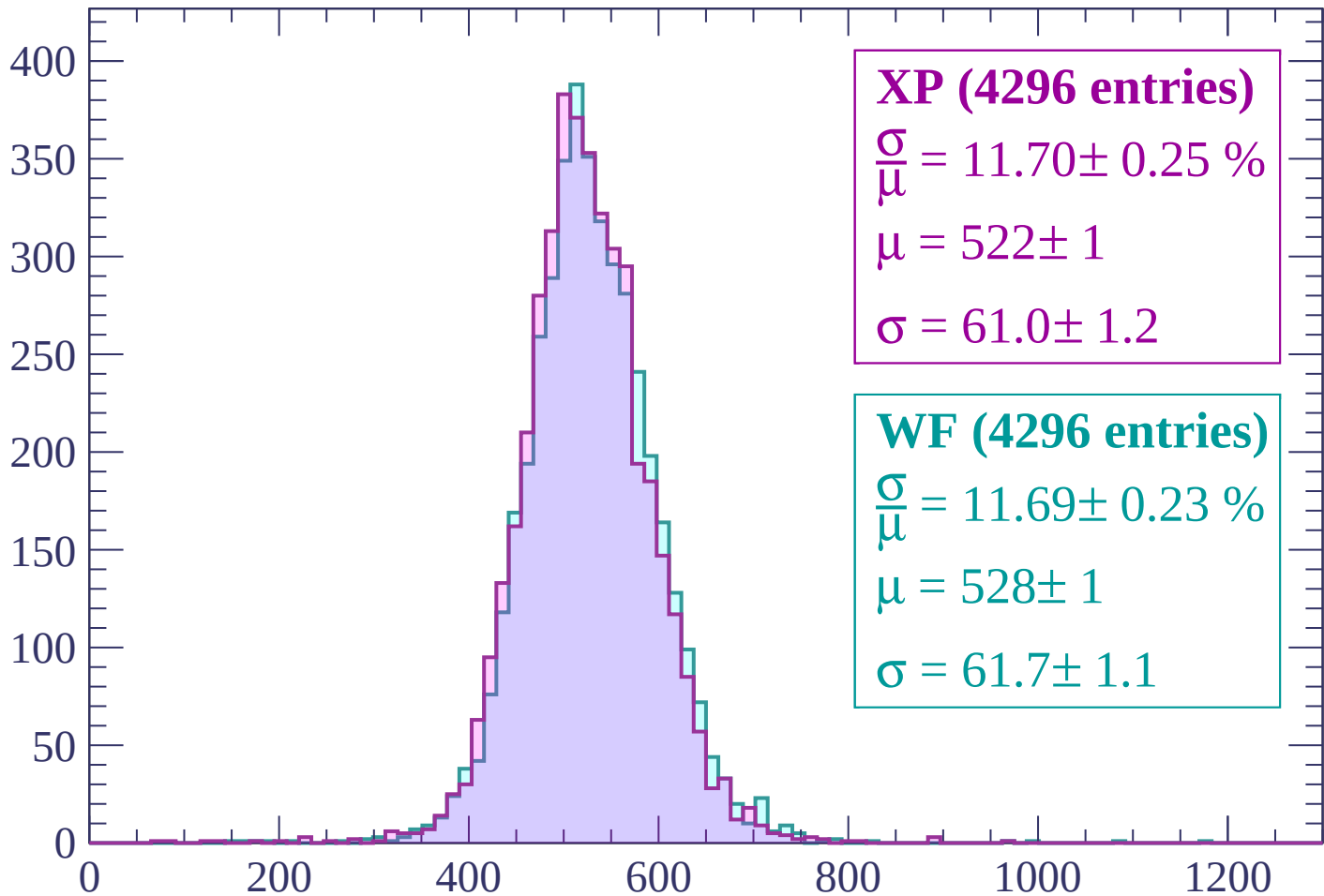


Energy loss | $-2940 < p < -2880$



Energy loss | $-2880 < p < -2820$

Count



XP (4296 entries)

$$\frac{\sigma}{\mu} = 11.70 \pm 0.25 \%$$

$$\mu = 522 \pm 1$$

$$\sigma = 61.0 \pm 1.2$$

WF (4296 entries)

$$\frac{\sigma}{\mu} = 11.69 \pm 0.23 \%$$

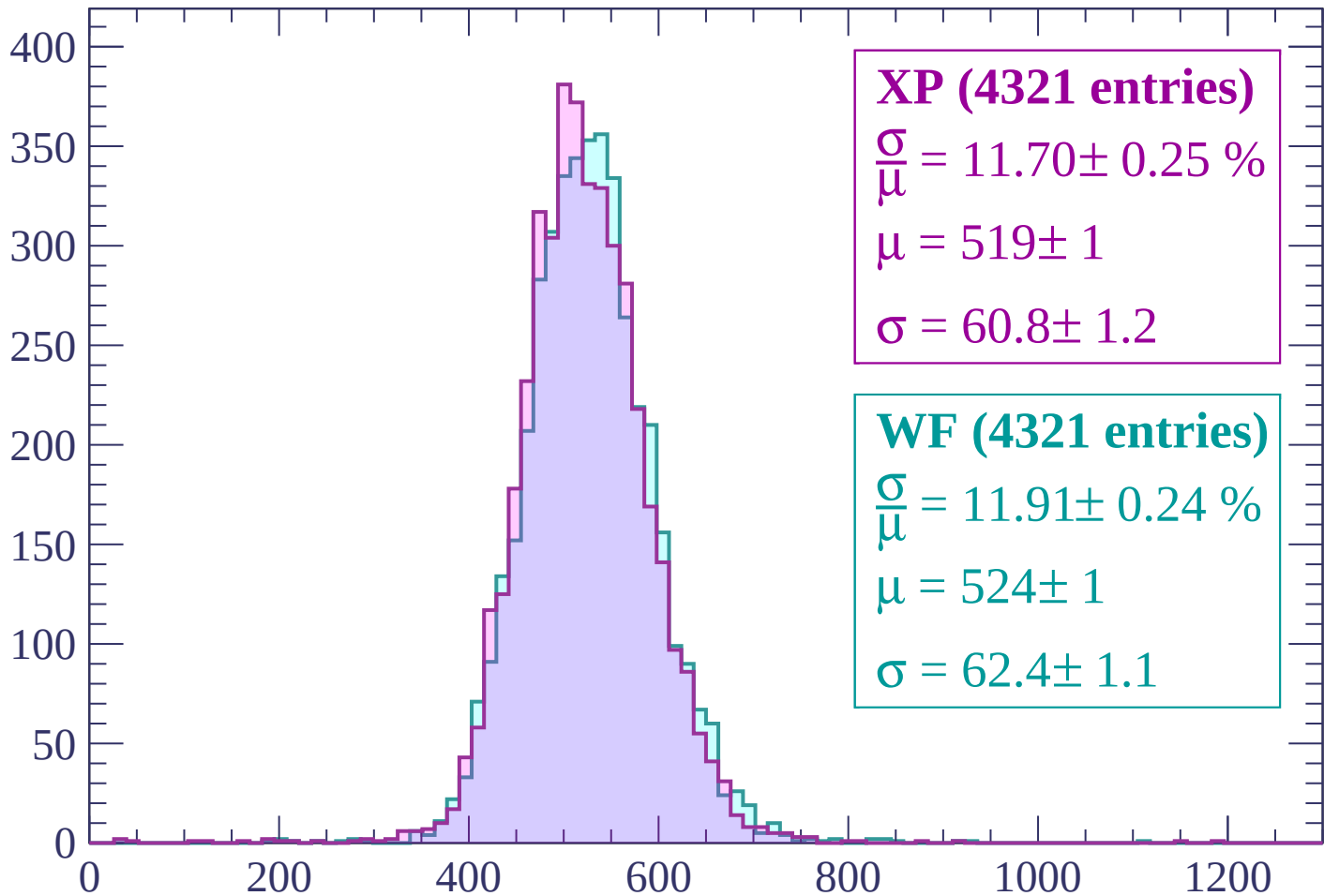
$$\mu = 528 \pm 1$$

$$\sigma = 61.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -2820 < p < -2760

Count



XP (4321 entries)

$$\frac{\sigma}{\mu} = 11.70 \pm 0.25 \%$$

$$\mu = 519 \pm 1$$

$$\sigma = 60.8 \pm 1.2$$

WF (4321 entries)

$$\frac{\sigma}{\mu} = 11.91 \pm 0.24 \%$$

$$\mu = 524 \pm 1$$

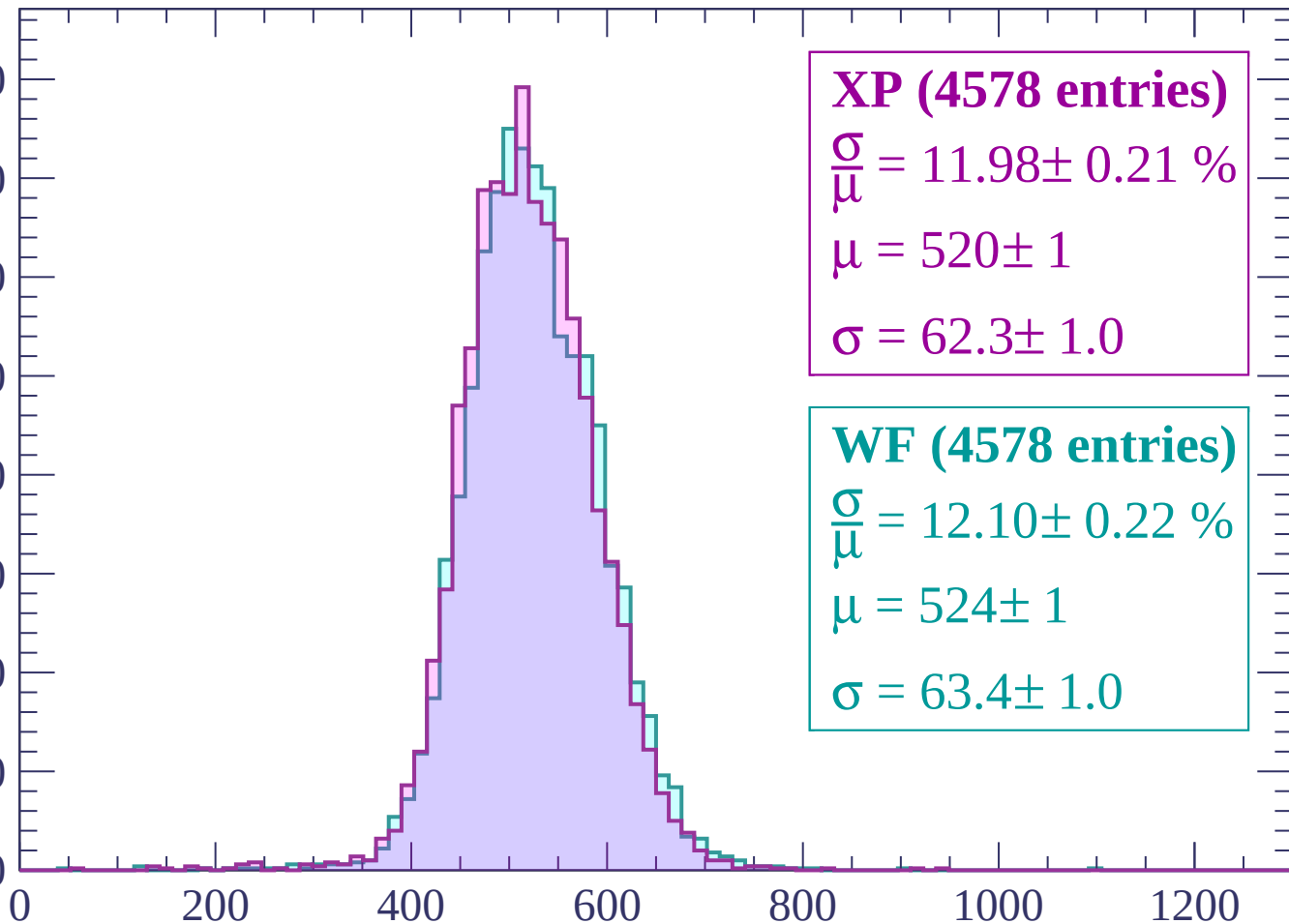
$$\sigma = 62.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -2760 < p < -2700

Count

400
350
300
250
200
150
100
50
0



XP (4578 entries)

$\frac{\sigma}{\mu} = 11.98 \pm 0.21 \%$

$\mu = 520 \pm 1$

$\sigma = 62.3 \pm 1.0$

WF (4578 entries)

$\frac{\sigma}{\mu} = 12.10 \pm 0.22 \%$

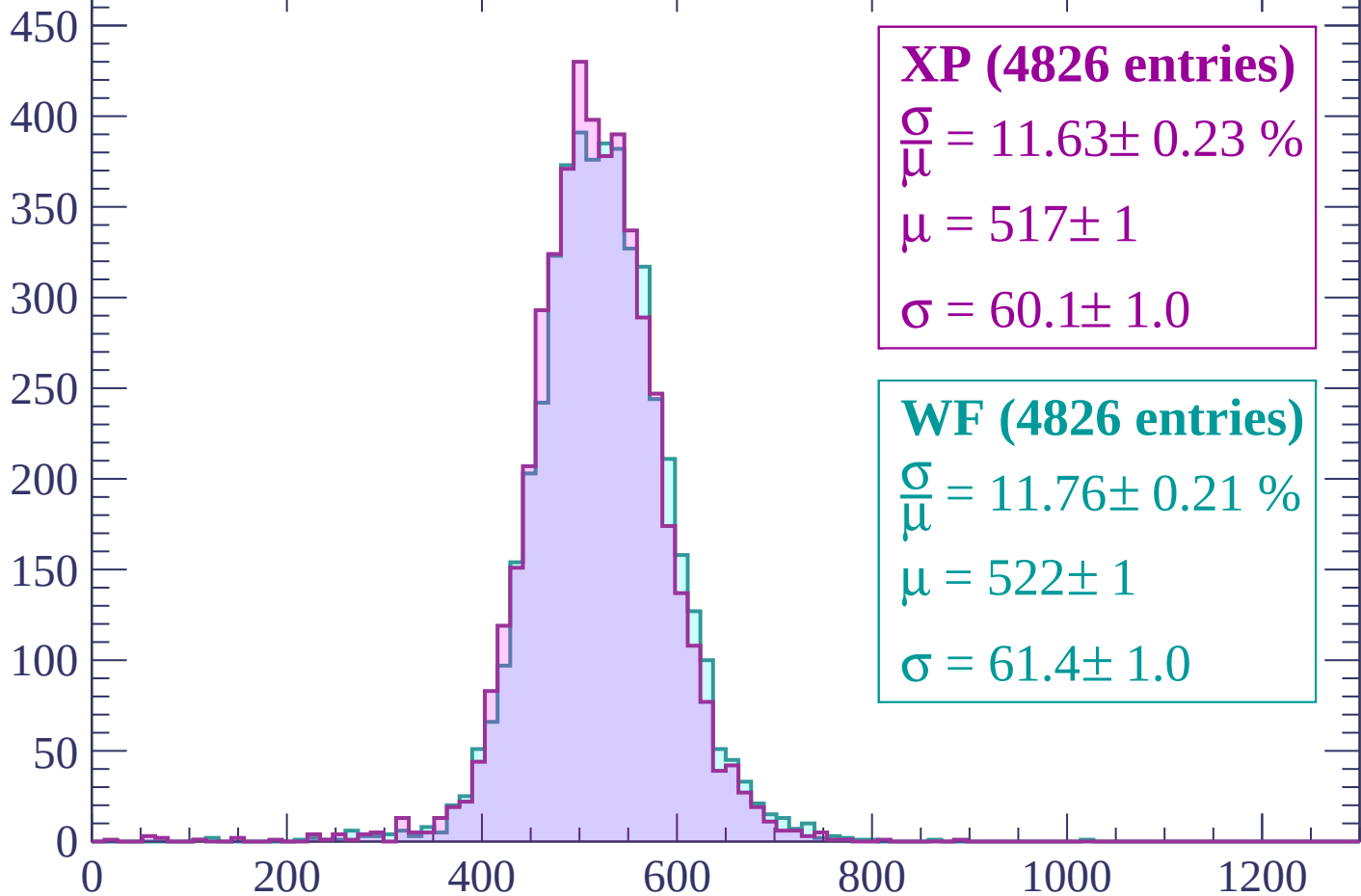
$\mu = 524 \pm 1$

$\sigma = 63.4 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -2700 < p < -2640

Count



XP (4826 entries)

$\frac{\sigma}{\mu} = 11.63 \pm 0.23 \%$

$\mu = 517 \pm 1$

$\sigma = 60.1 \pm 1.0$

WF (4826 entries)

$\frac{\sigma}{\mu} = 11.76 \pm 0.21 \%$

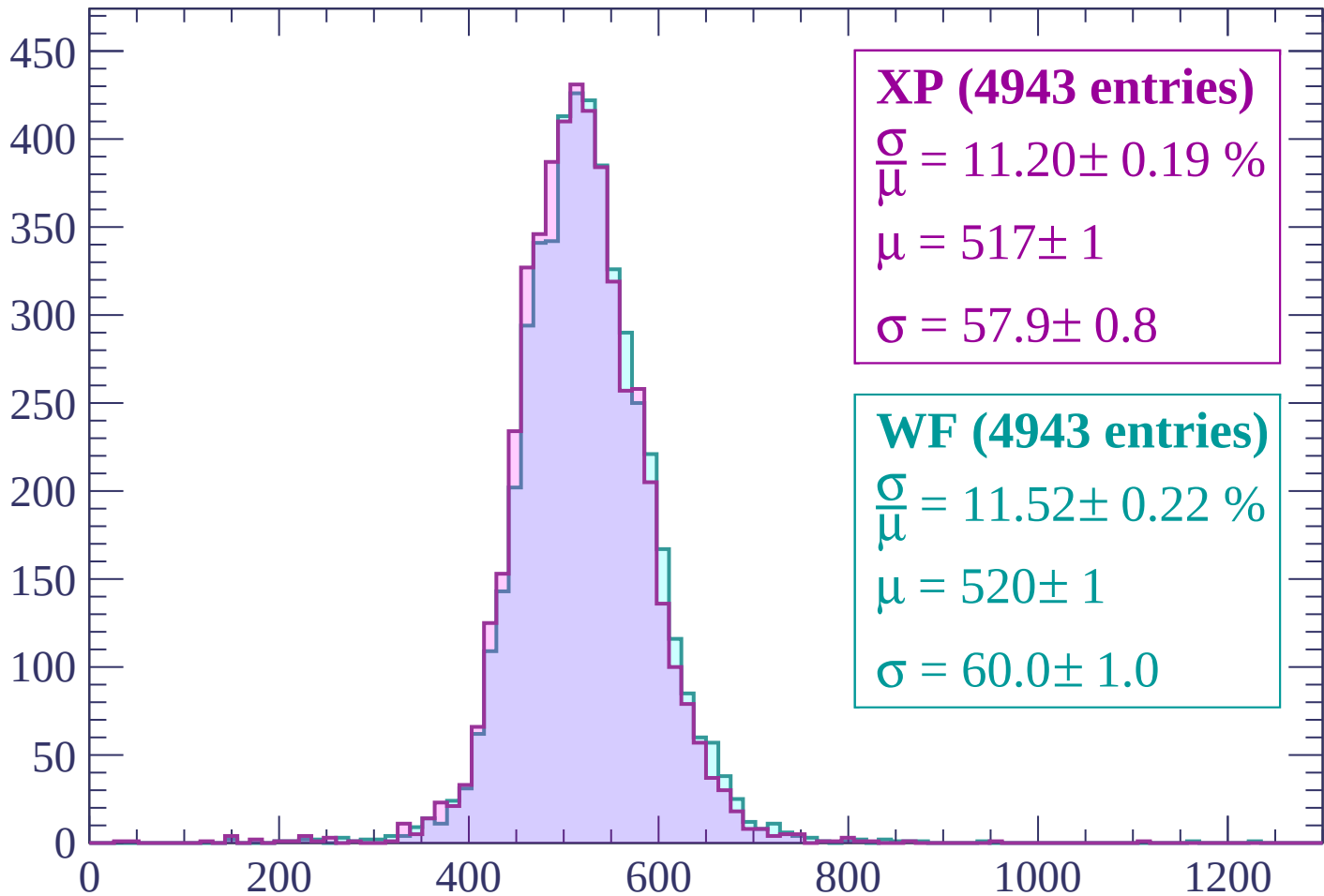
$\mu = 522 \pm 1$

$\sigma = 61.4 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $-2640 < p < -2580$

Count



XP (4943 entries)

$$\frac{\sigma}{\mu} = 11.20 \pm 0.19 \%$$

$$\mu = 517 \pm 1$$

$$\sigma = 57.9 \pm 0.8$$

WF (4943 entries)

$$\frac{\sigma}{\mu} = 11.52 \pm 0.22 \%$$

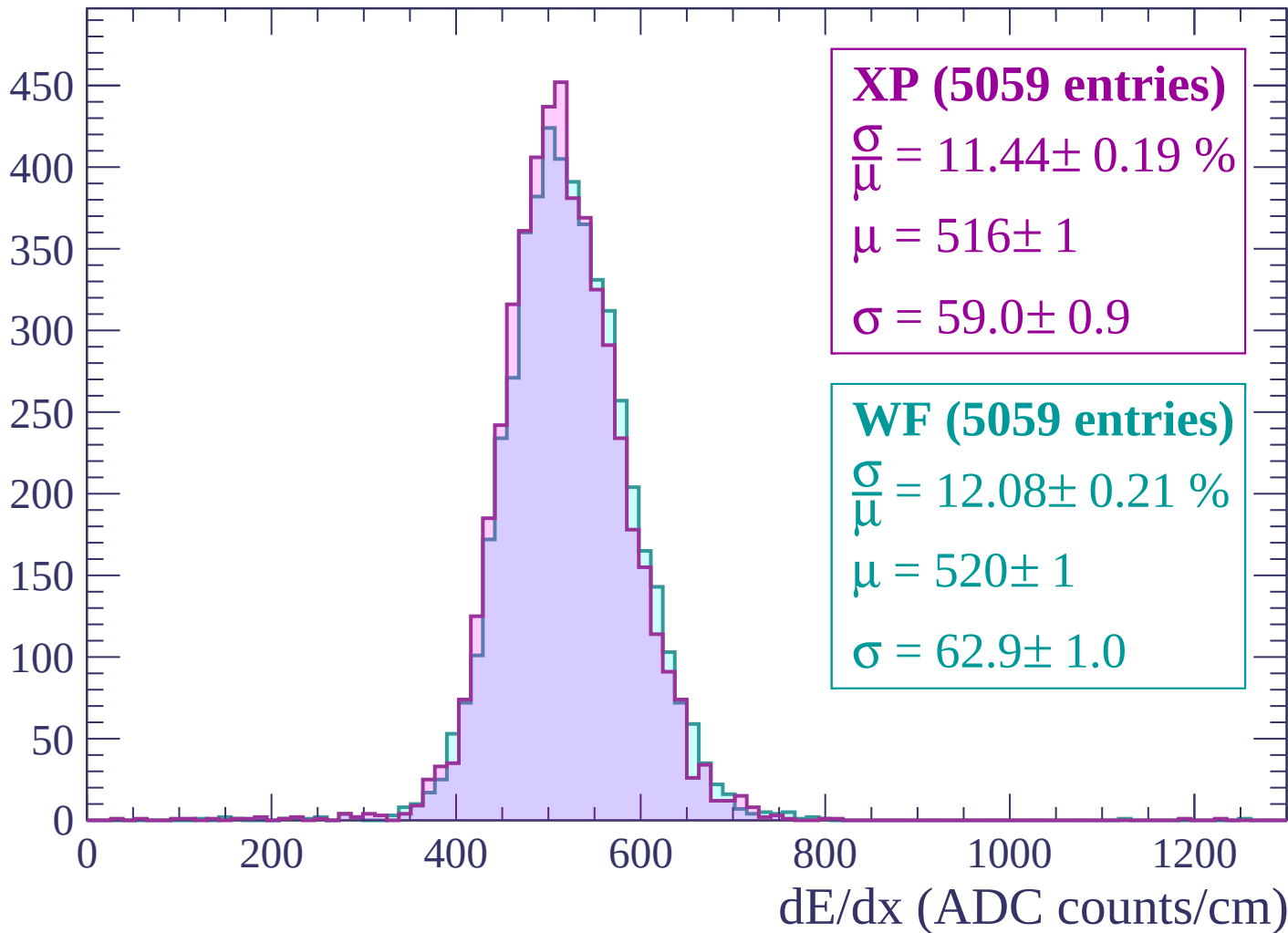
$$\mu = 520 \pm 1$$

$$\sigma = 60.0 \pm 1.0$$

dE/dx (ADC counts/cm)

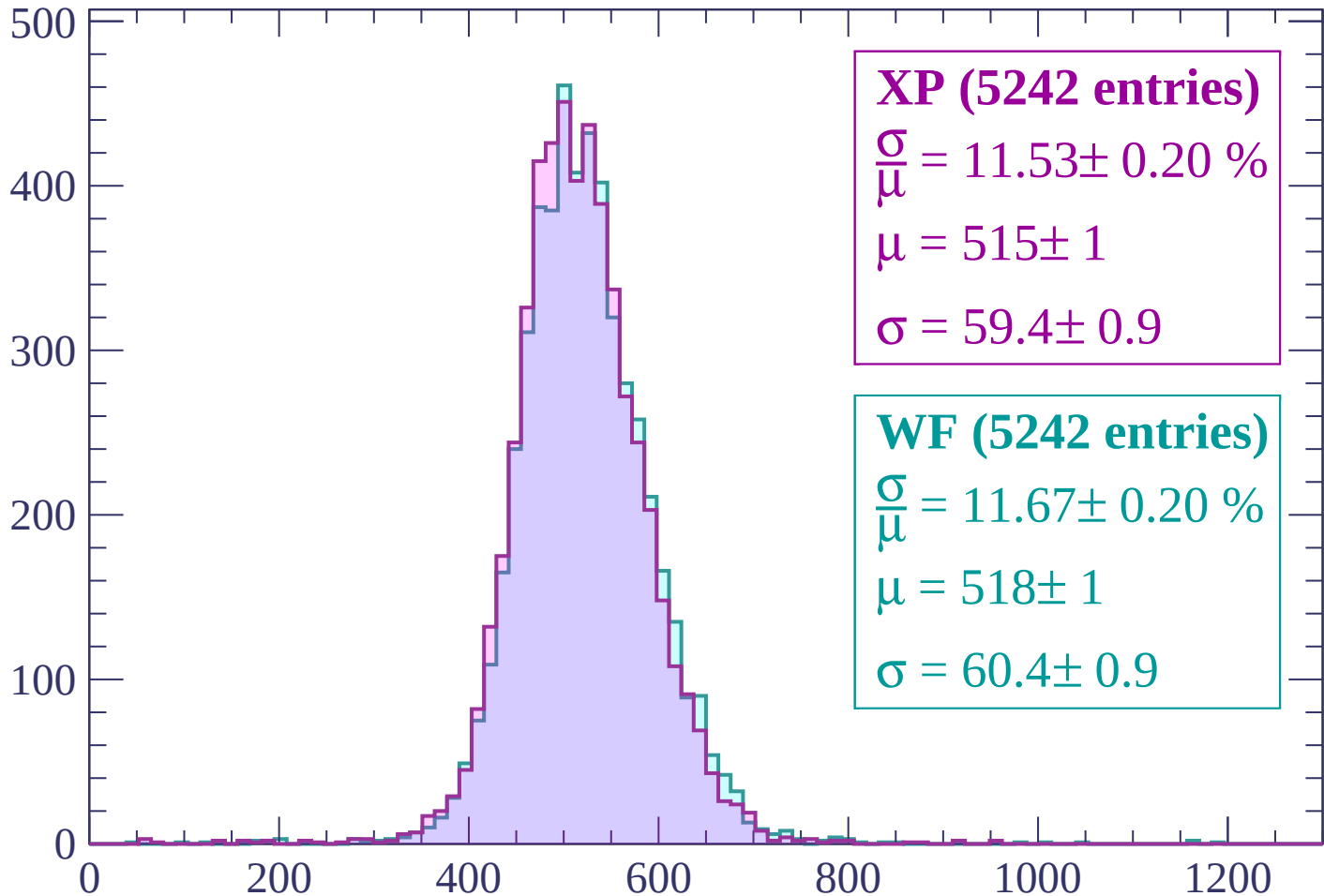
Energy loss | $-2580 < p < -2520$

Count



Energy loss | $-2520 < p < -2460$

Count



XP (5242 entries)

$$\frac{\sigma}{\mu} = 11.53 \pm 0.20 \%$$

$$\mu = 515 \pm 1$$

$$\sigma = 59.4 \pm 0.9$$

WF (5242 entries)

$$\frac{\sigma}{\mu} = 11.67 \pm 0.20 \%$$

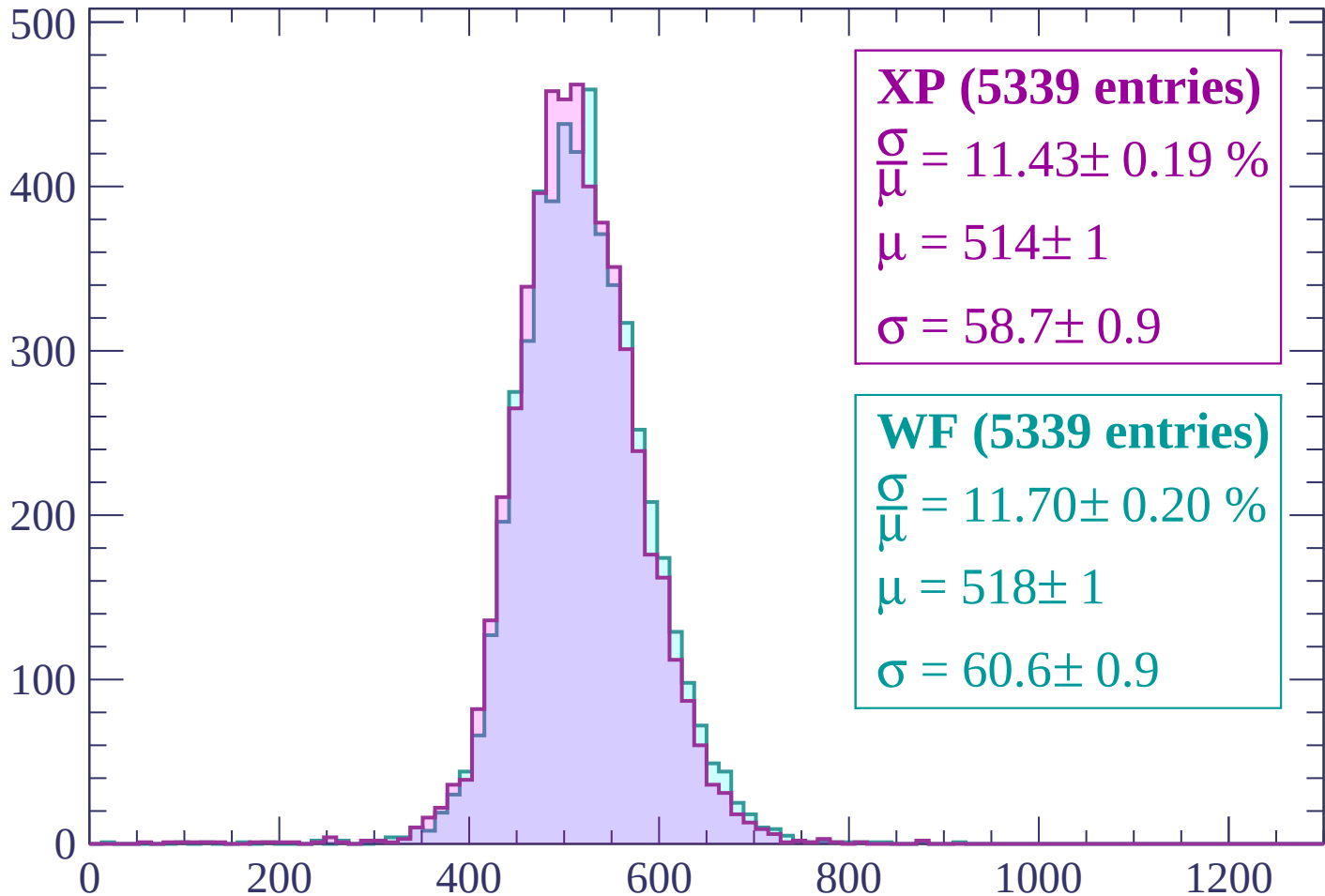
$$\mu = 518 \pm 1$$

$$\sigma = 60.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-2460 < p < -2400$

Count



XP (5339 entries)

$\frac{\sigma}{\mu} = 11.43 \pm 0.19 \%$

$\mu = 514 \pm 1$

$\sigma = 58.7 \pm 0.9$

WF (5339 entries)

$\frac{\sigma}{\mu} = 11.70 \pm 0.20 \%$

$\mu = 518 \pm 1$

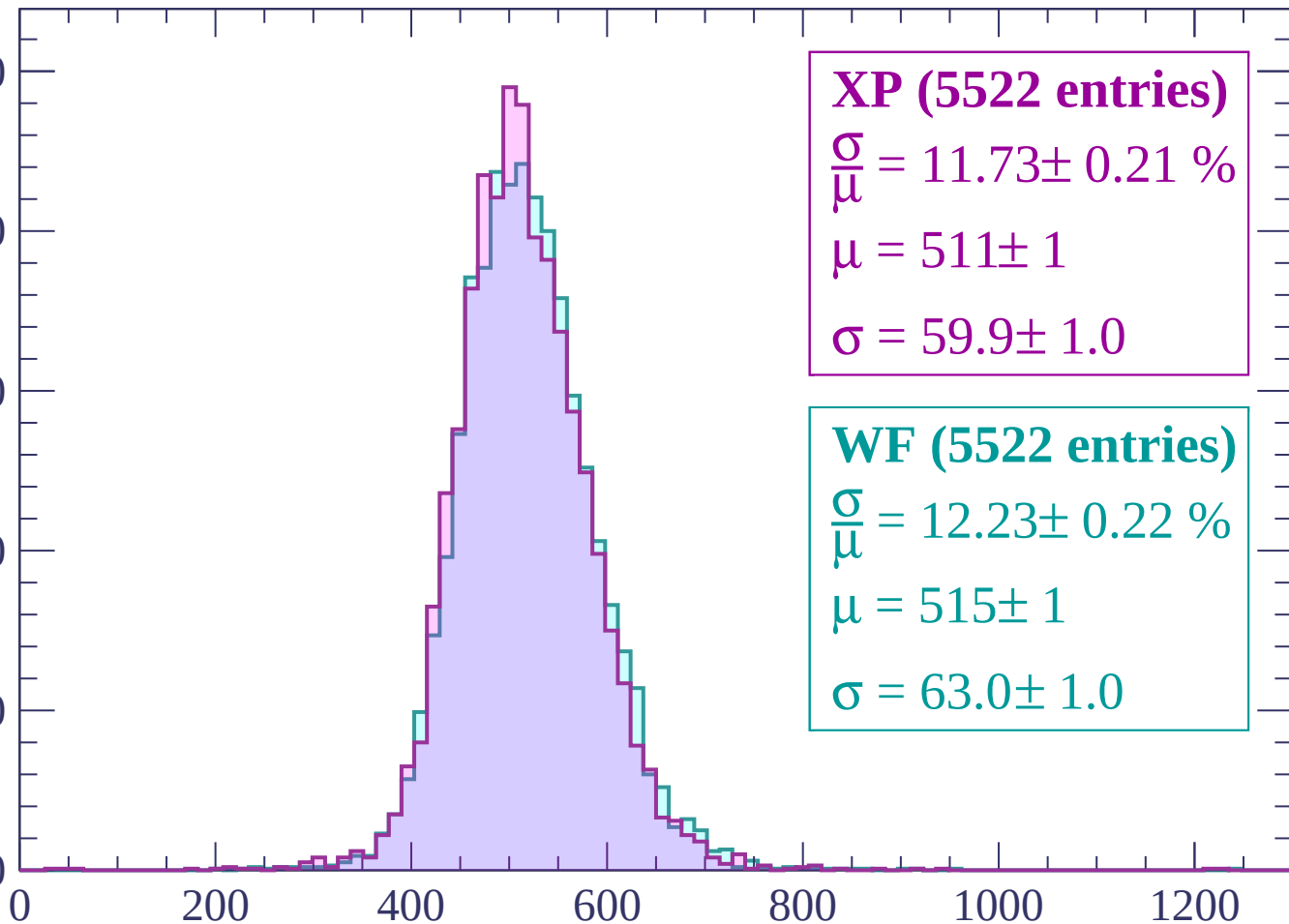
$\sigma = 60.6 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $-2400 < p < -2340$

Count

500
400
300
200
100
0



XP (5522 entries)

$\frac{\sigma}{\mu} = 11.73 \pm 0.21 \%$

$\mu = 511 \pm 1$

$\sigma = 59.9 \pm 1.0$

WF (5522 entries)

$\frac{\sigma}{\mu} = 12.23 \pm 0.22 \%$

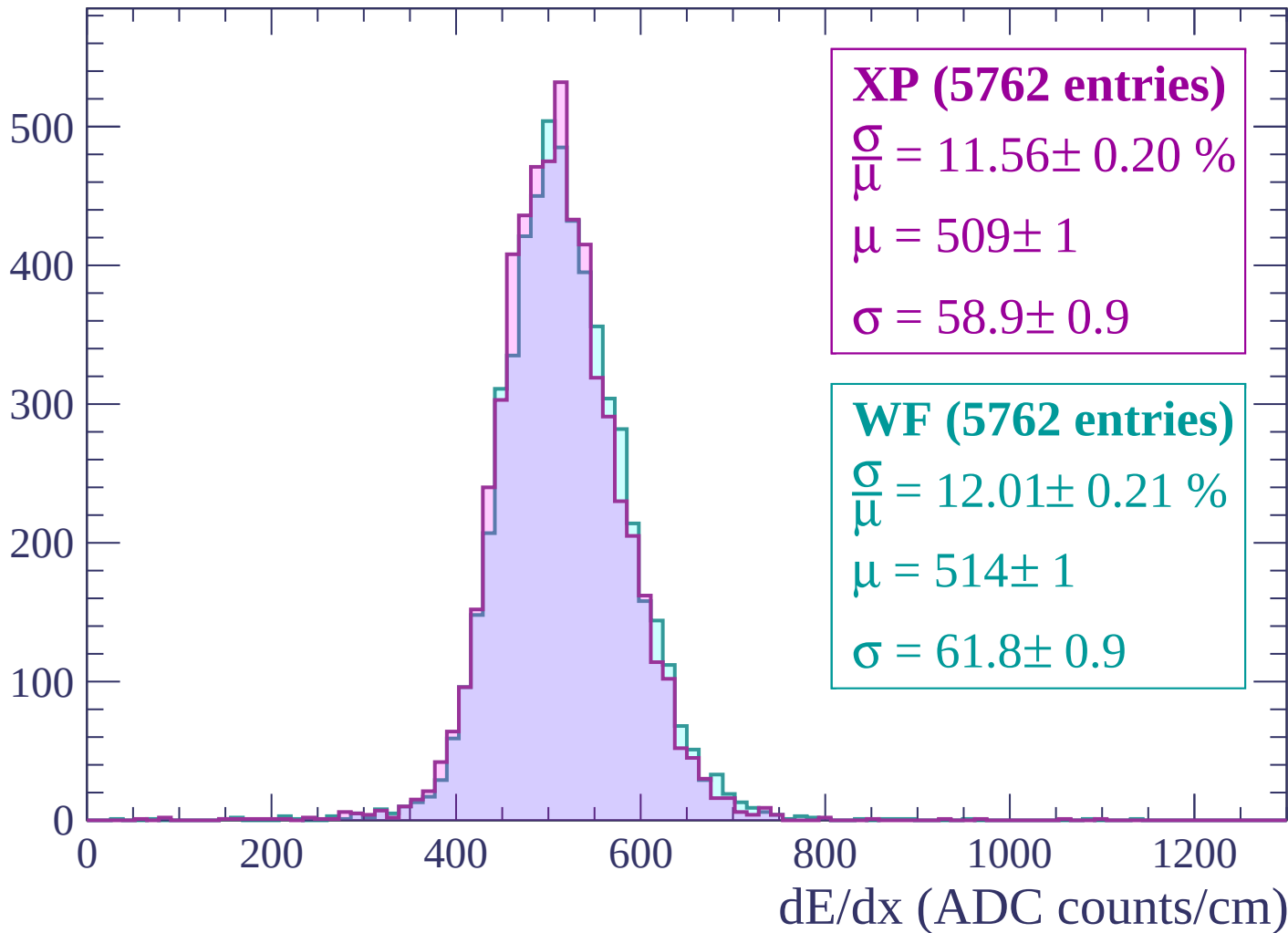
$\mu = 515 \pm 1$

$\sigma = 63.0 \pm 1.0$

dE/dx (ADC counts/cm)

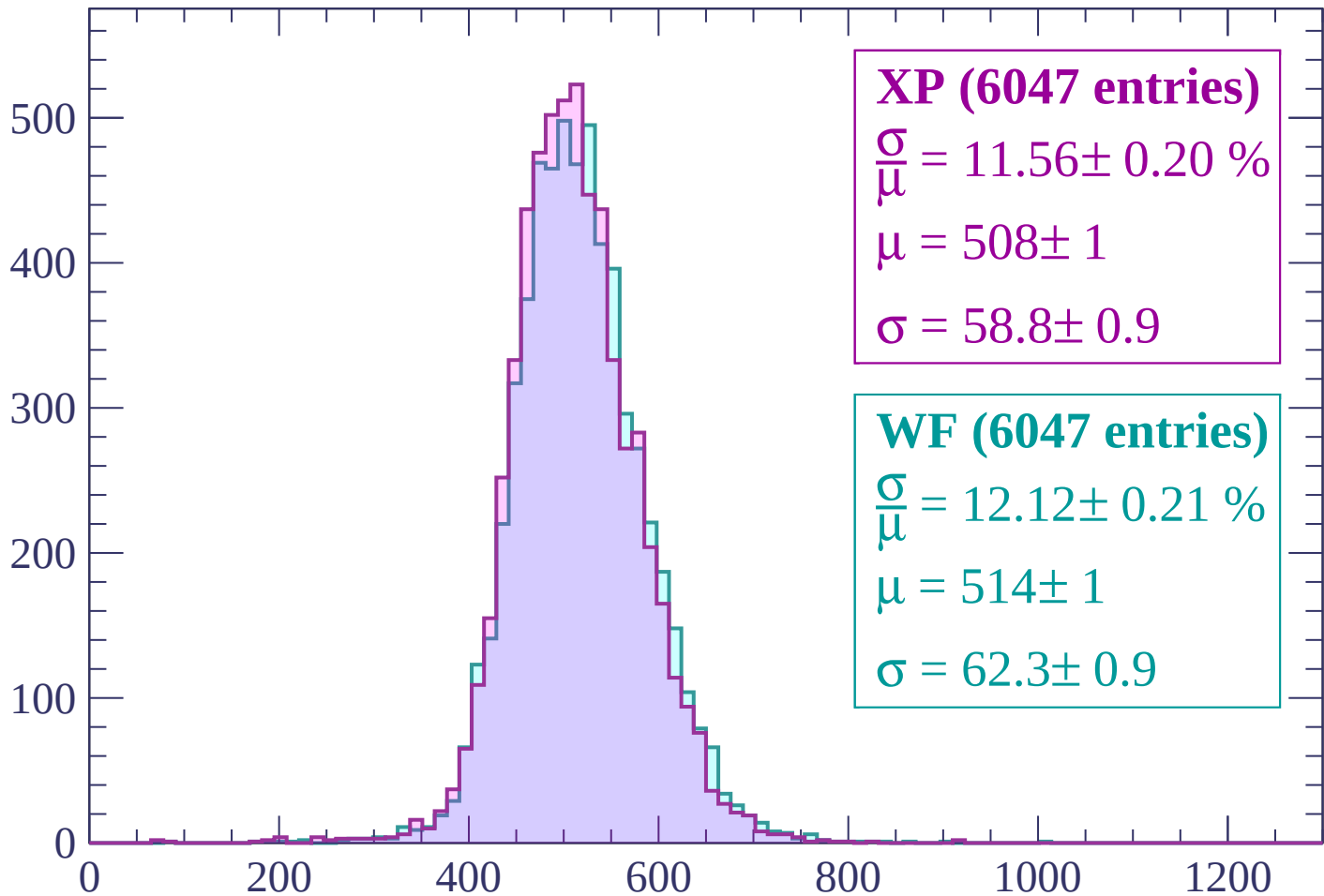
Energy loss | $-2340 < p < -2280$

Count



Energy loss | $-2280 < p < -2220$

Count



XP (6047 entries)

$$\frac{\sigma}{\mu} = 11.56 \pm 0.20 \%$$

$$\mu = 508 \pm 1$$

$$\sigma = 58.8 \pm 0.9$$

WF (6047 entries)

$$\frac{\sigma}{\mu} = 12.12 \pm 0.21 \%$$

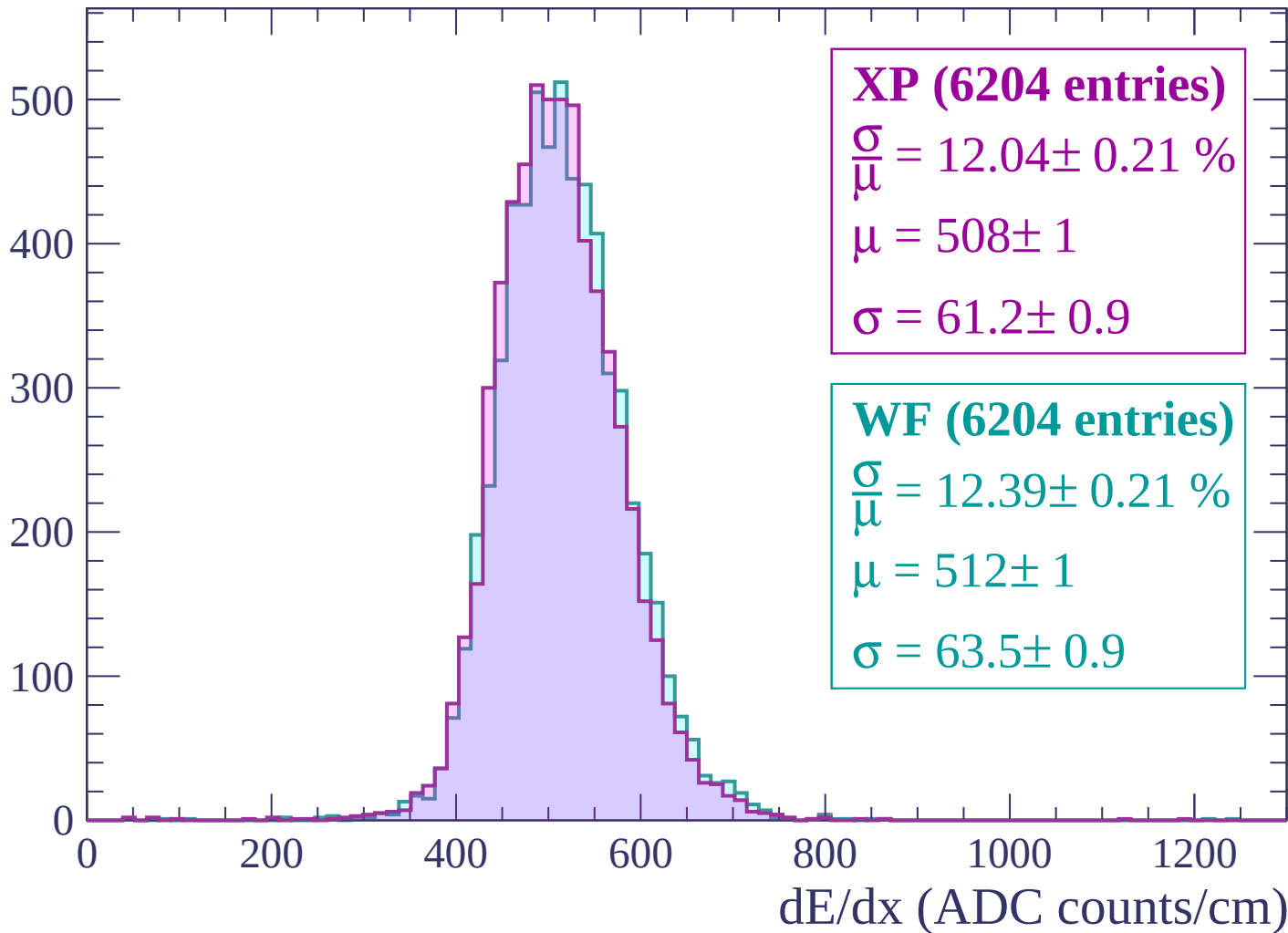
$$\mu = 514 \pm 1$$

$$\sigma = 62.3 \pm 0.9$$

dE/dx (ADC counts/cm)

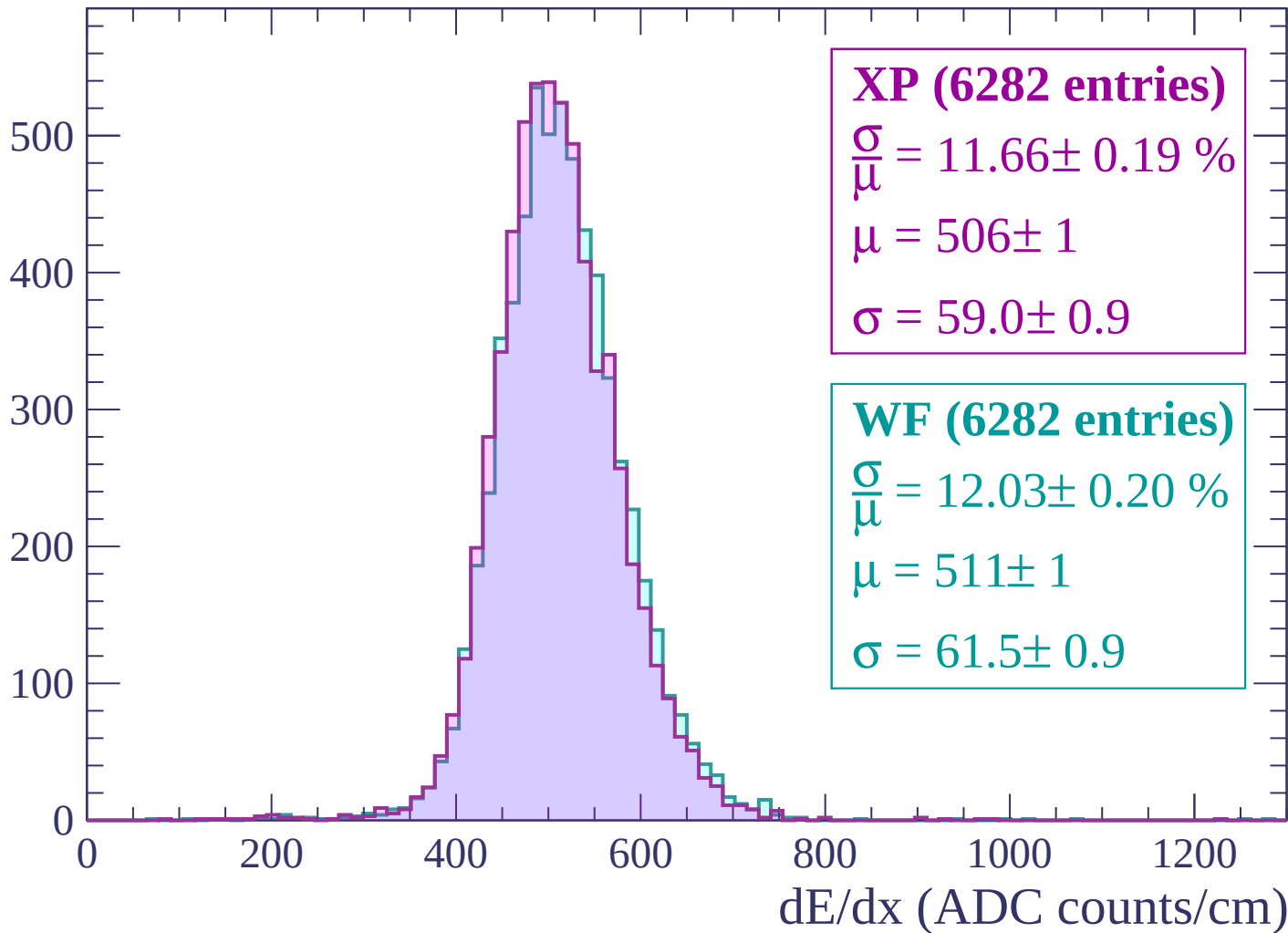
Energy loss | -2220 < p < -2160

Count



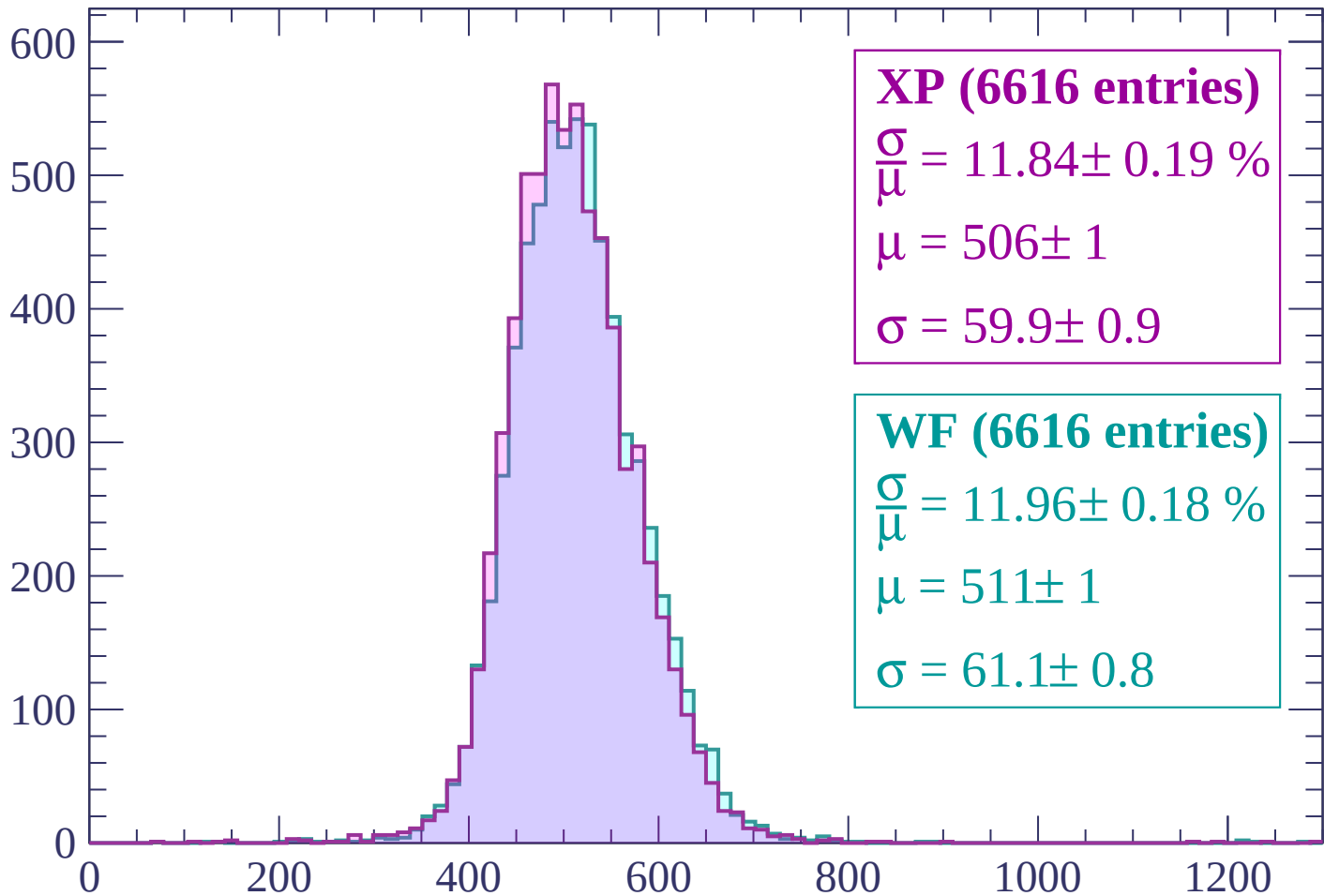
Energy loss | -2160 < p < -2100

Count



Energy loss | -2100 < p < -2040

Count



XP (6616 entries)

$$\frac{\sigma}{\mu} = 11.84 \pm 0.19 \%$$

$$\mu = 506 \pm 1$$

$$\sigma = 59.9 \pm 0.9$$

WF (6616 entries)

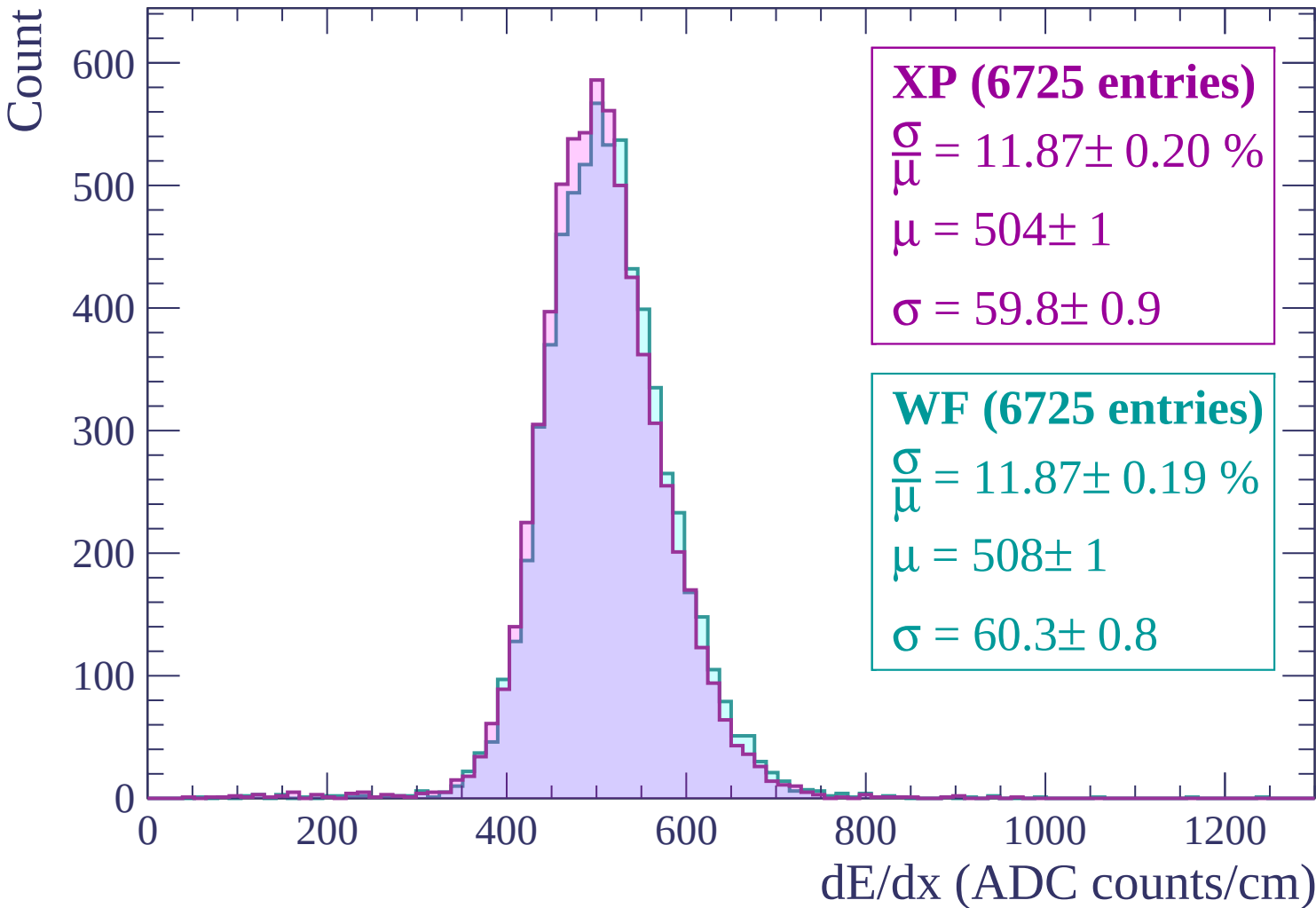
$$\frac{\sigma}{\mu} = 11.96 \pm 0.18 \%$$

$$\mu = 511 \pm 1$$

$$\sigma = 61.1 \pm 0.8$$

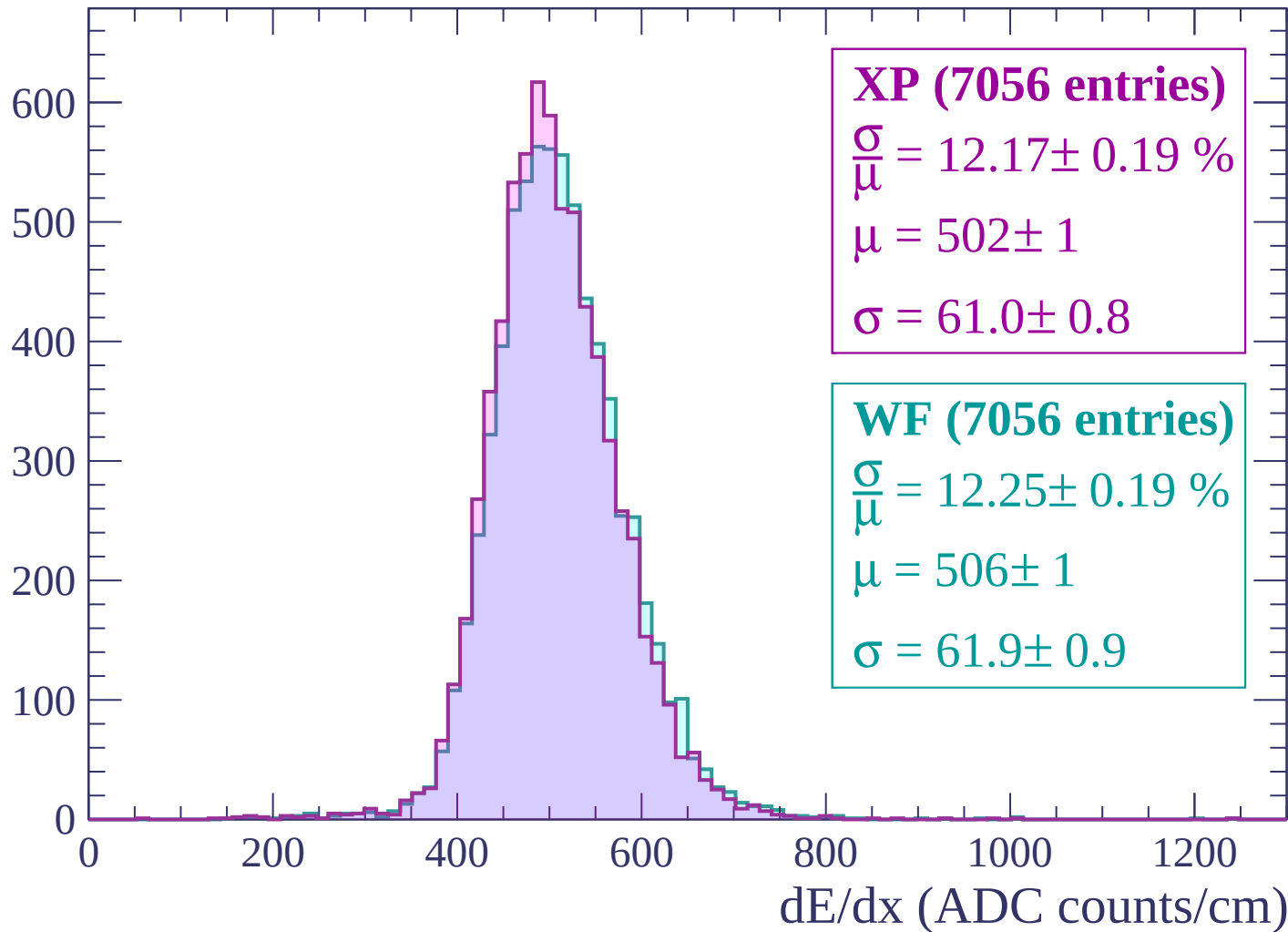
dE/dx (ADC counts/cm)

Energy loss | -2040 < p < -1980



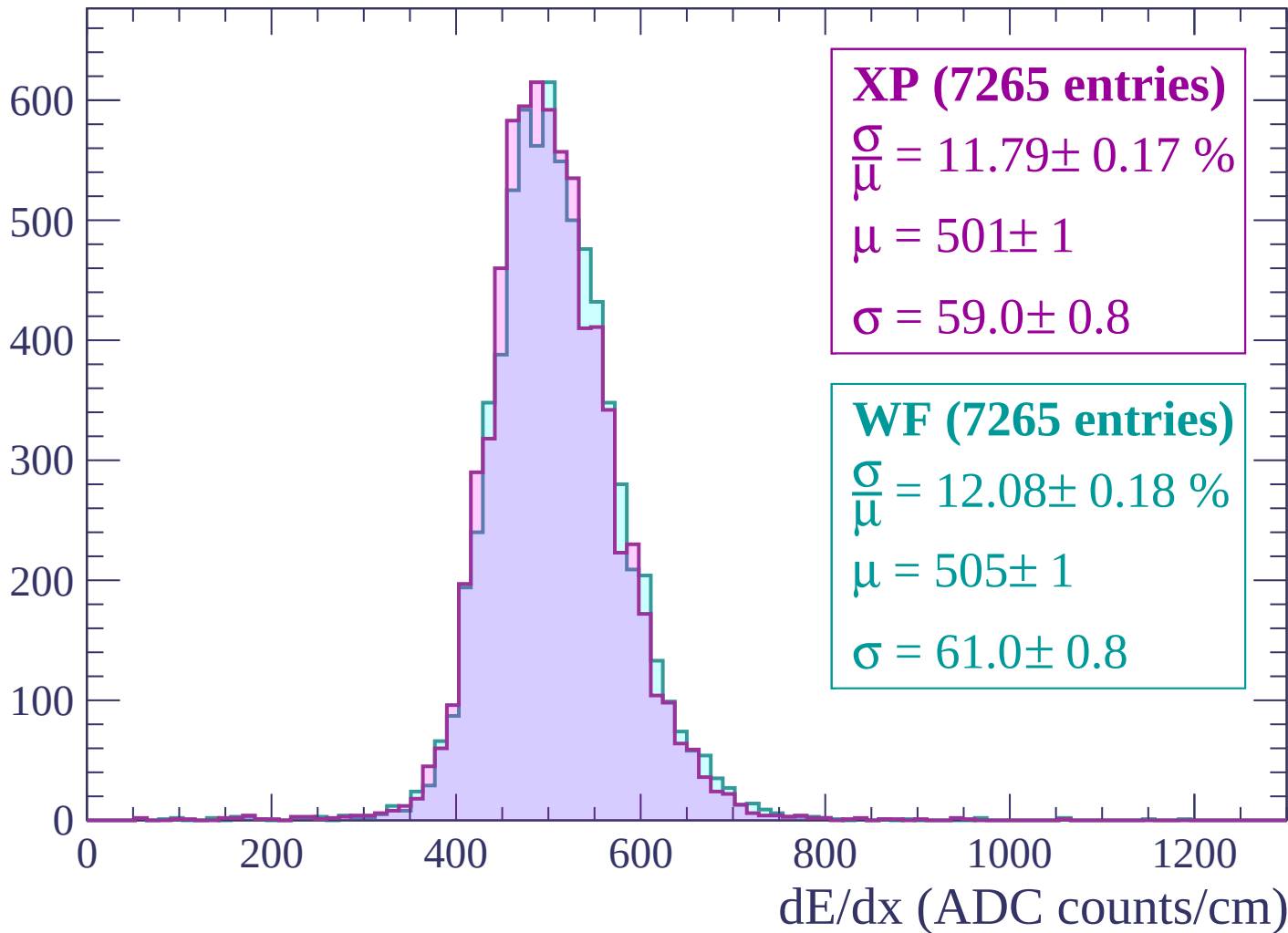
Energy loss | -1980 < p < -1920

Count



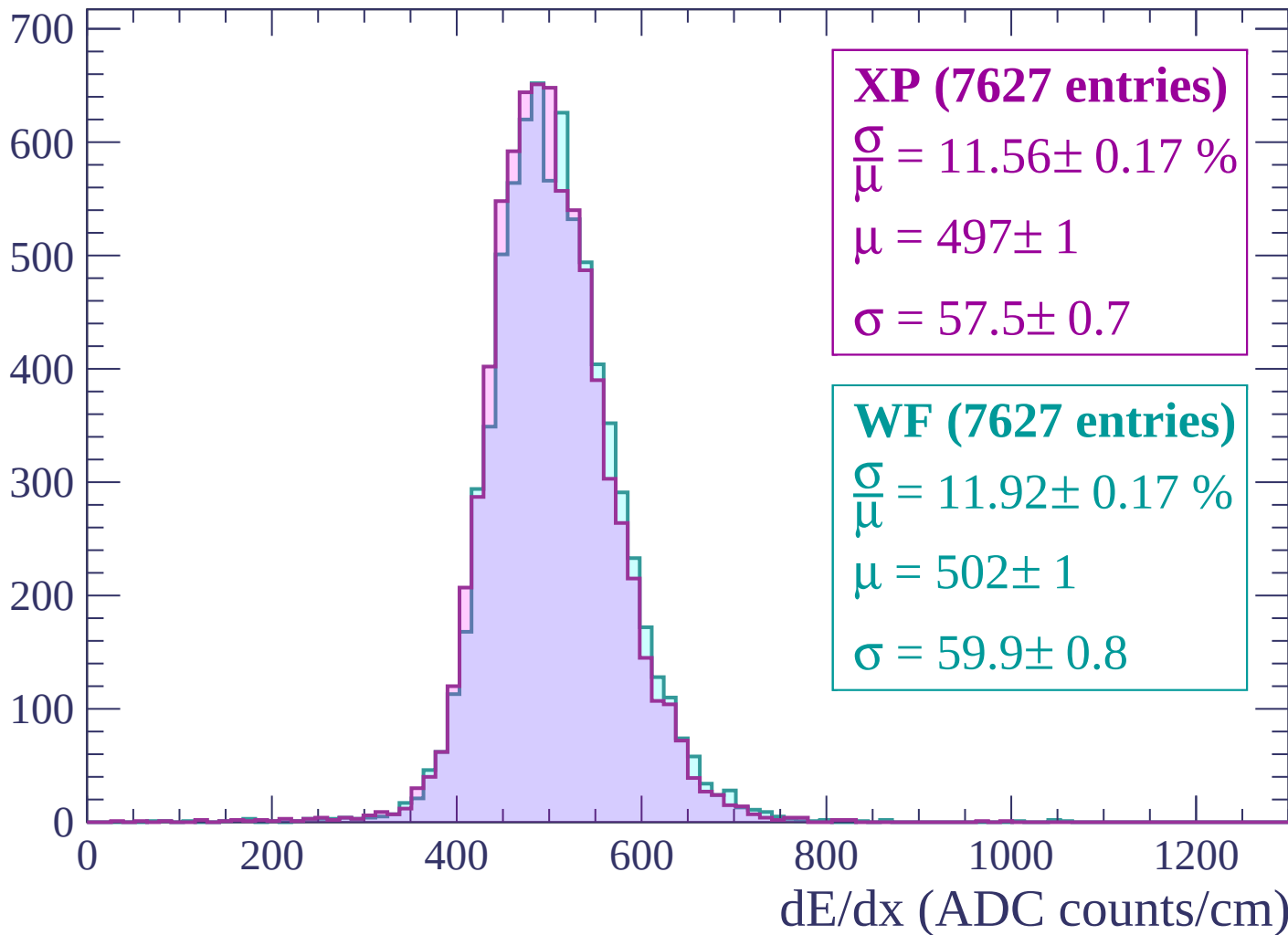
Energy loss | -1920 < p < -1860

Count



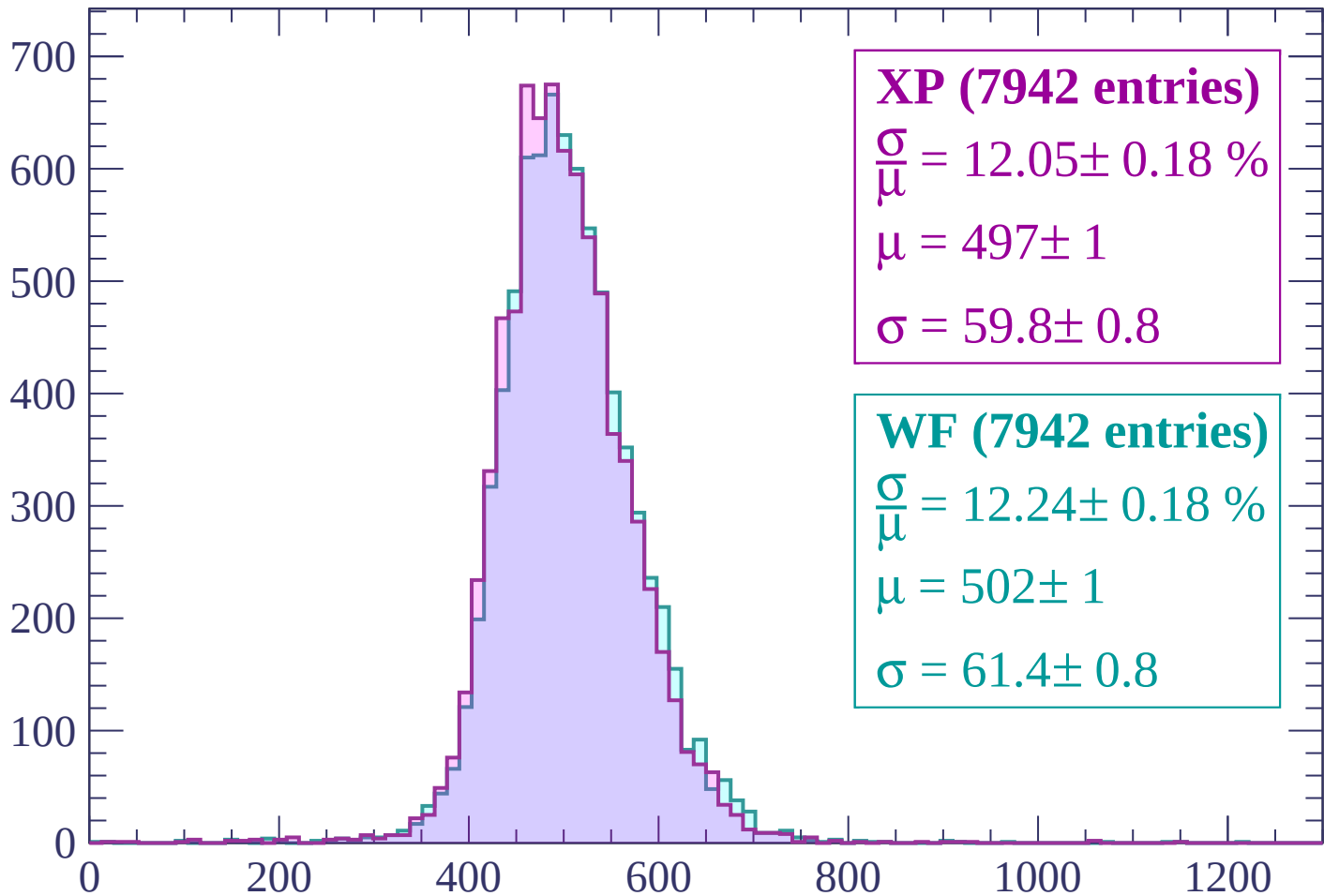
Energy loss | -1860 < p < -1800

Count



Energy loss | -1800 < p < -1740

Count



XP (7942 entries)

$$\frac{\sigma}{\mu} = 12.05 \pm 0.18 \%$$

$$\mu = 497 \pm 1$$

$$\sigma = 59.8 \pm 0.8$$

WF (7942 entries)

$$\frac{\sigma}{\mu} = 12.24 \pm 0.18 \%$$

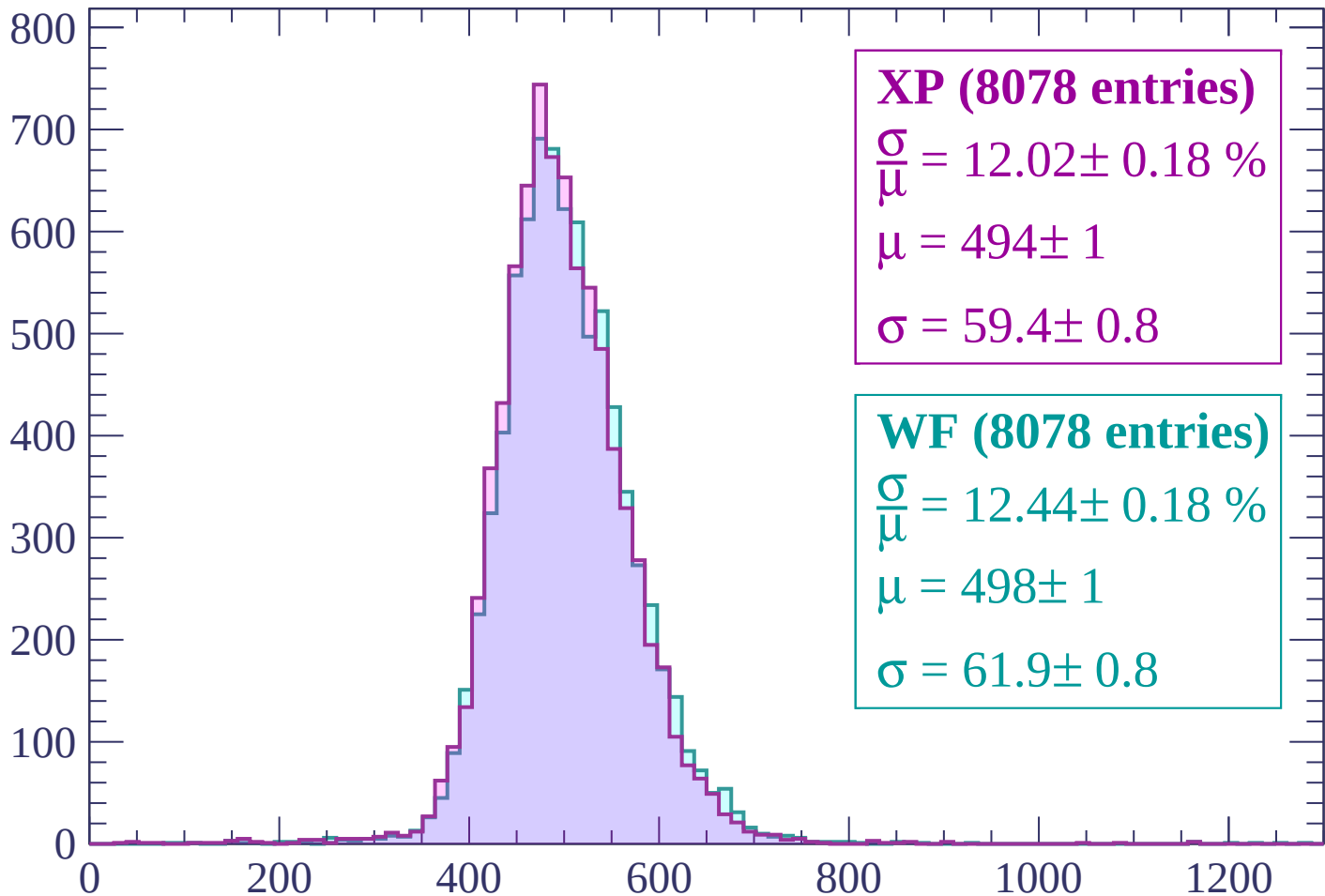
$$\mu = 502 \pm 1$$

$$\sigma = 61.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (8078 entries)

$\frac{\sigma}{\mu} = 12.02 \pm 0.18 \%$

$\mu = 494 \pm 1$

$\sigma = 59.4 \pm 0.8$

WF (8078 entries)

$\frac{\sigma}{\mu} = 12.44 \pm 0.18 \%$

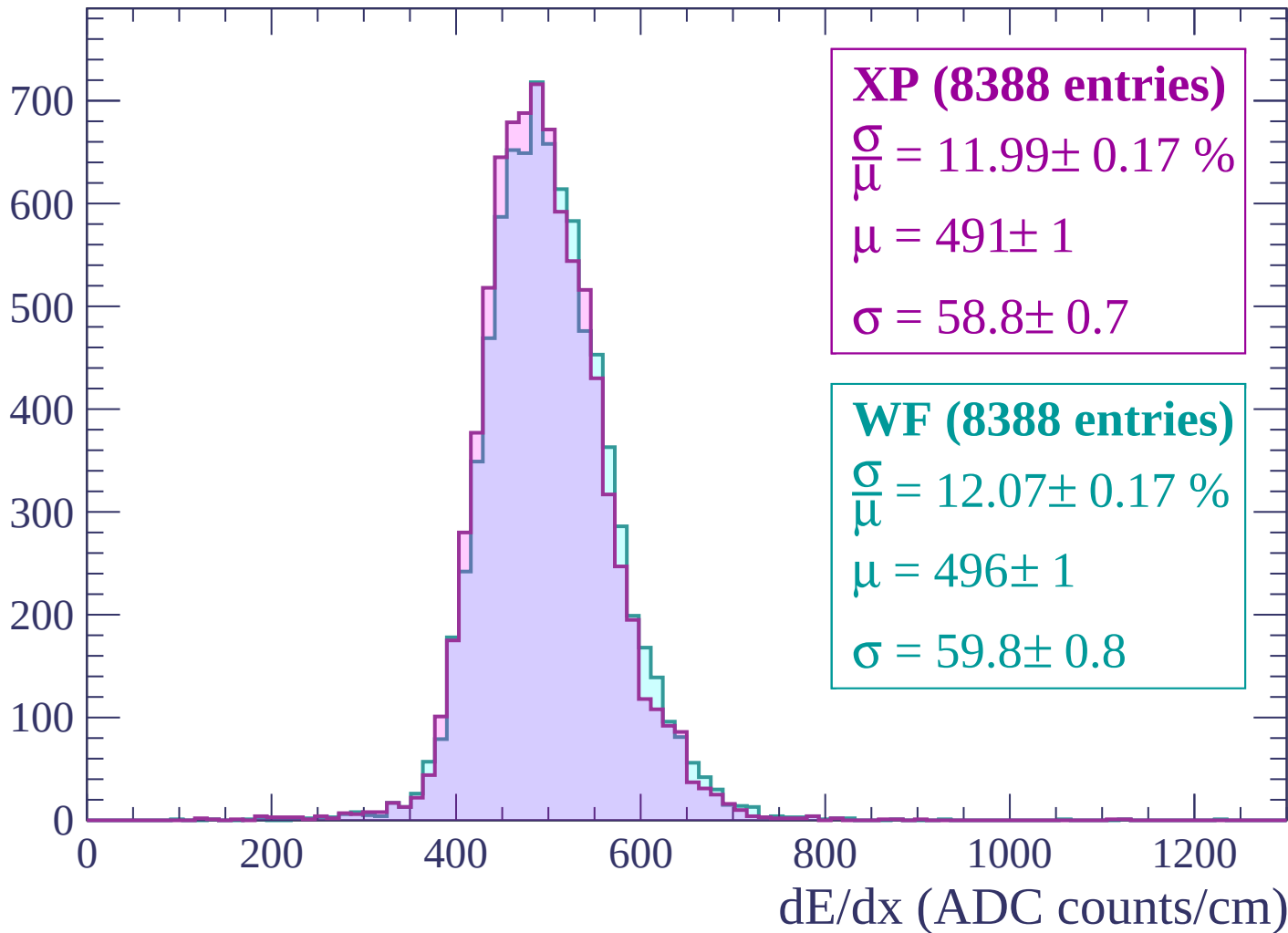
$\mu = 498 \pm 1$

$\sigma = 61.9 \pm 0.8$

dE/dx (ADC counts/cm)

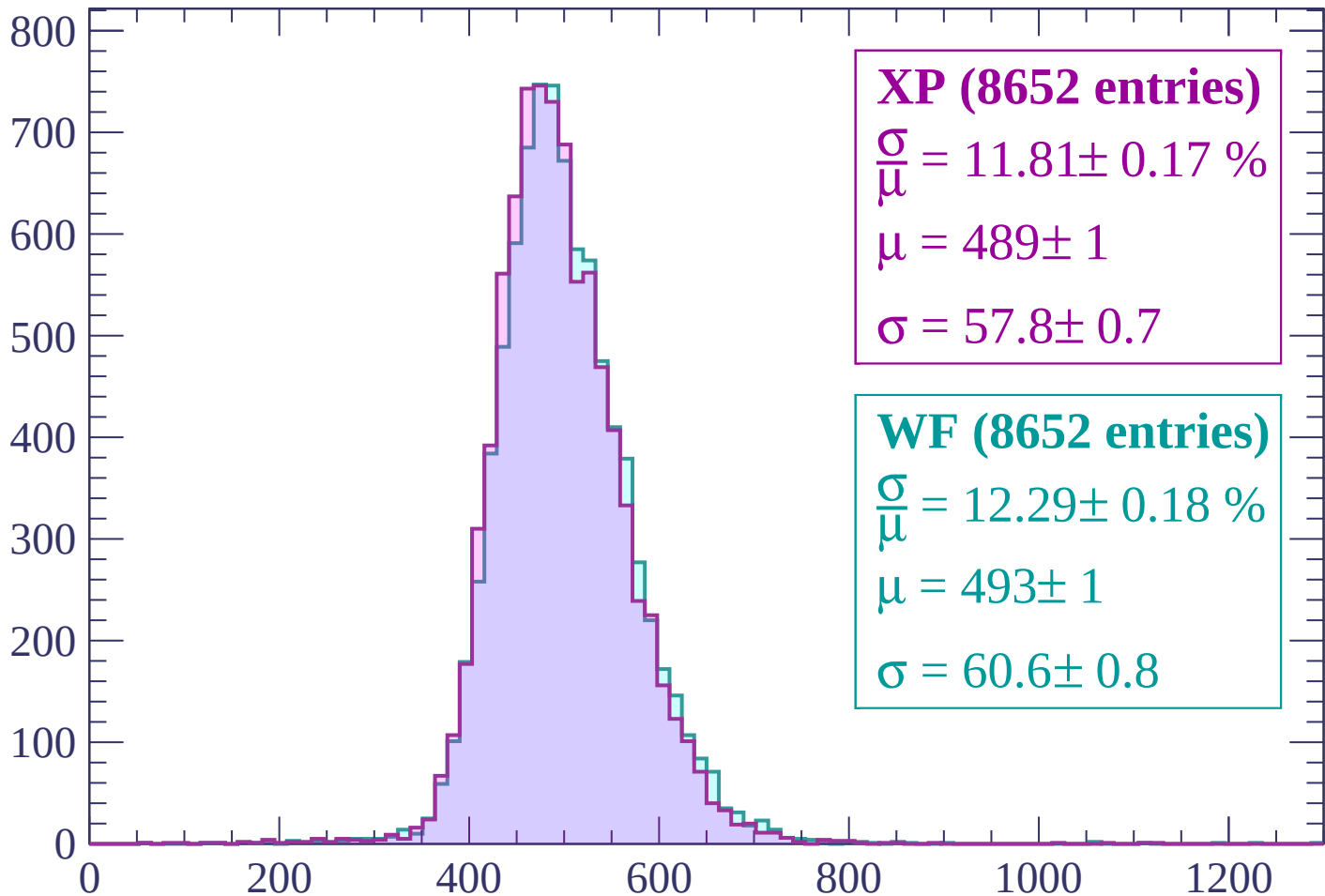
Energy loss | $-1680 < p < -1620$

Count



Energy loss | -1620 < p < -1560

Count



XP (8652 entries)

$$\frac{\sigma}{\mu} = 11.81 \pm 0.17 \%$$

$$\mu = 489 \pm 1$$

$$\sigma = 57.8 \pm 0.7$$

WF (8652 entries)

$$\frac{\sigma}{\mu} = 12.29 \pm 0.18 \%$$

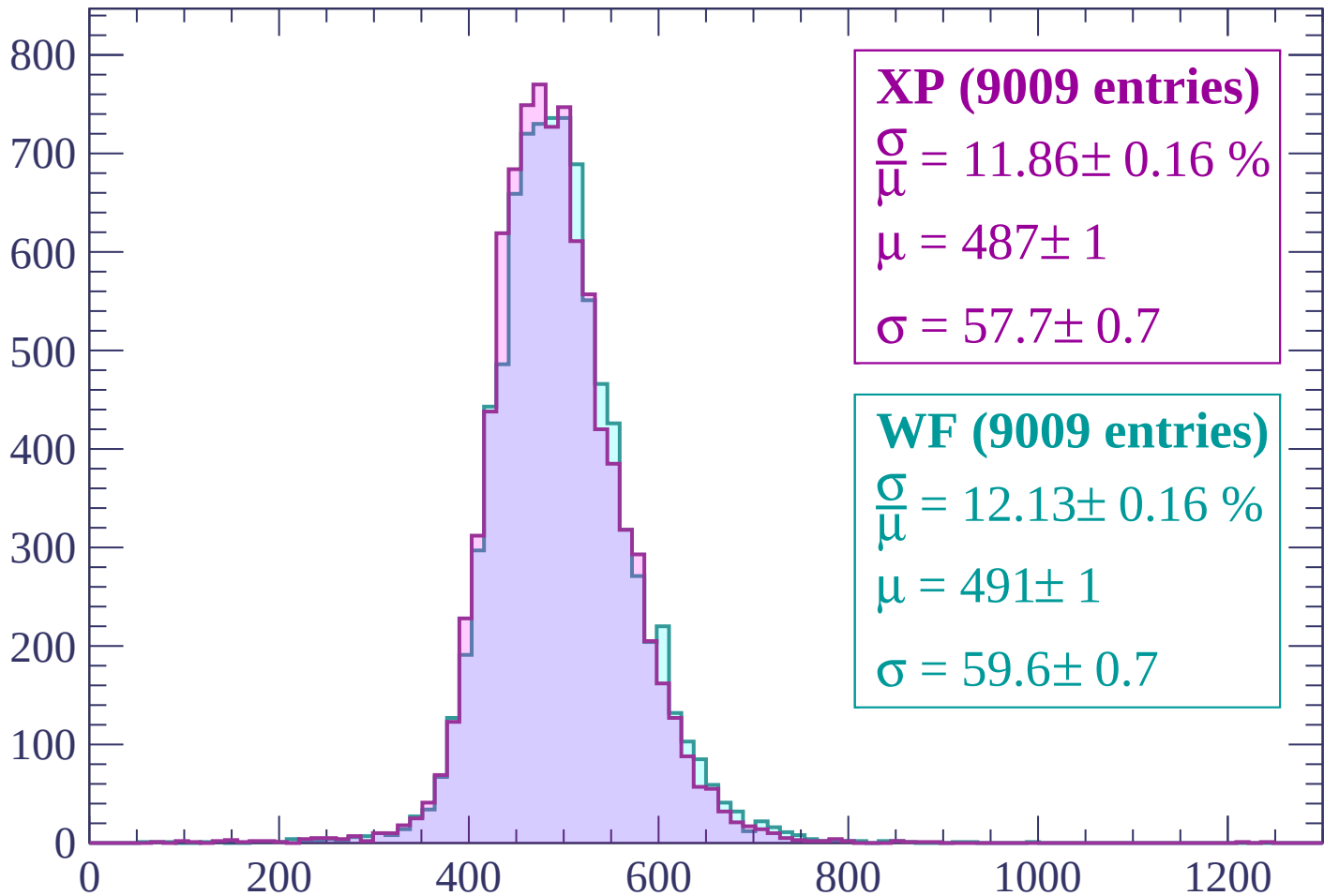
$$\mu = 493 \pm 1$$

$$\sigma = 60.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (9009 entries)

$$\frac{\sigma}{\mu} = 11.86 \pm 0.16 \%$$

$$\mu = 487 \pm 1$$

$$\sigma = 57.7 \pm 0.7$$

WF (9009 entries)

$$\frac{\sigma}{\mu} = 12.13 \pm 0.16 \%$$

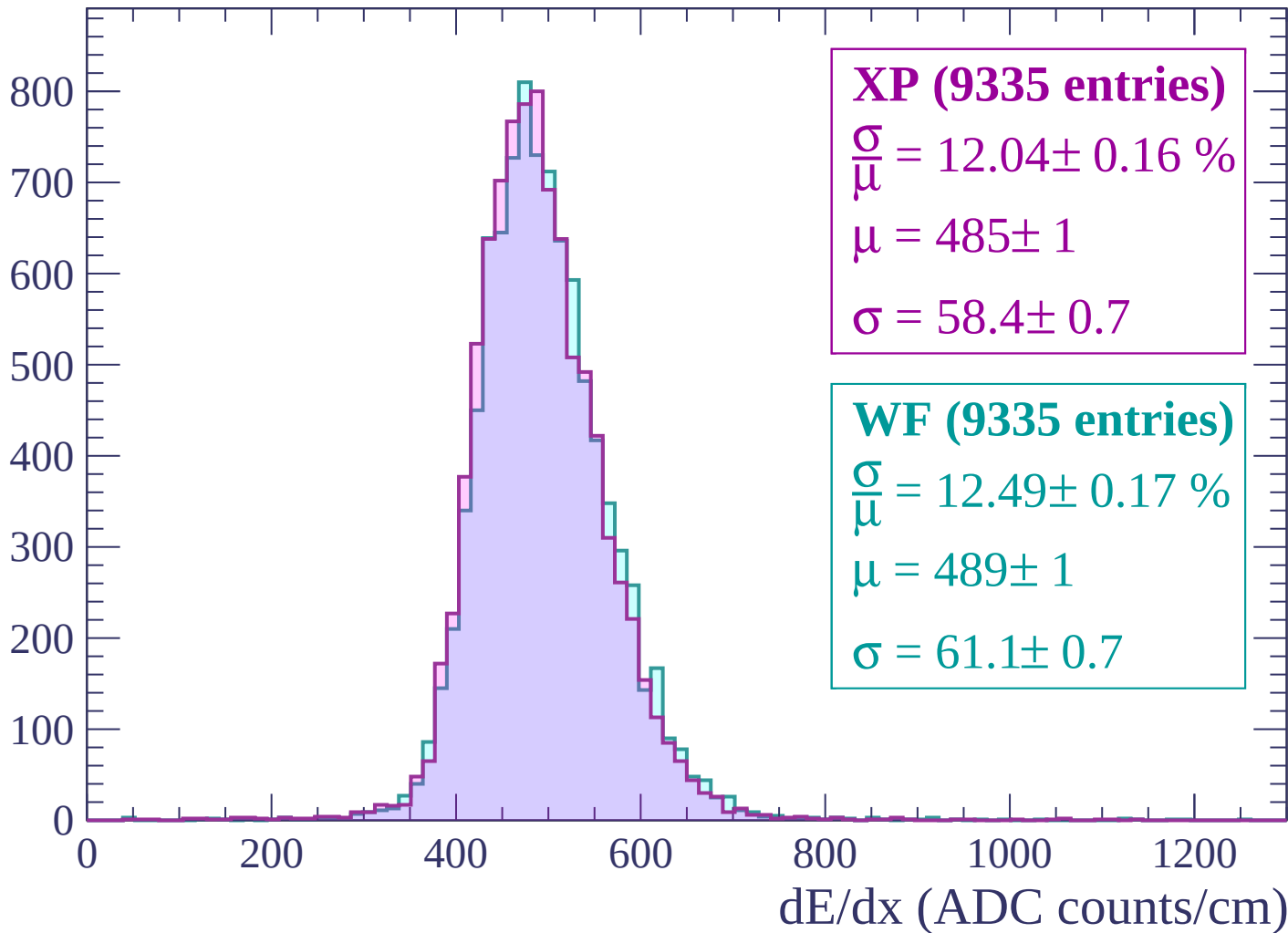
$$\mu = 491 \pm 1$$

$$\sigma = 59.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1500 < p < -1440$

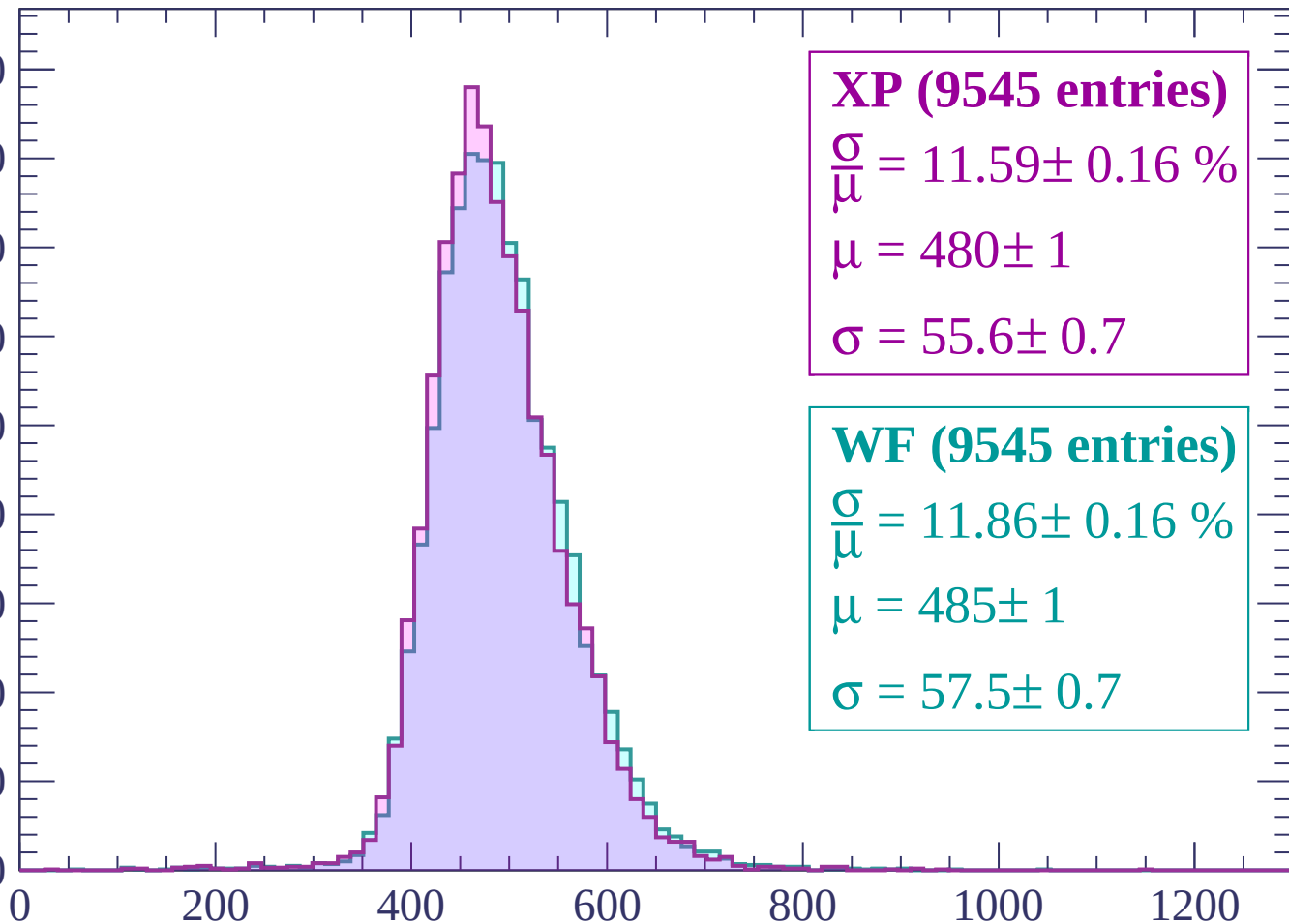
Count



Energy loss | $-1440 < p < -1380$

Count

900
800
700
600
500
400
300
200
100
0



XP (9545 entries)

$\frac{\sigma}{\mu} = 11.59 \pm 0.16 \%$

$\mu = 480 \pm 1$

$\sigma = 55.6 \pm 0.7$

WF (9545 entries)

$\frac{\sigma}{\mu} = 11.86 \pm 0.16 \%$

$\mu = 485 \pm 1$

$\sigma = 57.5 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (10038 entries)

$$\frac{\sigma}{\mu} = 11.54 \pm 0.16 \%$$

$$\mu = 477 \pm 1$$

$$\sigma = 55.0 \pm 0.7$$

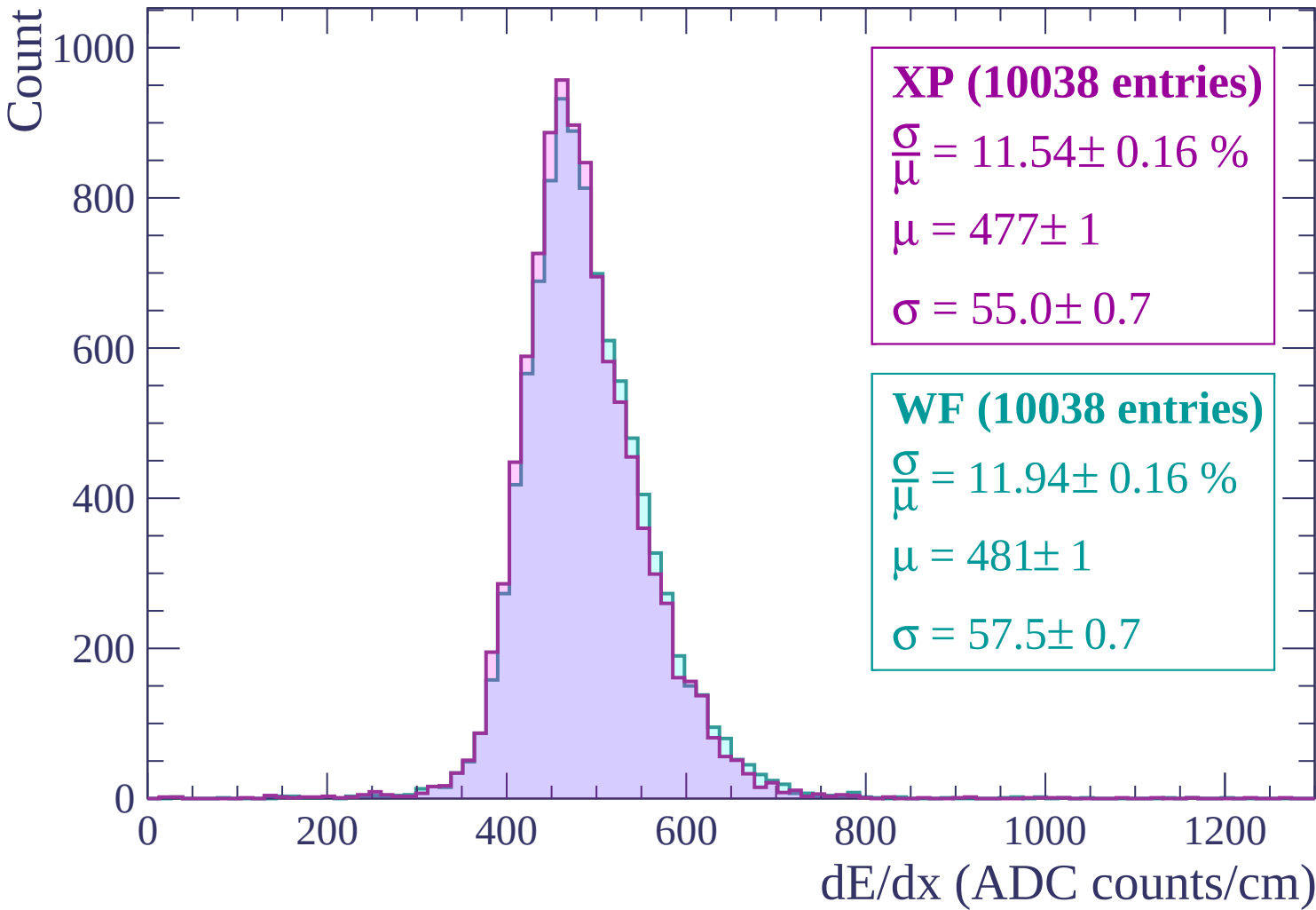
WF (10038 entries)

$$\frac{\sigma}{\mu} = 11.94 \pm 0.16 \%$$

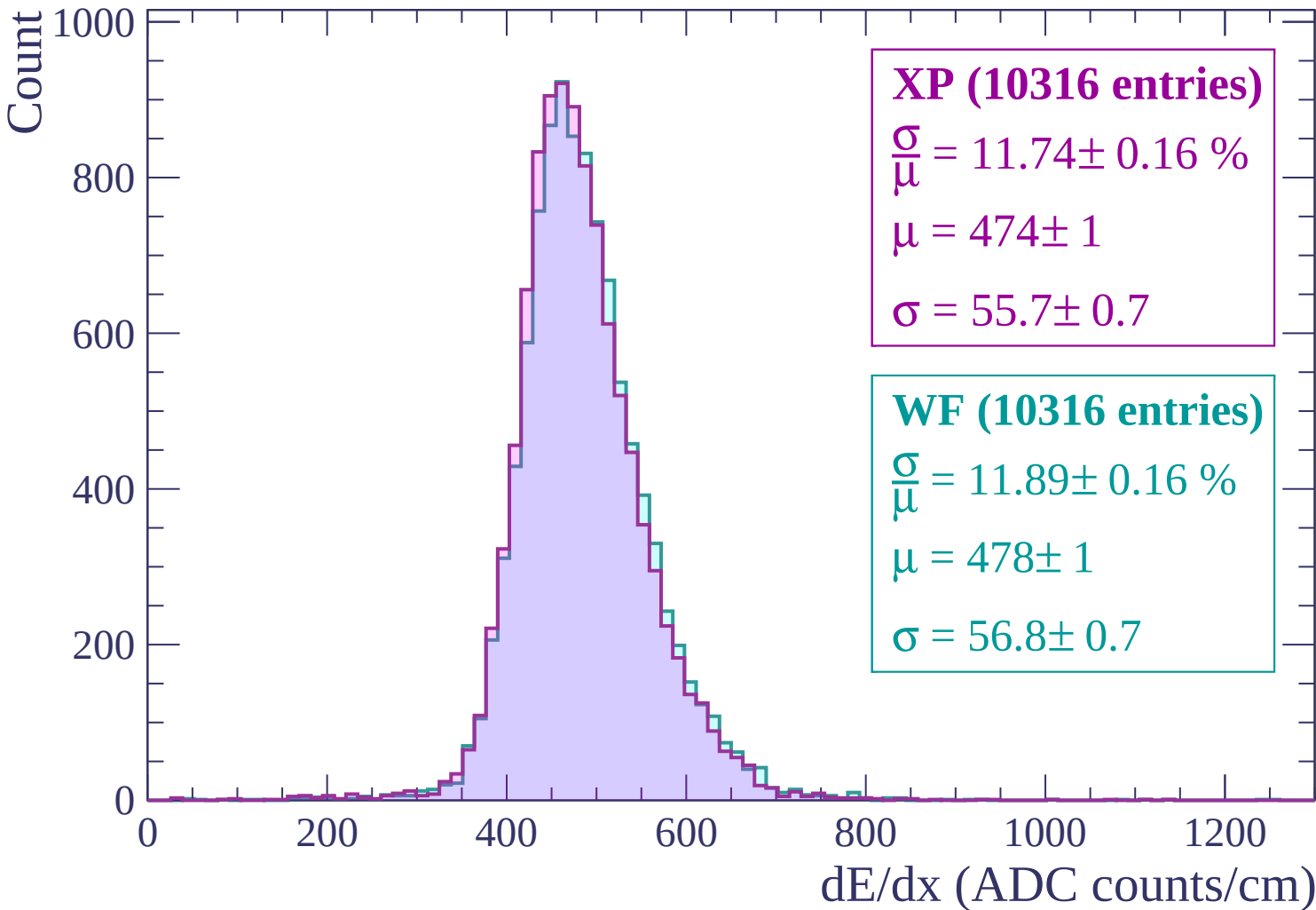
$$\mu = 481 \pm 1$$

$$\sigma = 57.5 \pm 0.7$$

dE/dx (ADC counts/cm)



Energy loss | -1320 < p < -1260



Energy loss | -1260 < p < -1200

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (10919 entries)

$\frac{\sigma}{\mu} = 11.67 \pm 0.16 \%$

$\mu = 470 \pm 1$

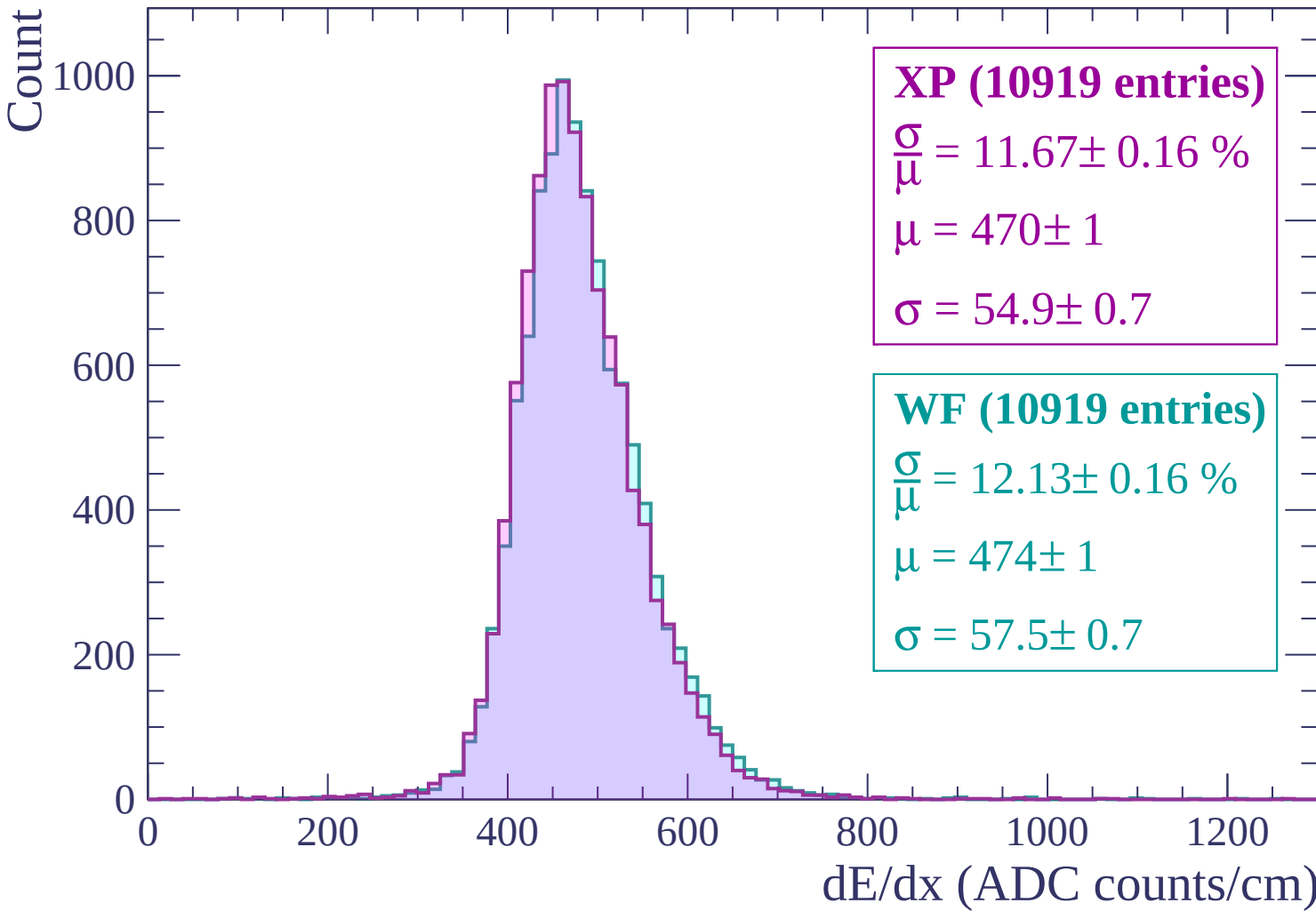
$\sigma = 54.9 \pm 0.7$

WF (10919 entries)

$\frac{\sigma}{\mu} = 12.13 \pm 0.16 \%$

$\mu = 474 \pm 1$

$\sigma = 57.5 \pm 0.7$



Energy loss | $-1200 < p < -1140$

Count

1000

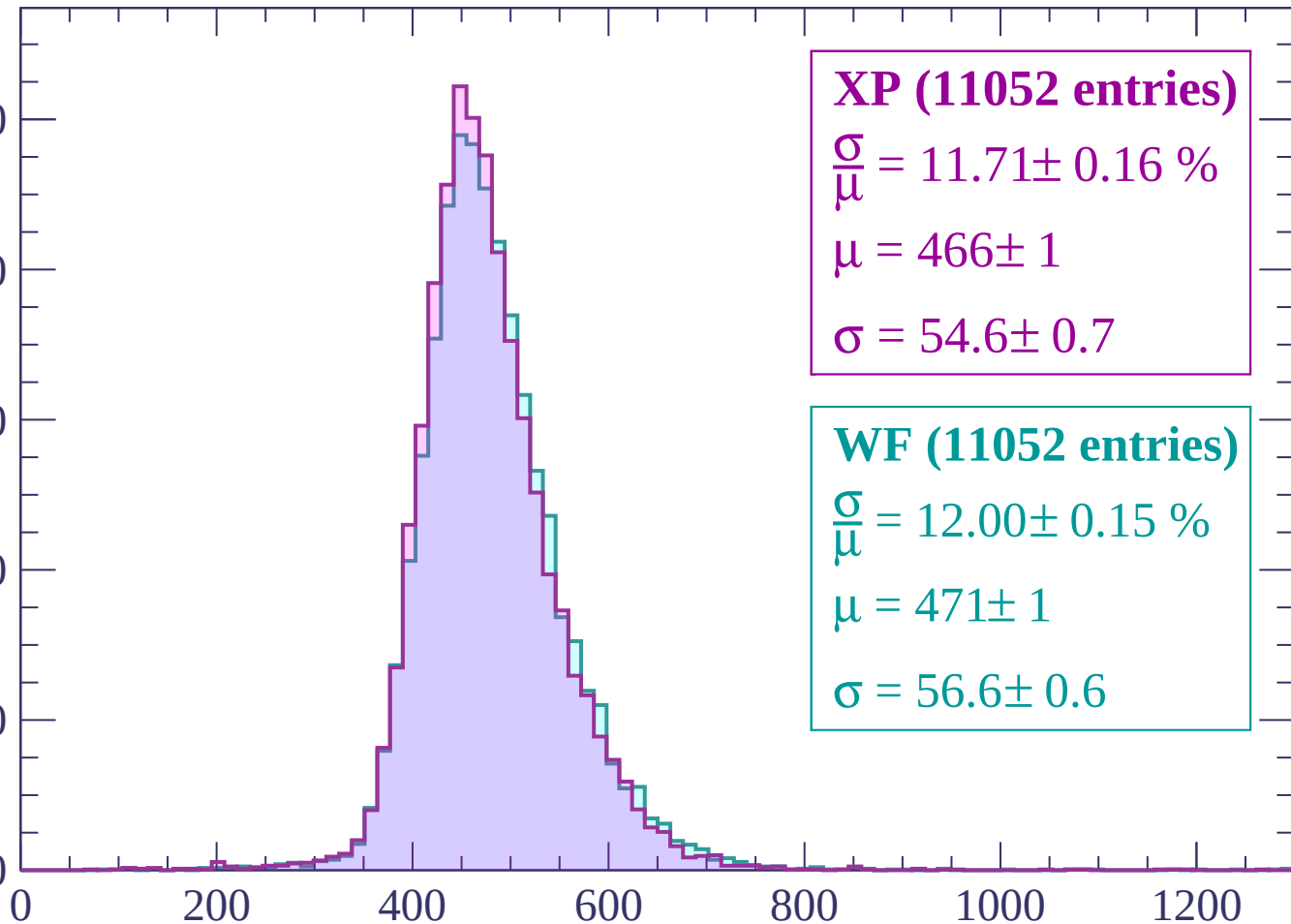
800

600

400

200

0



XP (11052 entries)

$\frac{\sigma}{\mu} = 11.71 \pm 0.16 \%$

$\mu = 466 \pm 1$

$\sigma = 54.6 \pm 0.7$

WF (11052 entries)

$\frac{\sigma}{\mu} = 12.00 \pm 0.15 \%$

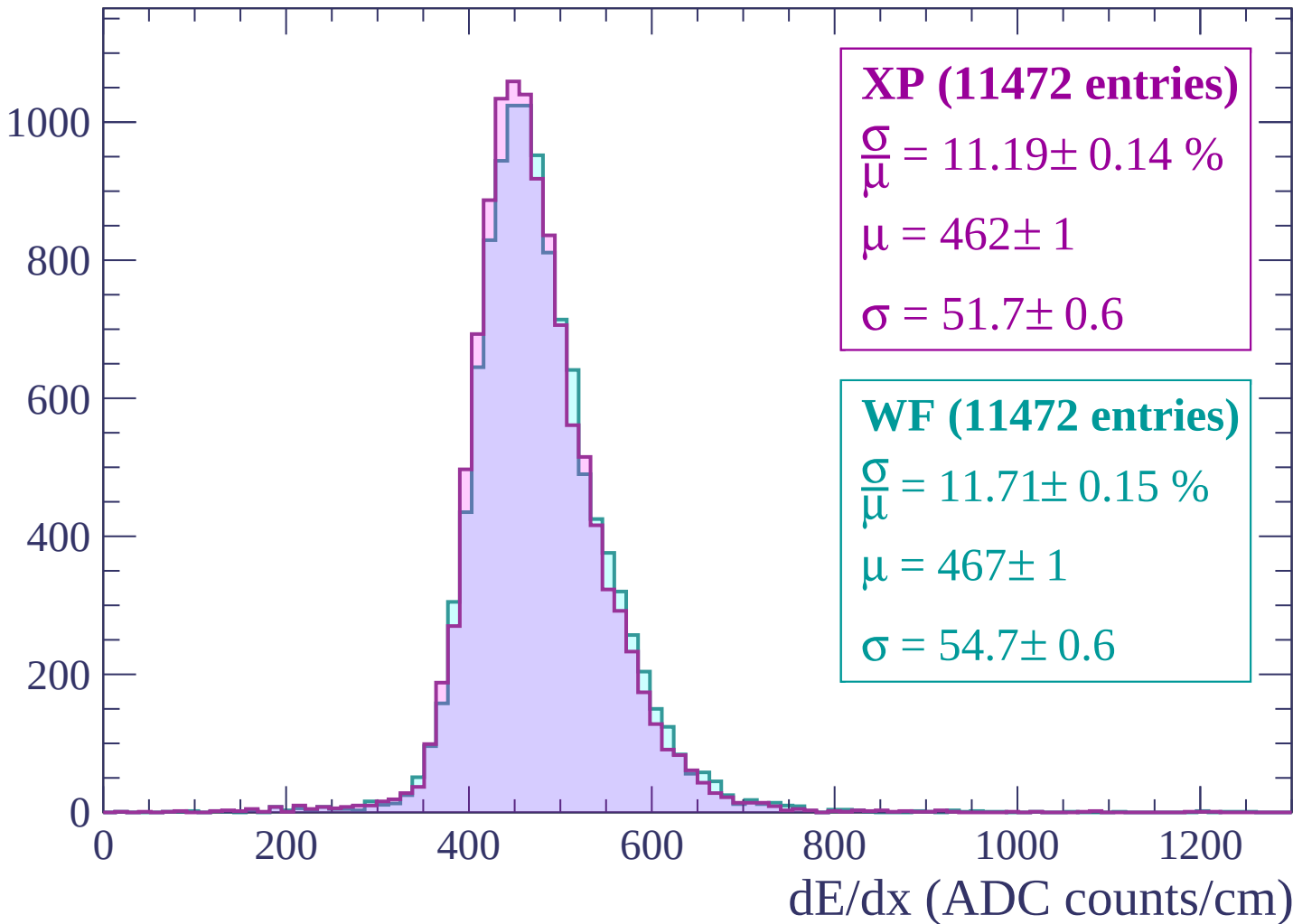
$\mu = 471 \pm 1$

$\sigma = 56.6 \pm 0.6$

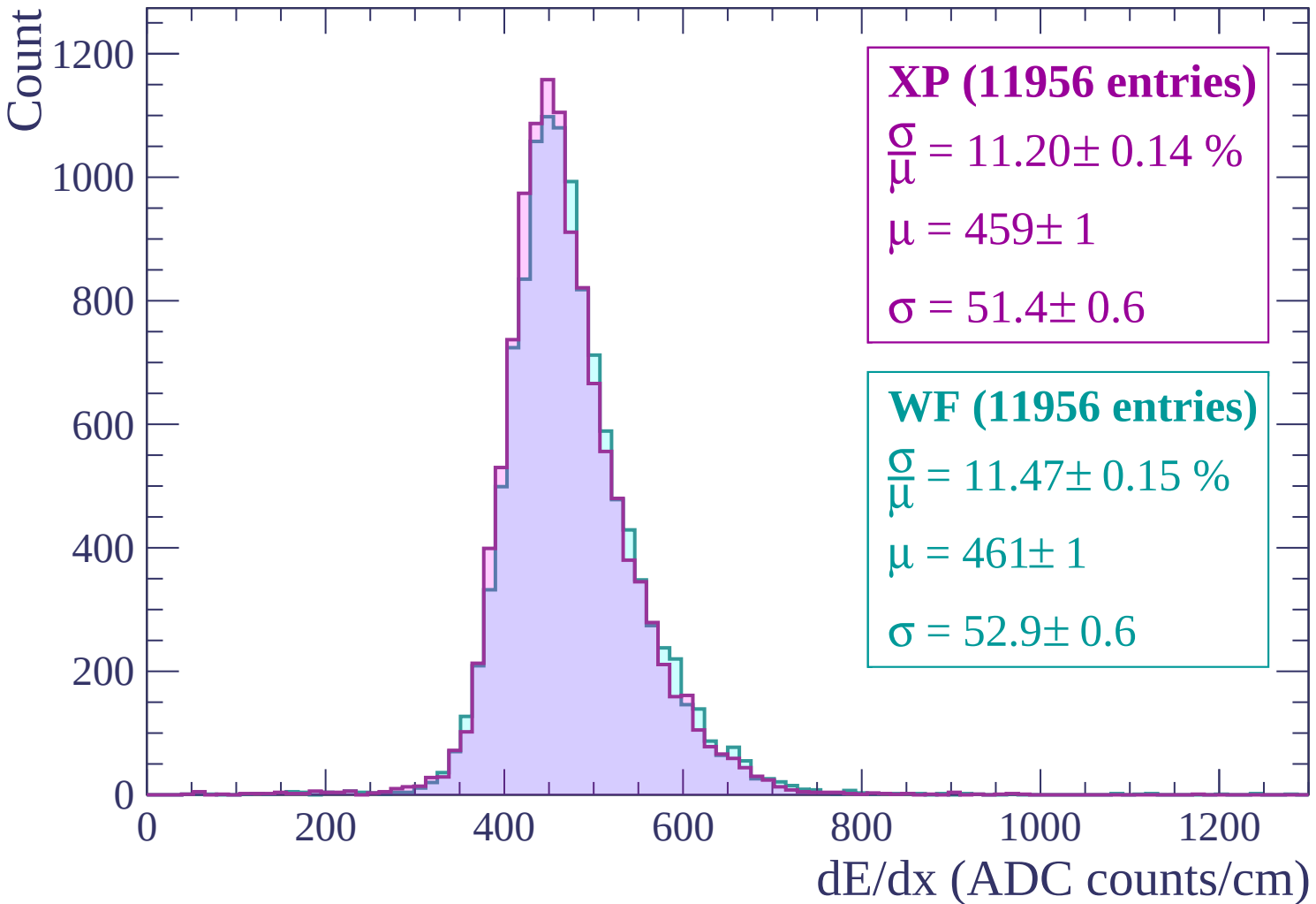
dE/dx (ADC counts/cm)

Energy loss | -1140 < p < -1080

Count



Energy loss | -1080 < p < -1020



Energy loss | $-1020 < p < -960$

Count

1200
1000
800
600
400
200
0

XP (12448 entries)

$$\frac{\sigma}{\mu} = 10.85 \pm 0.14 \%$$

$$\mu = 452 \pm 1$$

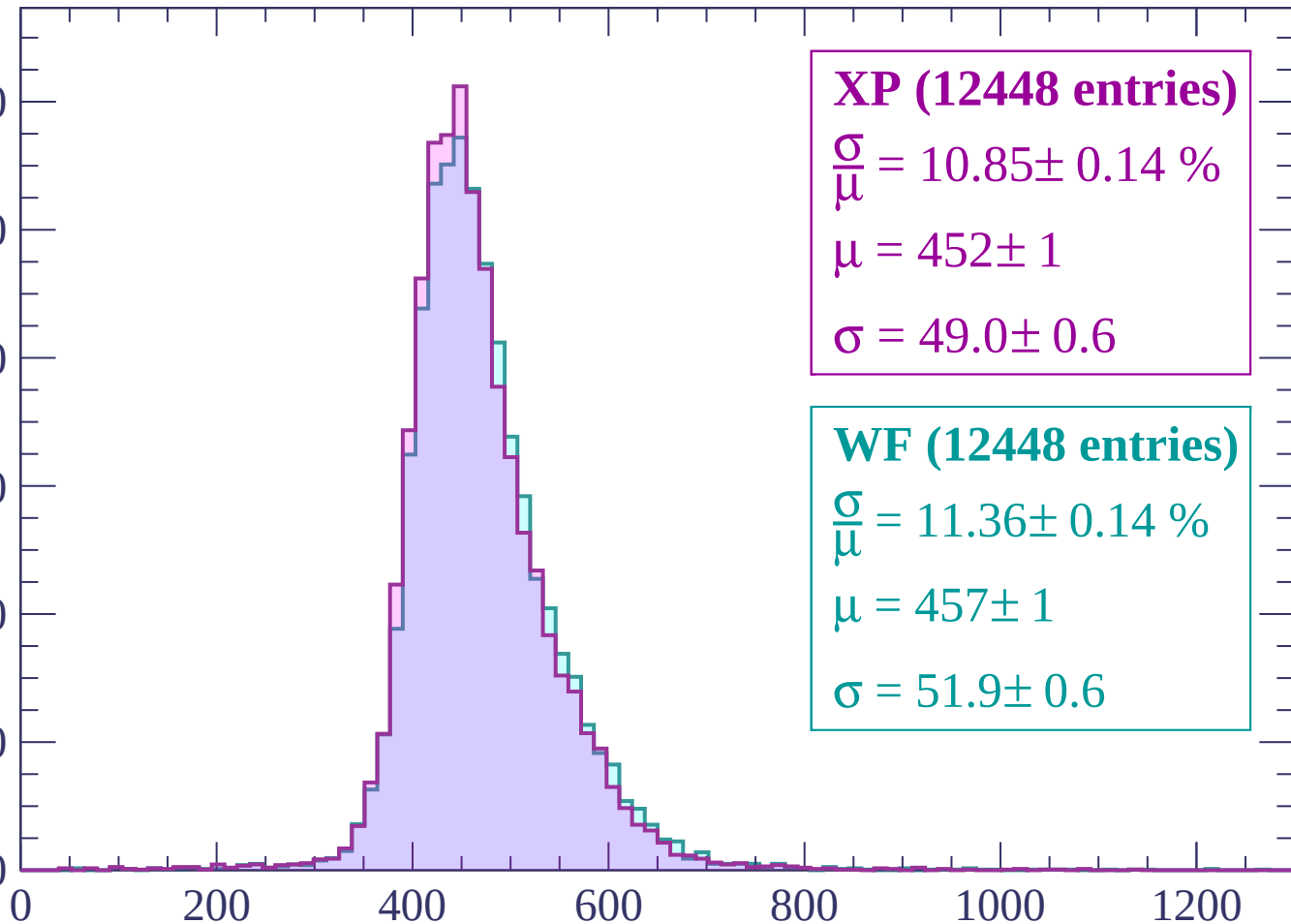
$$\sigma = 49.0 \pm 0.6$$

WF (12448 entries)

$$\frac{\sigma}{\mu} = 11.36 \pm 0.14 \%$$

$$\mu = 457 \pm 1$$

$$\sigma = 51.9 \pm 0.6$$



dE/dx (ADC counts/cm)

Energy loss | -960 < p < -900

Count

1200

1000

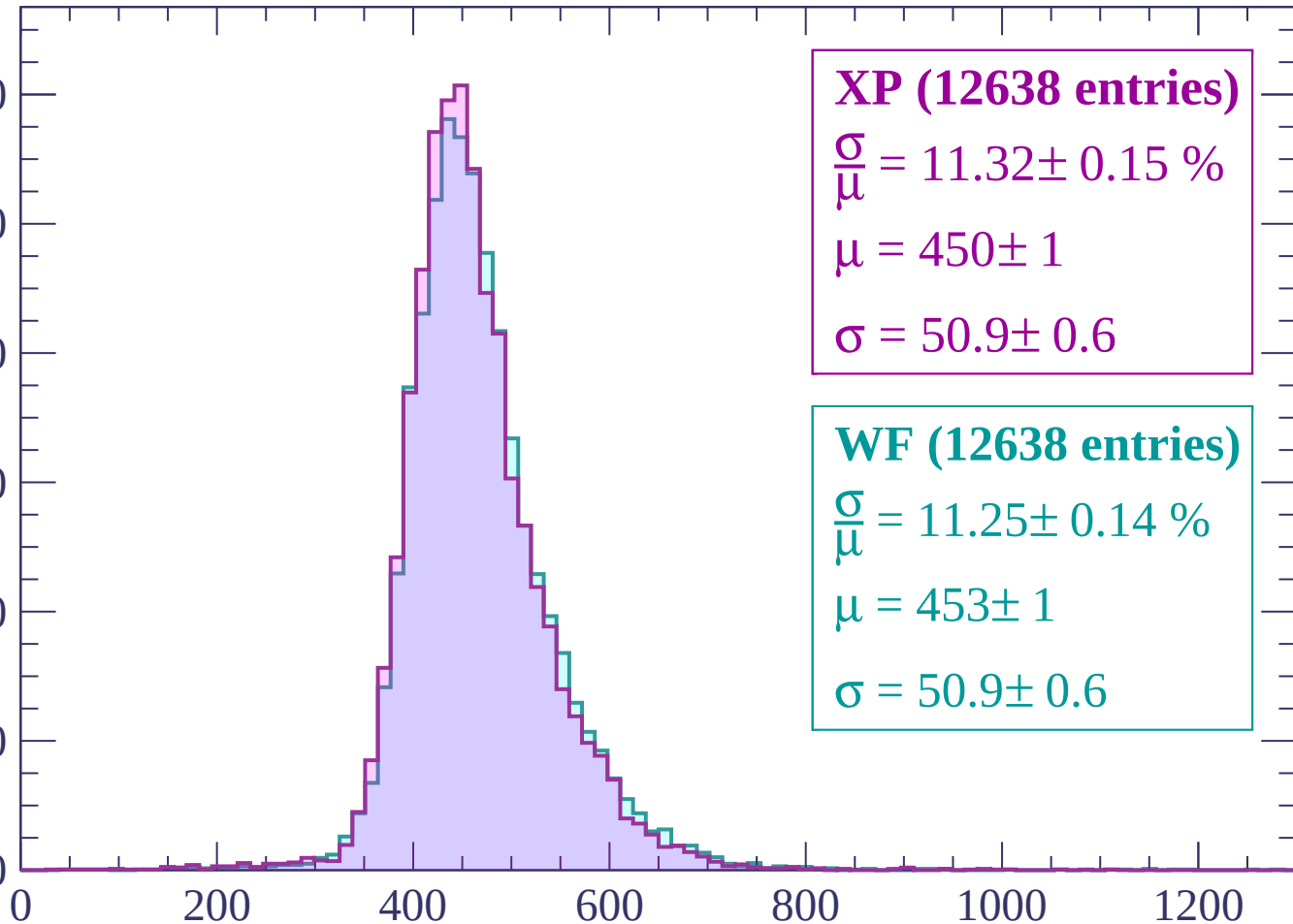
800

600

400

200

0



XP (12638 entries)

$\frac{\sigma}{\mu} = 11.32 \pm 0.15 \%$

$\mu = 450 \pm 1$

$\sigma = 50.9 \pm 0.6$

WF (12638 entries)

$\frac{\sigma}{\mu} = 11.25 \pm 0.14 \%$

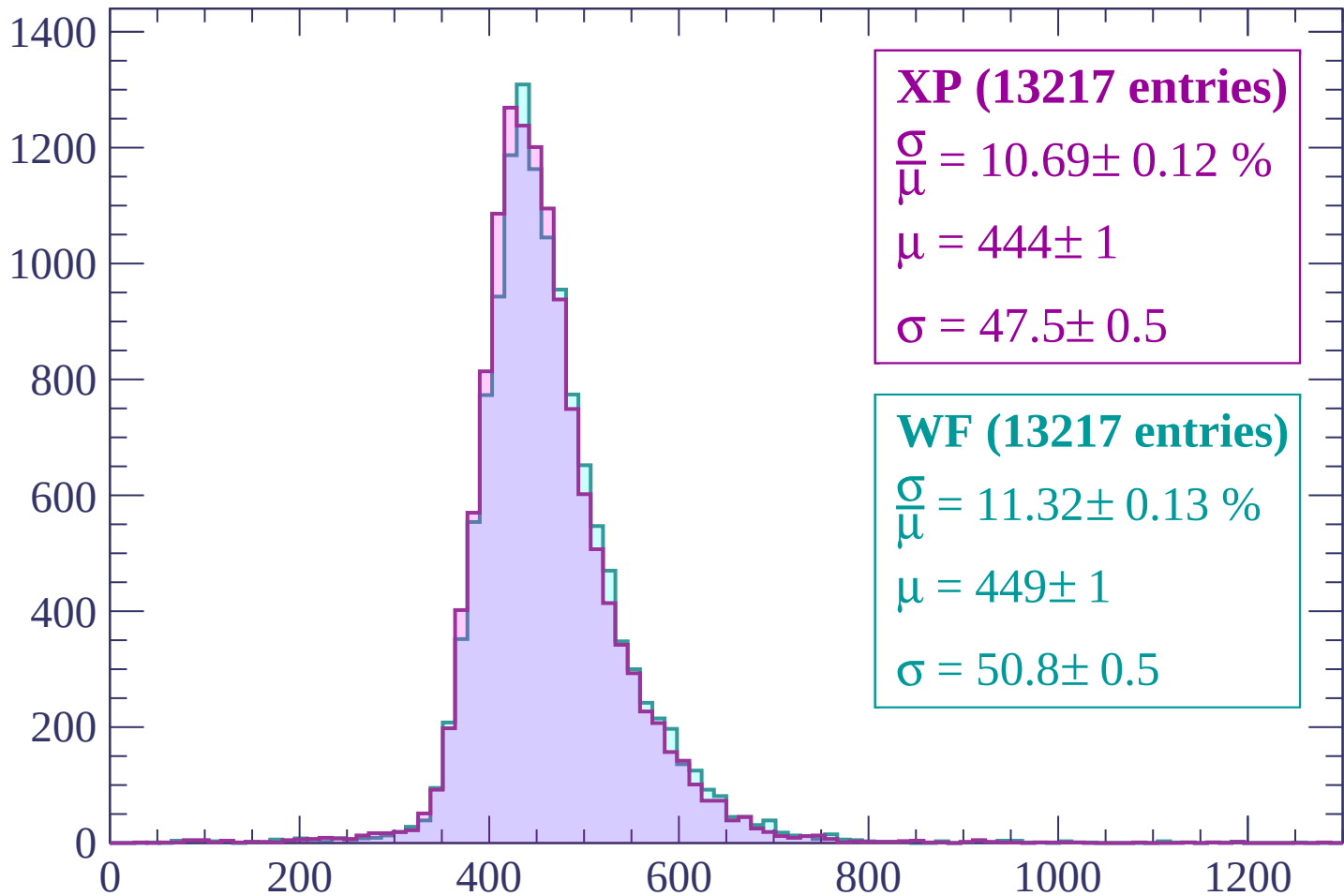
$\mu = 453 \pm 1$

$\sigma = 50.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (13217 entries)

$$\frac{\sigma}{\mu} = 10.69 \pm 0.12 \%$$

$$\mu = 444 \pm 1$$

$$\sigma = 47.5 \pm 0.5$$

WF (13217 entries)

$$\frac{\sigma}{\mu} = 11.32 \pm 0.13 \%$$

$$\mu = 449 \pm 1$$

$$\sigma = 50.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count

1400
1200
1000
800
600
400
200
0

XP (13801 entries)

$\frac{\sigma}{\mu} = 10.75 \pm 0.13 \%$

$\mu = 440 \pm 1$

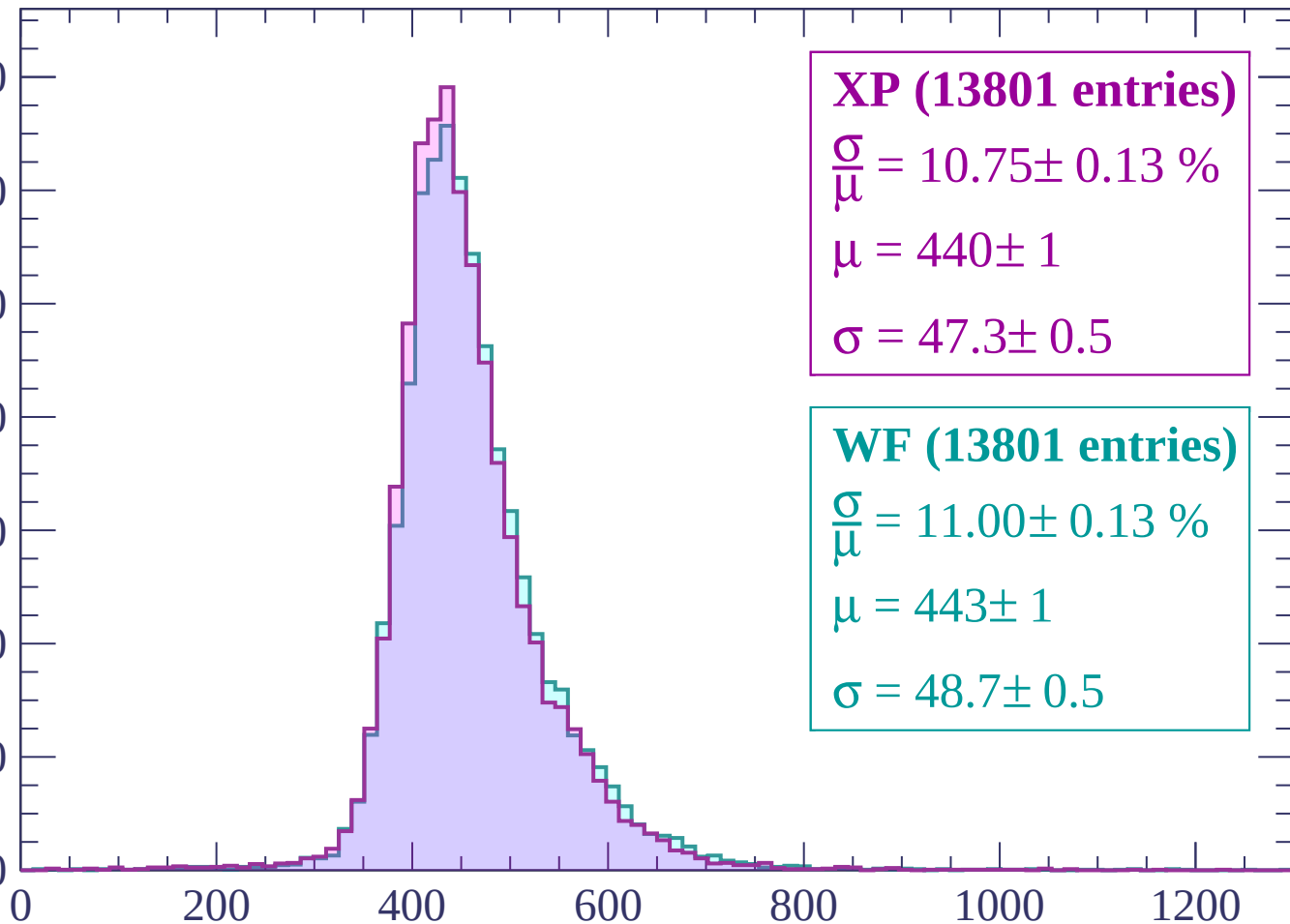
$\sigma = 47.3 \pm 0.5$

WF (13801 entries)

$\frac{\sigma}{\mu} = 11.00 \pm 0.13 \%$

$\mu = 443 \pm 1$

$\sigma = 48.7 \pm 0.5$



dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count

1400
1200
1000
800
600
400
200
0

XP (13982 entries)

$$\frac{\sigma}{\mu} = 10.53 \pm 0.12 \%$$

$$\mu = 434 \pm 1$$

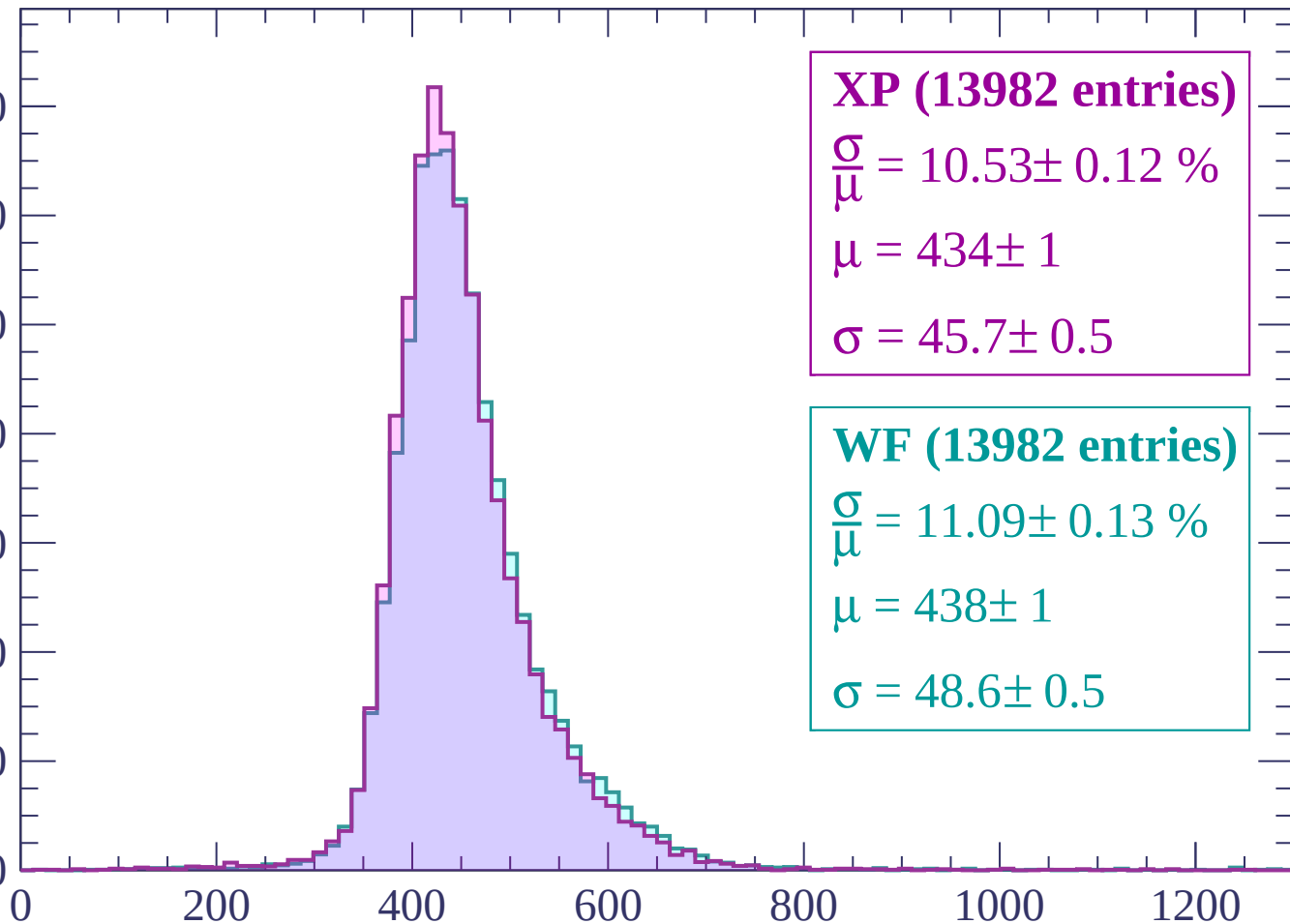
$$\sigma = 45.7 \pm 0.5$$

WF (13982 entries)

$$\frac{\sigma}{\mu} = 11.09 \pm 0.13 \%$$

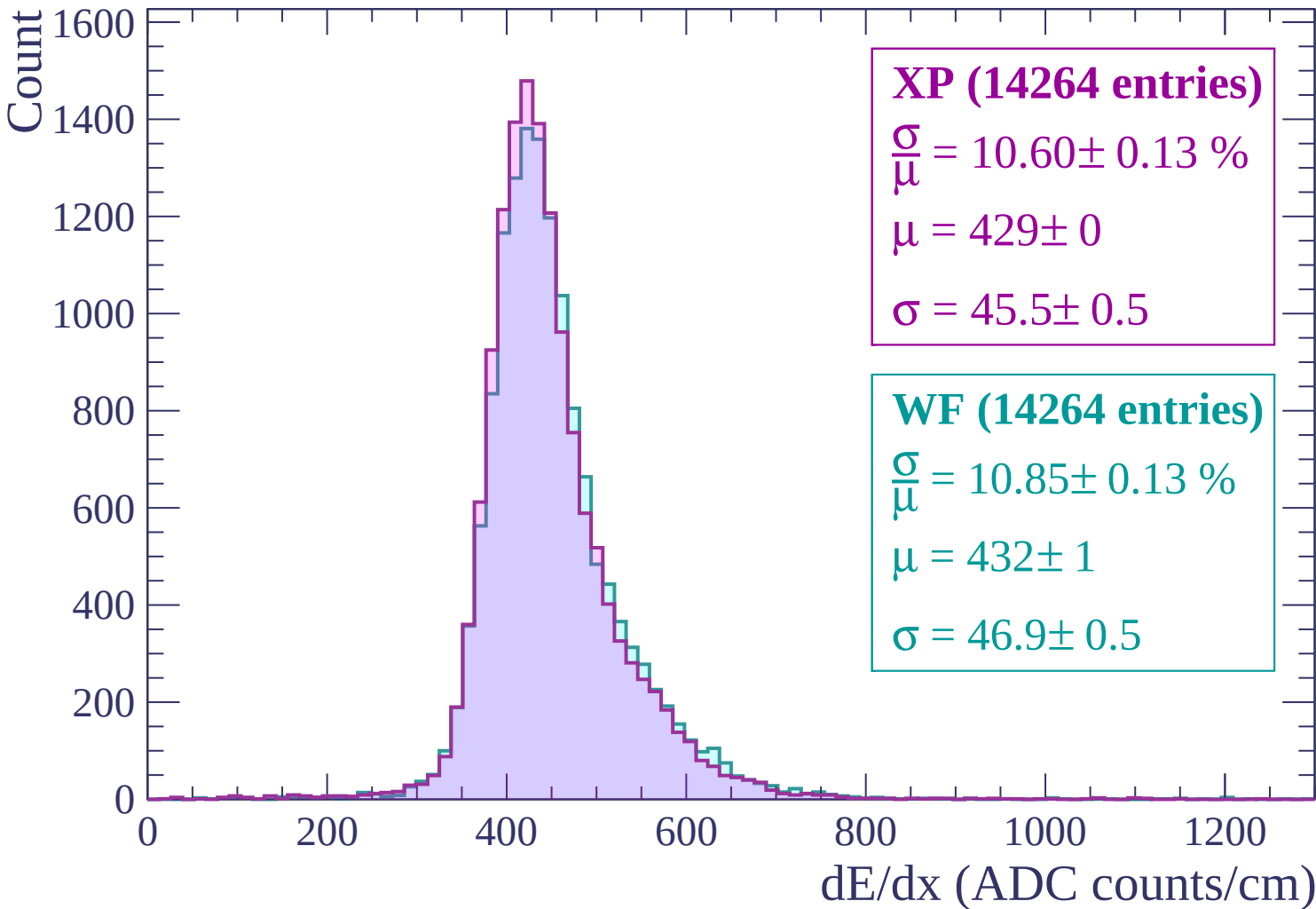
$$\mu = 438 \pm 1$$

$$\sigma = 48.6 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$



Energy loss | $-660 < p < -600$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14581 entries)

$$\frac{\sigma}{\mu} = 10.23 \pm 0.12 \%$$

$$\mu = 423 \pm 0$$

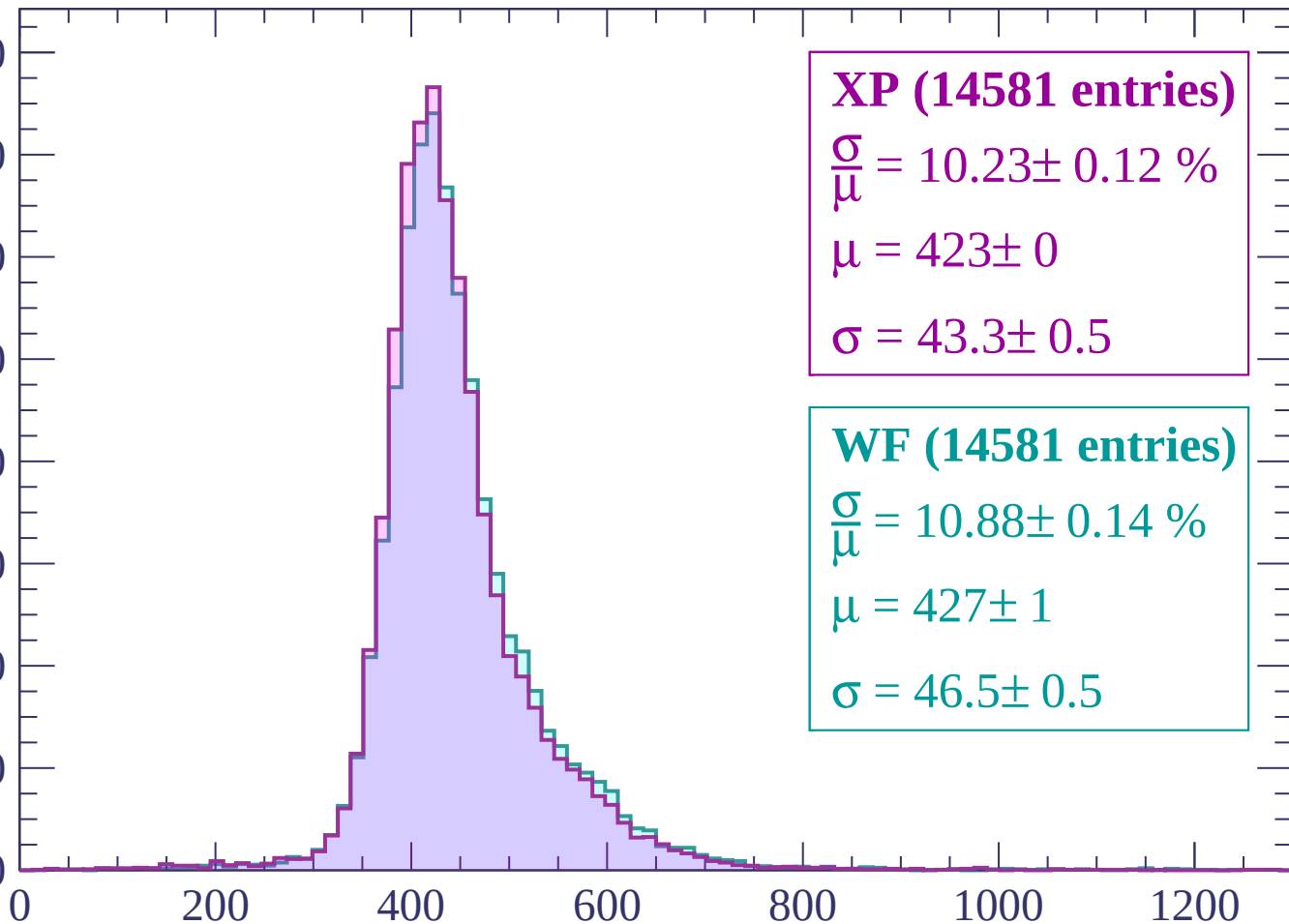
$$\sigma = 43.3 \pm 0.5$$

WF (14581 entries)

$$\frac{\sigma}{\mu} = 10.88 \pm 0.14 \%$$

$$\mu = 427 \pm 1$$

$$\sigma = 46.5 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14859 entries)

$$\frac{\sigma}{\mu} = 10.27 \pm 0.13 \%$$

$$\mu = 419 \pm 0$$

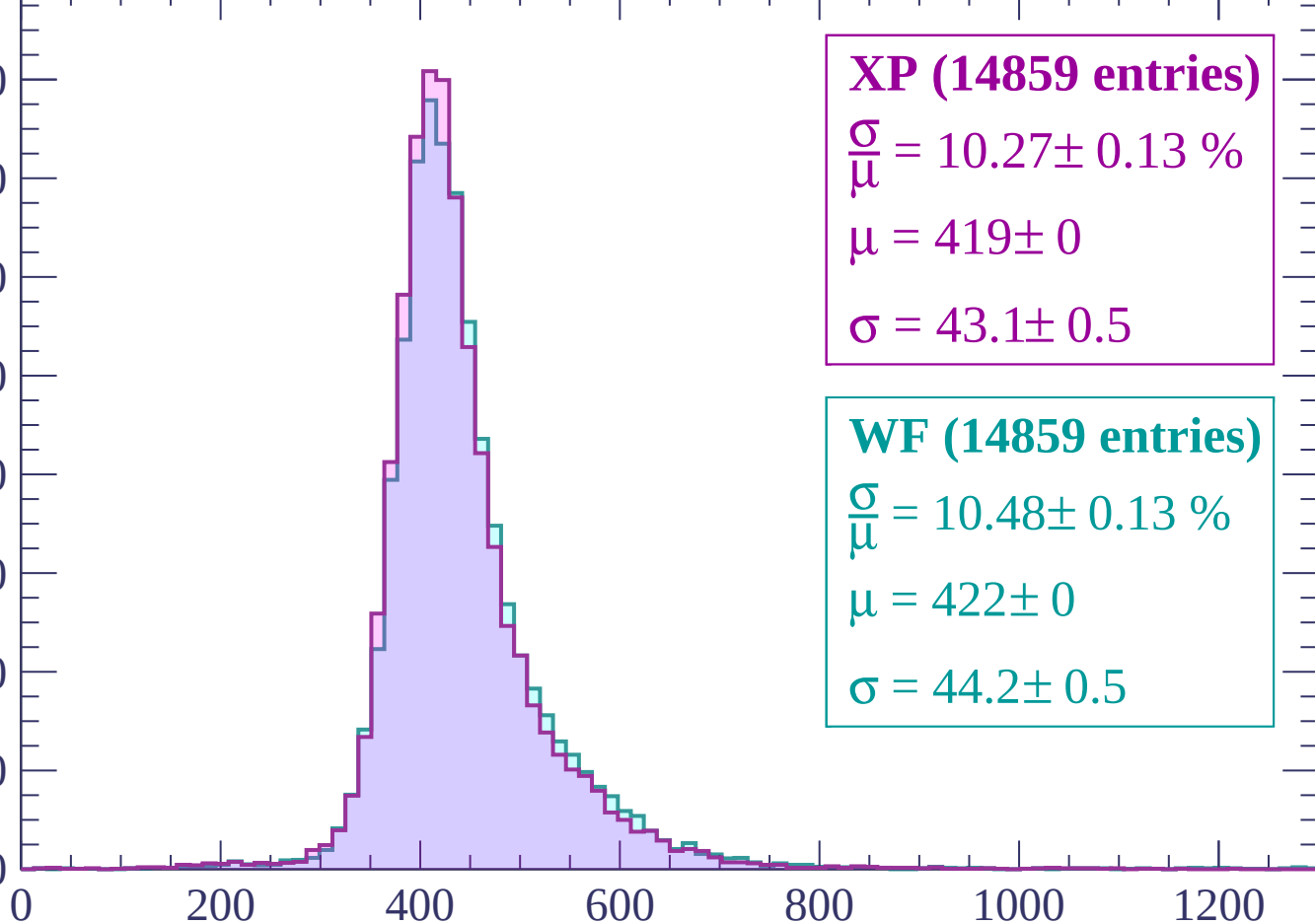
$$\sigma = 43.1 \pm 0.5$$

WF (14859 entries)

$$\frac{\sigma}{\mu} = 10.48 \pm 0.13 \%$$

$$\mu = 422 \pm 0$$

$$\sigma = 44.2 \pm 0.5$$

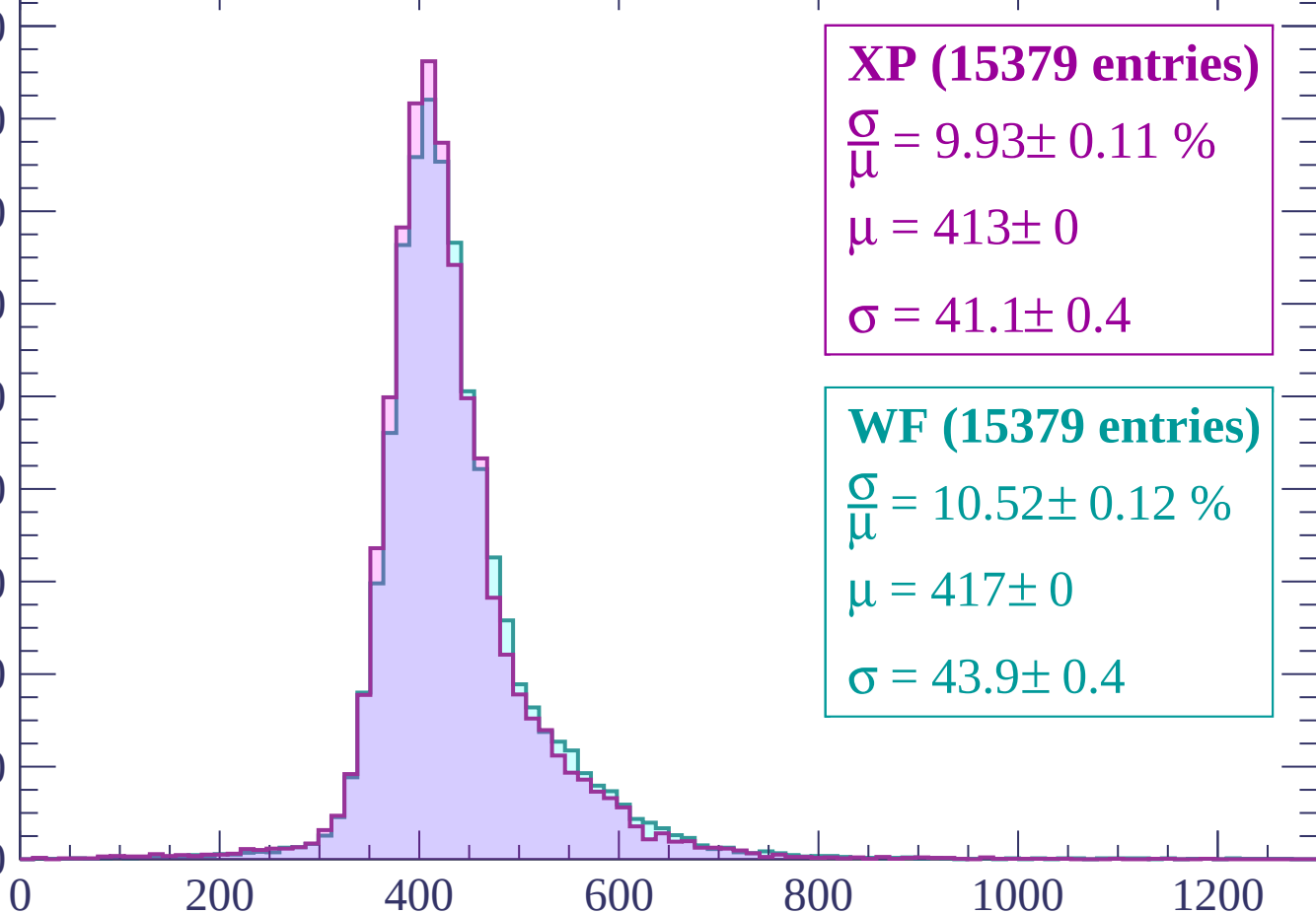


dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (15379 entries)

$$\frac{\sigma}{\mu} = 9.93 \pm 0.11 \%$$

$$\mu = 413 \pm 0$$

$$\sigma = 41.1 \pm 0.4$$

WF (15379 entries)

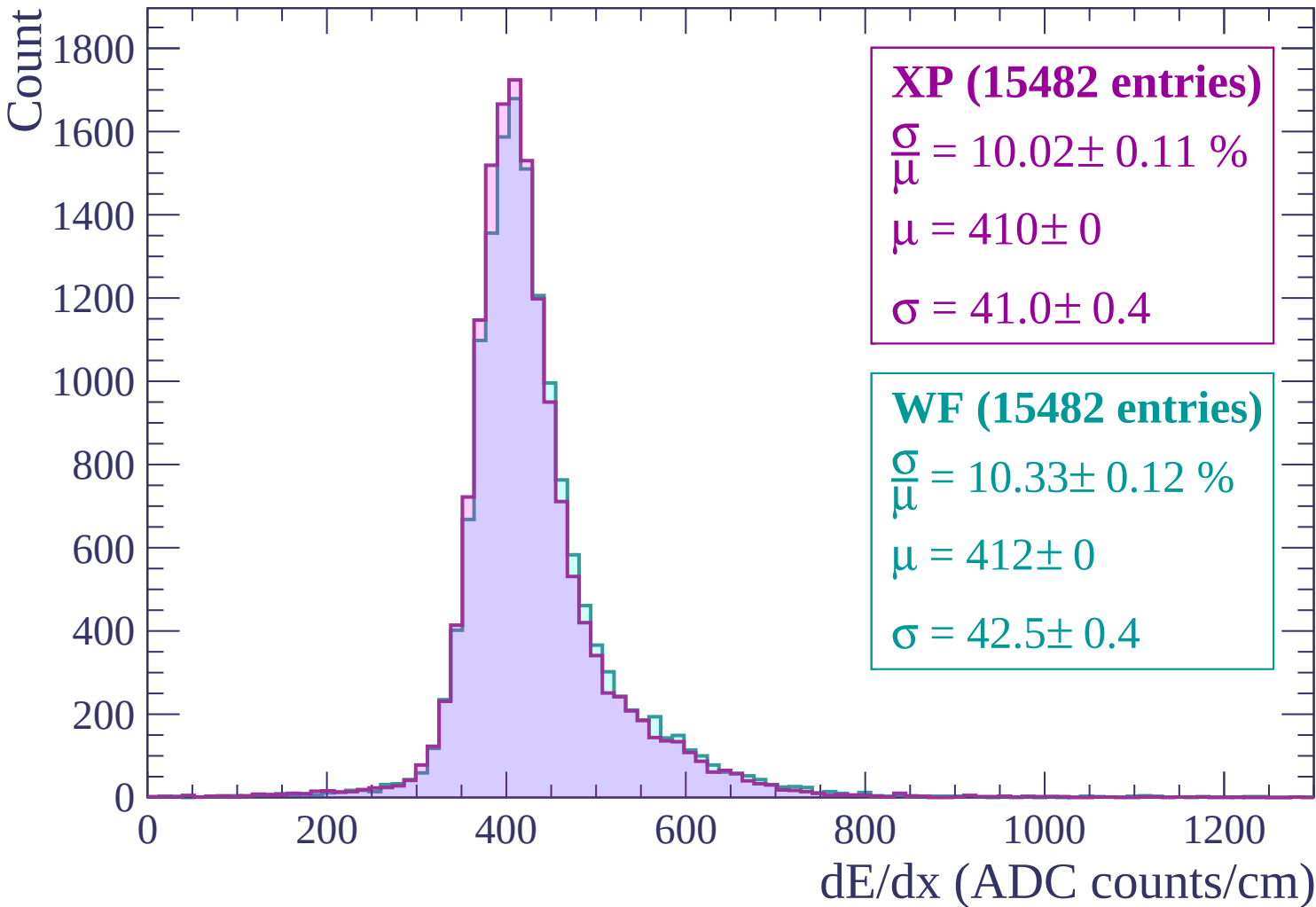
$$\frac{\sigma}{\mu} = 10.52 \pm 0.12 \%$$

$$\mu = 417 \pm 0$$

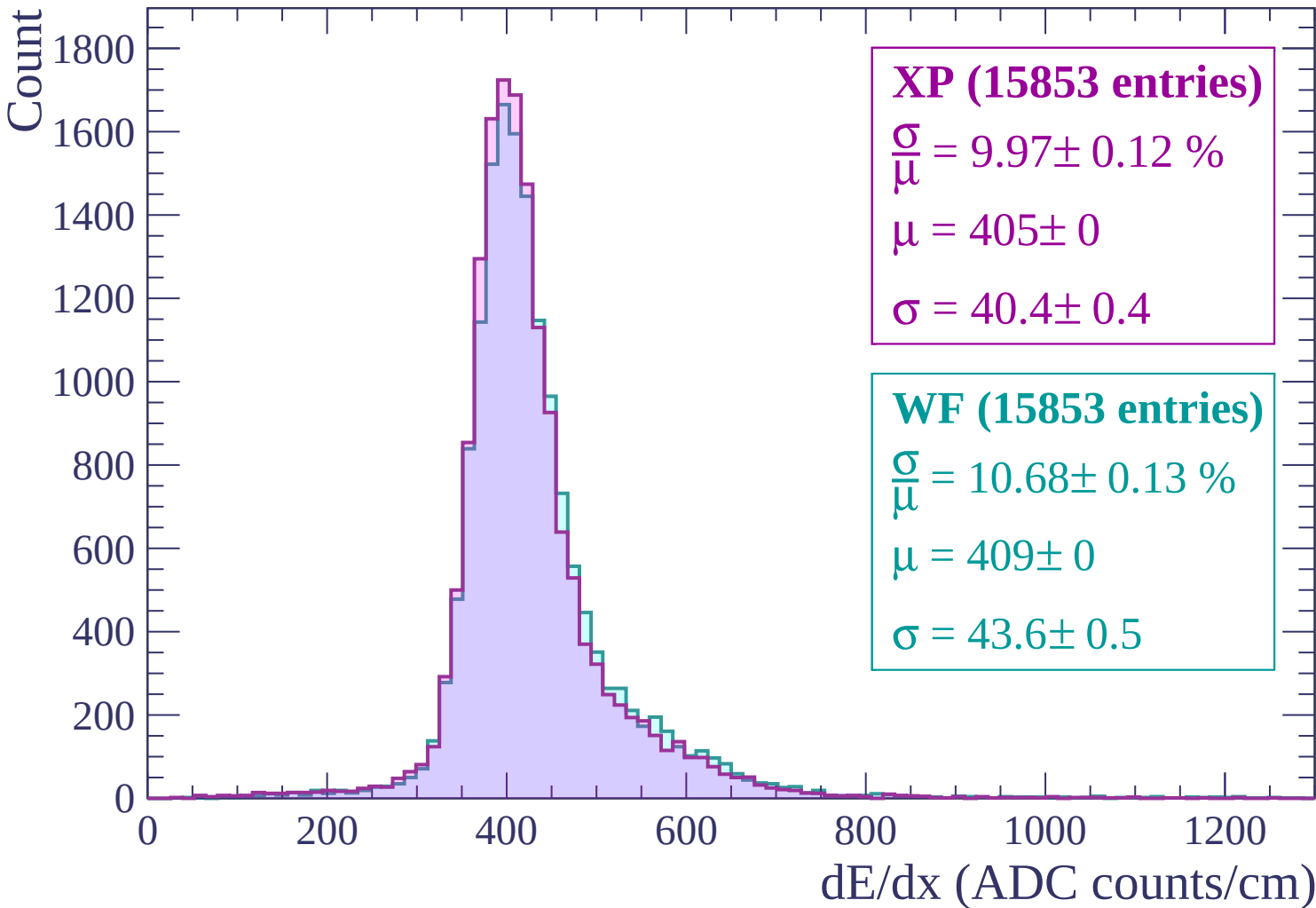
$$\sigma = 43.9 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$



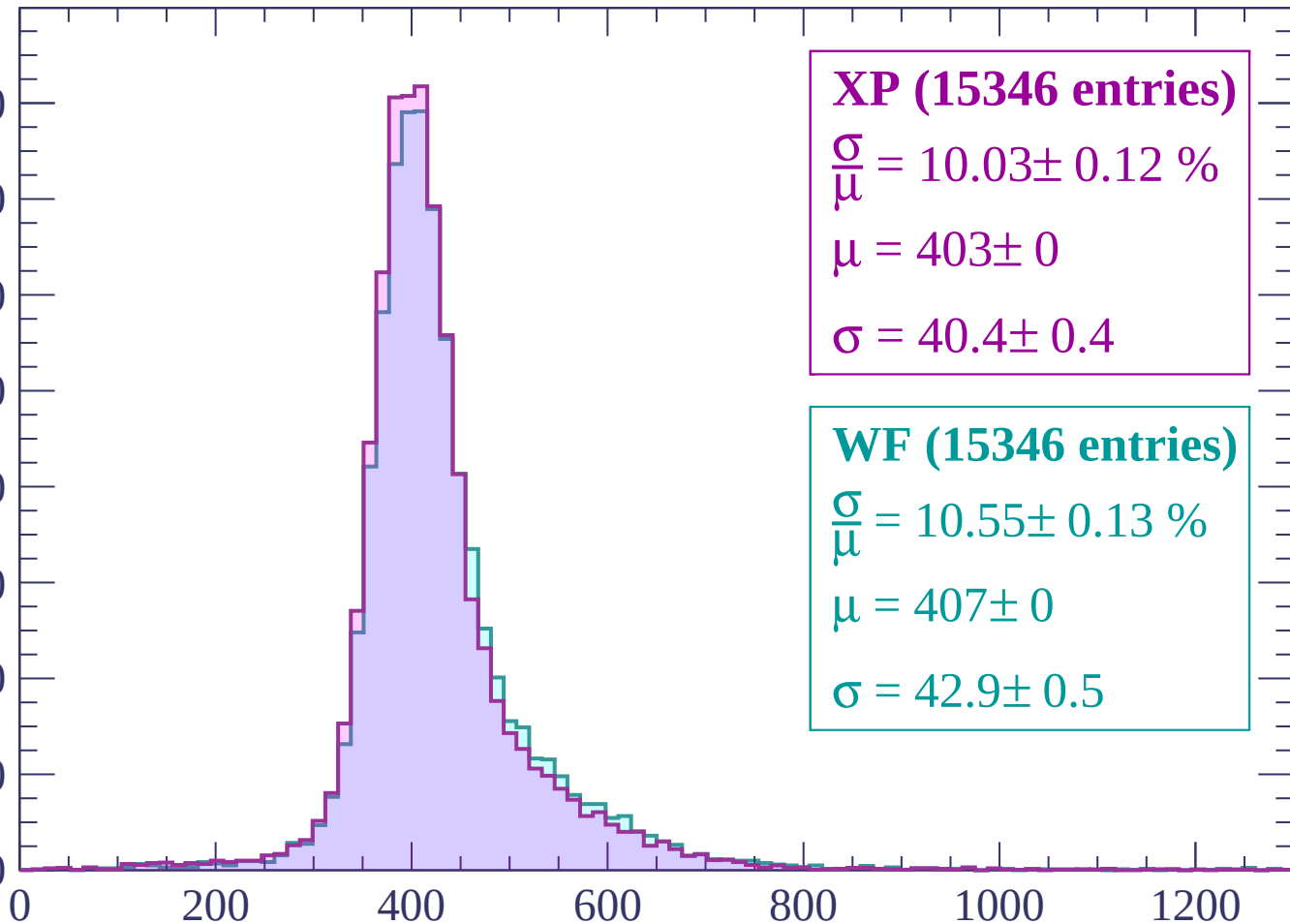
Energy loss | $-420 < p < -360$



Energy loss | $-360 < p < -300$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (15346 entries)

$\frac{\sigma}{\mu} = 10.03 \pm 0.12 \%$

$\mu = 403 \pm 0$

$\sigma = 40.4 \pm 0.4$

WF (15346 entries)

$\frac{\sigma}{\mu} = 10.55 \pm 0.13 \%$

$\mu = 407 \pm 0$

$\sigma = 42.9 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14996 entries)

$$\frac{\sigma}{\mu} = 10.08 \pm 0.13 \%$$

$$\mu = 403 \pm 0$$

$$\sigma = 40.6 \pm 0.5$$

WF (14996 entries)

$$\frac{\sigma}{\mu} = 10.67 \pm 0.13 \%$$

$$\mu = 407 \pm 0$$

$$\sigma = 43.4 \pm 0.5$$

0

200

400

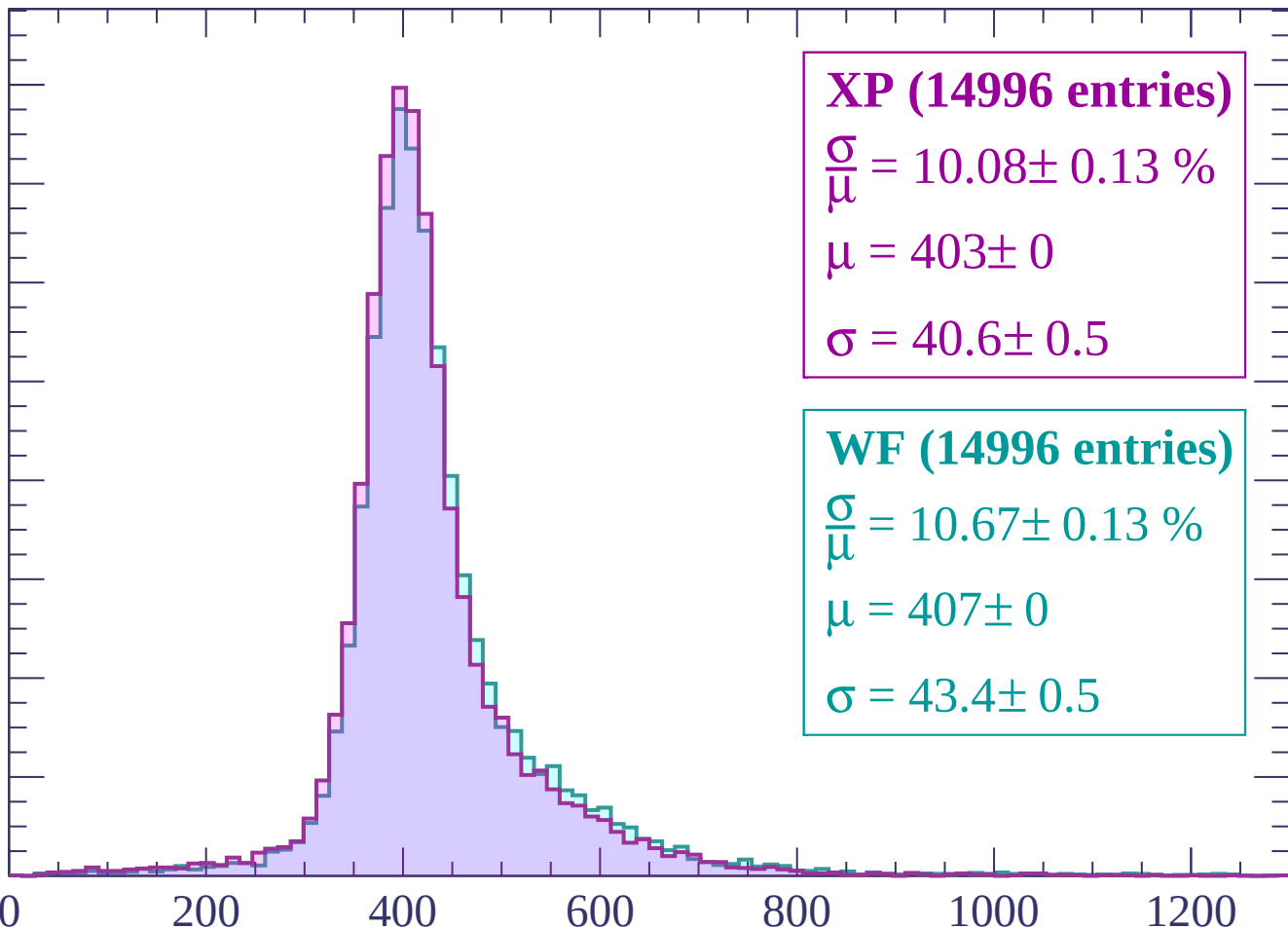
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | -240 < p < -180

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (13385 entries)

$\frac{\sigma}{\mu} = 11.46 \pm 0.16 \%$

$\mu = 415 \pm 1$

$\sigma = 47.5 \pm 0.6$

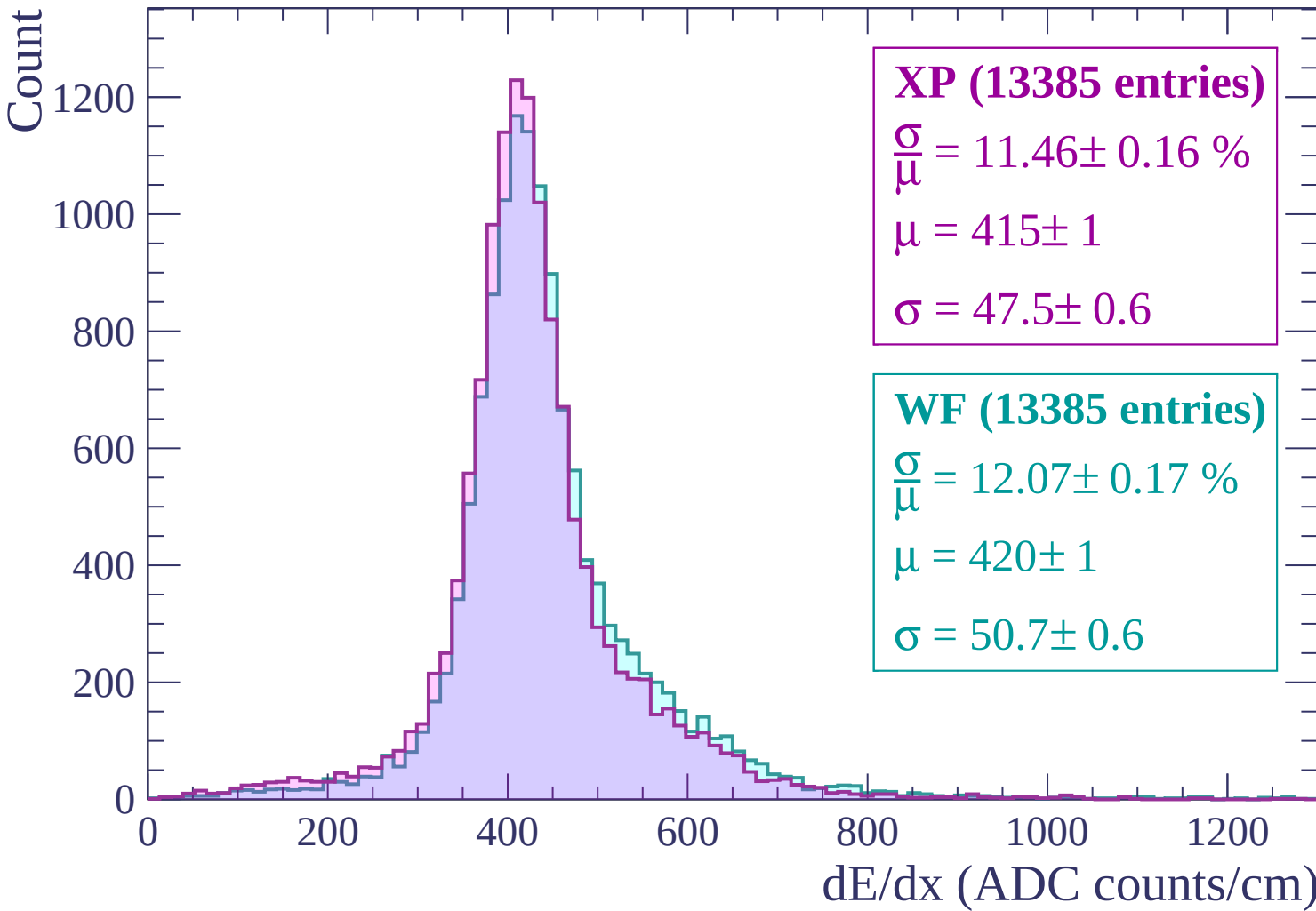
WF (13385 entries)

$\frac{\sigma}{\mu} = 12.07 \pm 0.17 \%$

$\mu = 420 \pm 1$

$\sigma = 50.7 \pm 0.6$

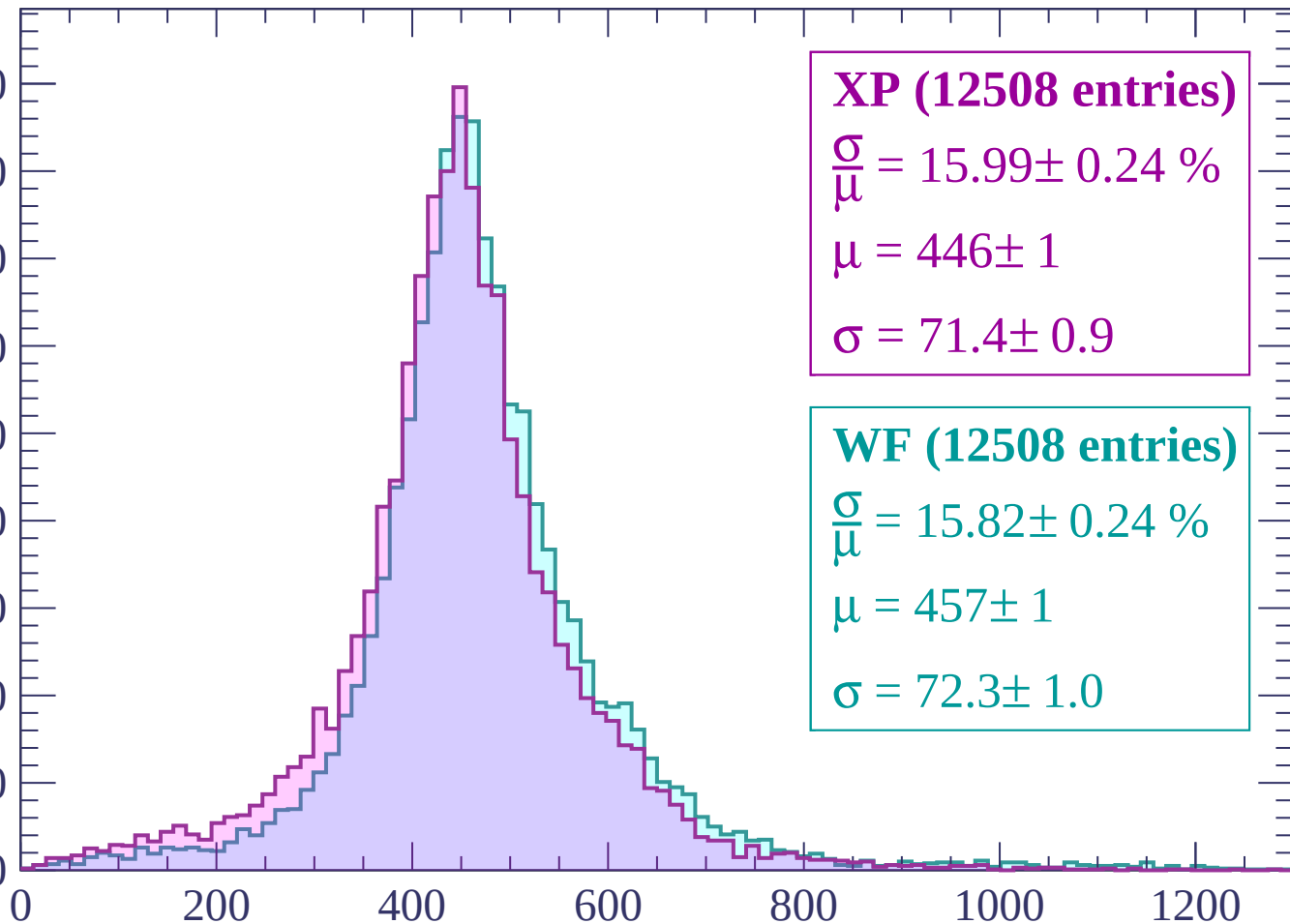
dE/dx (ADC counts/cm)



Energy loss | -180 < p < -120

Count

900
800
700
600
500
400
300
200
100
0



XP (12508 entries)

$$\frac{\sigma}{\mu} = 15.99 \pm 0.24 \%$$

$$\mu = 446 \pm 1$$

$$\sigma = 71.4 \pm 0.9$$

WF (12508 entries)

$$\frac{\sigma}{\mu} = 15.82 \pm 0.24 \%$$

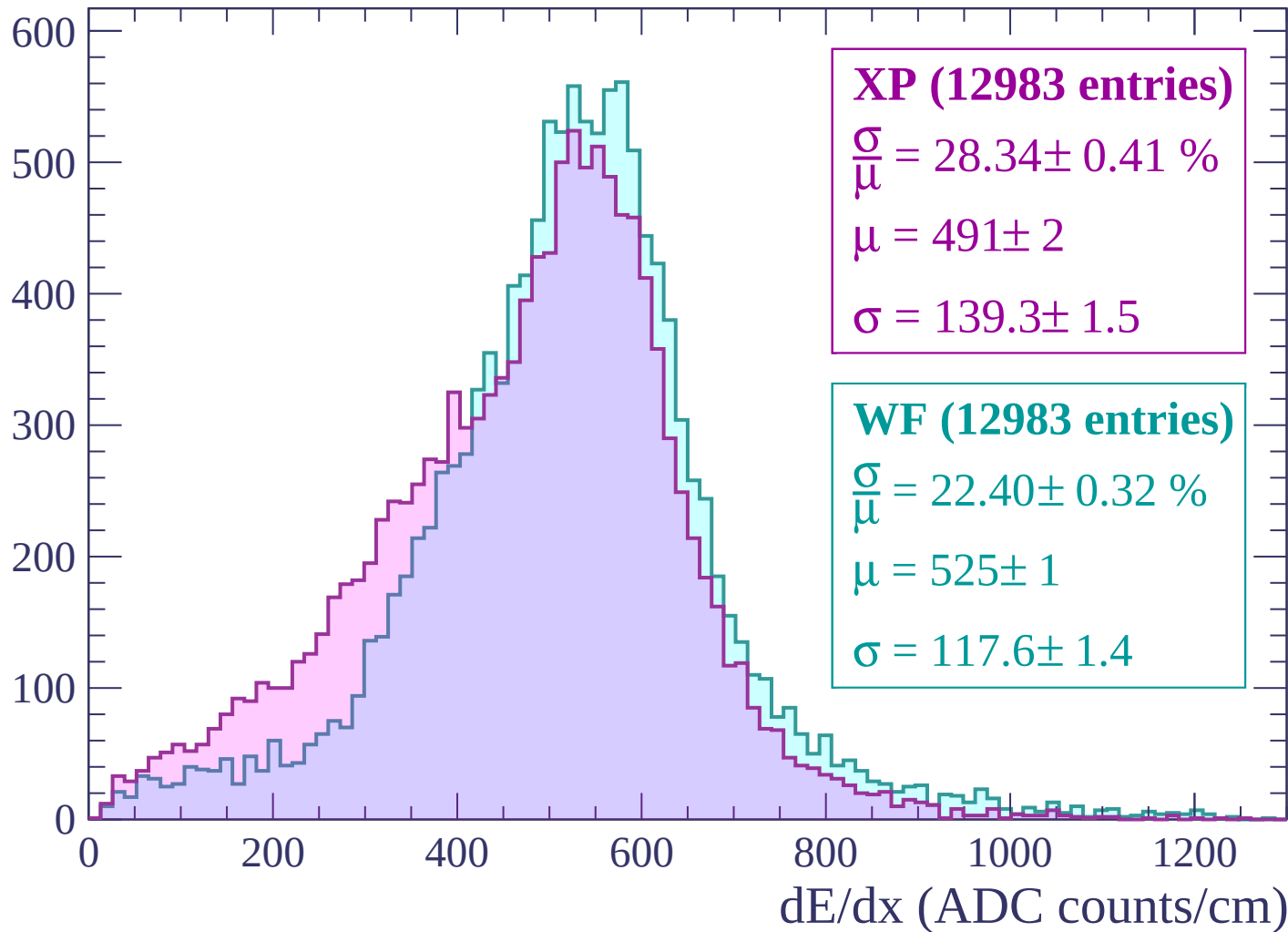
$$\mu = 457 \pm 1$$

$$\sigma = 72.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (26830 entries)

$$\frac{\sigma}{\mu} = 12.78 \pm 0.13 \%$$

$$\mu = 544 \pm 1$$

$$\sigma = 69.5 \pm 0.6$$

WF (26830 entries)

$$\frac{\sigma}{\mu} = 13.46 \pm 0.13 \%$$

$$\mu = 547 \pm 1$$

$$\sigma = 73.6 \pm 0.7$$

0

200

400

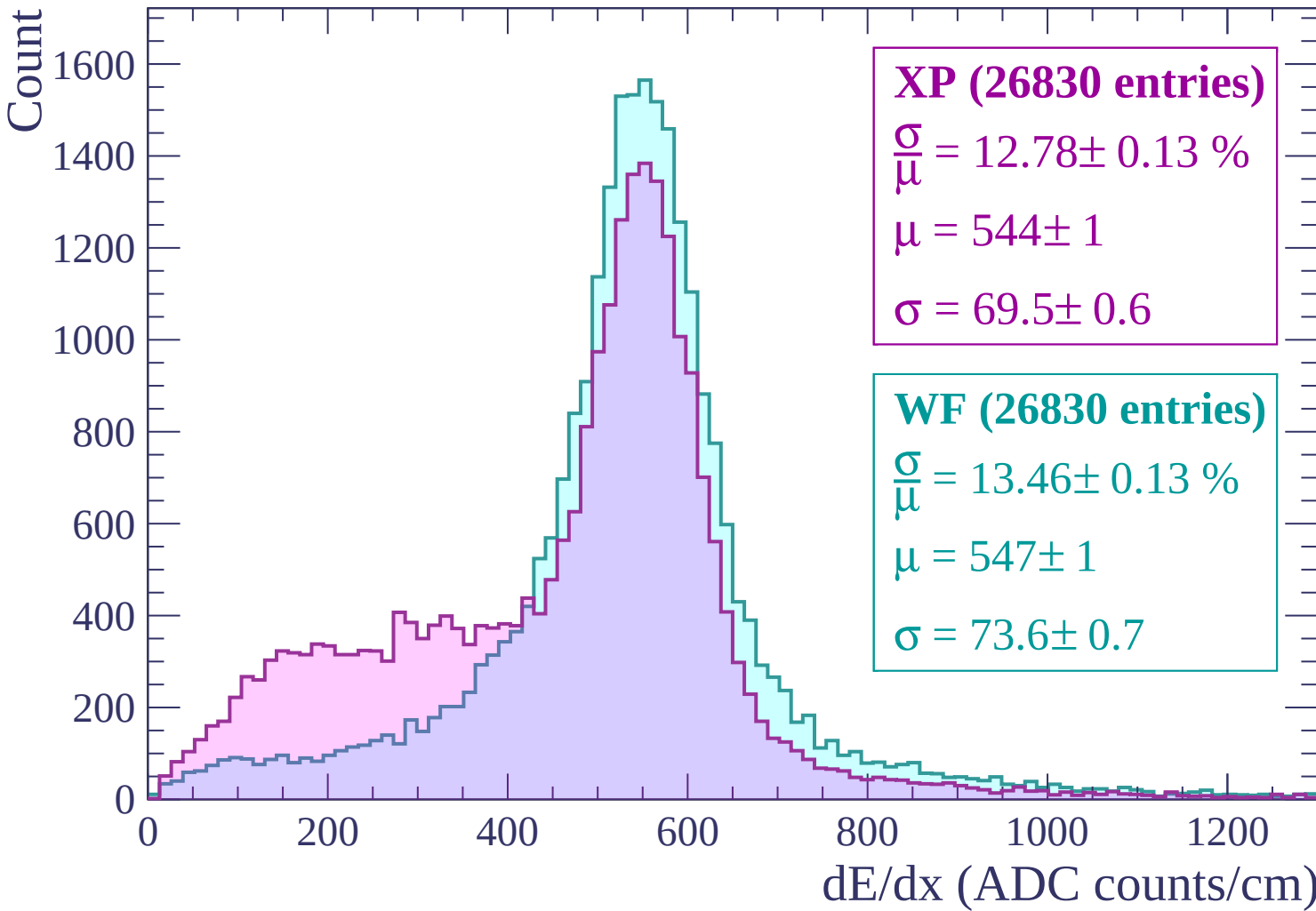
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $0 < p < 60$

Count

4000
3500
3000
2500
2000
1500
1000
500
0

XP (68308 entries)

$\frac{\sigma}{\mu} = 18.00 \pm 0.12 \%$

$\mu = 485 \pm 0$

$\sigma = 87.2 \pm 0.5$

WF (68308 entries)

$\frac{\sigma}{\mu} = 15.85 \pm 0.10 \%$

$\mu = 501 \pm 0$

$\sigma = 79.3 \pm 0.4$

0

200

400

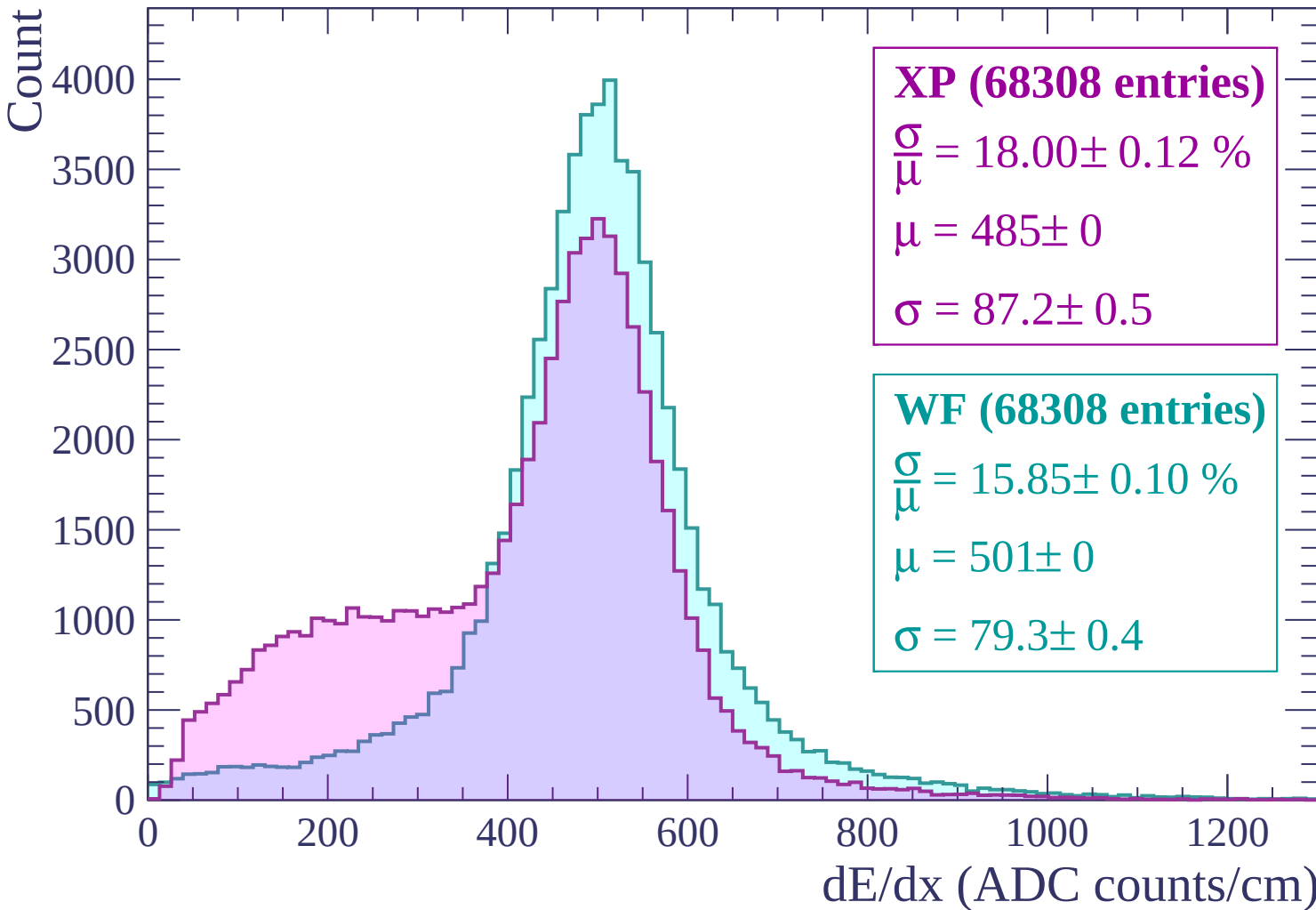
600

800

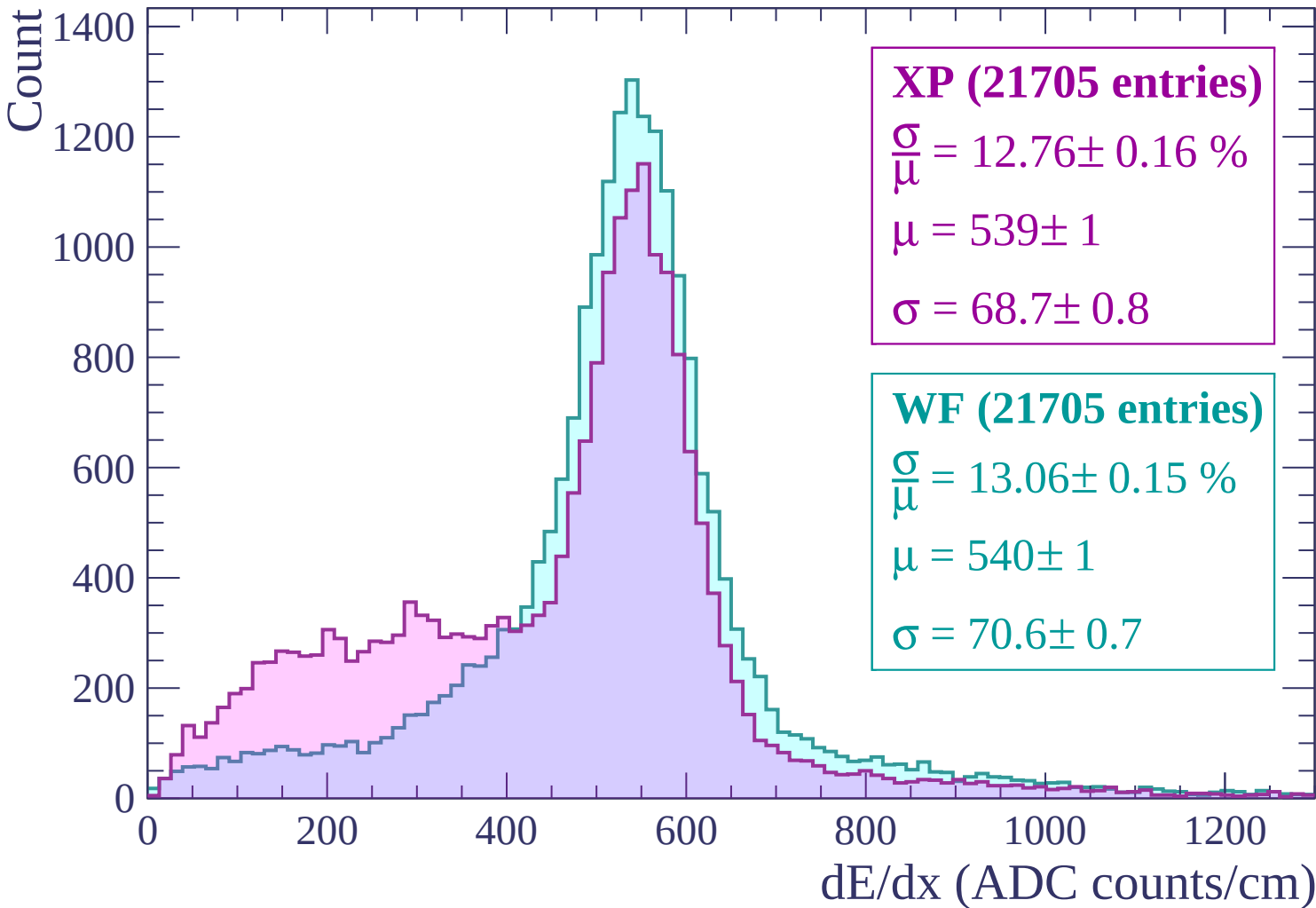
1000

1200

dE/dx (ADC counts/cm)

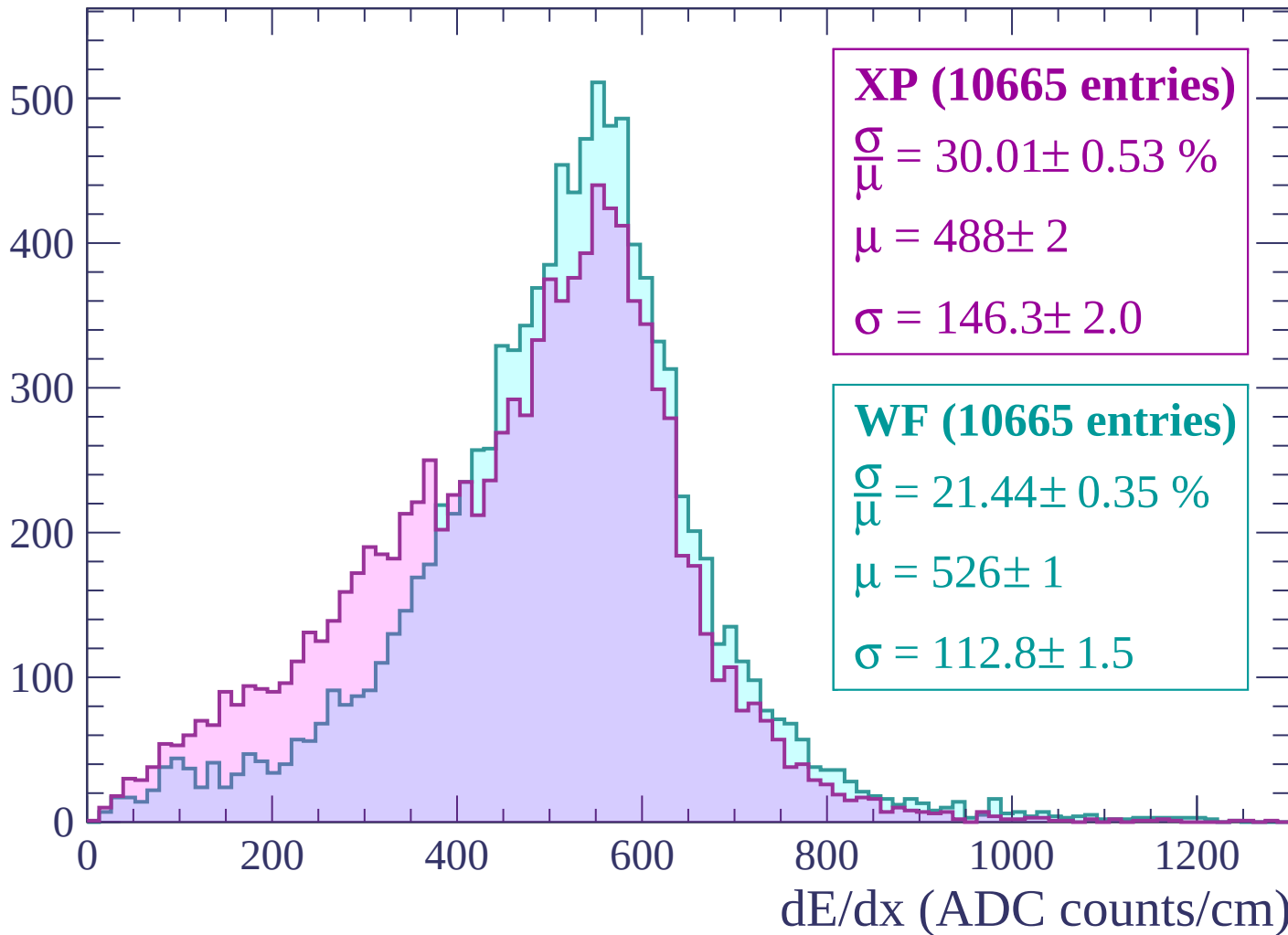


Energy loss | $60 < p < 120$



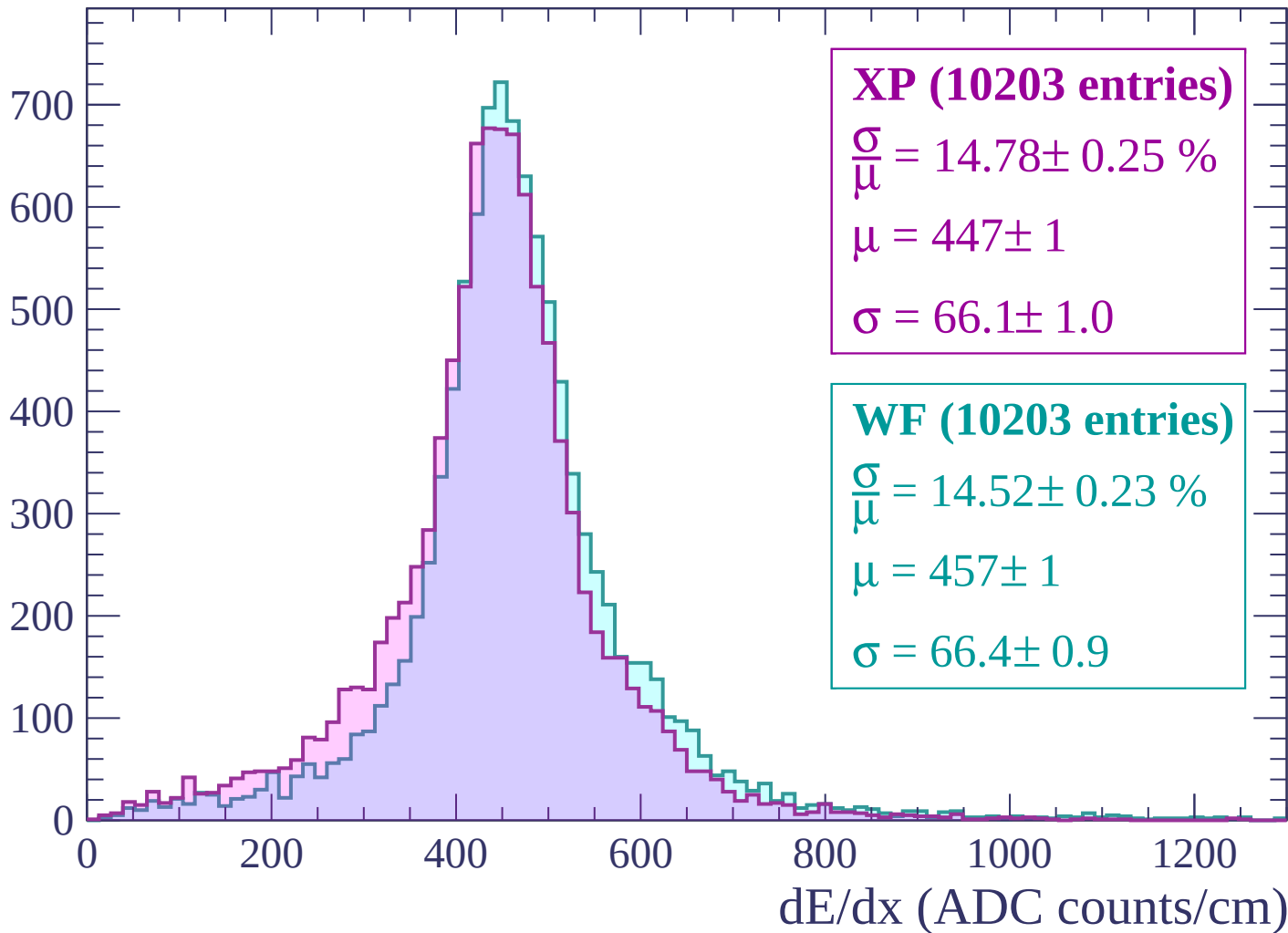
Energy loss | $120 < p < 180$

Count



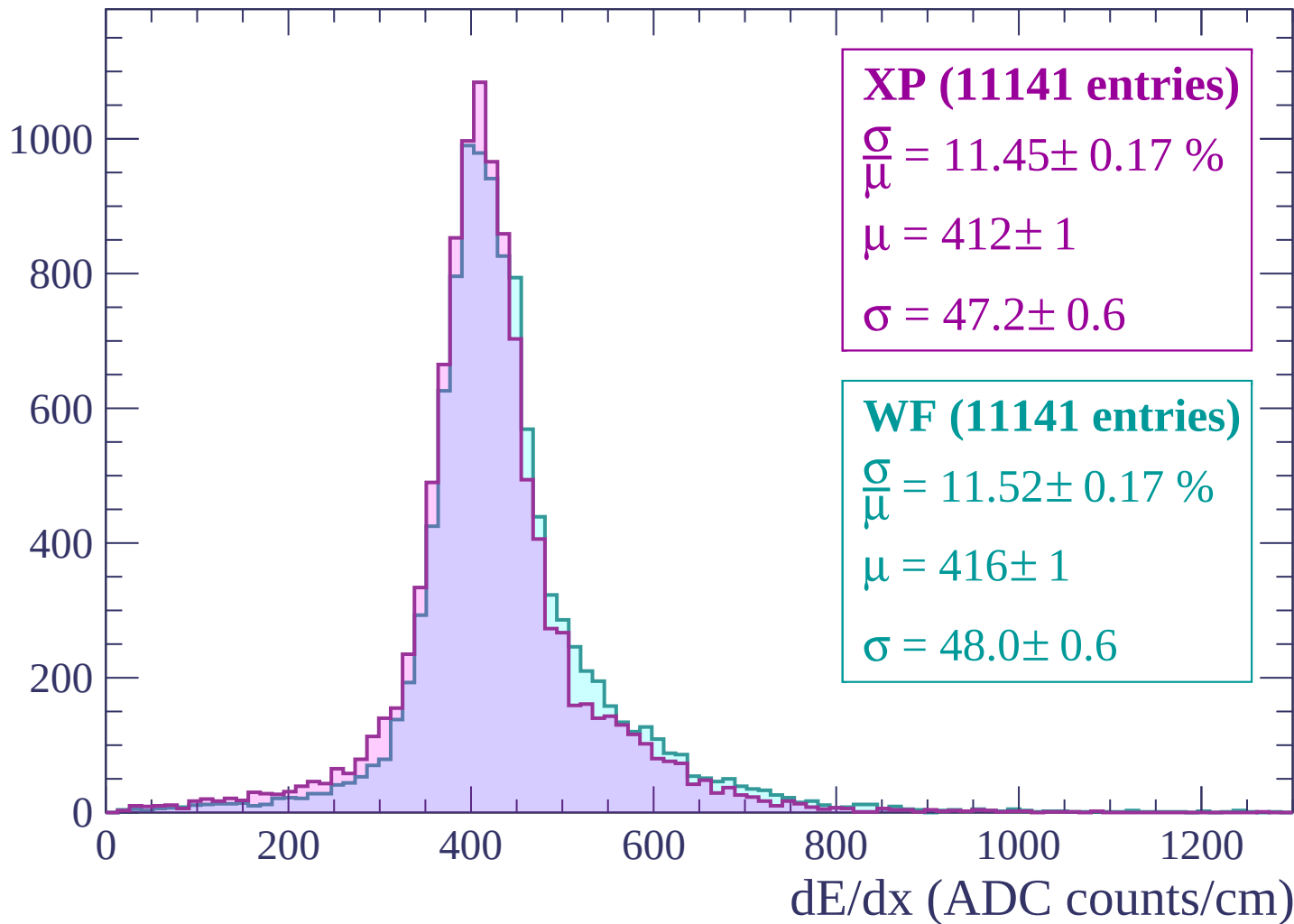
Energy loss | $180 < p < 240$

Count

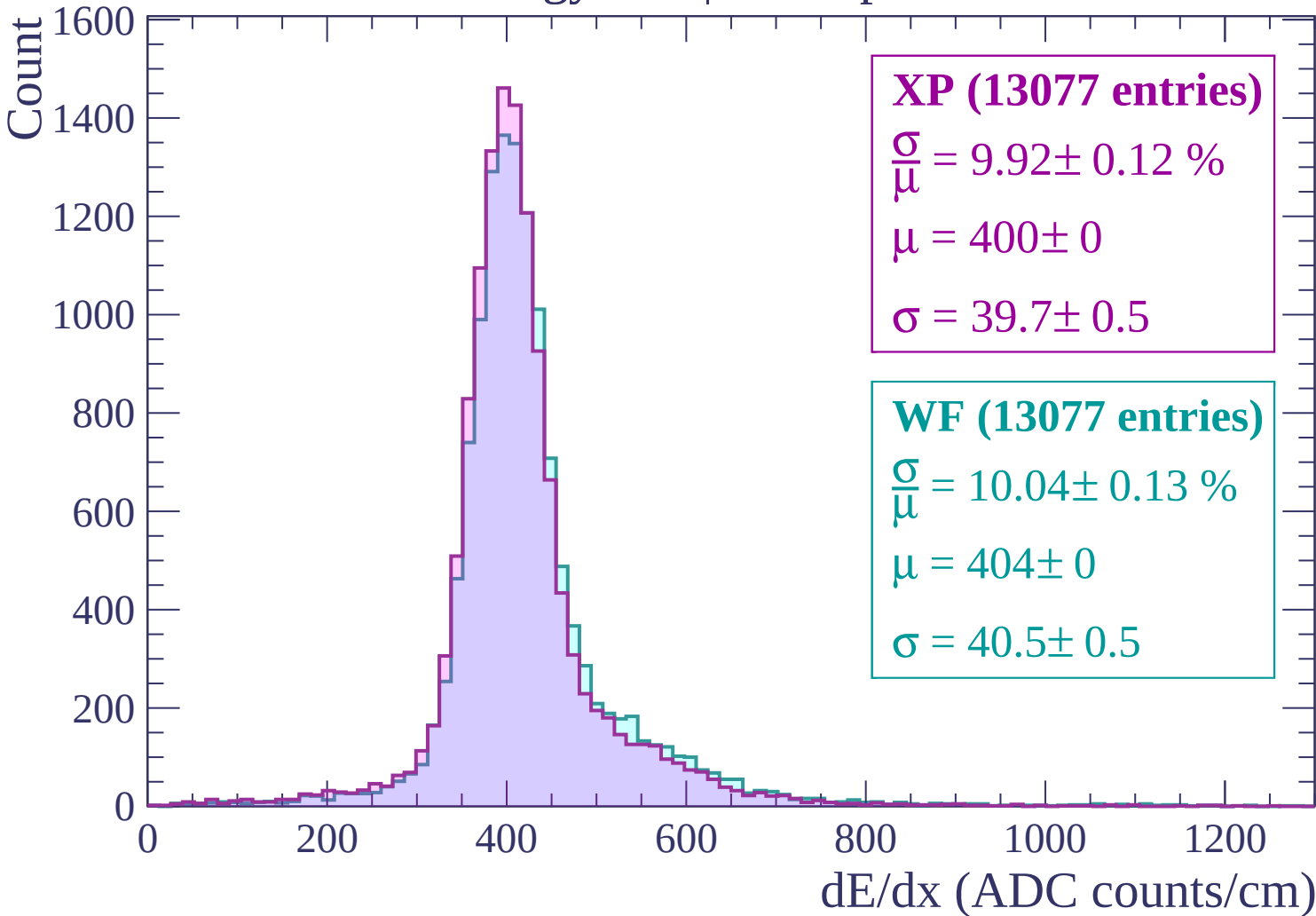


Energy loss | $240 < p < 300$

Count



Energy loss | $300 < p < 360$



Energy loss | $360 < p < 420$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13532 entries)

$$\frac{\sigma}{\mu} = 9.67 \pm 0.13 \%$$

$$\mu = 397 \pm 0$$

$$\sigma = 38.4 \pm 0.5$$

WF (13532 entries)

$$\frac{\sigma}{\mu} = 10.21 \pm 0.13 \%$$

$$\mu = 401 \pm 0$$

$$\sigma = 41.0 \pm 0.5$$

0

200

400

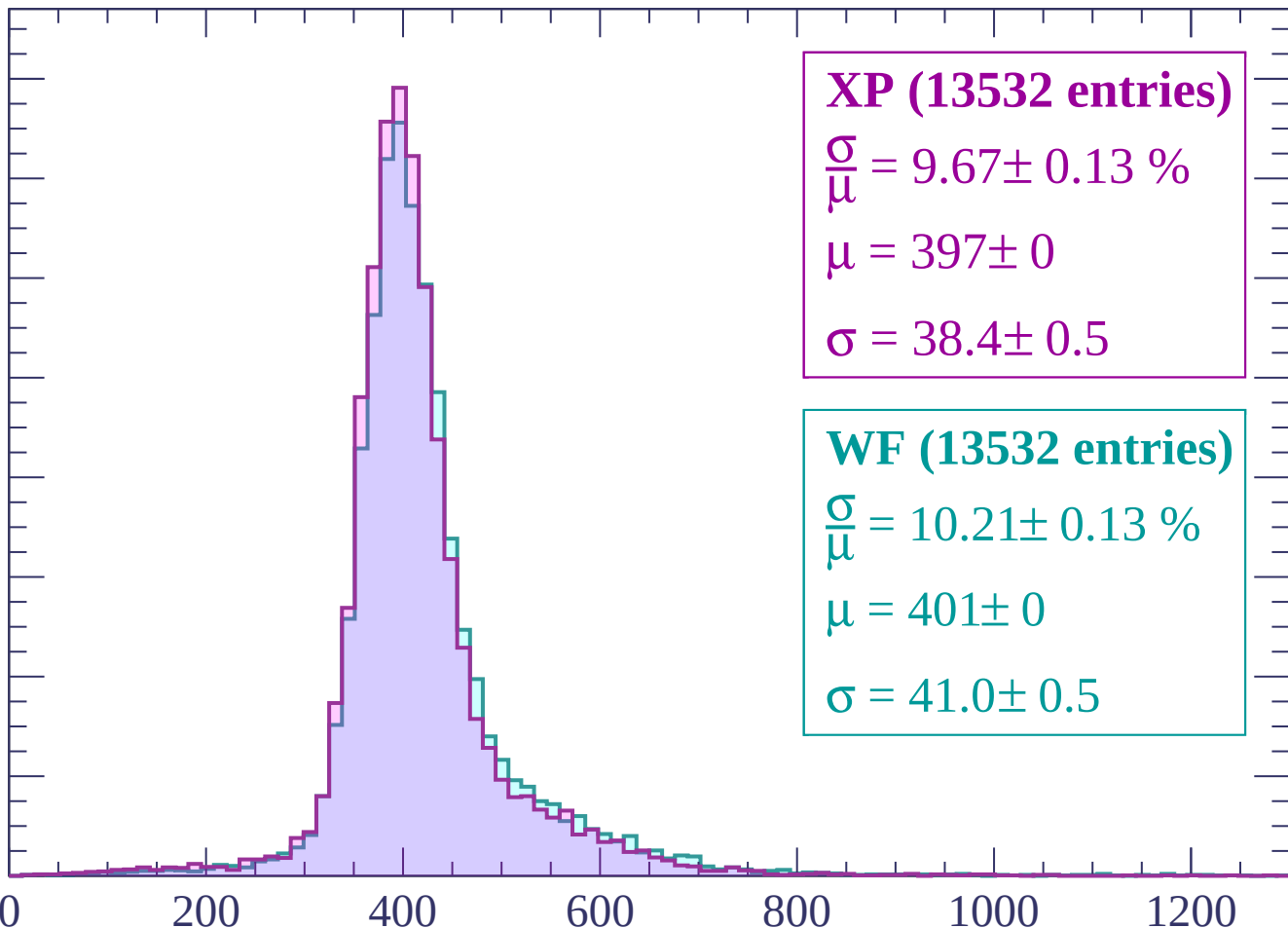
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $420 < p < 480$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13112 entries)

$$\frac{\sigma}{\mu} = 9.28 \pm 0.12 \%$$

$$\mu = 399 \pm 0$$

$$\sigma = 37.1 \pm 0.4$$

WF (13112 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 0.12 \%$$

$$\mu = 403 \pm 0$$

$$\sigma = 39.8 \pm 0.5$$

0

200

400

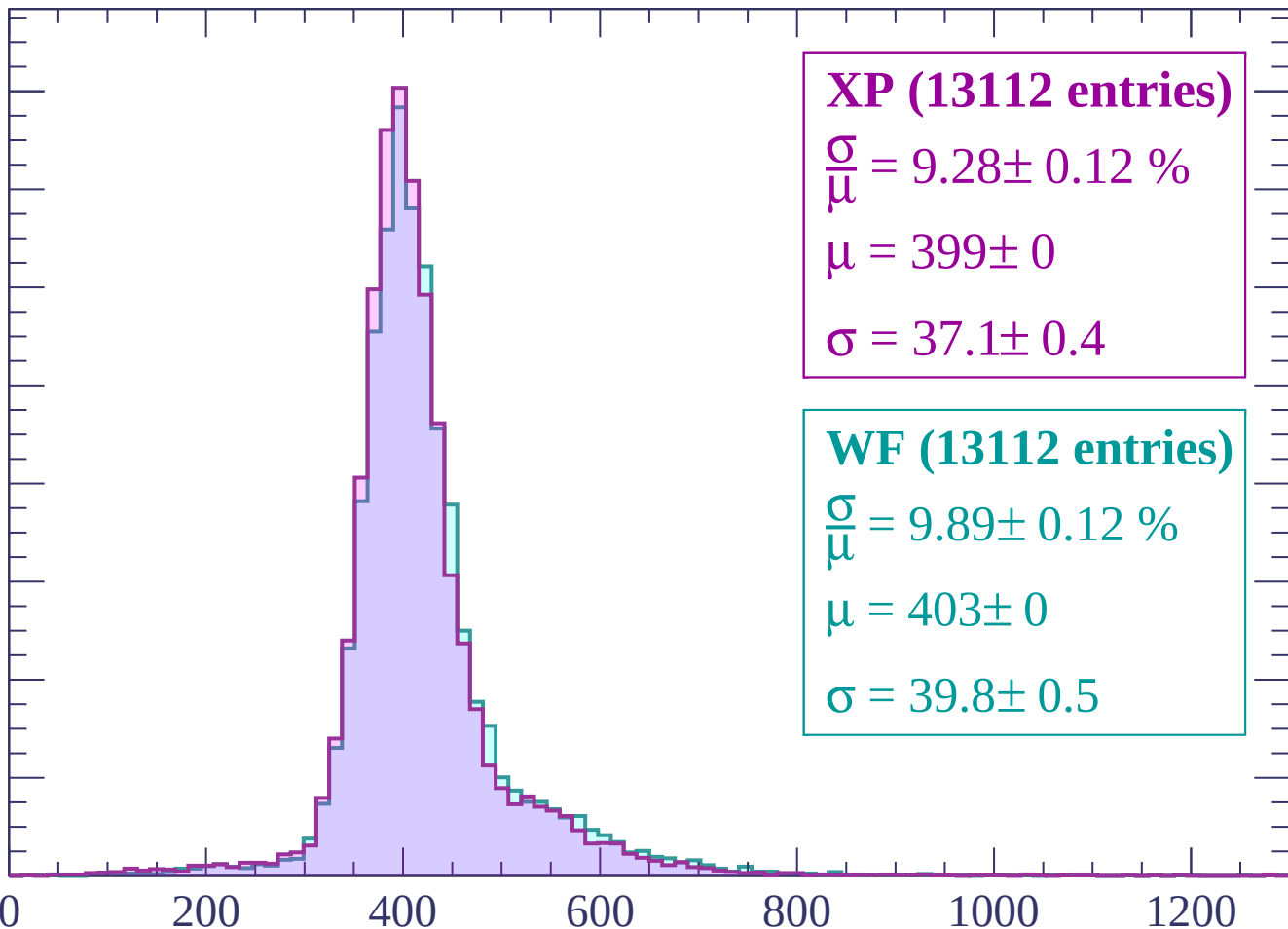
600

800

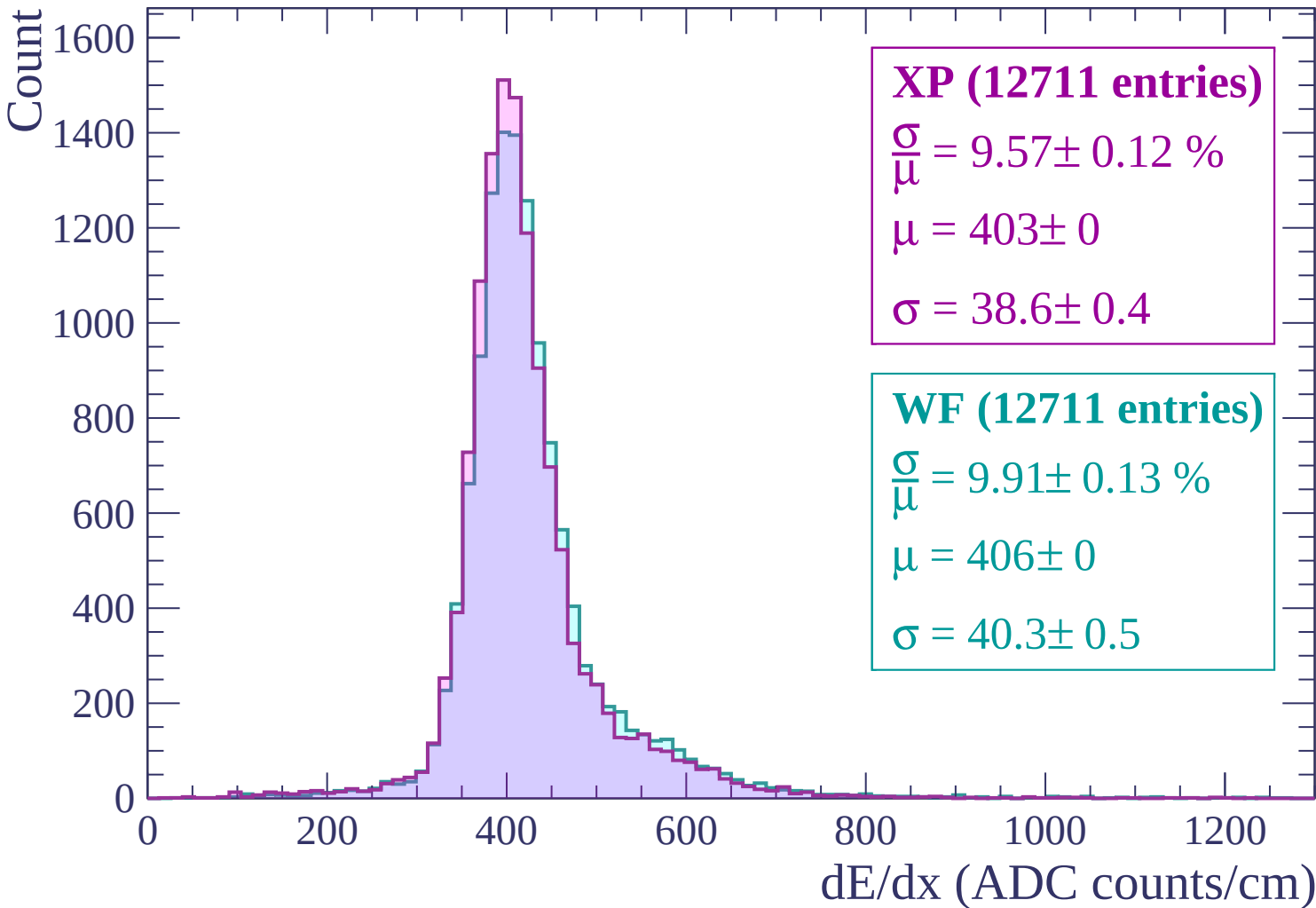
1000

1200

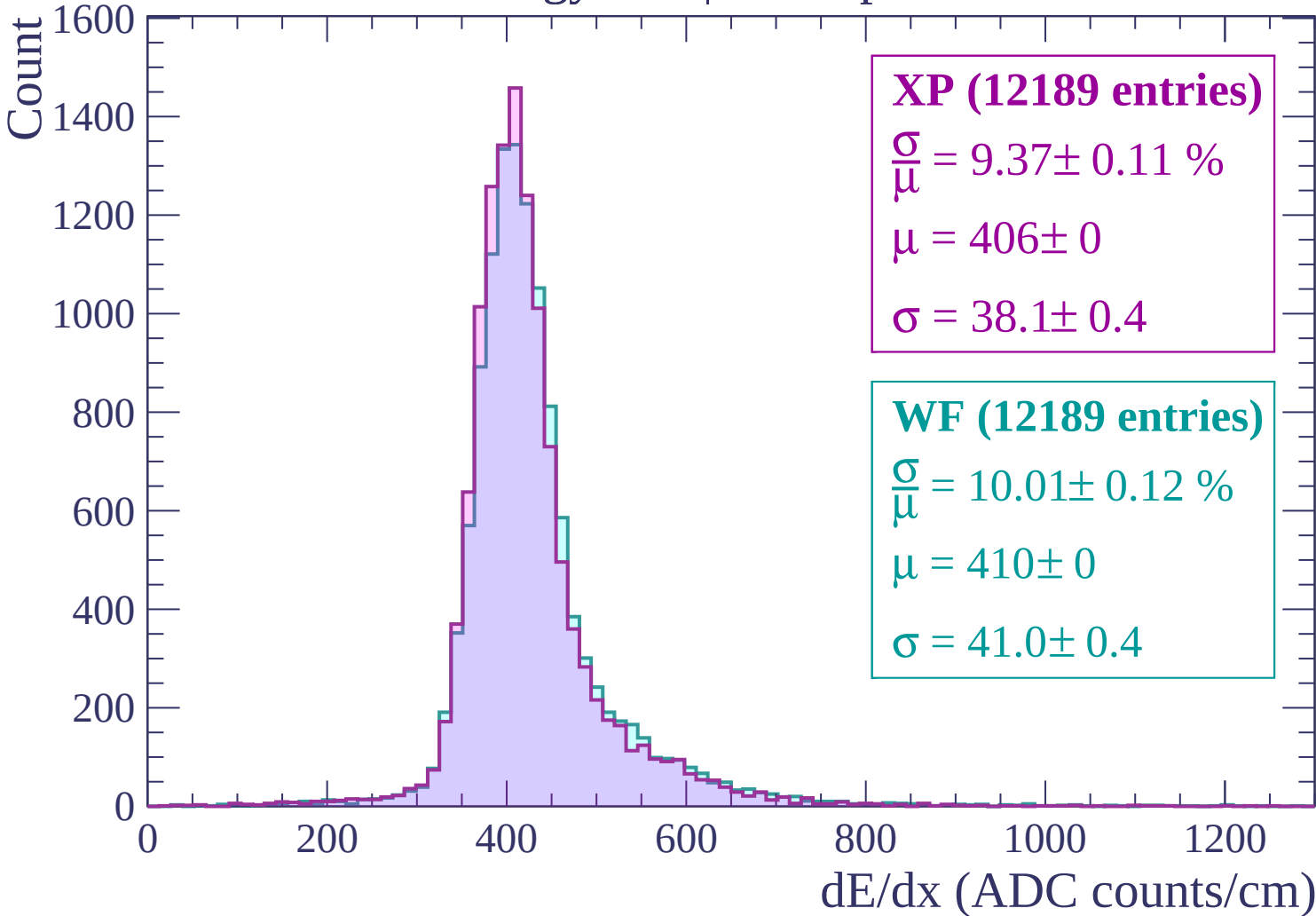
dE/dx (ADC counts/cm)



Energy loss | $480 < p < 540$

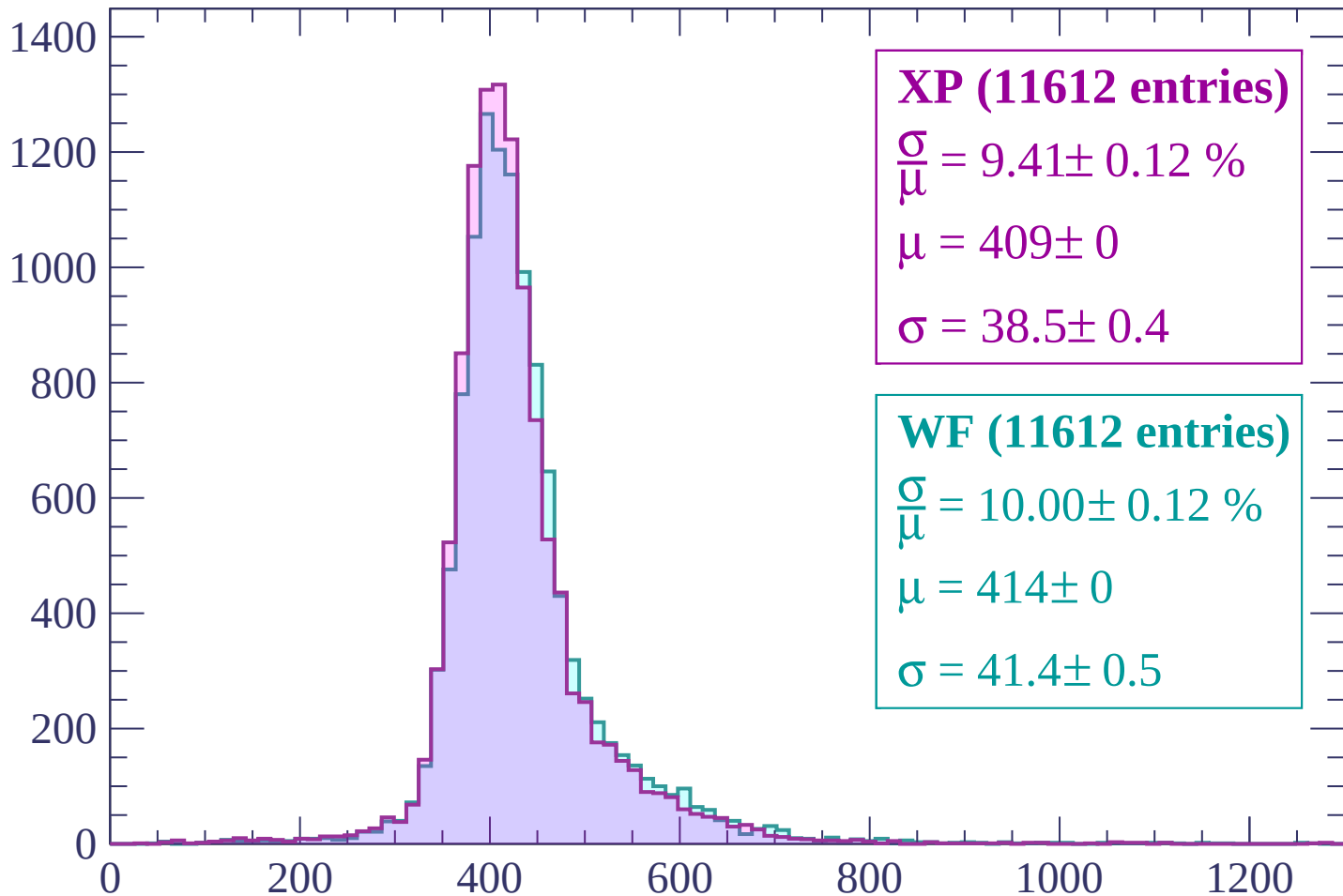


Energy loss | $540 < p < 600$



Energy loss | $600 < p < 660$

Count



XP (11612 entries)

$$\frac{\sigma}{\mu} = 9.41 \pm 0.12 \%$$

$$\mu = 409 \pm 0$$

$$\sigma = 38.5 \pm 0.4$$

WF (11612 entries)

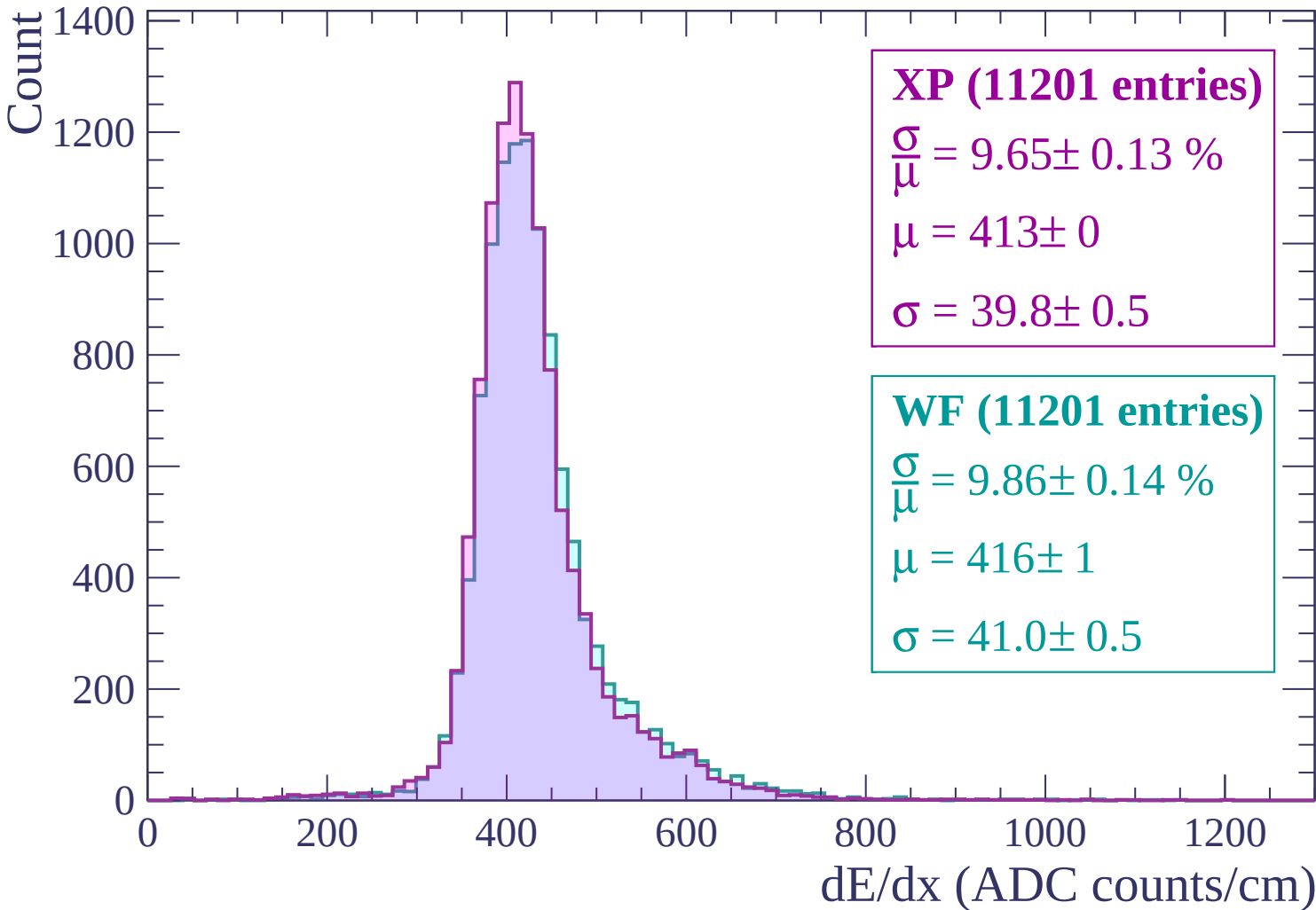
$$\frac{\sigma}{\mu} = 10.00 \pm 0.12 \%$$

$$\mu = 414 \pm 0$$

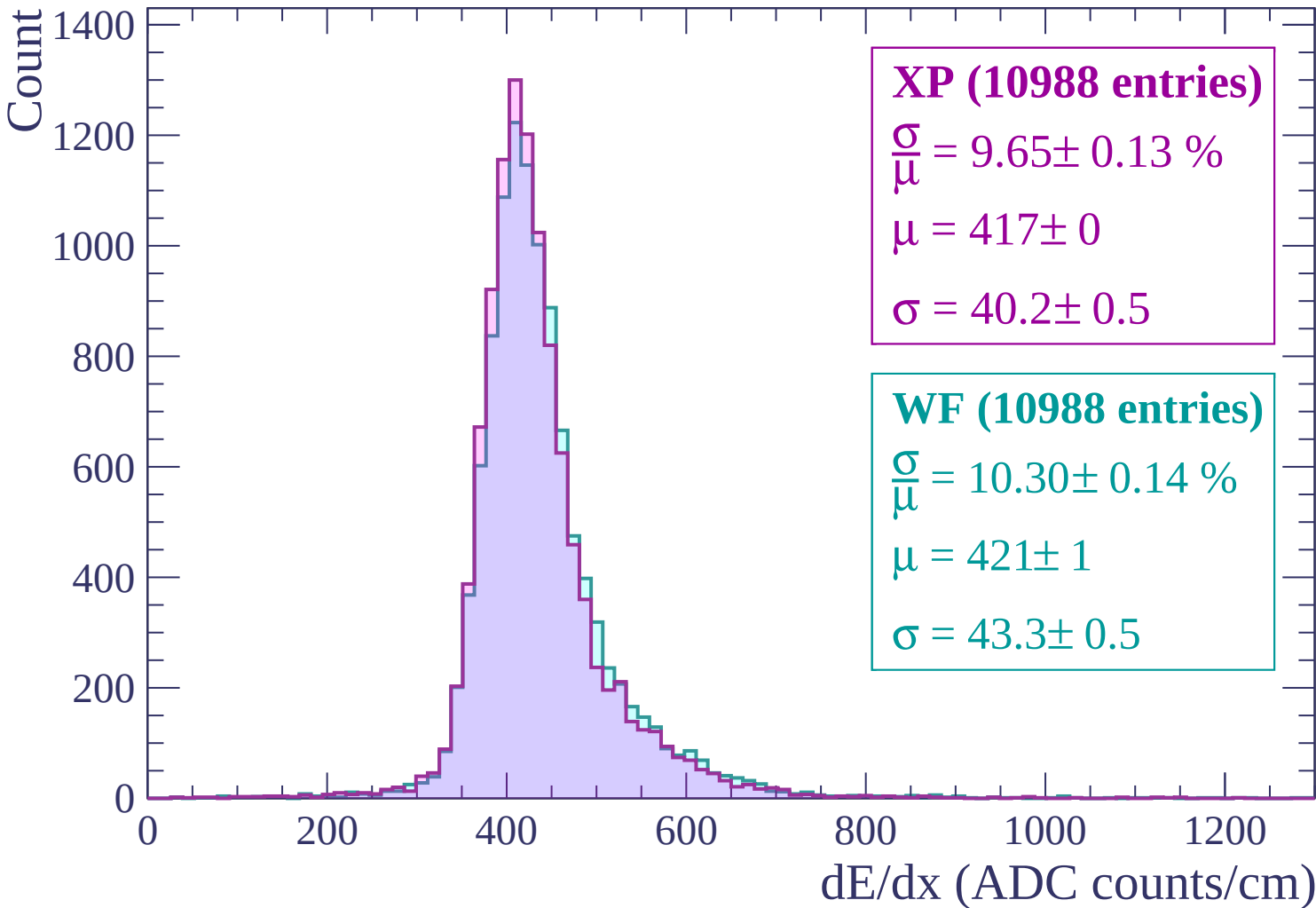
$$\sigma = 41.4 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$



Energy loss | $720 < p < 780$



Energy loss | $780 < p < 840$

Count

1200

1000

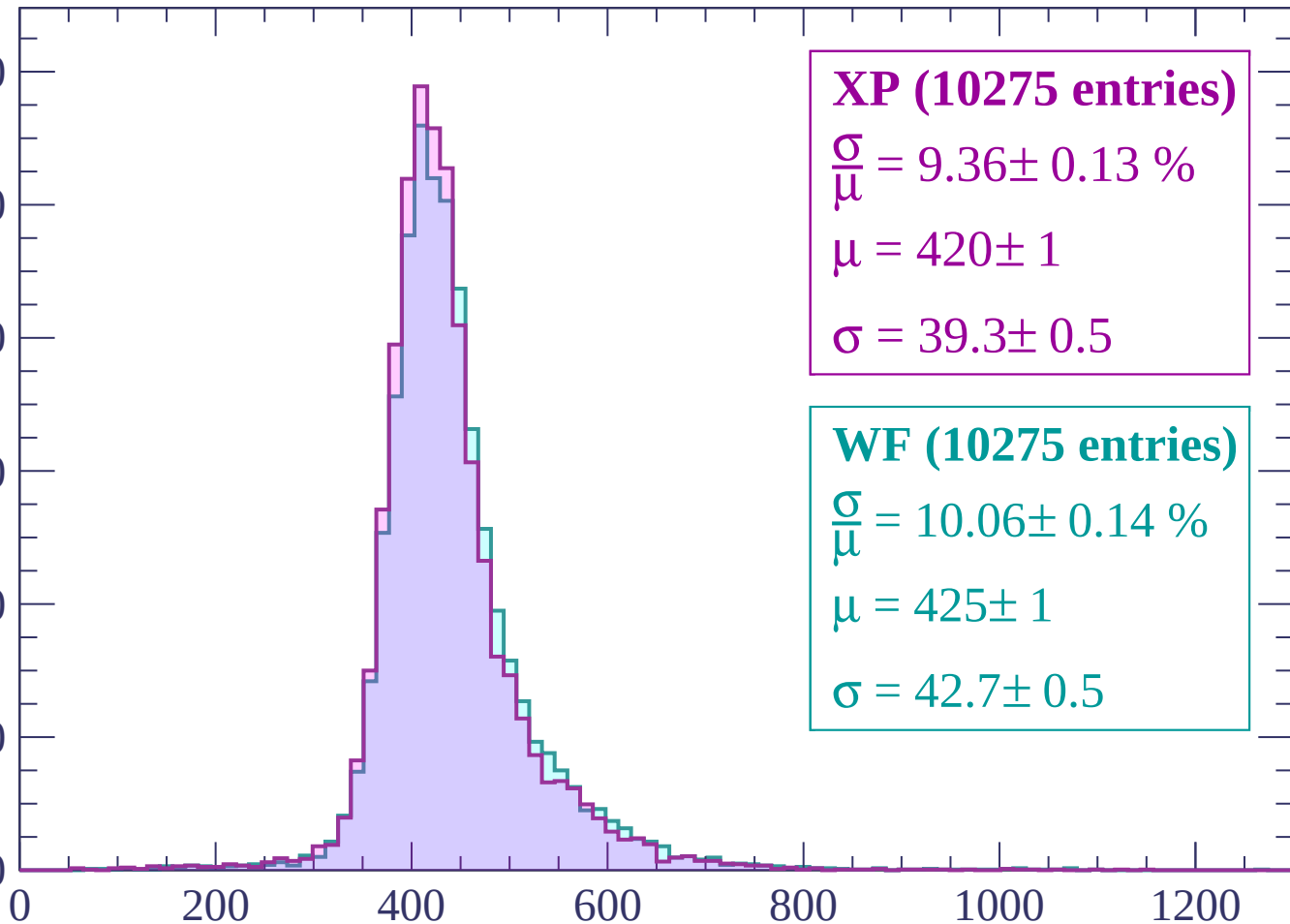
800

600

400

200

0



XP (10275 entries)

$\frac{\sigma}{\mu} = 9.36 \pm 0.13 \%$

$\mu = 420 \pm 1$

$\sigma = 39.3 \pm 0.5$

WF (10275 entries)

$\frac{\sigma}{\mu} = 10.06 \pm 0.14 \%$

$\mu = 425 \pm 1$

$\sigma = 42.7 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count

1200

1000

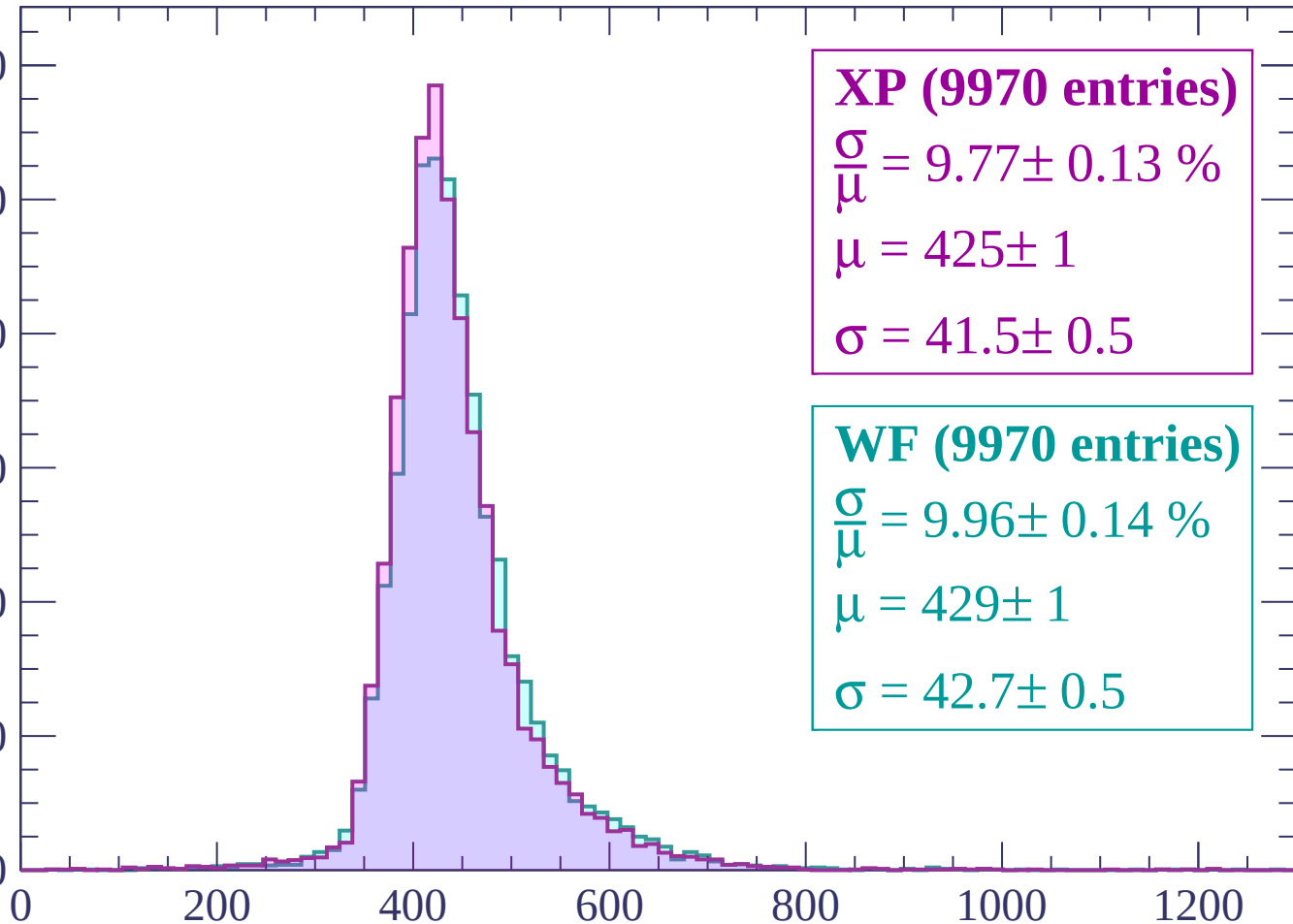
800

600

400

200

0



XP (9970 entries)

$\frac{\sigma}{\mu} = 9.77 \pm 0.13 \%$

$\mu = 425 \pm 1$

$\sigma = 41.5 \pm 0.5$

WF (9970 entries)

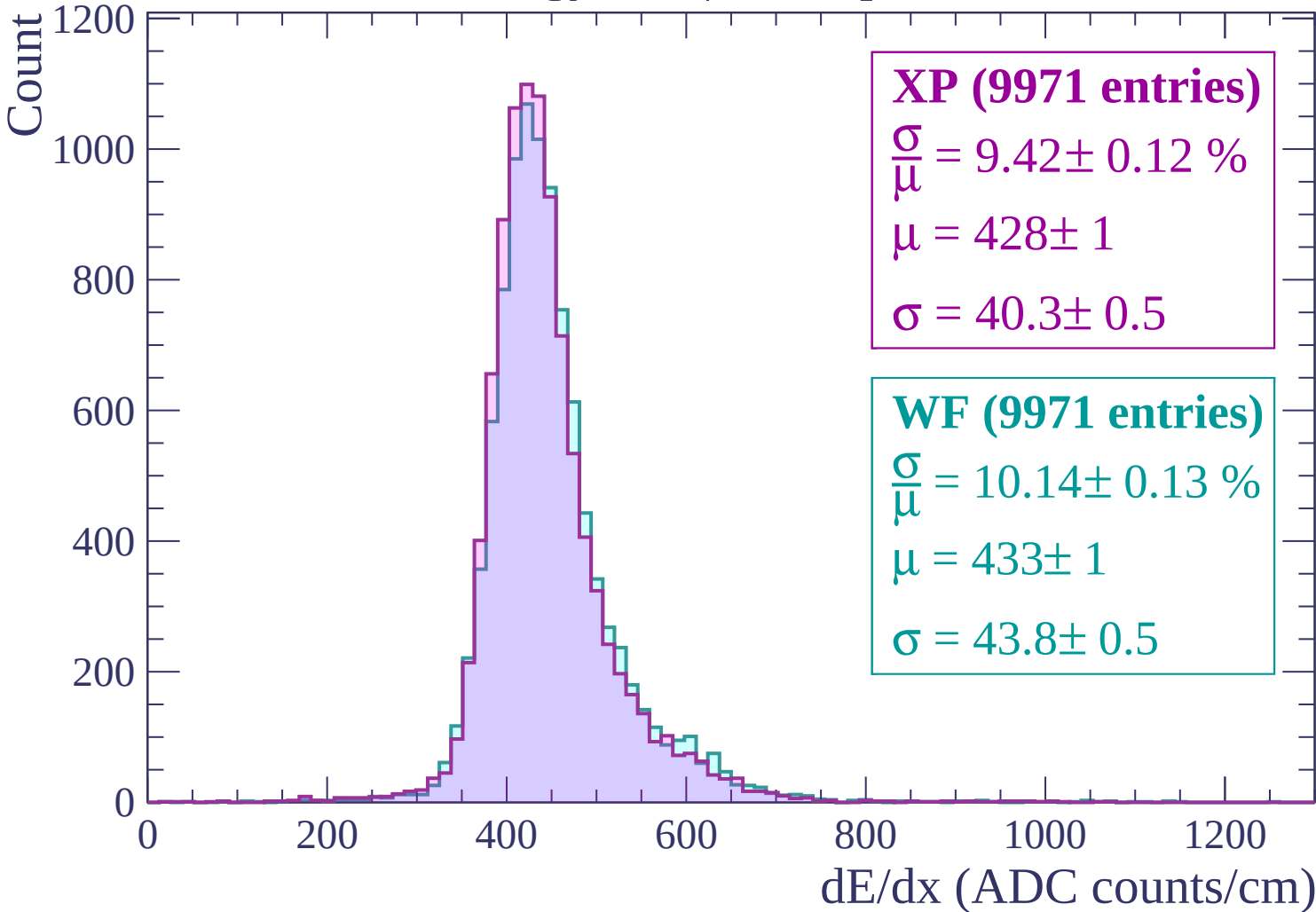
$\frac{\sigma}{\mu} = 9.96 \pm 0.14 \%$

$\mu = 429 \pm 1$

$\sigma = 42.7 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$



Energy loss | $960 < p < 1020$

Count

1000

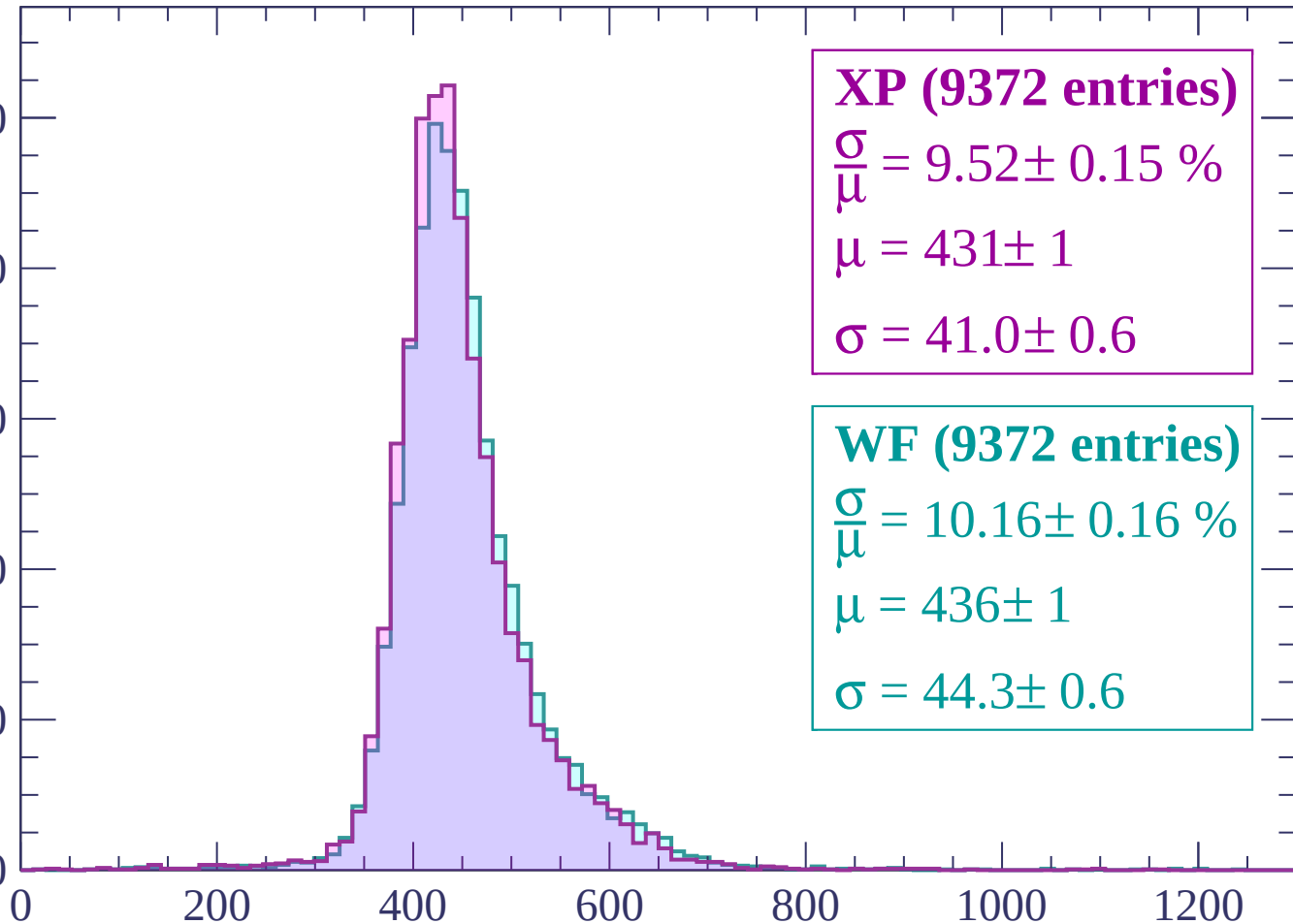
800

600

400

200

0



XP (9372 entries)

$\frac{\sigma}{\mu} = 9.52 \pm 0.15 \%$

$\mu = 431 \pm 1$

$\sigma = 41.0 \pm 0.6$

WF (9372 entries)

$\frac{\sigma}{\mu} = 10.16 \pm 0.16 \%$

$\mu = 436 \pm 1$

$\sigma = 44.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (9114 entries)

$\frac{\sigma}{\mu} = 9.76 \pm 0.14 \%$

$\mu = 436 \pm 1$

$\sigma = 42.6 \pm 0.6$

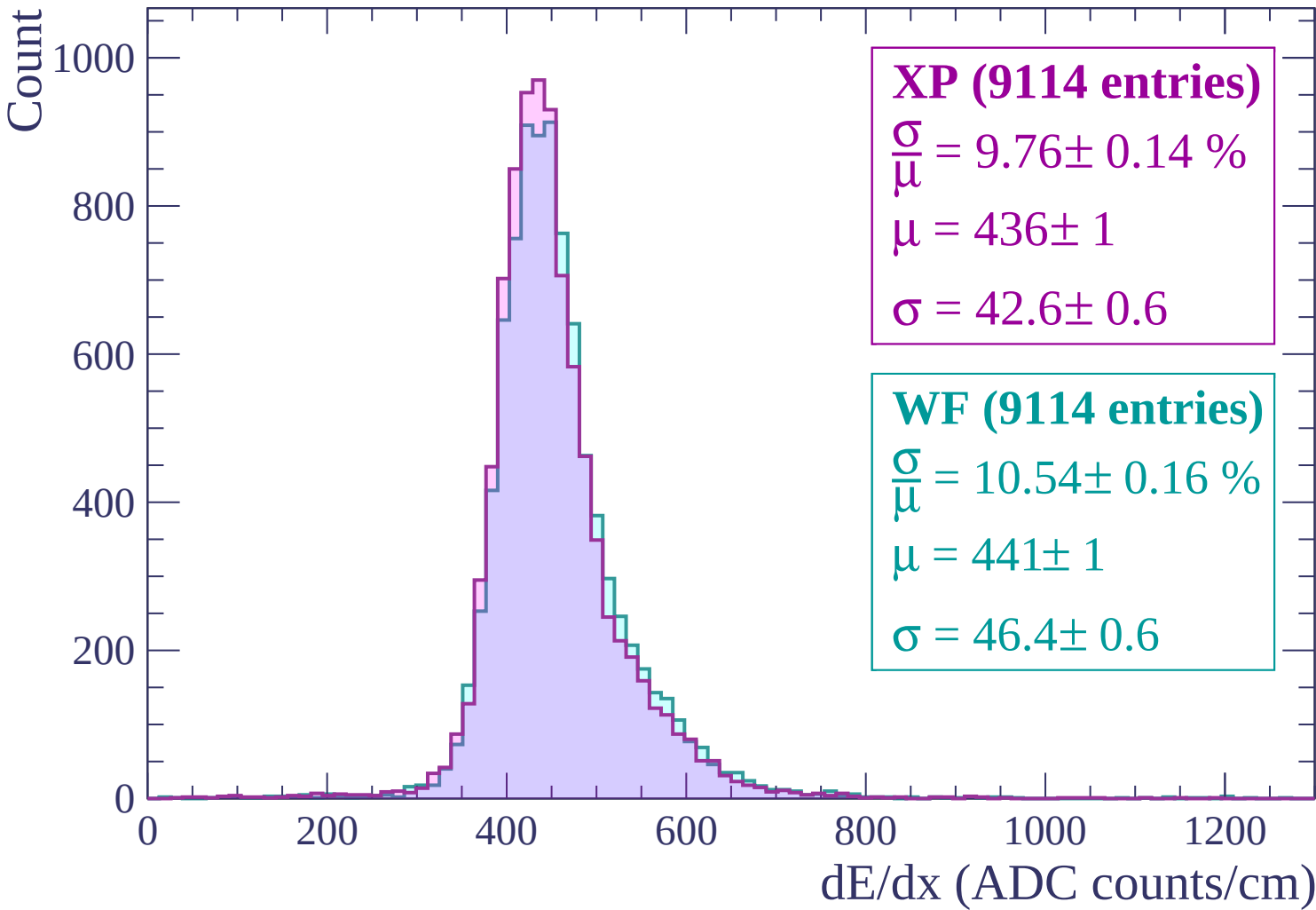
WF (9114 entries)

$\frac{\sigma}{\mu} = 10.54 \pm 0.16 \%$

$\mu = 441 \pm 1$

$\sigma = 46.4 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $1080 < p < 1140$

Count

1000

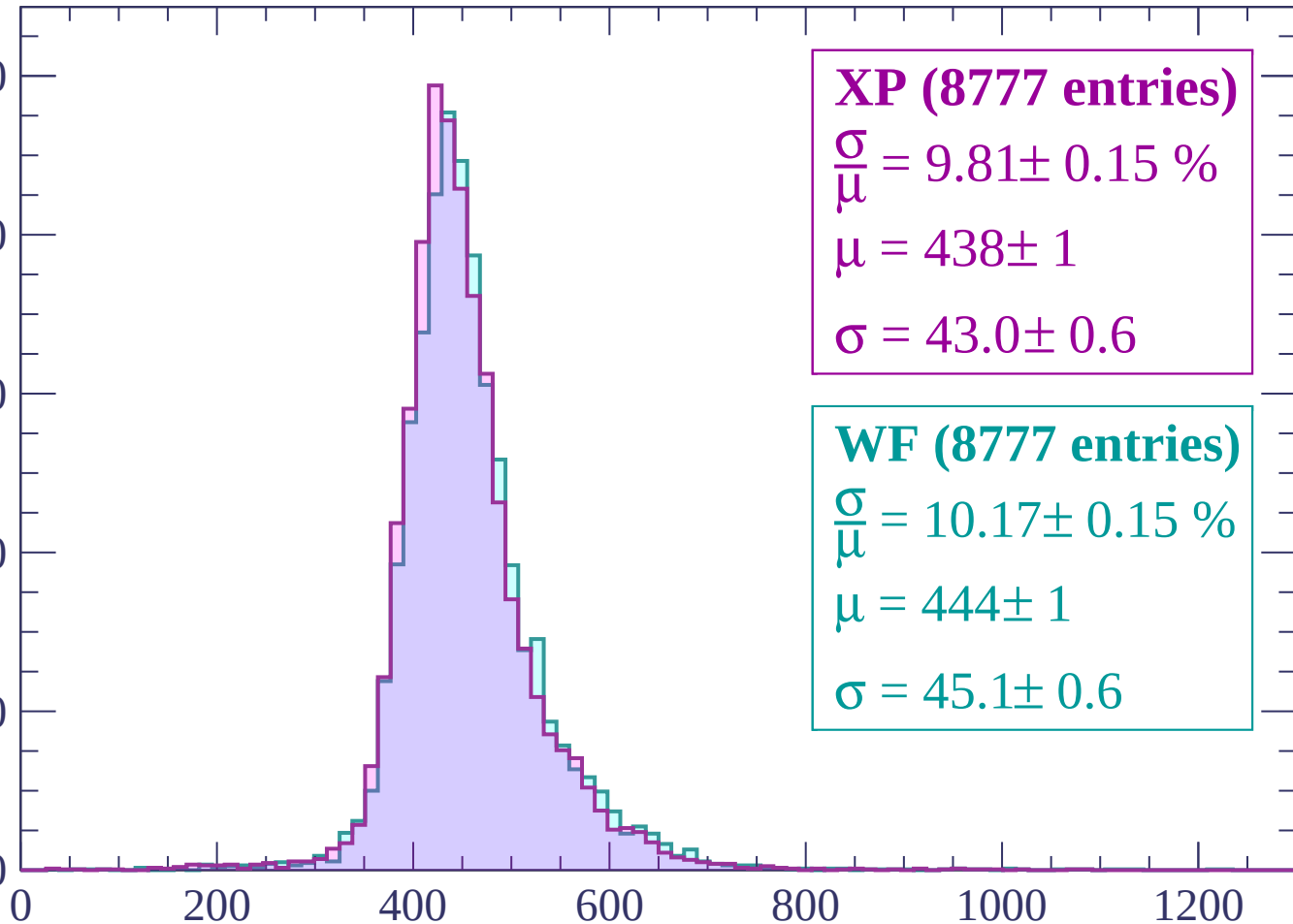
800

600

400

200

0



XP (8777 entries)

$\frac{\sigma}{\mu} = 9.81 \pm 0.15 \%$

$\mu = 438 \pm 1$

$\sigma = 43.0 \pm 0.6$

WF (8777 entries)

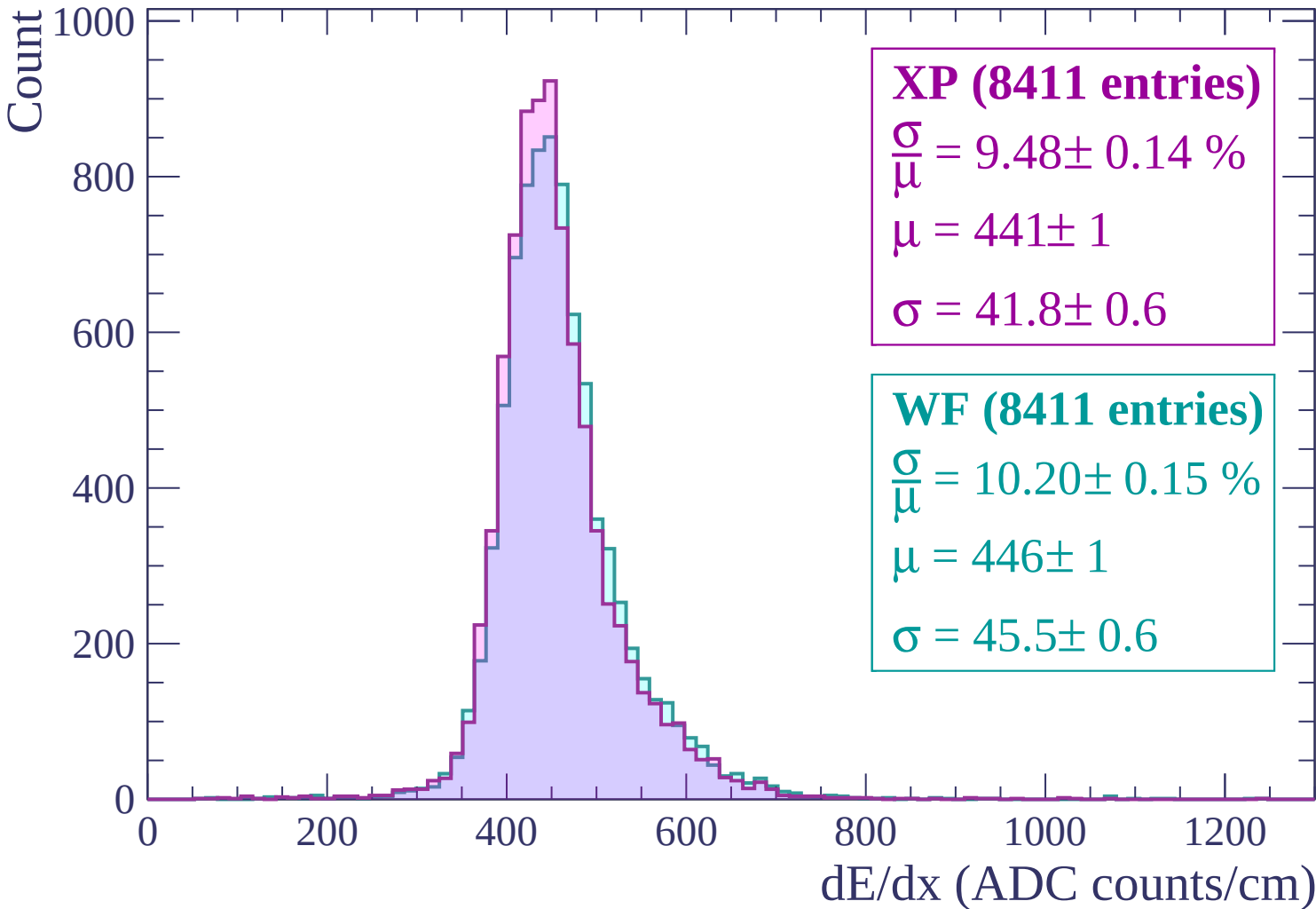
$\frac{\sigma}{\mu} = 10.17 \pm 0.15 \%$

$\mu = 444 \pm 1$

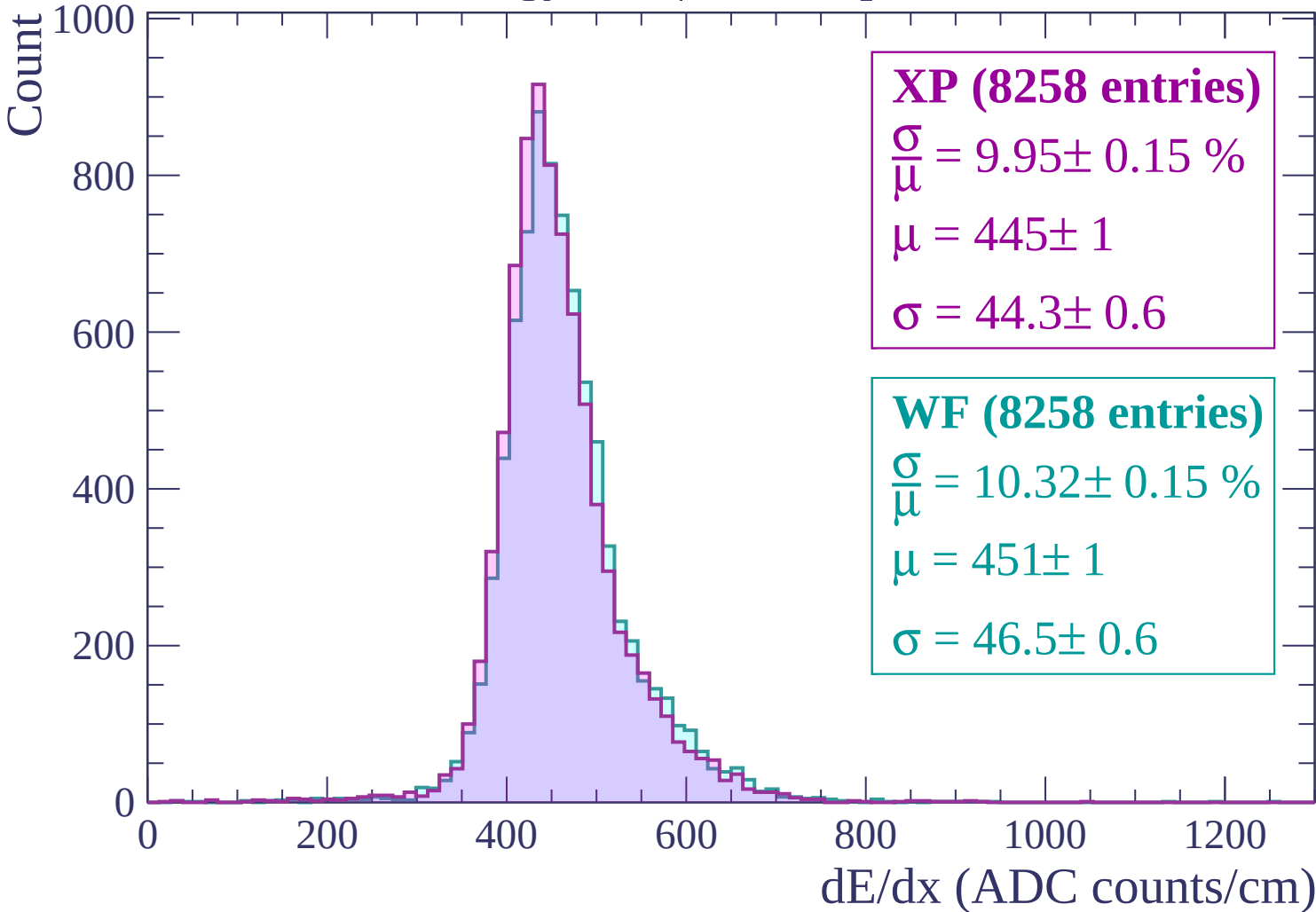
$\sigma = 45.1 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

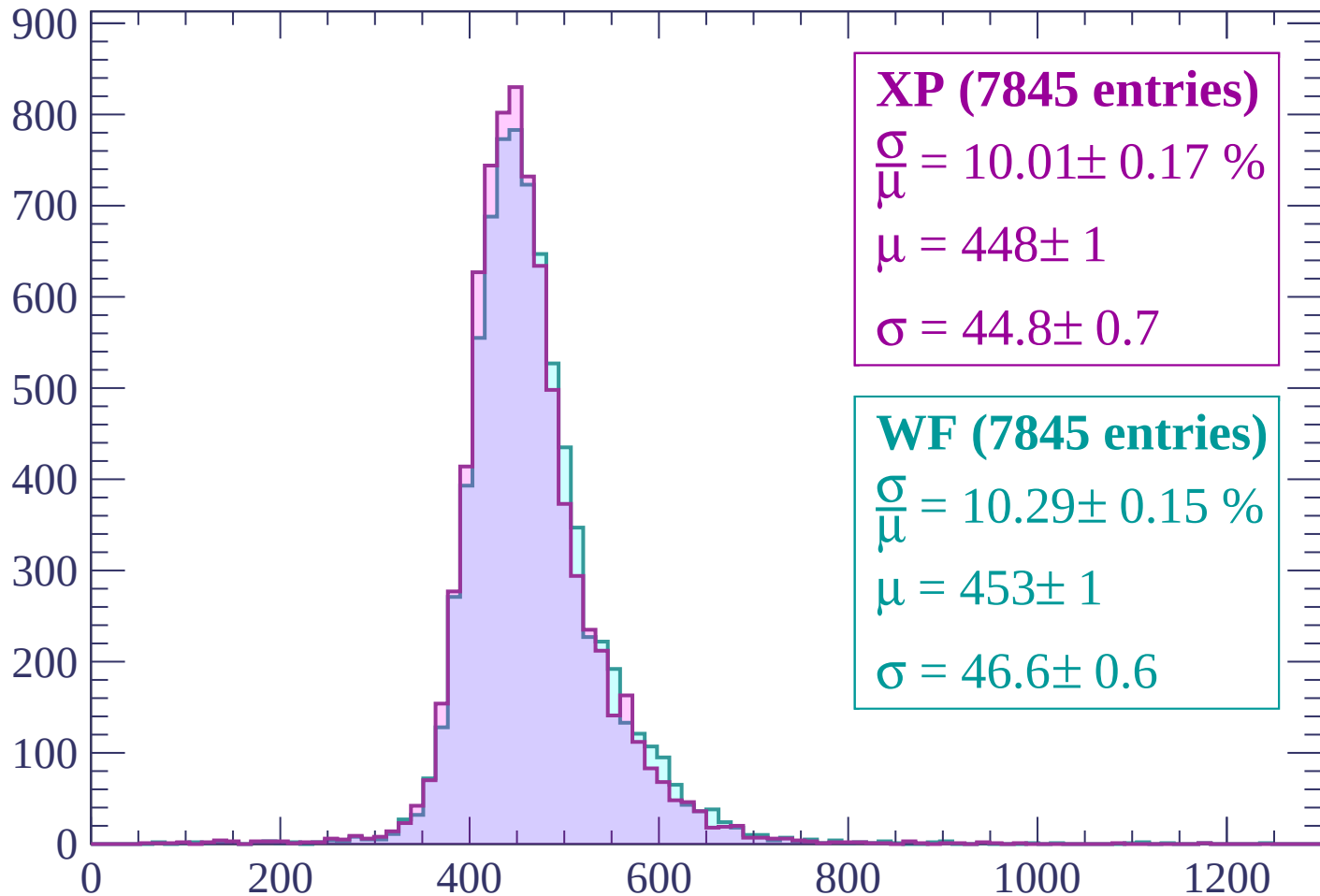


Energy loss | $1200 < p < 1260$



Energy loss | $1260 < p < 1320$

Count



XP (7845 entries)

$\frac{\sigma}{\mu} = 10.01 \pm 0.17 \%$

$\mu = 448 \pm 1$

$\sigma = 44.8 \pm 0.7$

WF (7845 entries)

$\frac{\sigma}{\mu} = 10.29 \pm 0.15 \%$

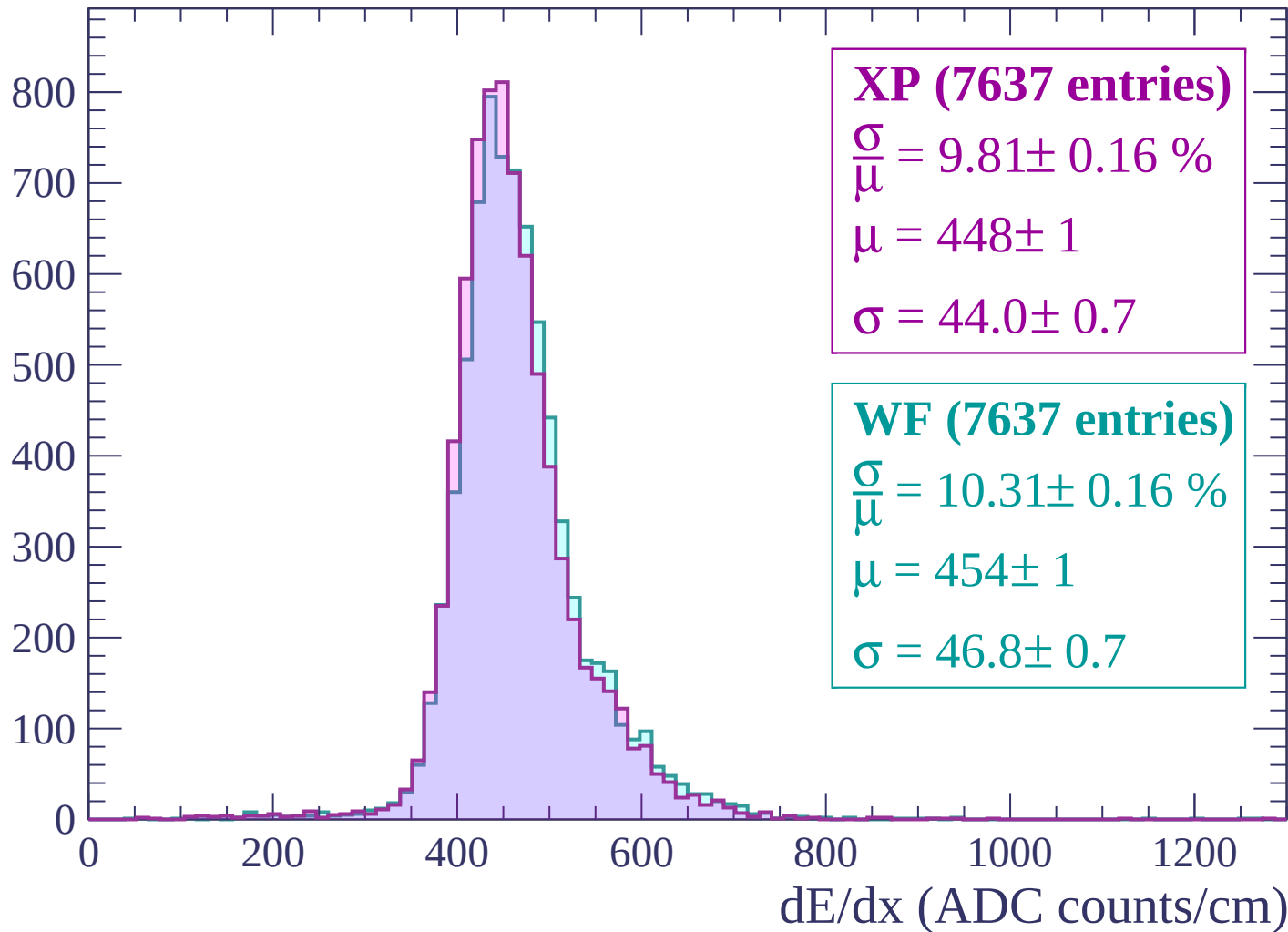
$\mu = 453 \pm 1$

$\sigma = 46.6 \pm 0.6$

dE/dx (ADC counts/cm)

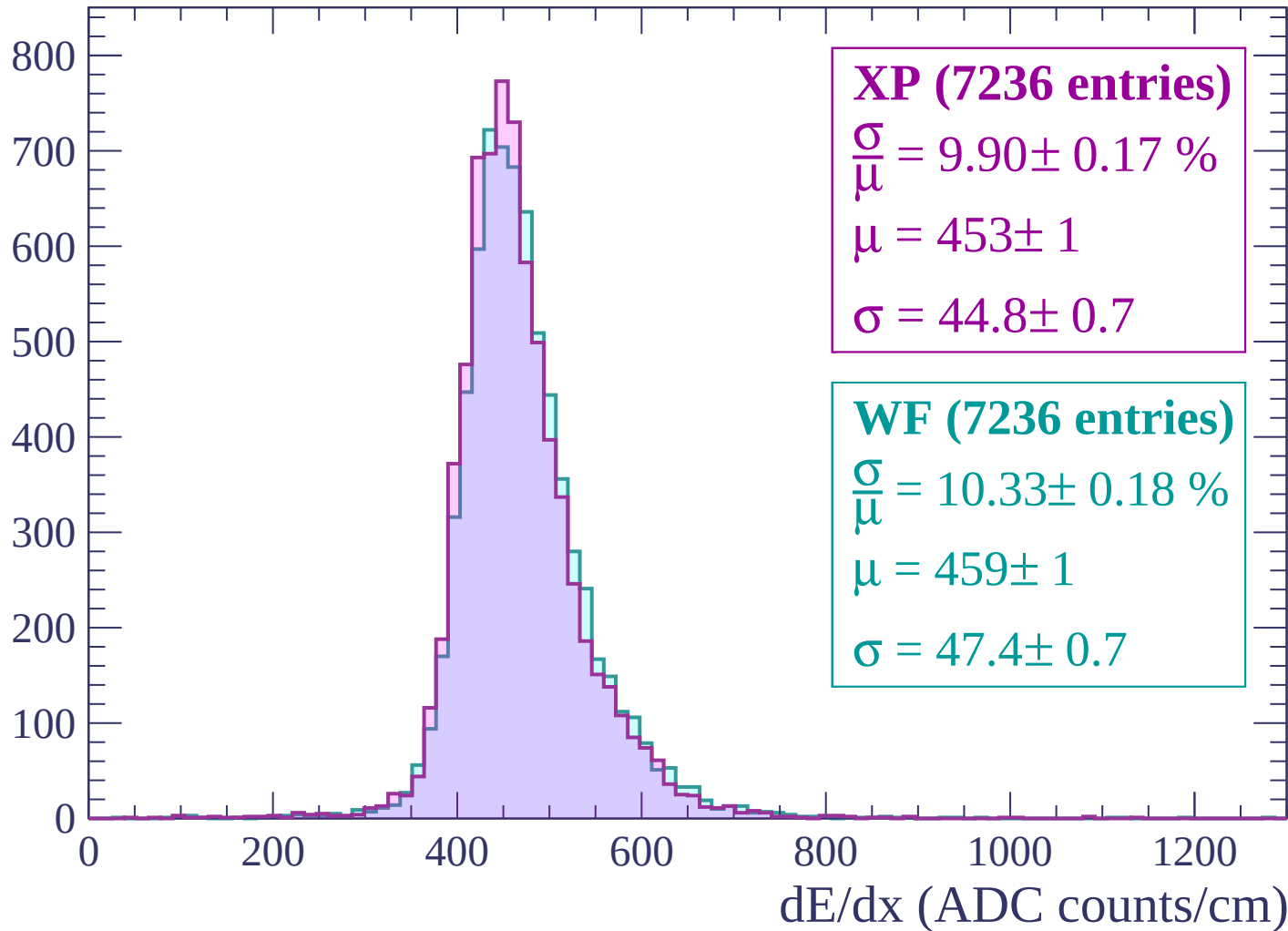
Energy loss | $1320 < p < 1380$

Count



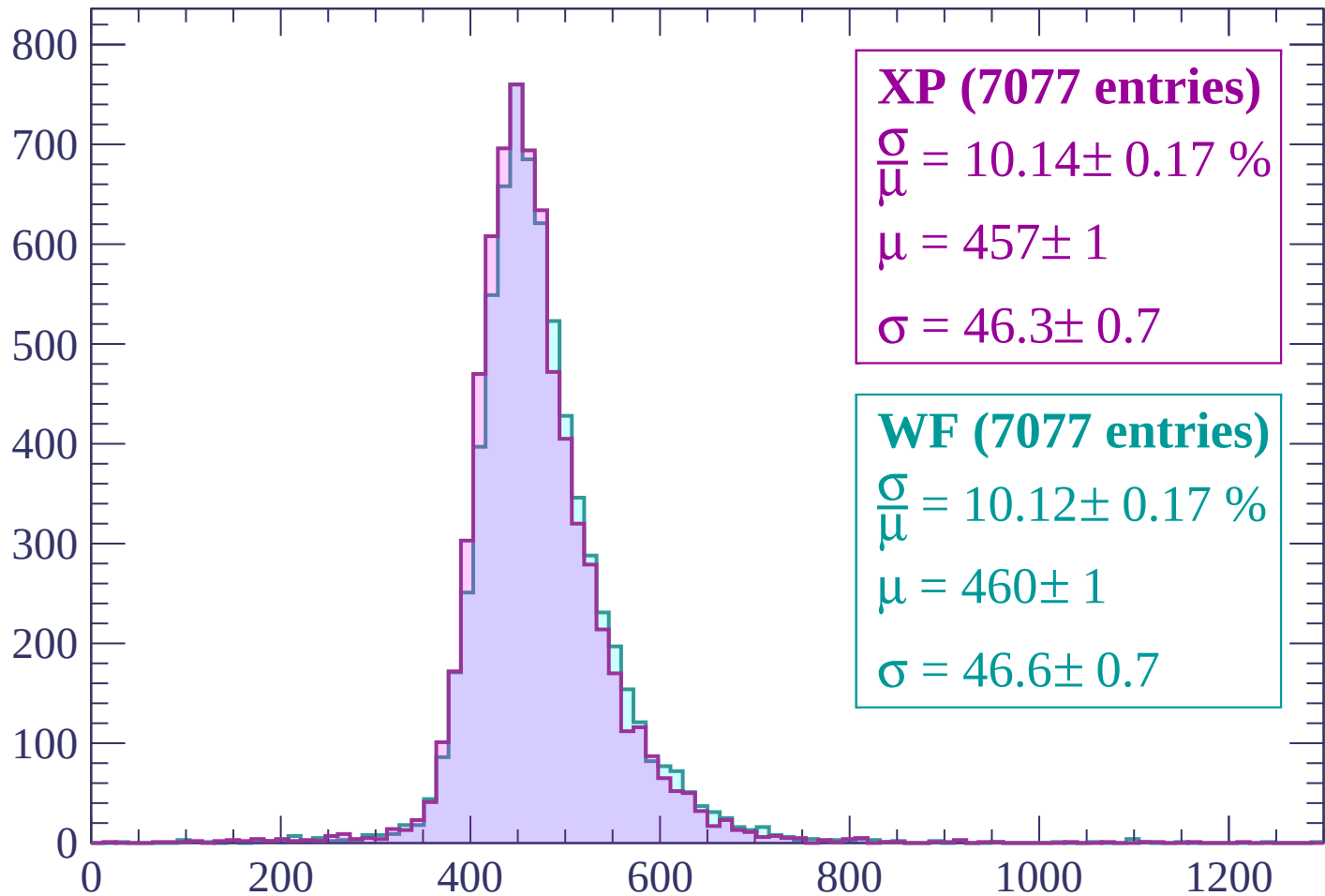
Energy loss | $1380 < p < 1440$

Count



Energy loss | $1440 < p < 1500$

Count



XP (7077 entries)

$$\frac{\sigma}{\mu} = 10.14 \pm 0.17 \%$$

$$\mu = 457 \pm 1$$

$$\sigma = 46.3 \pm 0.7$$

WF (7077 entries)

$$\frac{\sigma}{\mu} = 10.12 \pm 0.17 \%$$

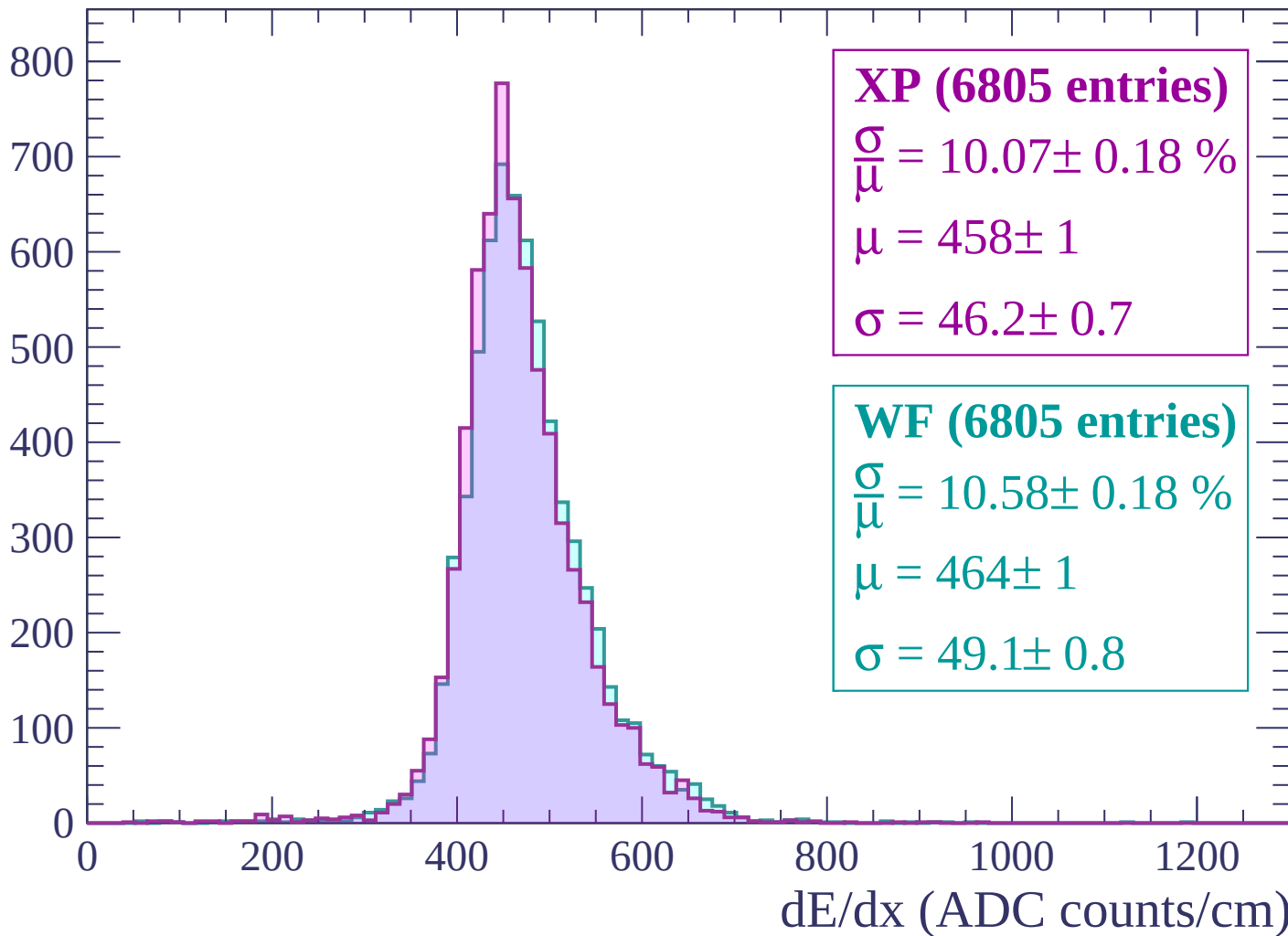
$$\mu = 460 \pm 1$$

$$\sigma = 46.6 \pm 0.7$$

dE/dx (ADC counts/cm)

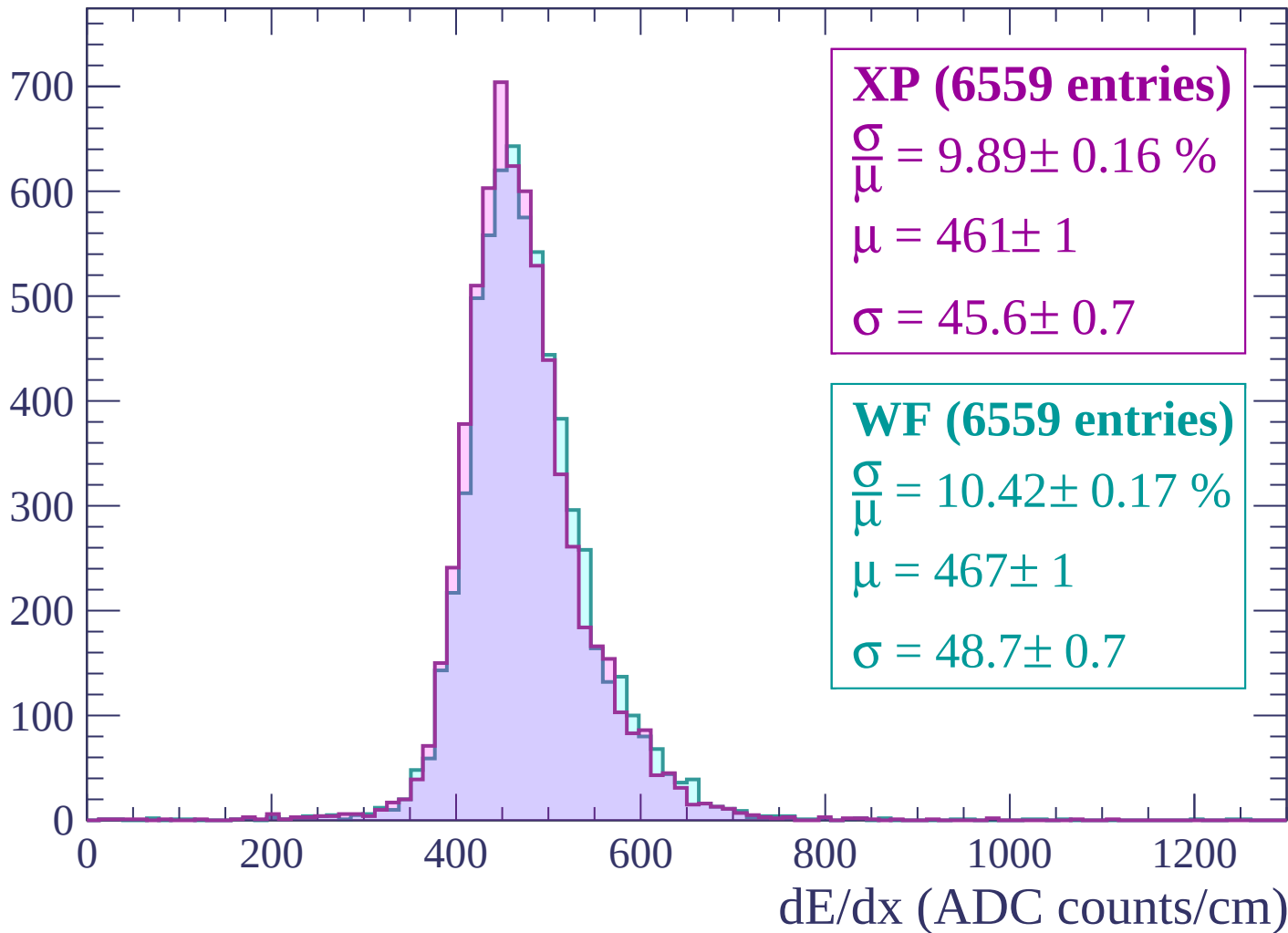
Energy loss | $1500 < p < 1560$

Count



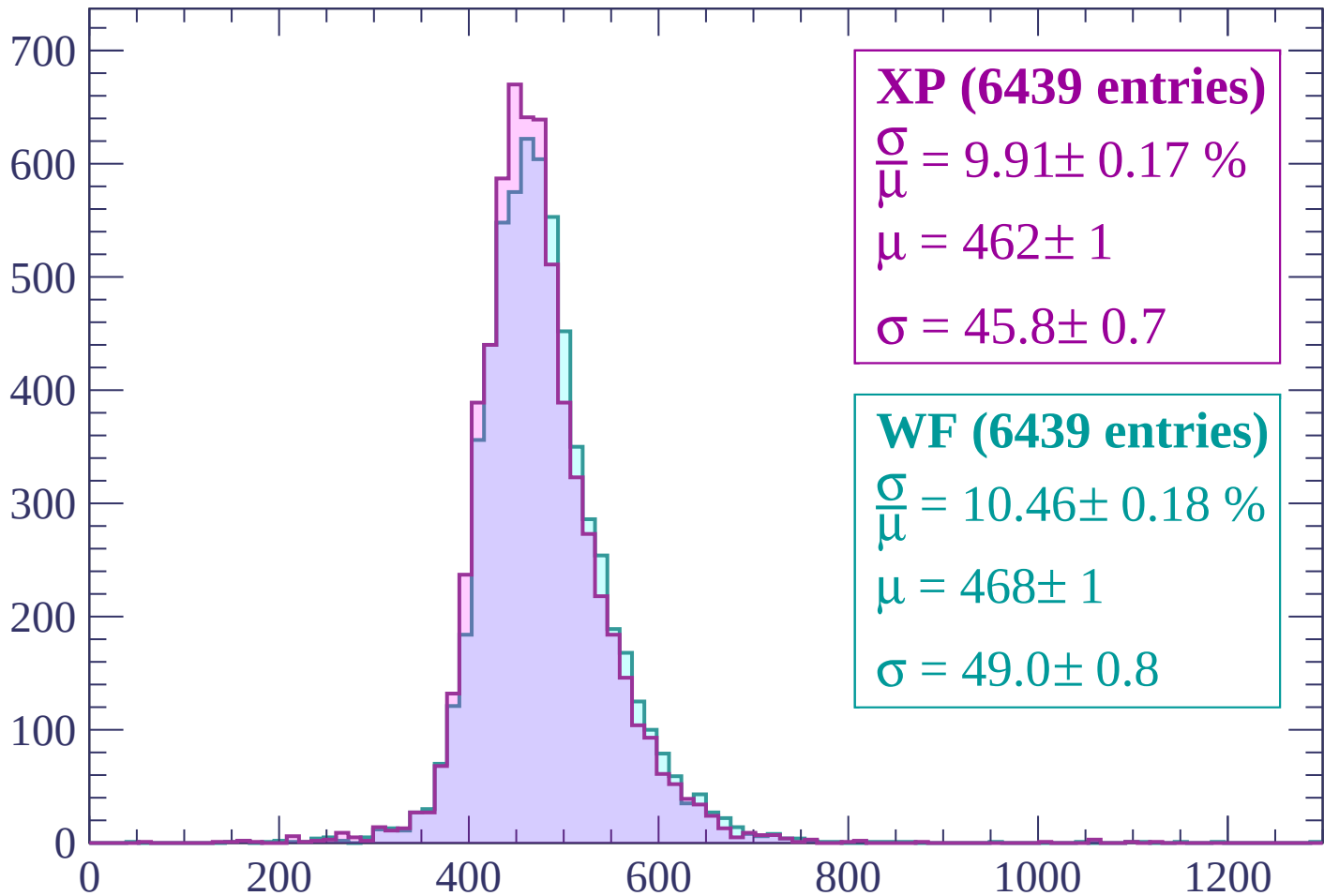
Energy loss | $1560 < p < 1620$

Count



Energy loss | $1620 < p < 1680$

Count



XP (6439 entries)

$$\frac{\sigma}{\mu} = 9.91 \pm 0.17 \%$$

$$\mu = 462 \pm 1$$

$$\sigma = 45.8 \pm 0.7$$

WF (6439 entries)

$$\frac{\sigma}{\mu} = 10.46 \pm 0.18 \%$$

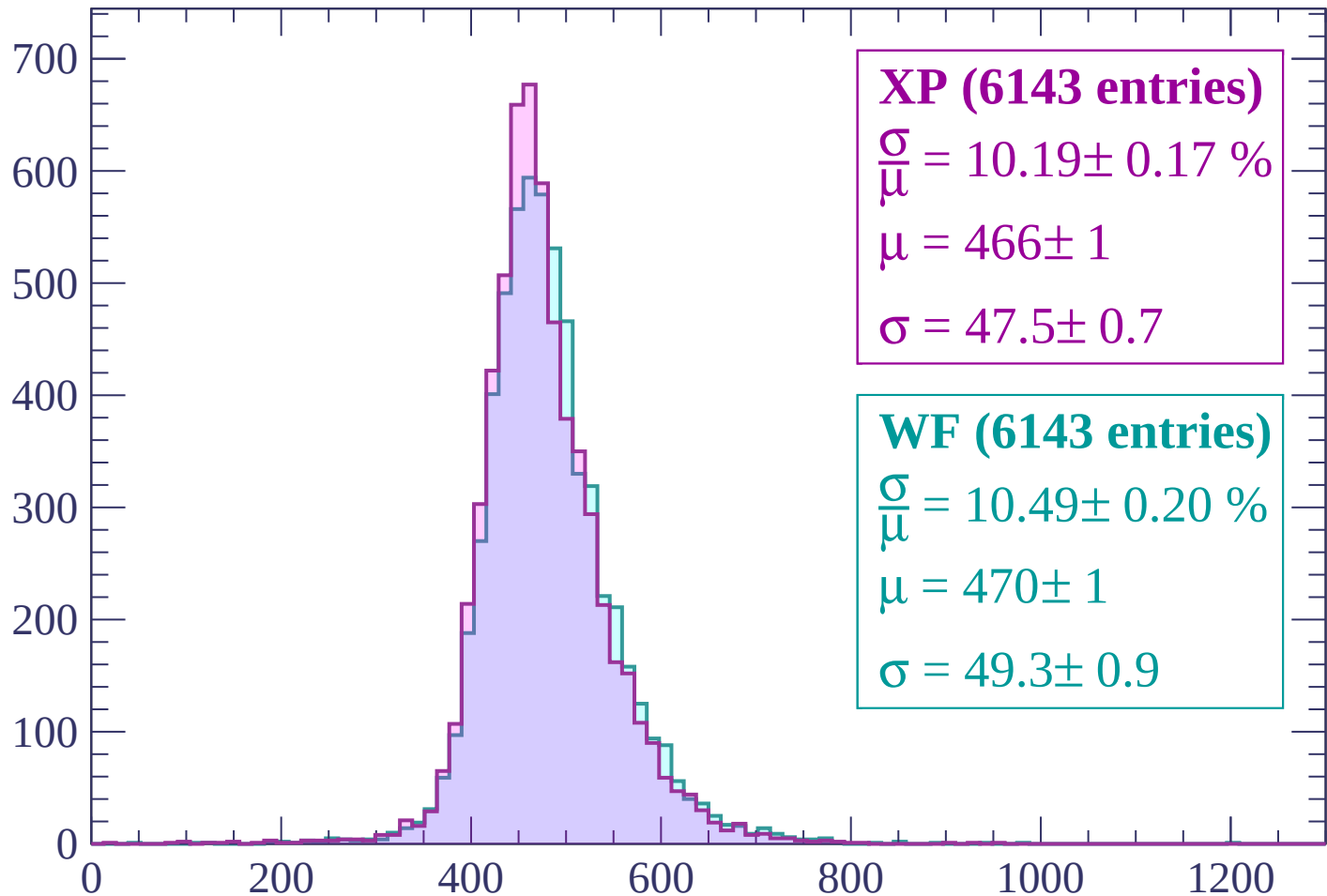
$$\mu = 468 \pm 1$$

$$\sigma = 49.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (6143 entries)

$$\frac{\sigma}{\mu} = 10.19 \pm 0.17 \%$$

$$\mu = 466 \pm 1$$

$$\sigma = 47.5 \pm 0.7$$

WF (6143 entries)

$$\frac{\sigma}{\mu} = 10.49 \pm 0.20 \%$$

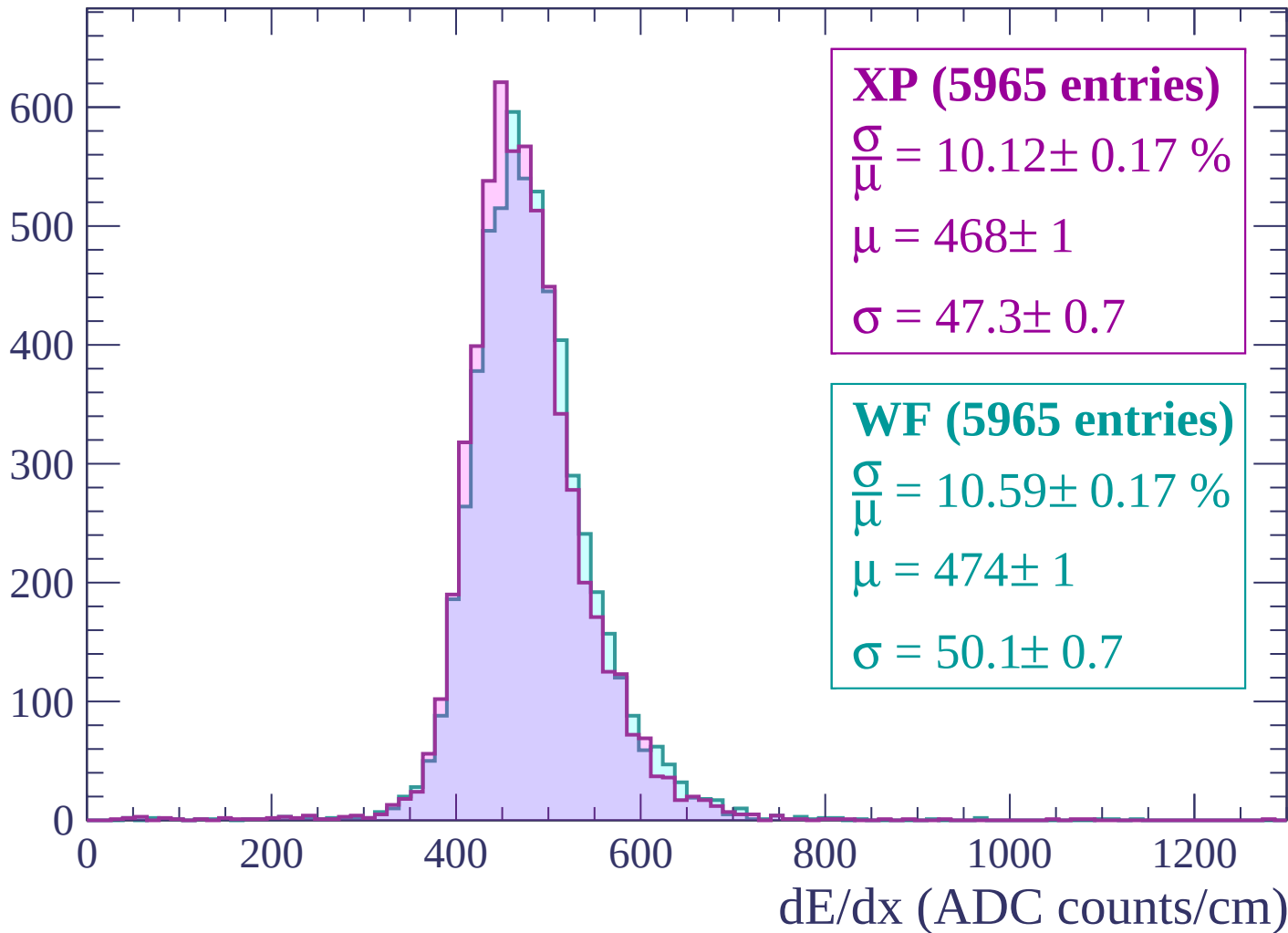
$$\mu = 470 \pm 1$$

$$\sigma = 49.3 \pm 0.9$$

dE/dx (ADC counts/cm)

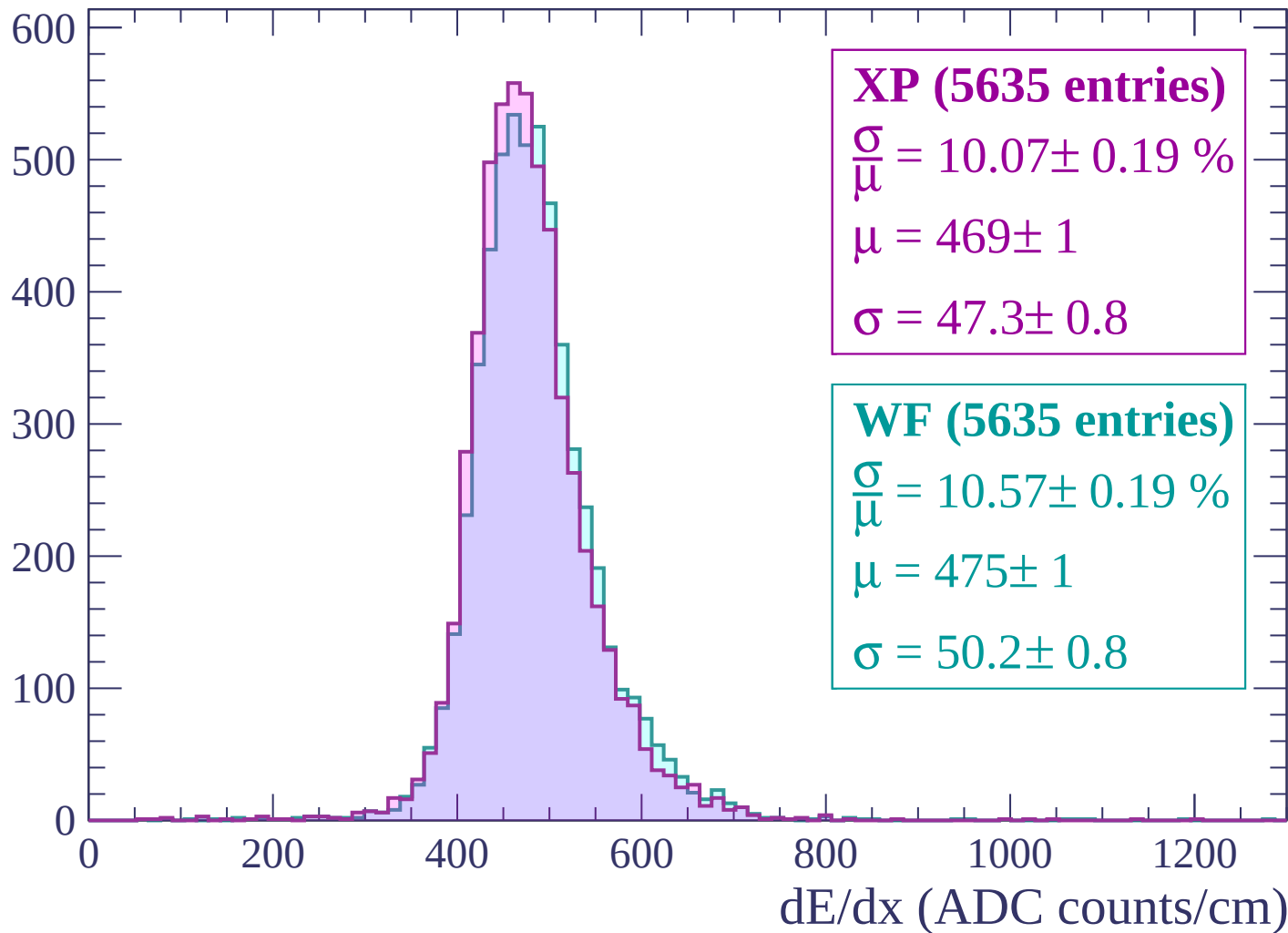
Energy loss | $1740 < p < 1800$

Count



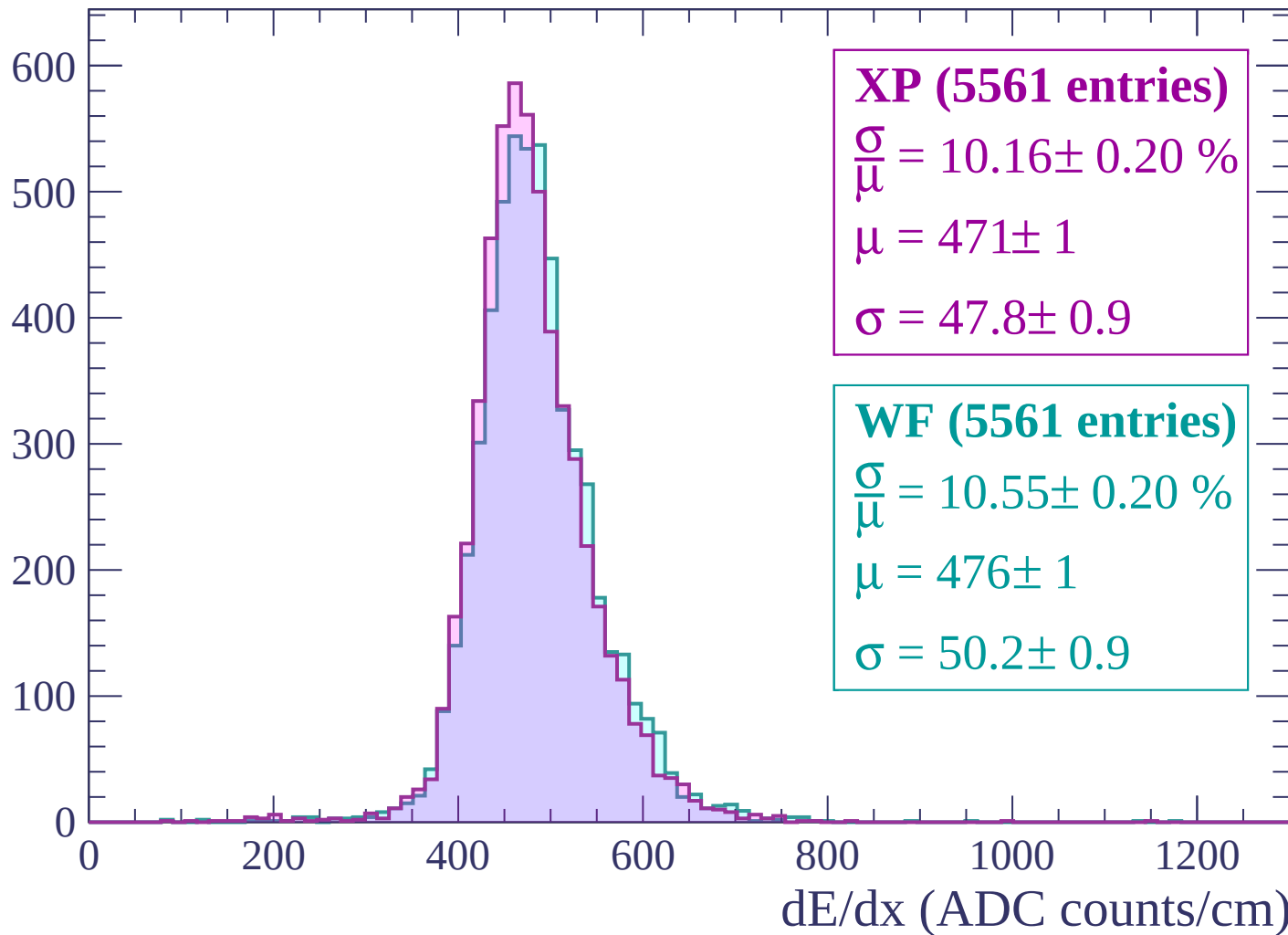
Energy loss | $1800 < p < 1860$

Count

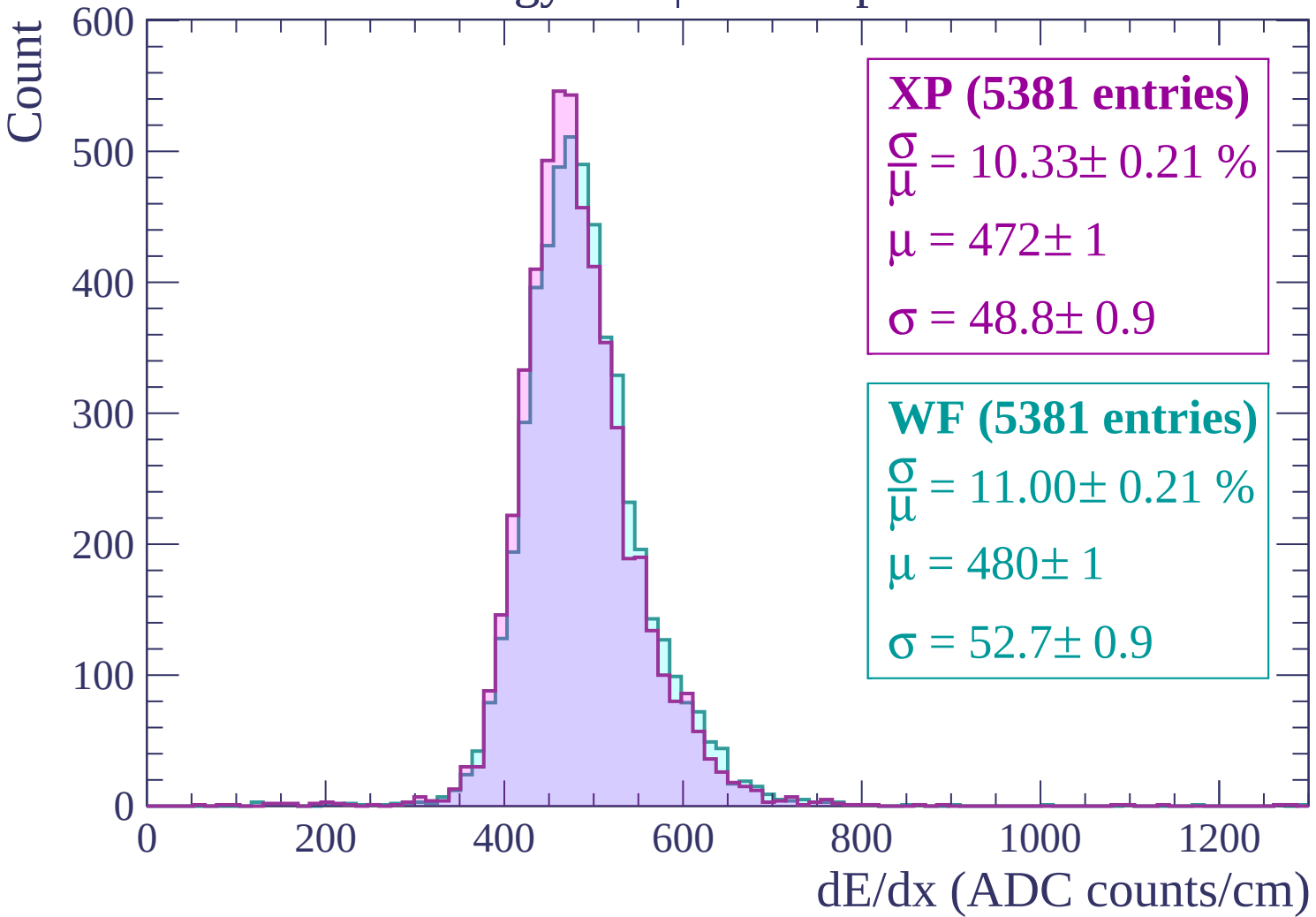


Energy loss | $1860 < p < 1920$

Count

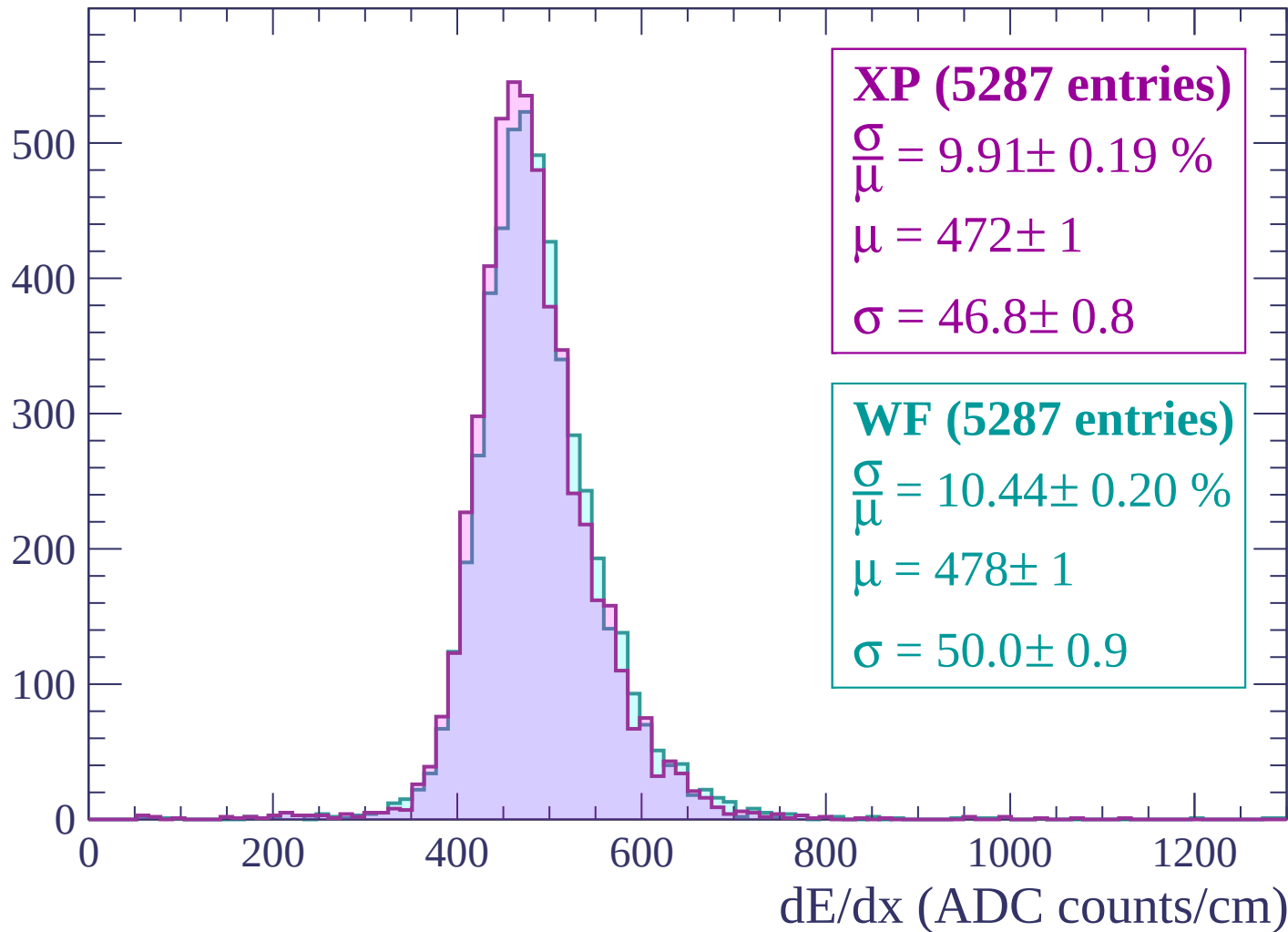


Energy loss | $1920 < p < 1980$



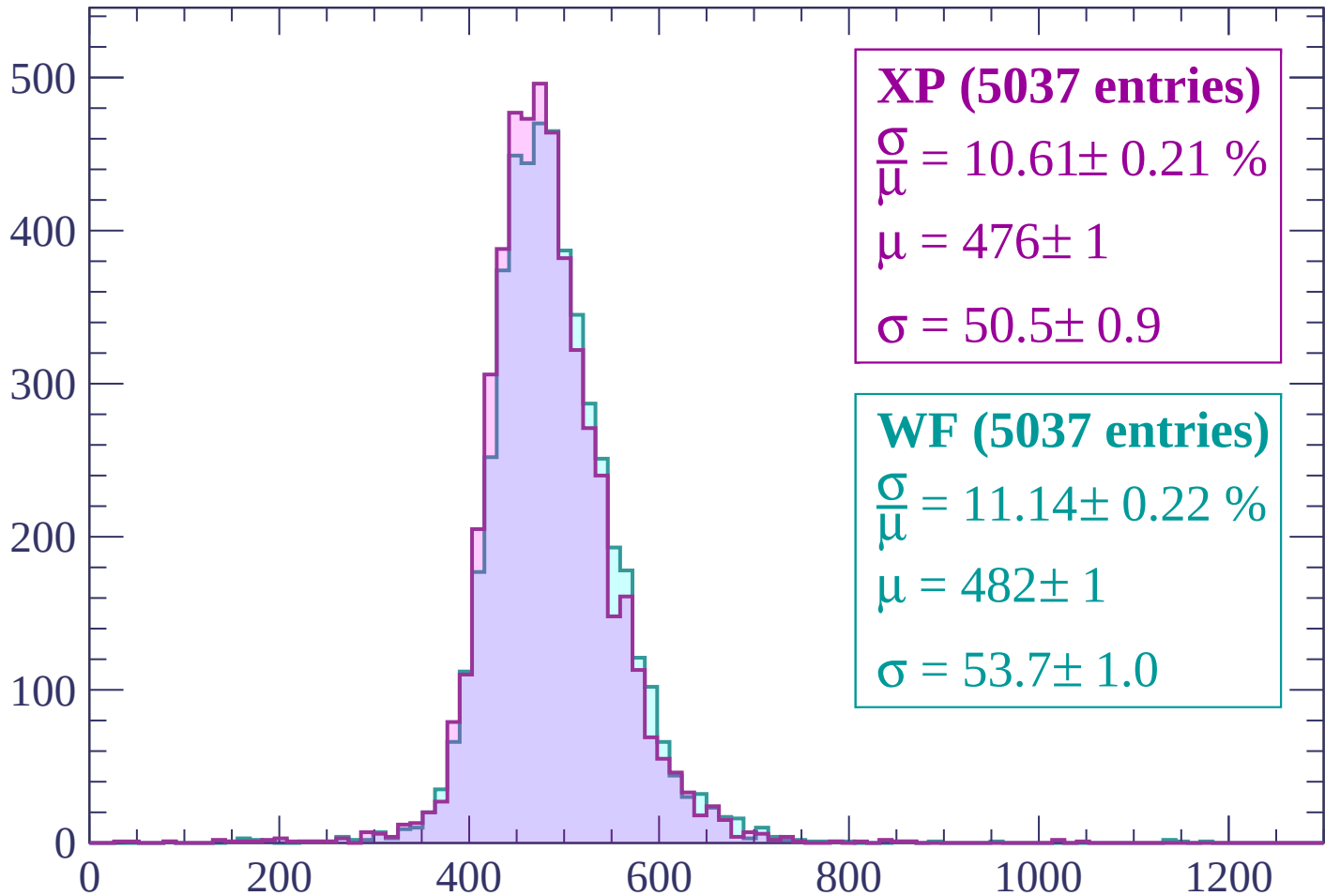
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP (5037 entries)

$$\frac{\sigma}{\mu} = 10.61 \pm 0.21 \%$$

$$\mu = 476 \pm 1$$

$$\sigma = 50.5 \pm 0.9$$

WF (5037 entries)

$$\frac{\sigma}{\mu} = 11.14 \pm 0.22 \%$$

$$\mu = 482 \pm 1$$

$$\sigma = 53.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

XP (4988 entries)

$\frac{\sigma}{\mu} = 10.43 \pm 0.21 \%$

$\mu = 480 \pm 1$

$\sigma = 50.1 \pm 0.9$

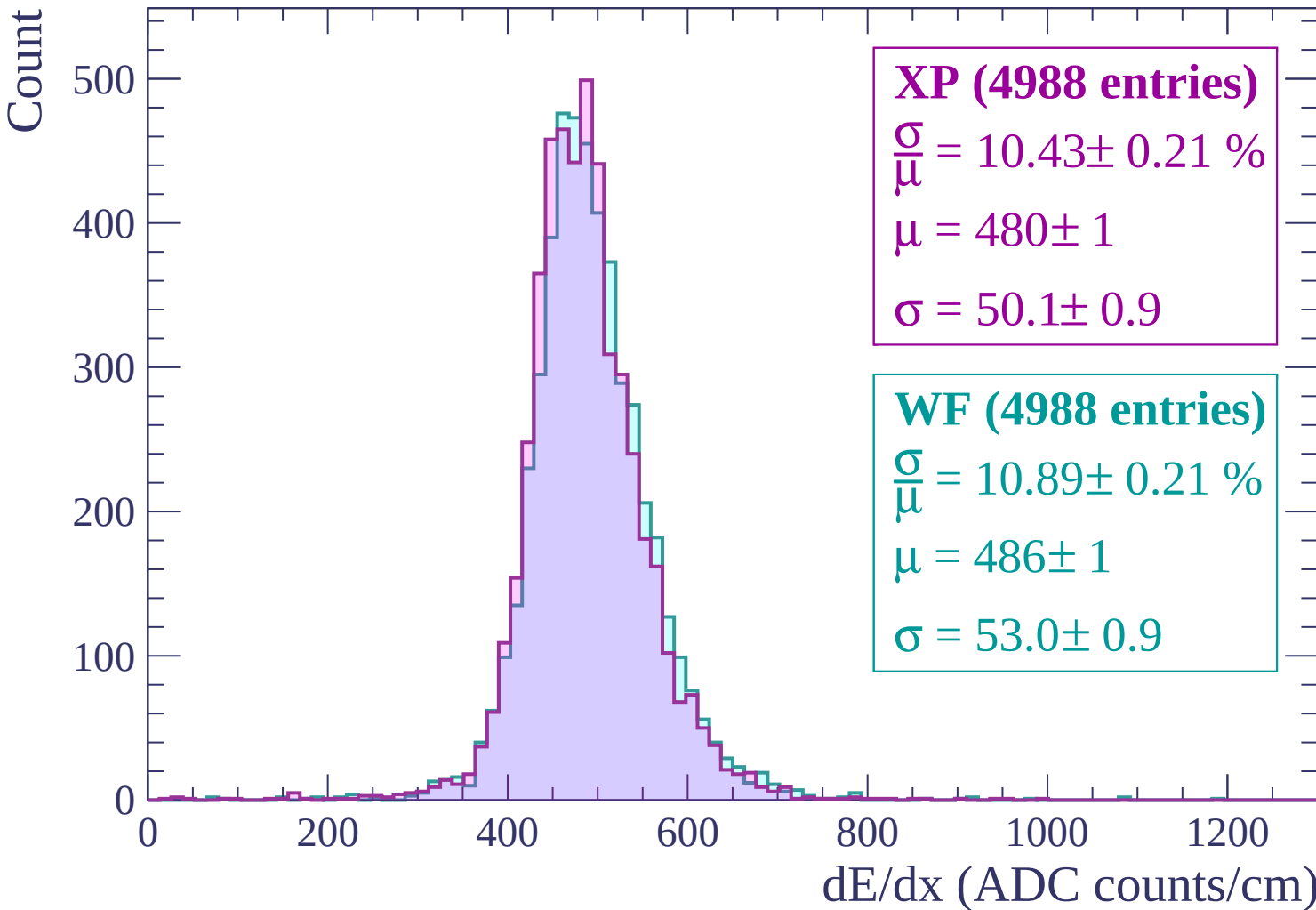
WF (4988 entries)

$\frac{\sigma}{\mu} = 10.89 \pm 0.21 \%$

$\mu = 486 \pm 1$

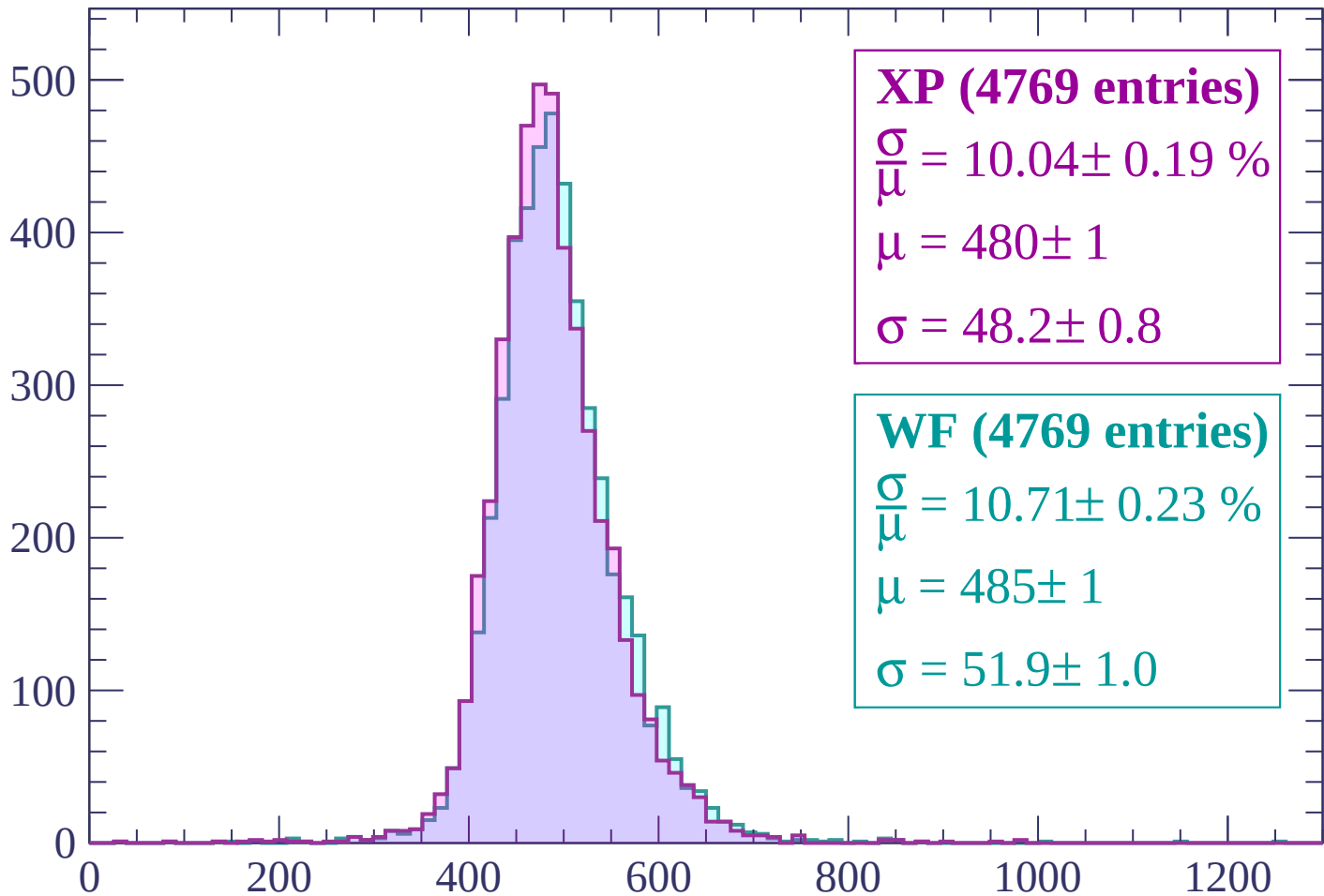
$\sigma = 53.0 \pm 0.9$

dE/dx (ADC counts/cm)



Energy loss | $2160 < p < 2220$

Count



XP (4769 entries)

$\frac{\sigma}{\mu} = 10.04 \pm 0.19 \%$

$\mu = 480 \pm 1$

$\sigma = 48.2 \pm 0.8$

WF (4769 entries)

$\frac{\sigma}{\mu} = 10.71 \pm 0.23 \%$

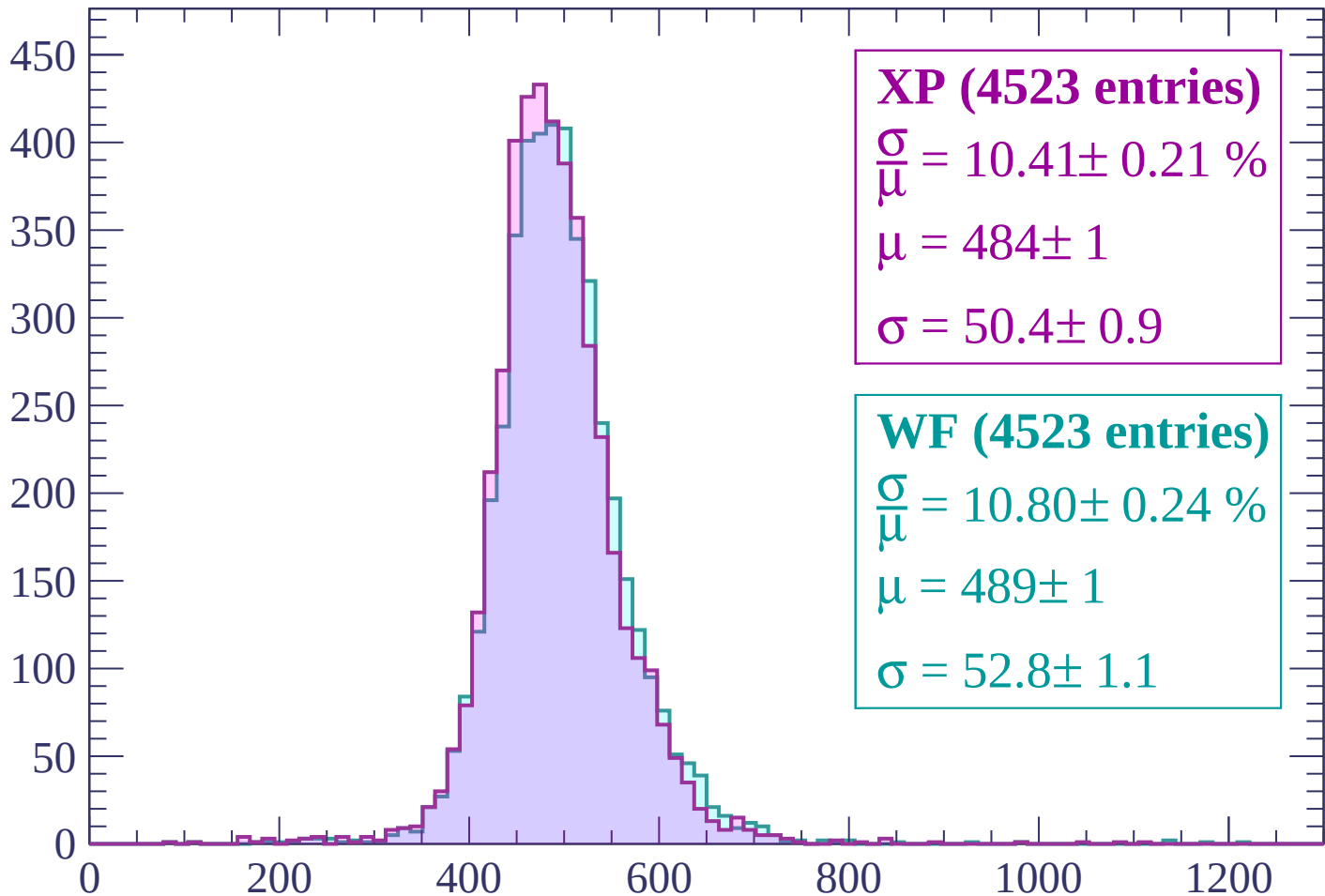
$\mu = 485 \pm 1$

$\sigma = 51.9 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (4523 entries)

$$\frac{\sigma}{\mu} = 10.41 \pm 0.21 \%$$

$$\mu = 484 \pm 1$$

$$\sigma = 50.4 \pm 0.9$$

WF (4523 entries)

$$\frac{\sigma}{\mu} = 10.80 \pm 0.24 \%$$

$$\mu = 489 \pm 1$$

$$\sigma = 52.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $2280 < p < 2340$

Count

450
400
350
300
250
200
150
100
50
0

XP (4415 entries)

$\frac{\sigma}{\mu} = 10.13 \pm 0.23 \%$

$\mu = 484 \pm 1$

$\sigma = 49.0 \pm 1.0$

WF (4415 entries)

$\frac{\sigma}{\mu} = 10.67 \pm 0.24 \%$

$\mu = 490 \pm 1$

$\sigma = 52.2 \pm 1.1$

0

200

400

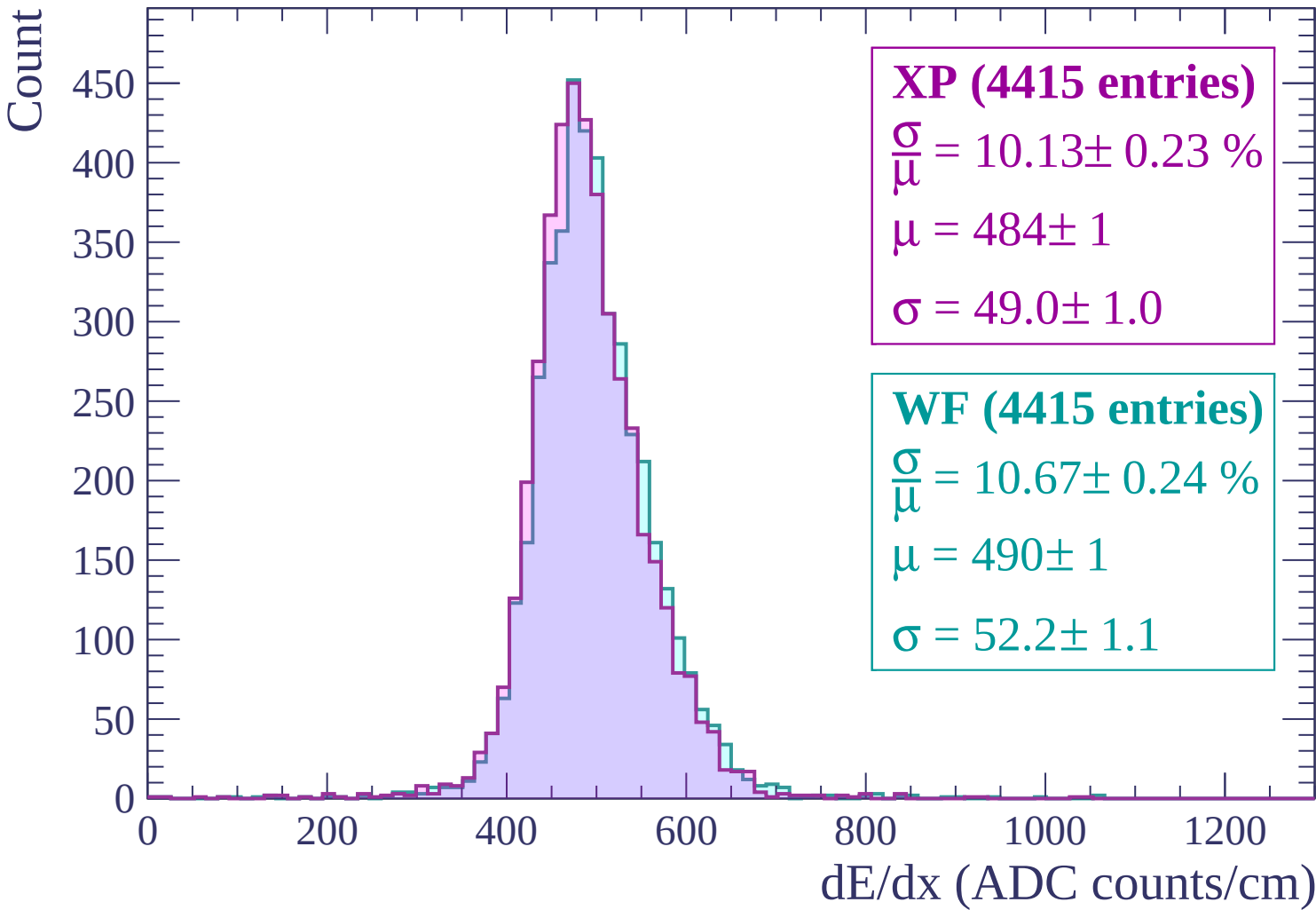
600

800

1000

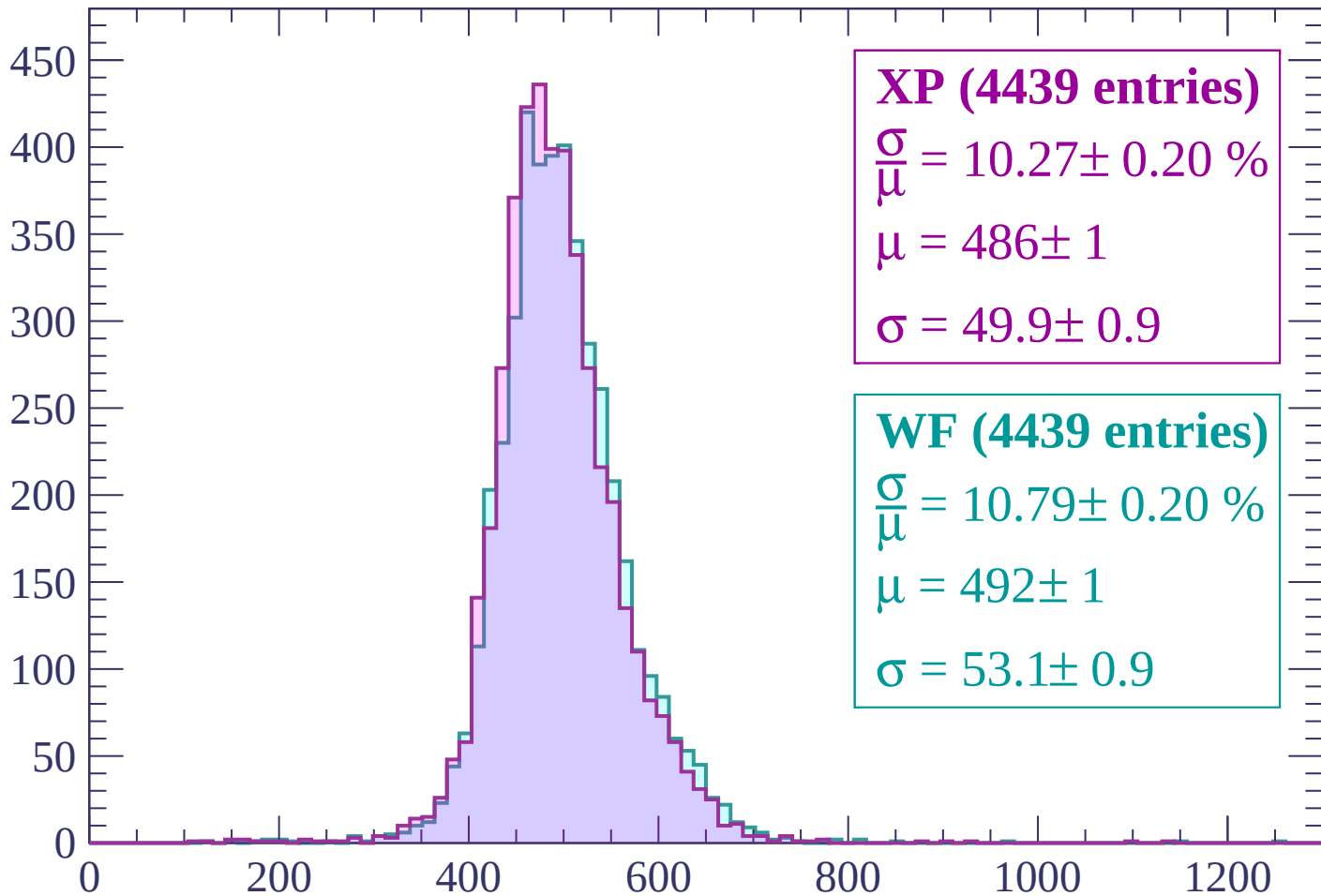
1200

dE/dx (ADC counts/cm)



Energy loss | $2340 < p < 2400$

Count



XP (4439 entries)

$$\frac{\sigma}{\mu} = 10.27 \pm 0.20 \%$$

$$\mu = 486 \pm 1$$

$$\sigma = 49.9 \pm 0.9$$

WF (4439 entries)

$$\frac{\sigma}{\mu} = 10.79 \pm 0.20 \%$$

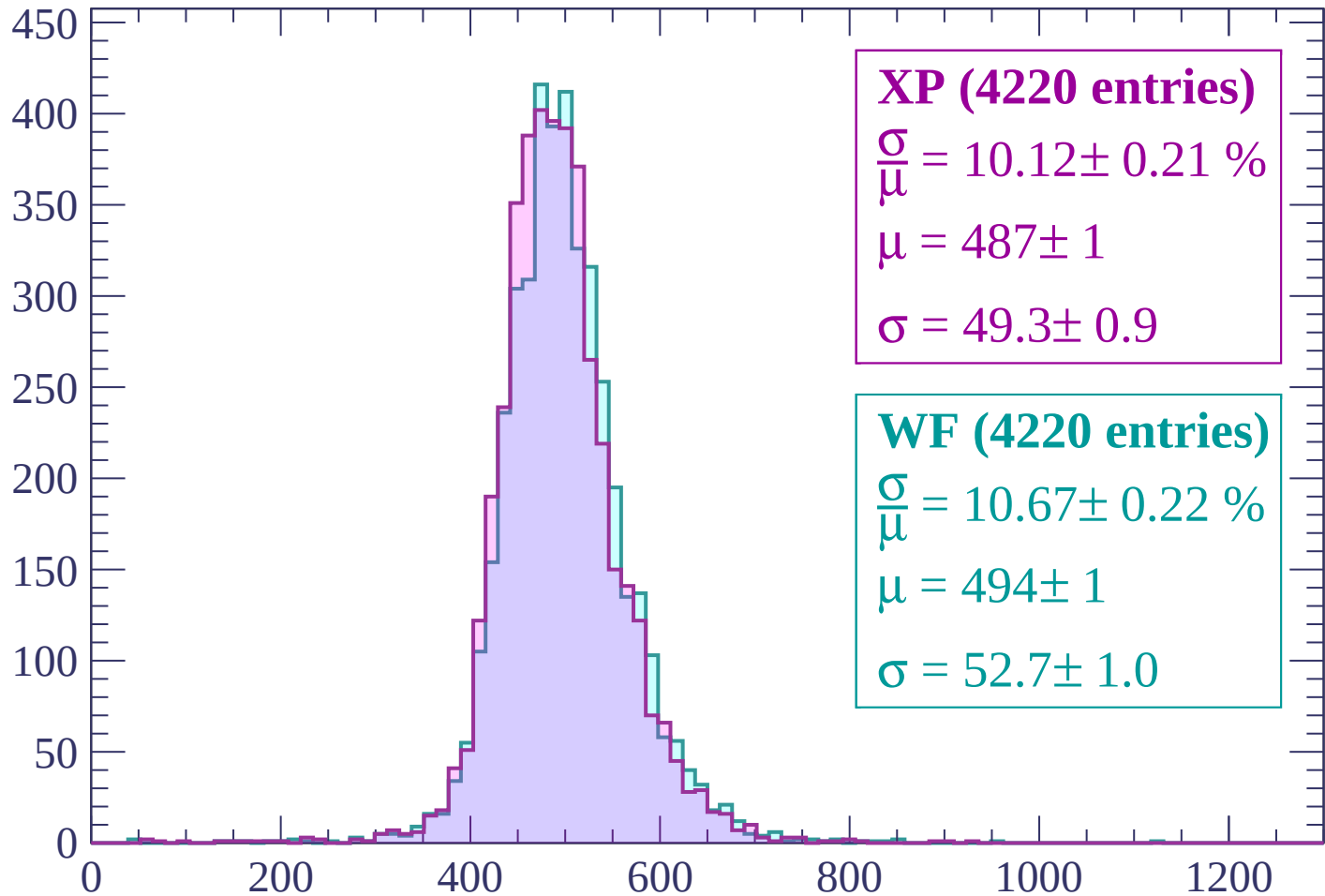
$$\mu = 492 \pm 1$$

$$\sigma = 53.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2400 < p < 2460$

Count



XP (4220 entries)

$\frac{\sigma}{\mu} = 10.12 \pm 0.21 \%$

$\mu = 487 \pm 1$

$\sigma = 49.3 \pm 0.9$

WF (4220 entries)

$\frac{\sigma}{\mu} = 10.67 \pm 0.22 \%$

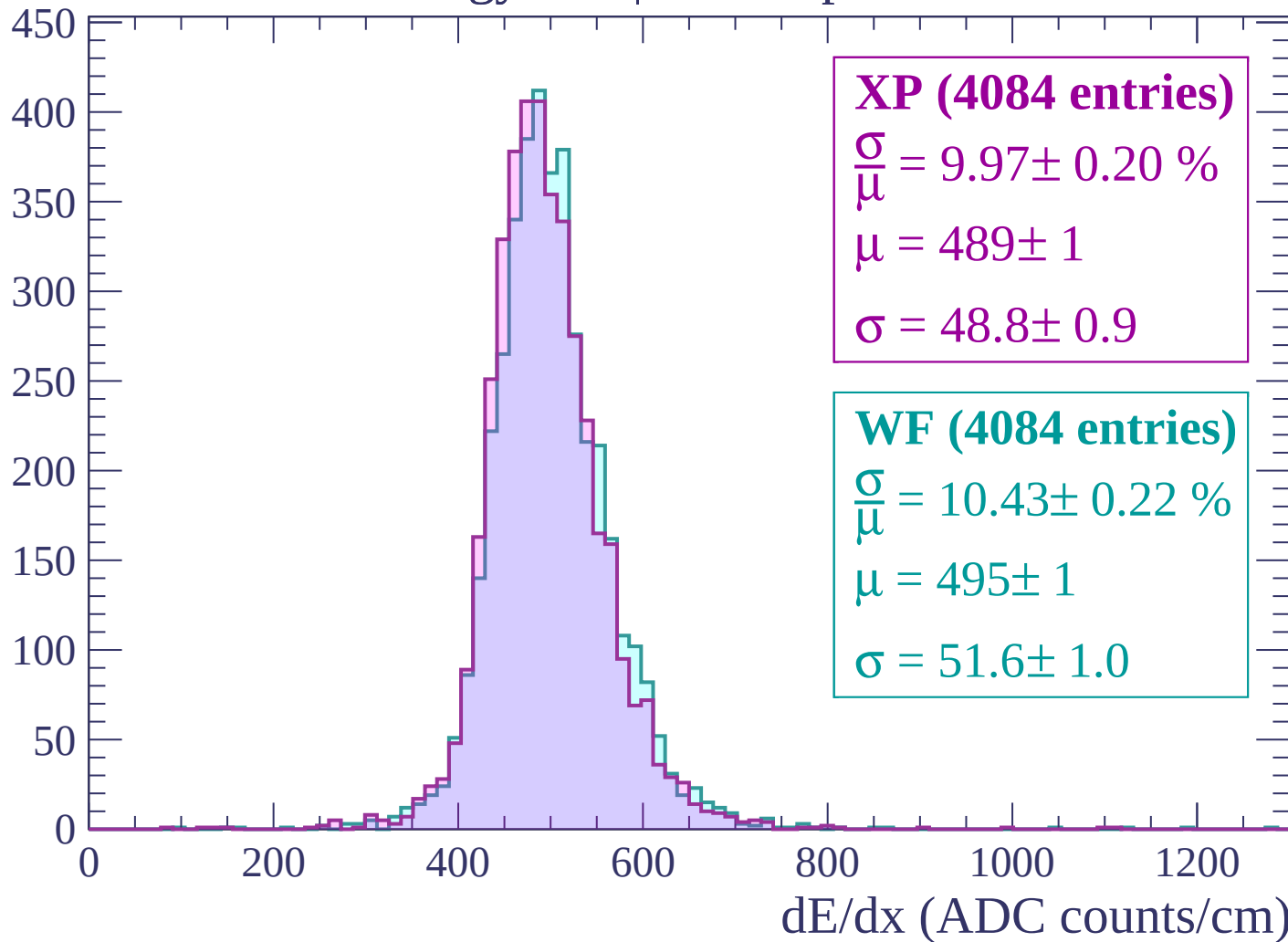
$\mu = 494 \pm 1$

$\sigma = 52.7 \pm 1.0$

dE/dx (ADC counts/cm)

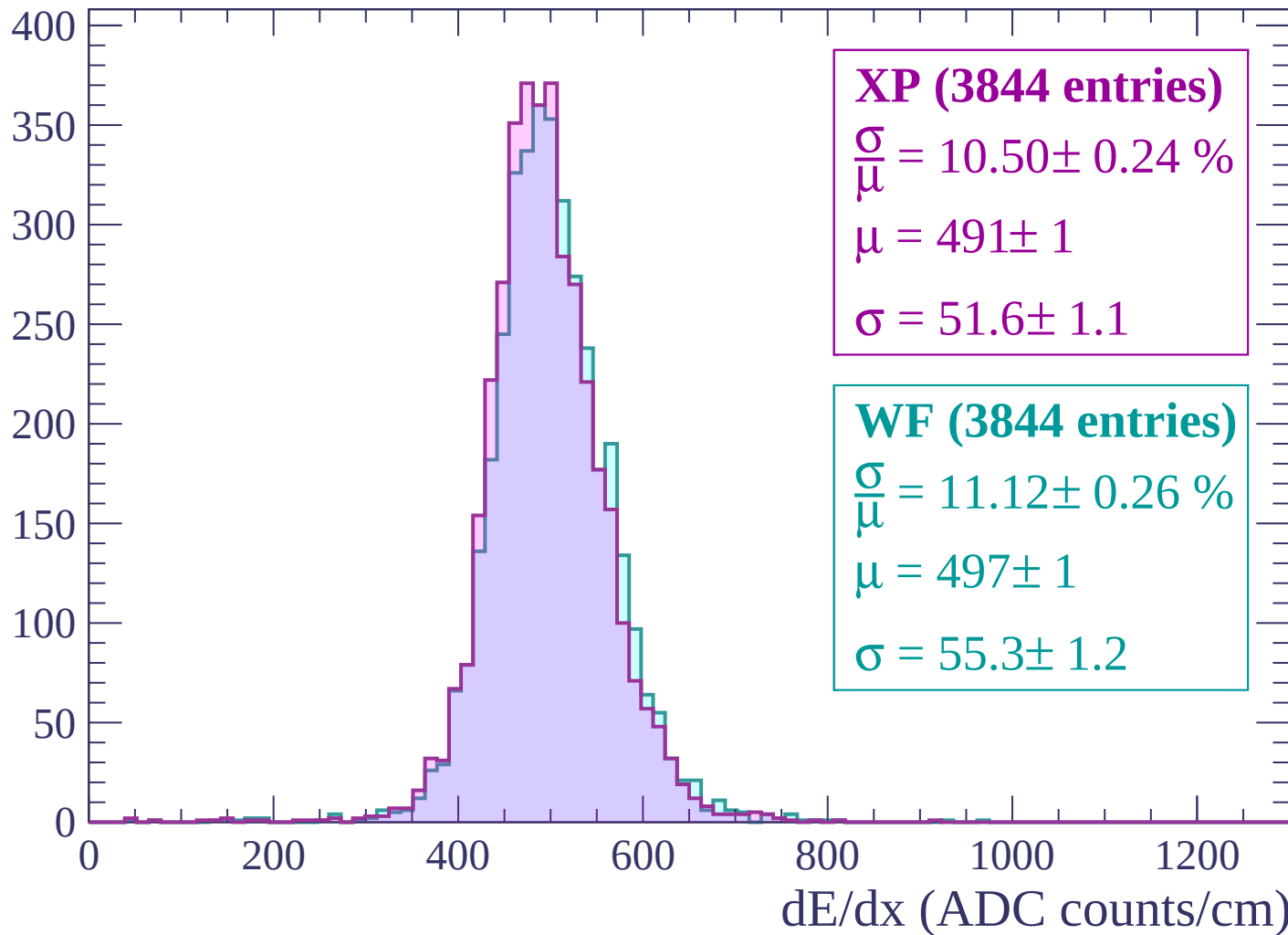
Energy loss | $2460 < p < 2520$

Count



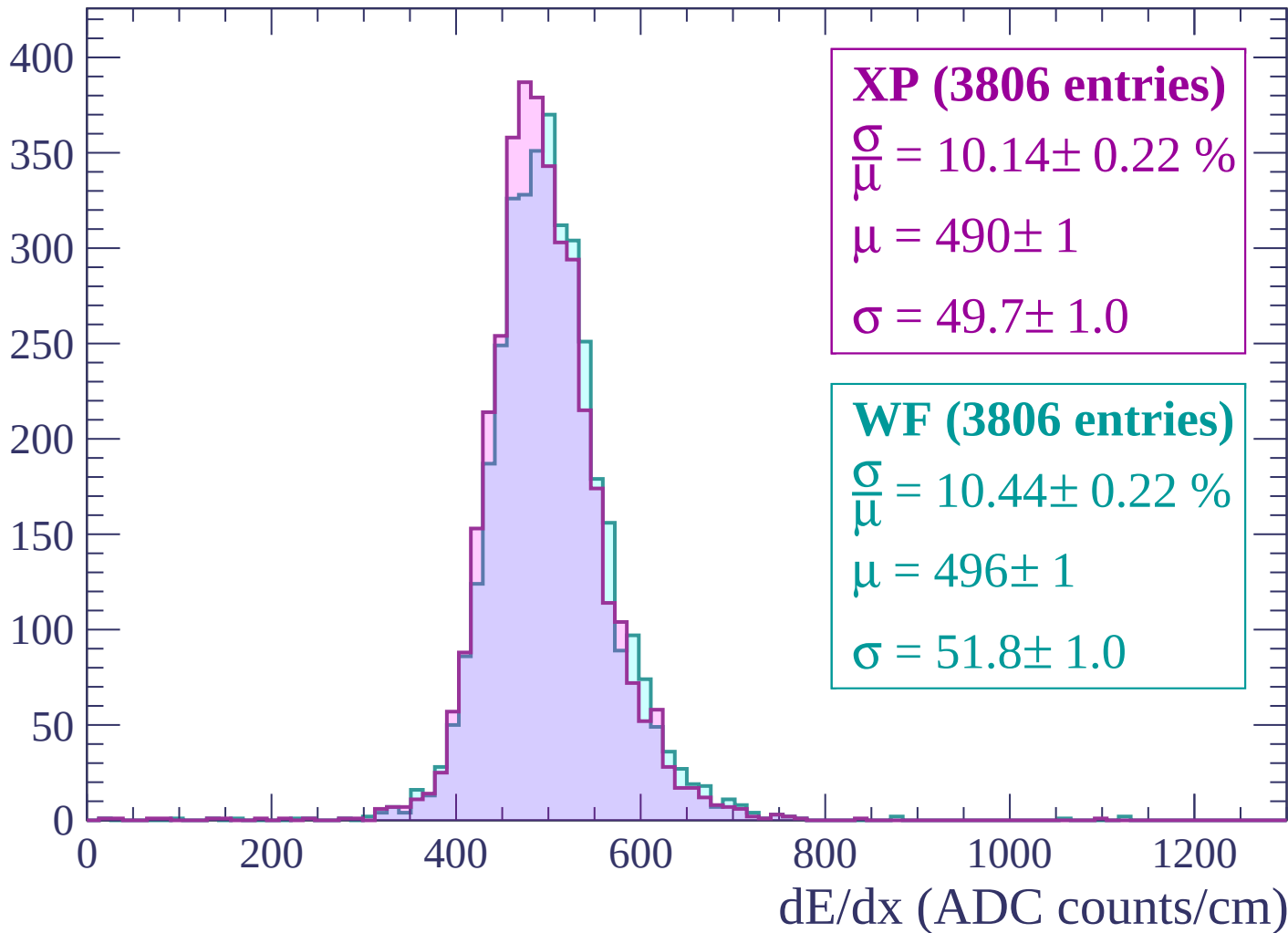
Energy loss | $2520 < p < 2580$

Count



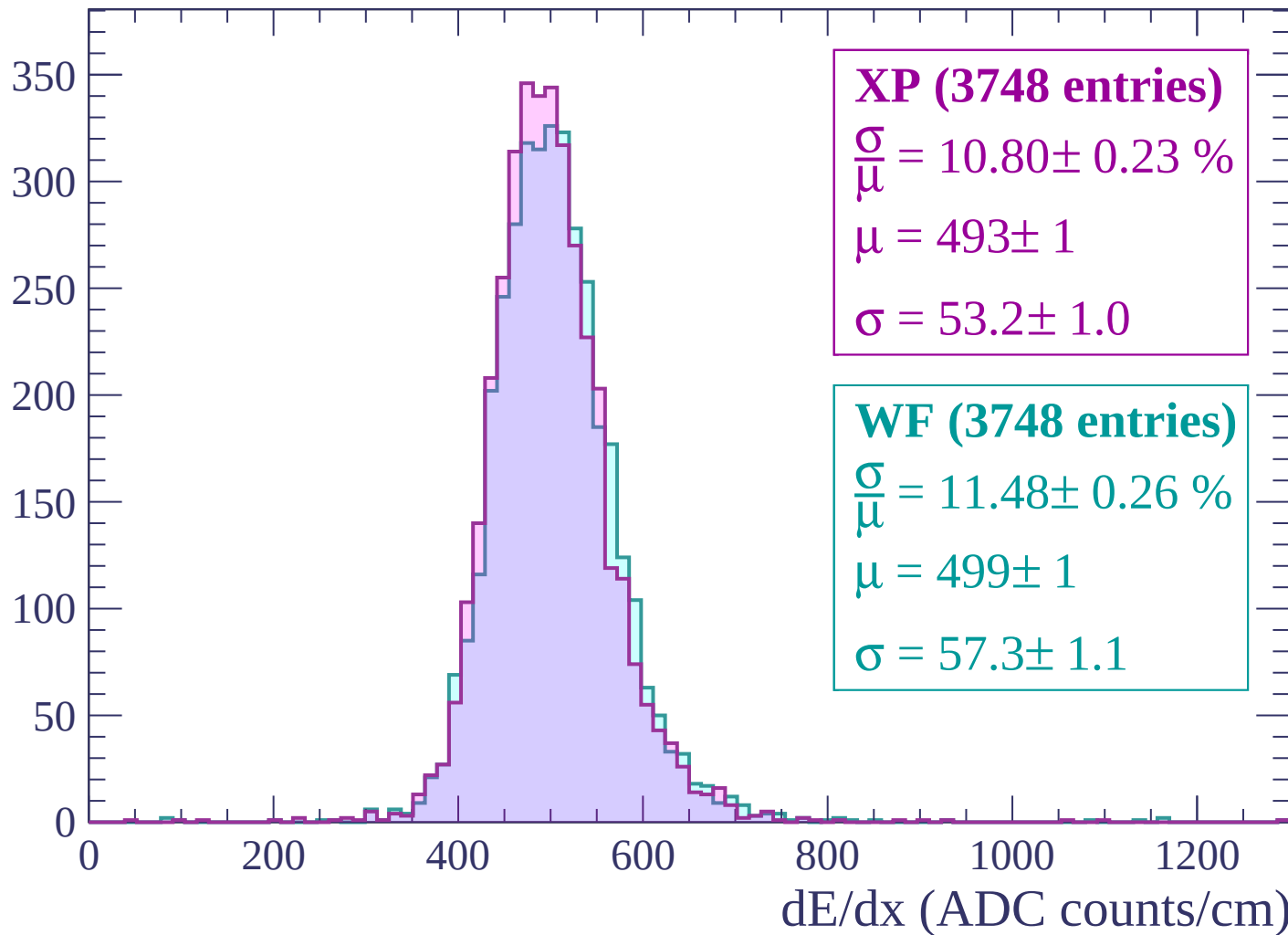
Energy loss | $2580 < p < 2640$

Count



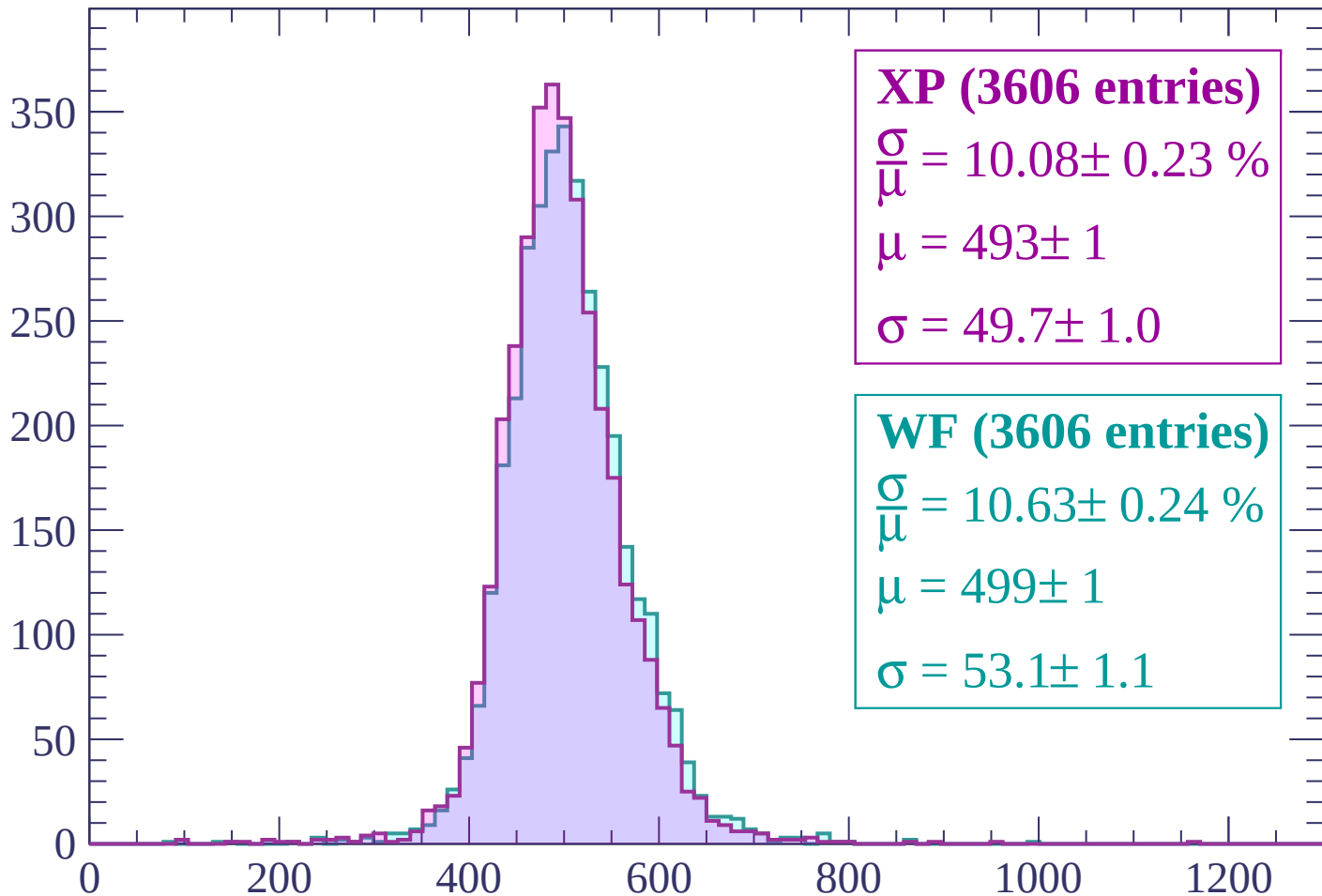
Energy loss | $2640 < p < 2700$

Count



Energy loss | $2700 < p < 2760$

Count



XP (3606 entries)

$\frac{\sigma}{\mu} = 10.08 \pm 0.23 \%$

$\mu = 493 \pm 1$

$\sigma = 49.7 \pm 1.0$

WF (3606 entries)

$\frac{\sigma}{\mu} = 10.63 \pm 0.24 \%$

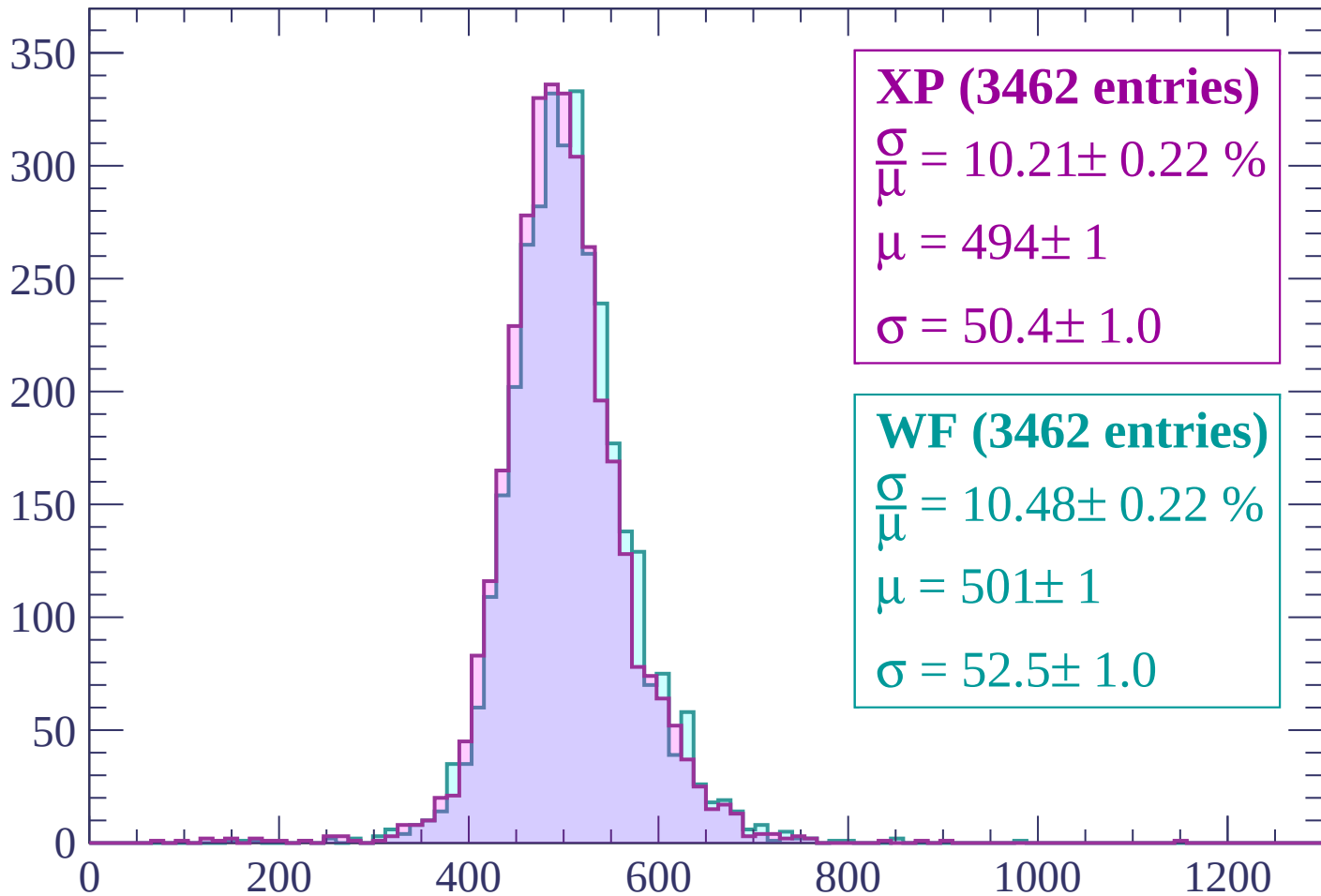
$\mu = 499 \pm 1$

$\sigma = 53.1 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $2760 < p < 2820$

Count



XP (3462 entries)

$\frac{\sigma}{\mu} = 10.21 \pm 0.22 \%$

$\mu = 494 \pm 1$

$\sigma = 50.4 \pm 1.0$

WF (3462 entries)

$\frac{\sigma}{\mu} = 10.48 \pm 0.22 \%$

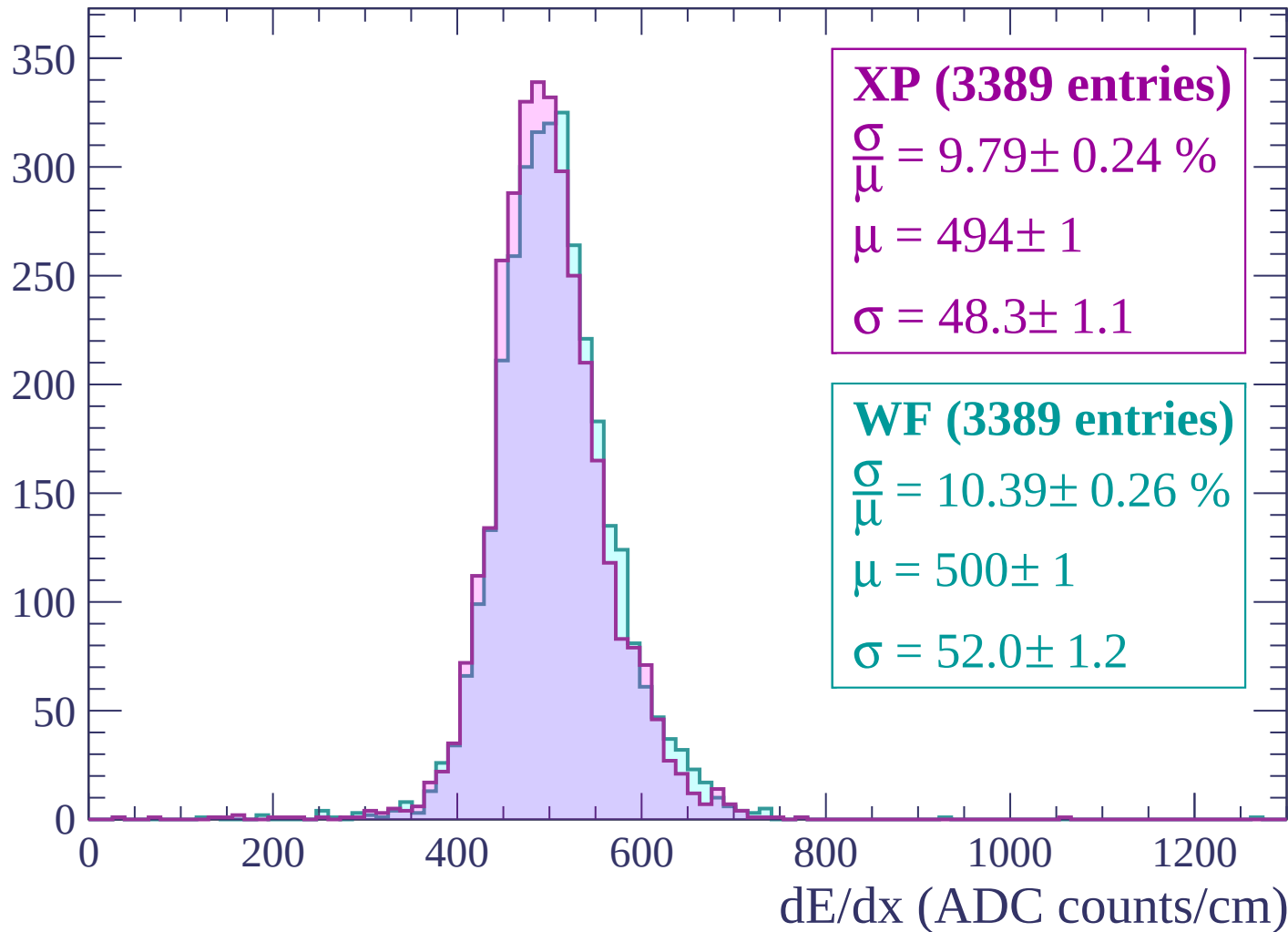
$\mu = 501 \pm 1$

$\sigma = 52.5 \pm 1.0$

dE/dx (ADC counts/cm)

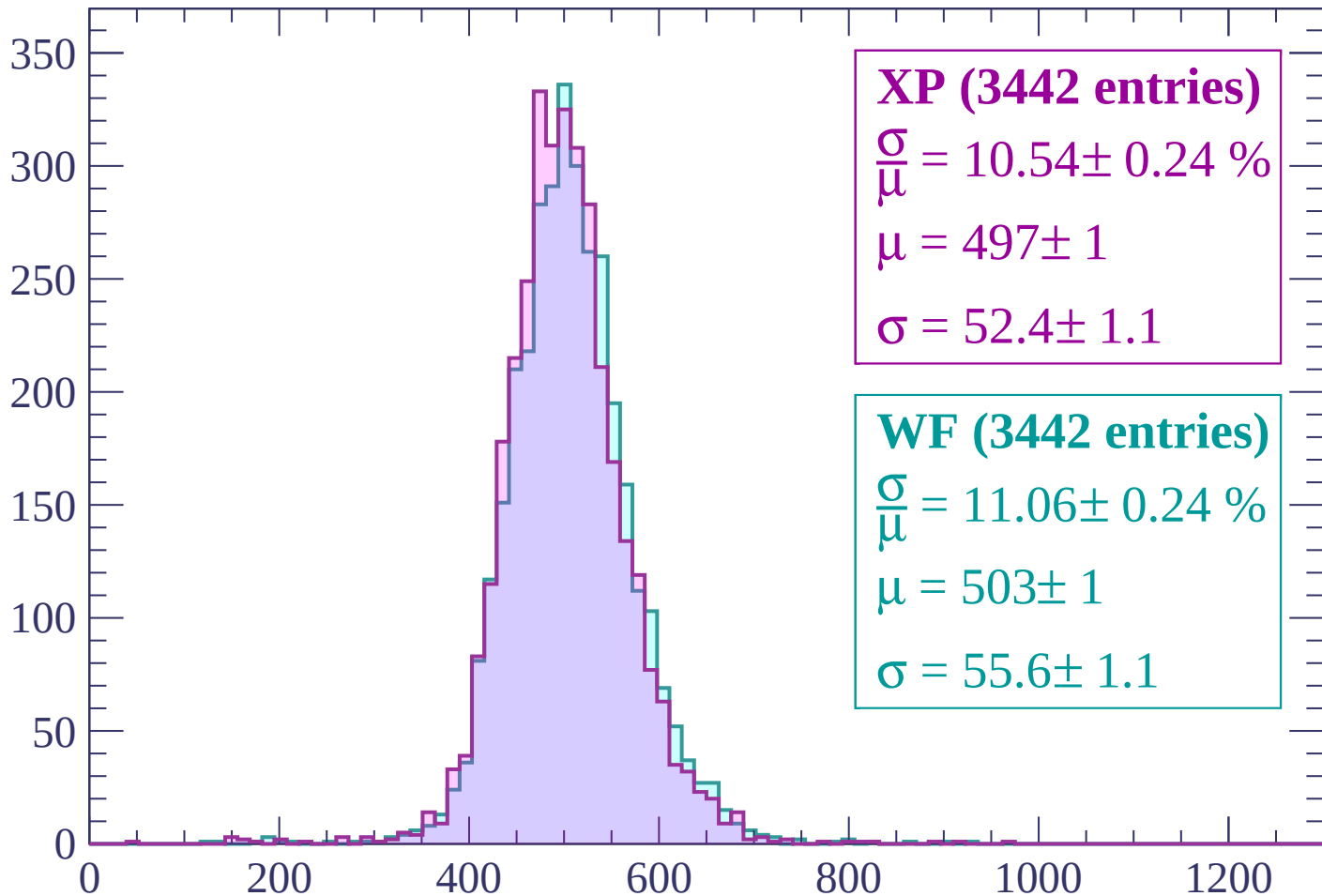
Energy loss | $2820 < p < 2880$

Count



Energy loss | $2880 < p < 2940$

Count



XP (3442 entries)

$\frac{\sigma}{\mu} = 10.54 \pm 0.24 \%$

$\mu = 497 \pm 1$

$\sigma = 52.4 \pm 1.1$

WF (3442 entries)

$\frac{\sigma}{\mu} = 11.06 \pm 0.24 \%$

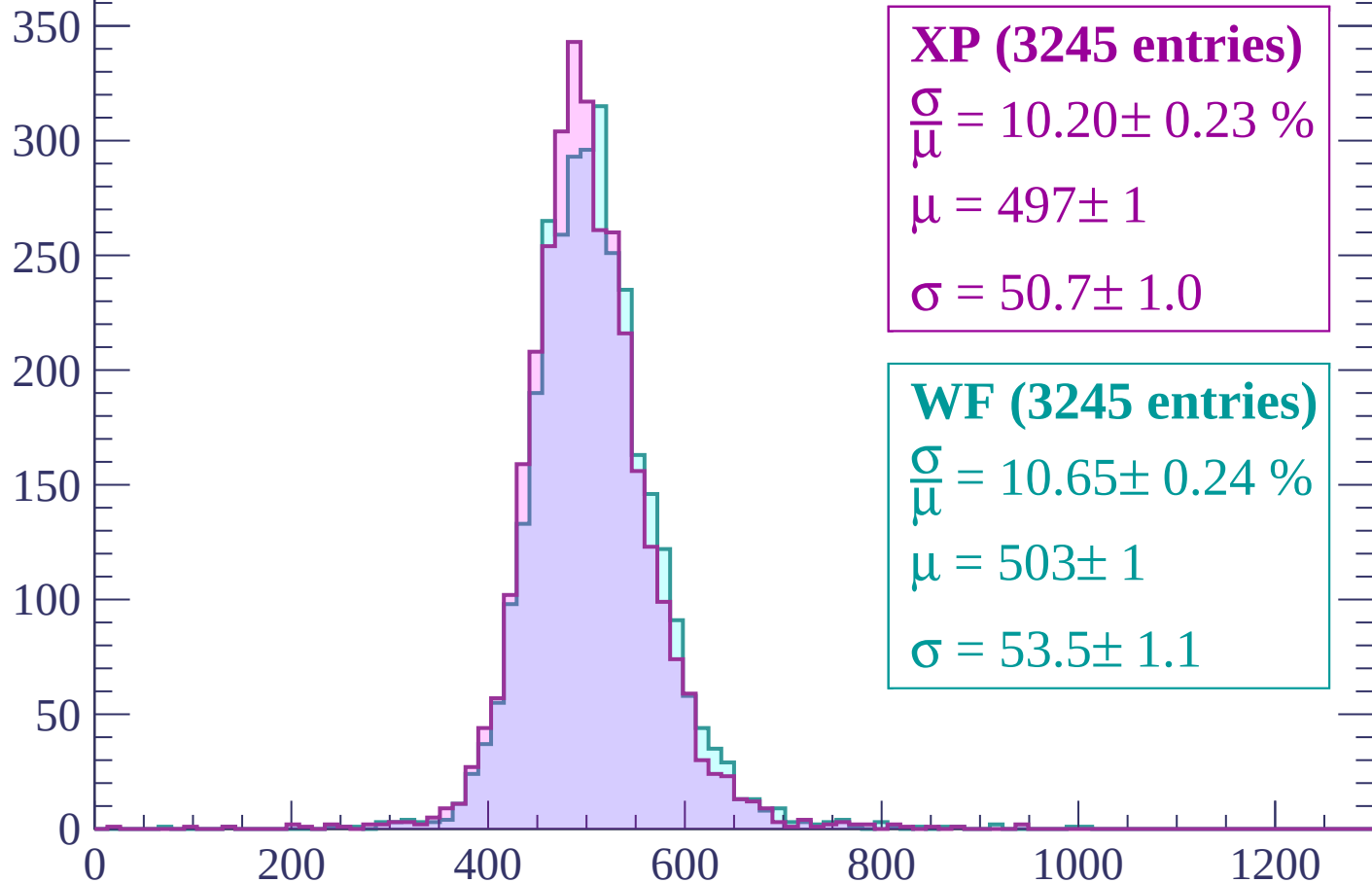
$\mu = 503 \pm 1$

$\sigma = 55.6 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$

Count



XP (3245 entries)

$\frac{\sigma}{\mu} = 10.20 \pm 0.23 \%$

$\mu = 497 \pm 1$

$\sigma = 50.7 \pm 1.0$

WF (3245 entries)

$\frac{\sigma}{\mu} = 10.65 \pm 0.24 \%$

$\mu = 503 \pm 1$

$\sigma = 53.5 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count

